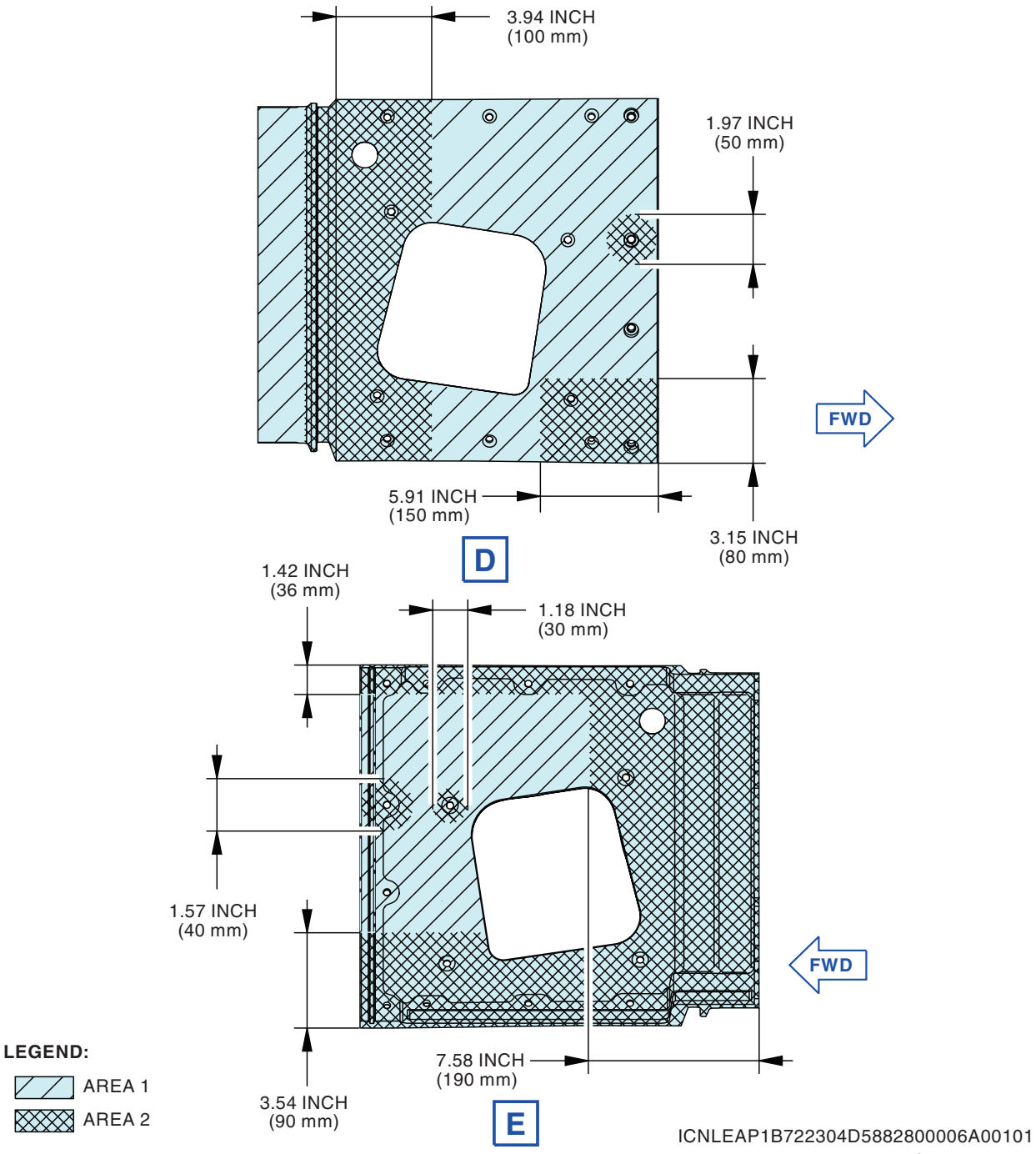
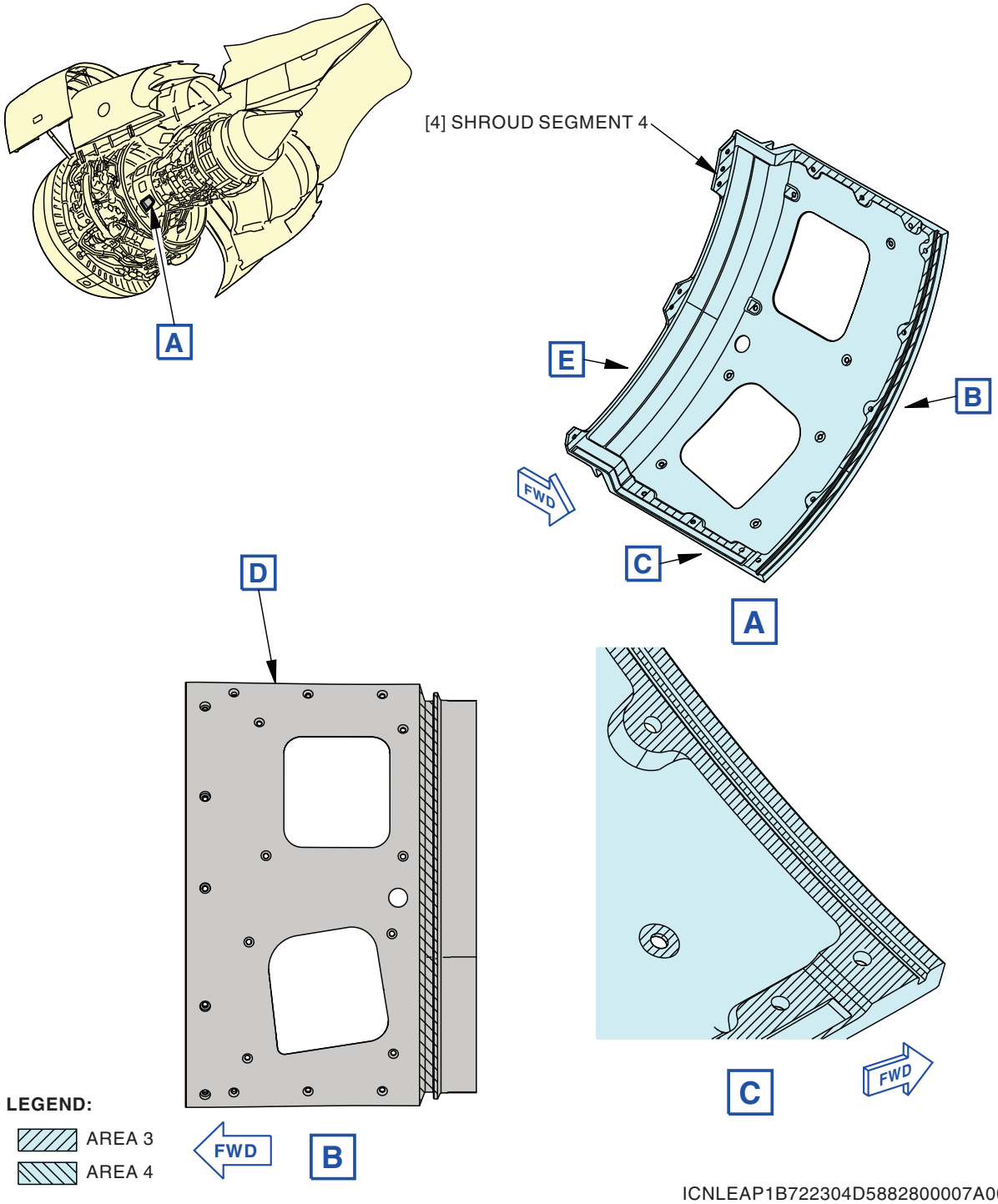


Fan Frame Module Segment 3 Shroud Inspection
Figure 603/72-23-04-990-818-G00 (Sheet 2 of 2)

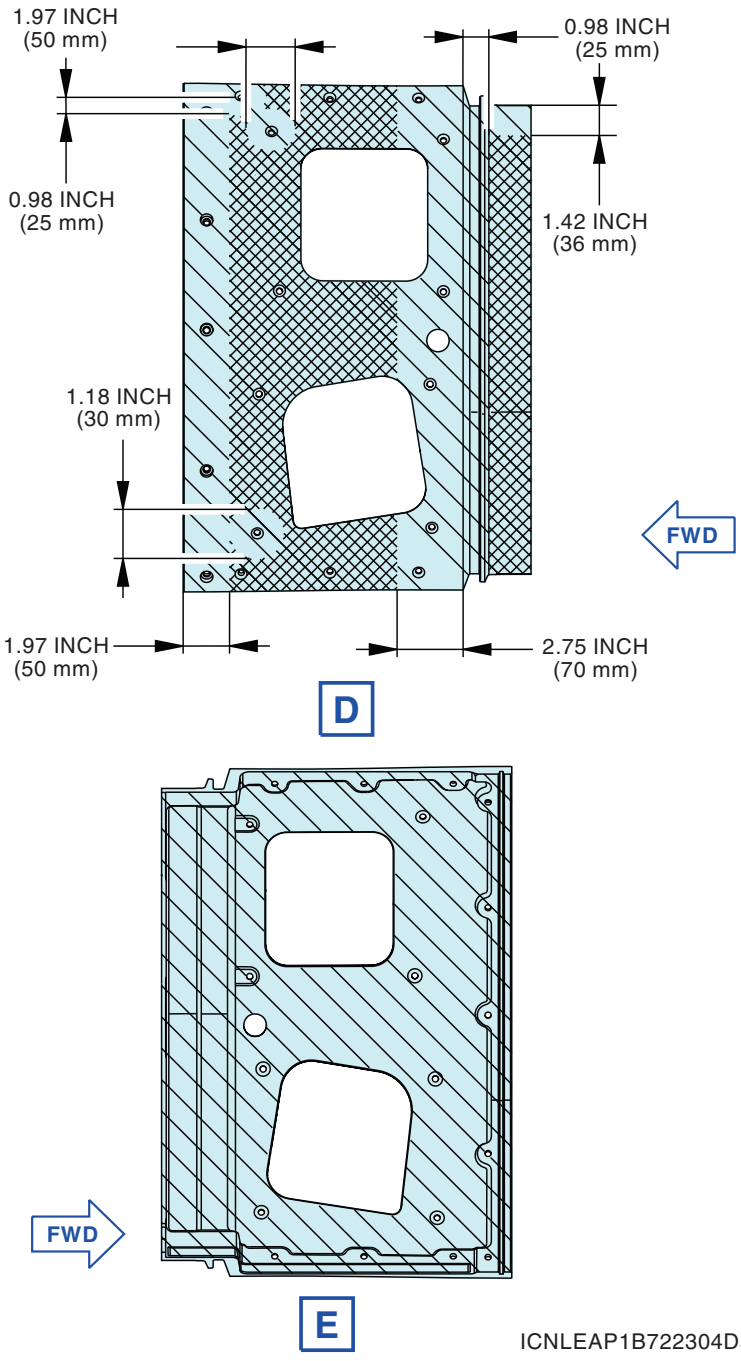
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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2552918 S0000607572_V2

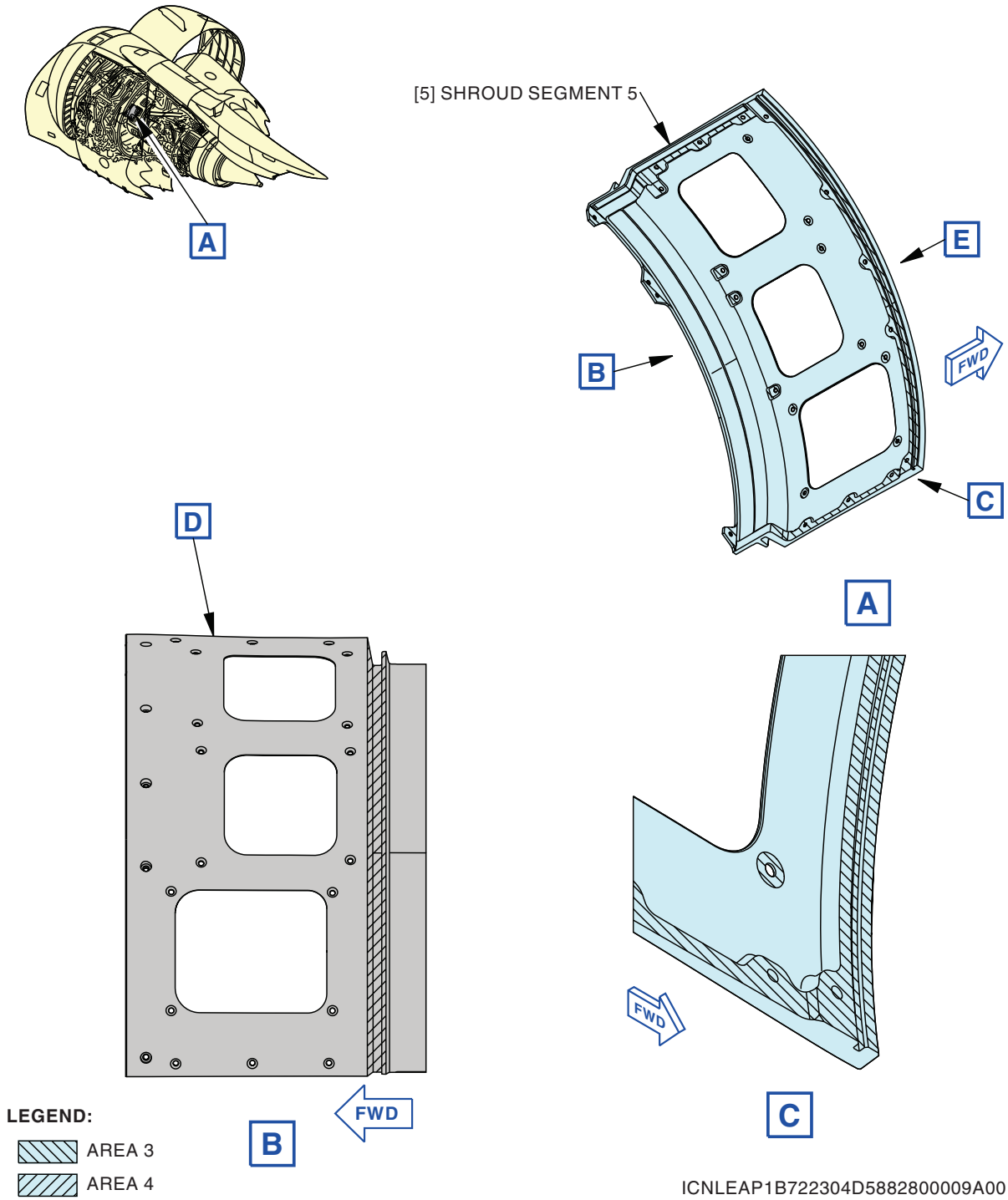
Fan Frame Module Segment 4 Shroud Inspection
Figure 604/72-23-04-990-819-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



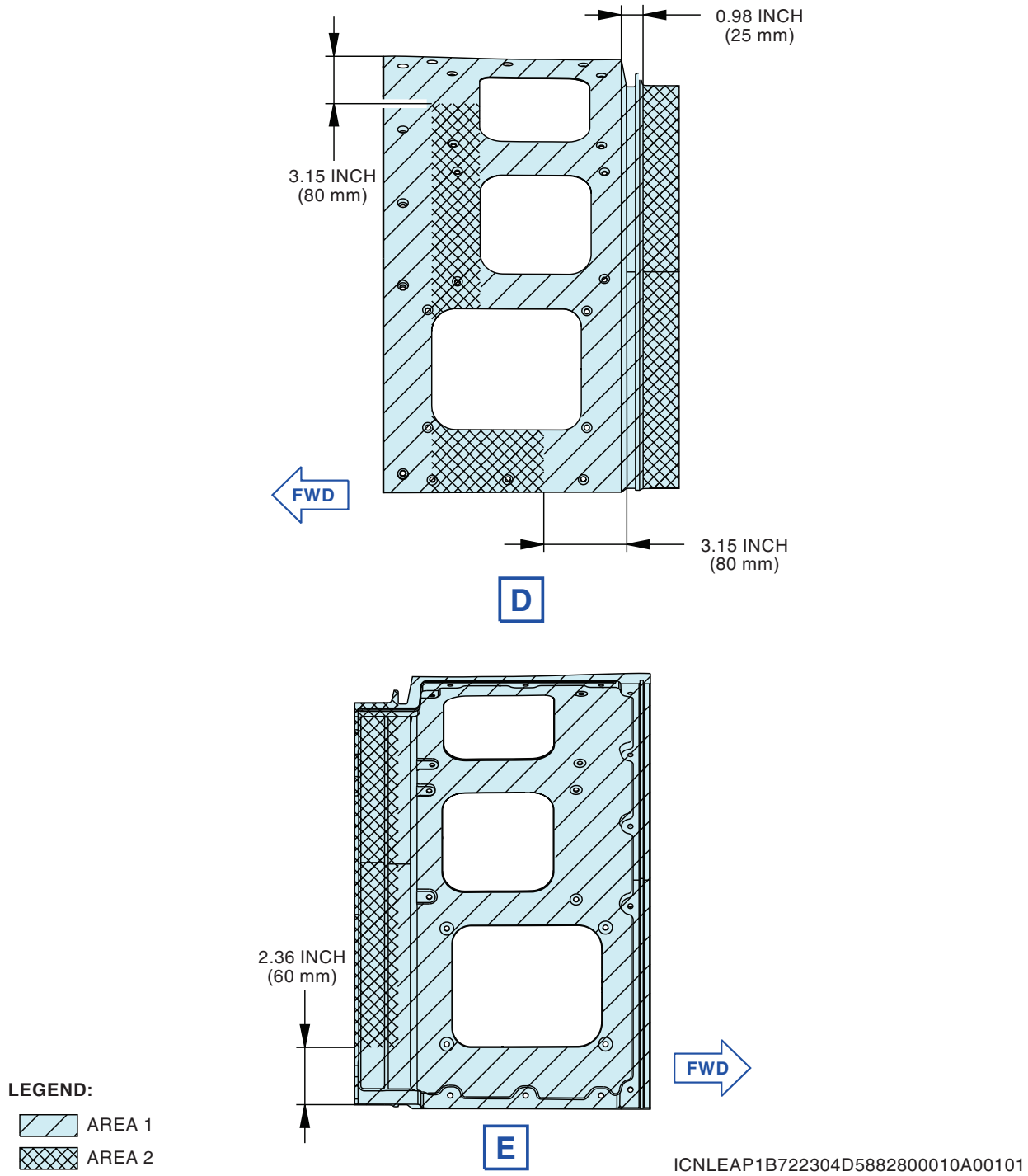
Fan Frame Module Segment 4 Shroud Inspection
Figure 604/72-23-04-990-819-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



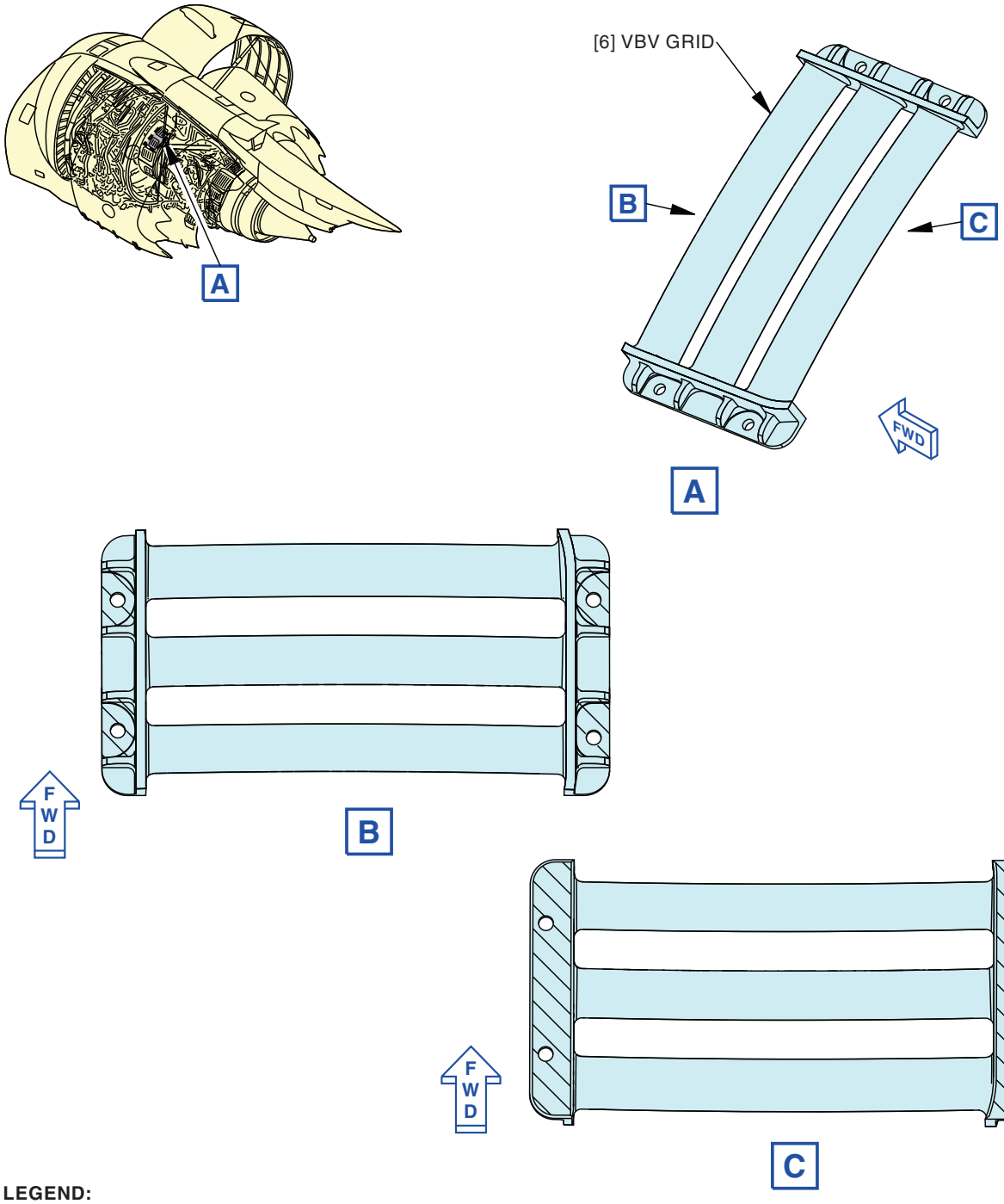
Fan Frame Module Segment 5 Shroud Inspection
Figure 605/72-23-04-990-820-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



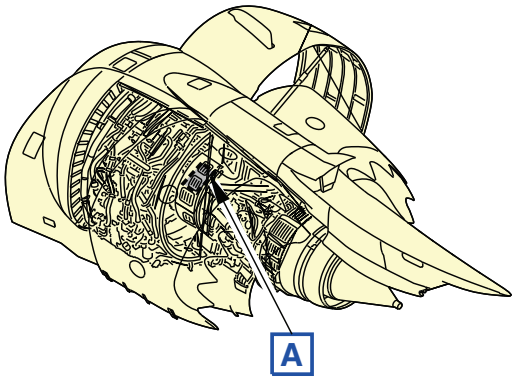
Fan Frame Module Segment 5 Shroud Inspection
Figure 605/72-23-04-990-820-G00 (Sheet 2 of 2)

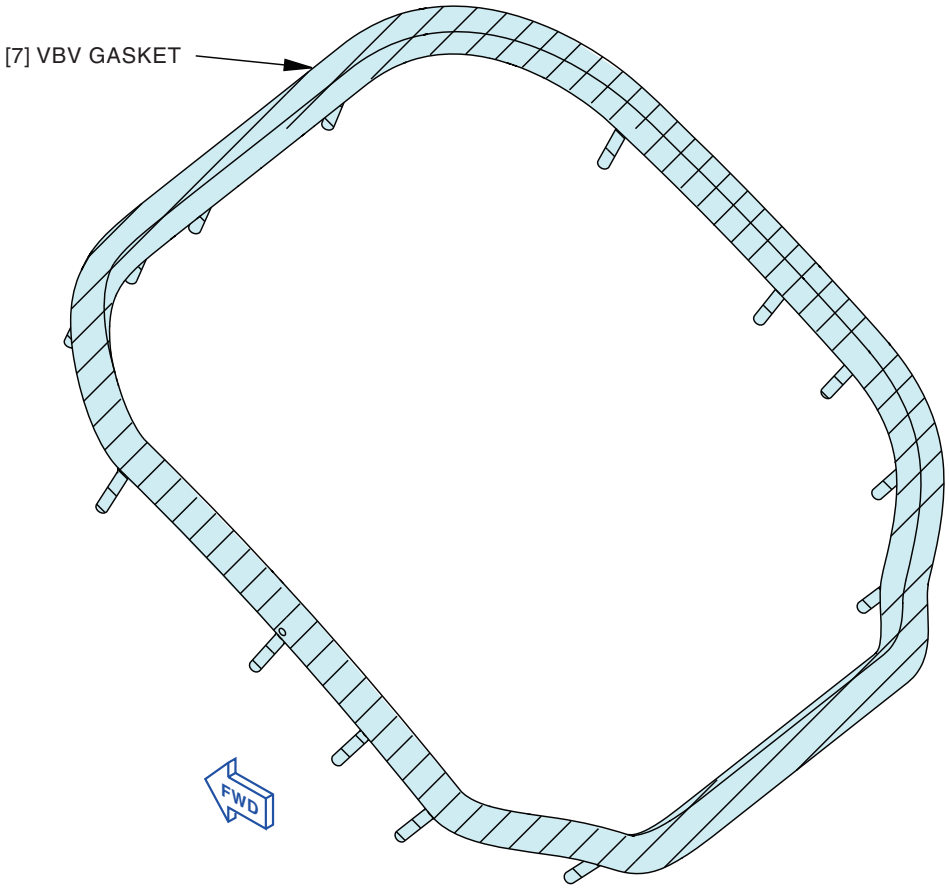
ECCN 9E991 BOEING PROPRIETARY - See title page for details



VBV Grid Inspection
Figure 606/72-23-04-990-821-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details





LEGEND:

AREA A (SEAL PROFILE)

AREA B (SEAL RUBBER PIN)

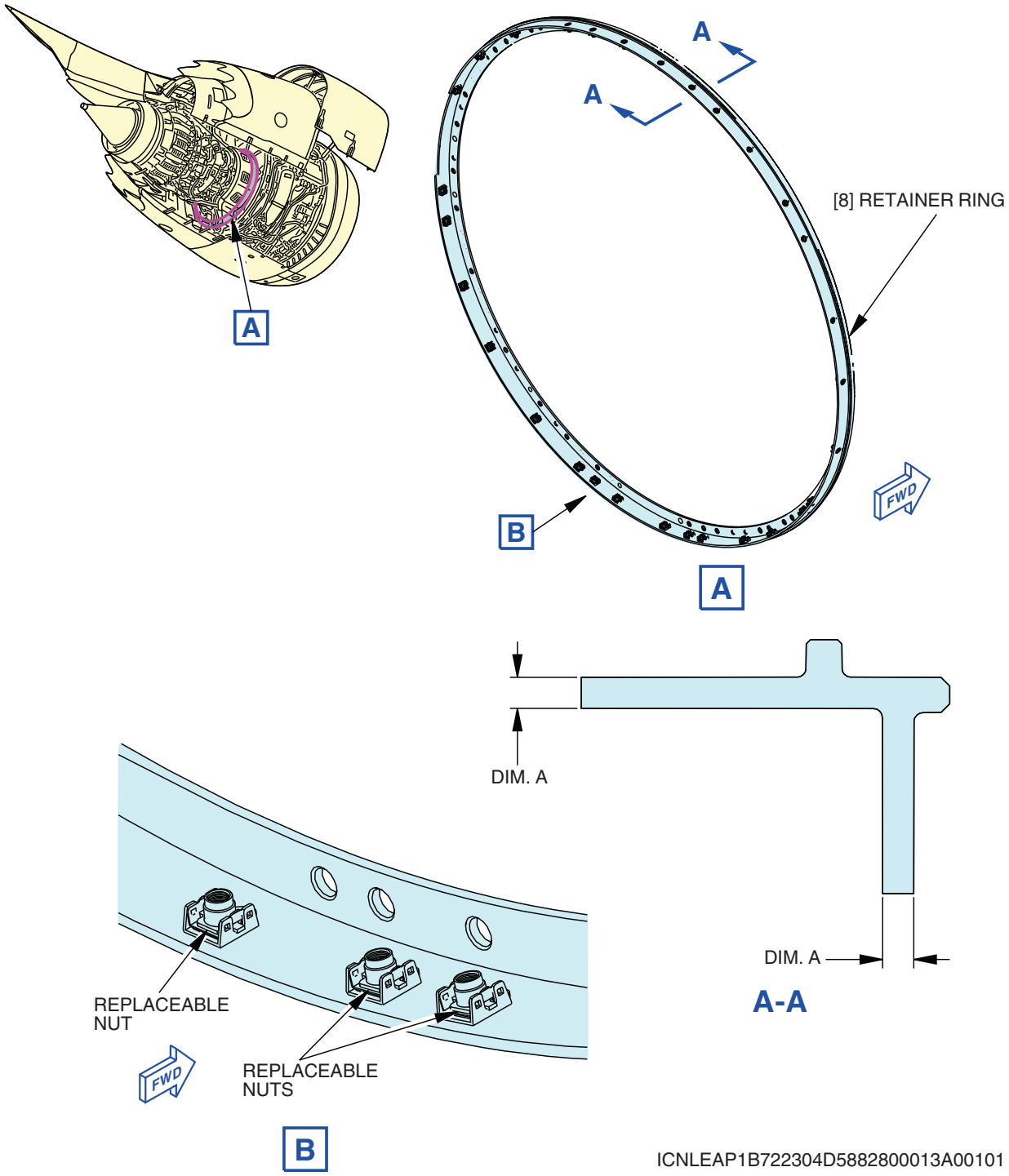
A

ICNLEAP1B722304D5882800012A00101

2552924 S0000607577_V2

VBV Gasket Inspection
Figure 607/72-23-04-990-822-G00

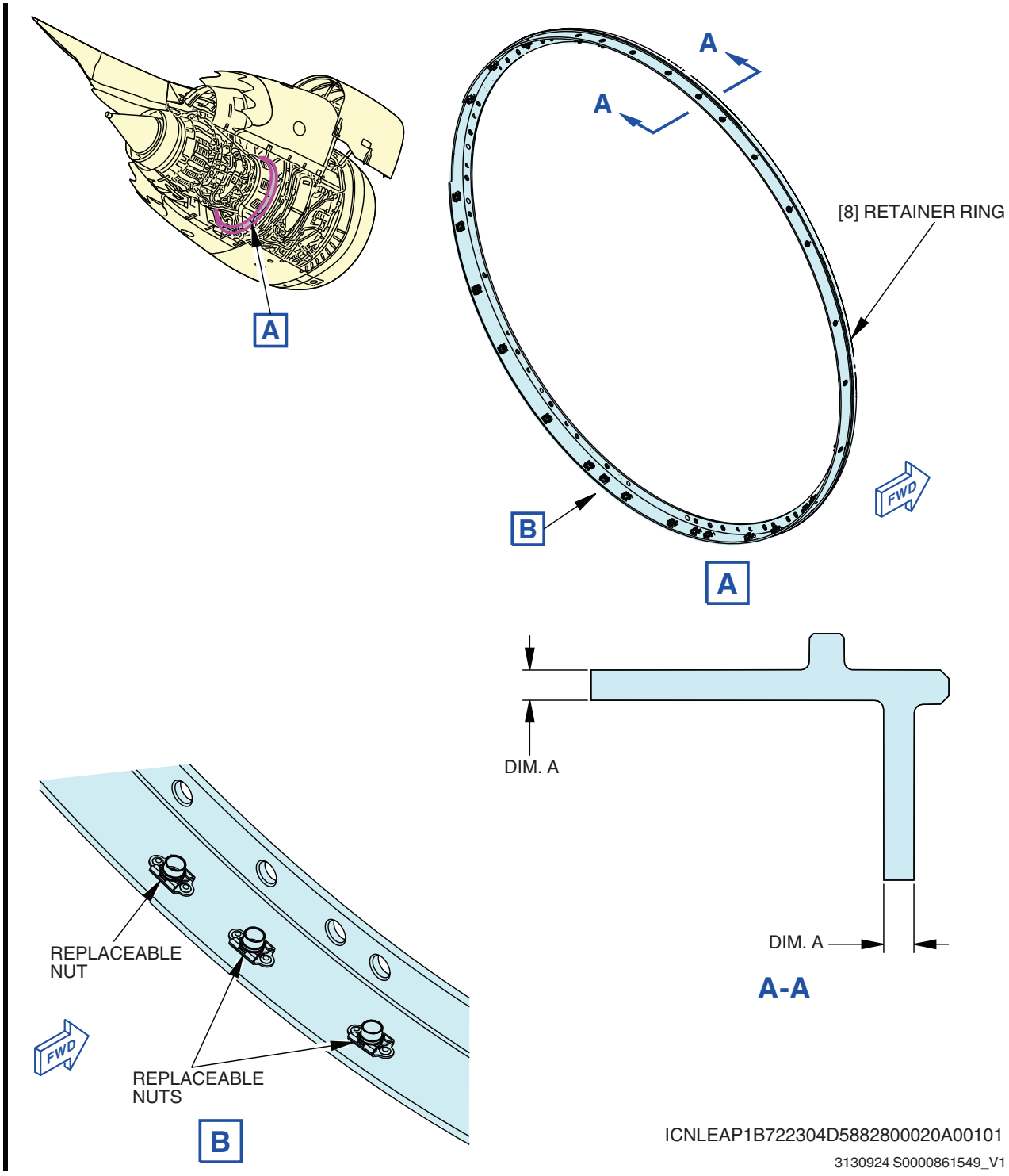
ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICNLEAP1B722304D5882800013A00101
2552925 S0000607578_V3

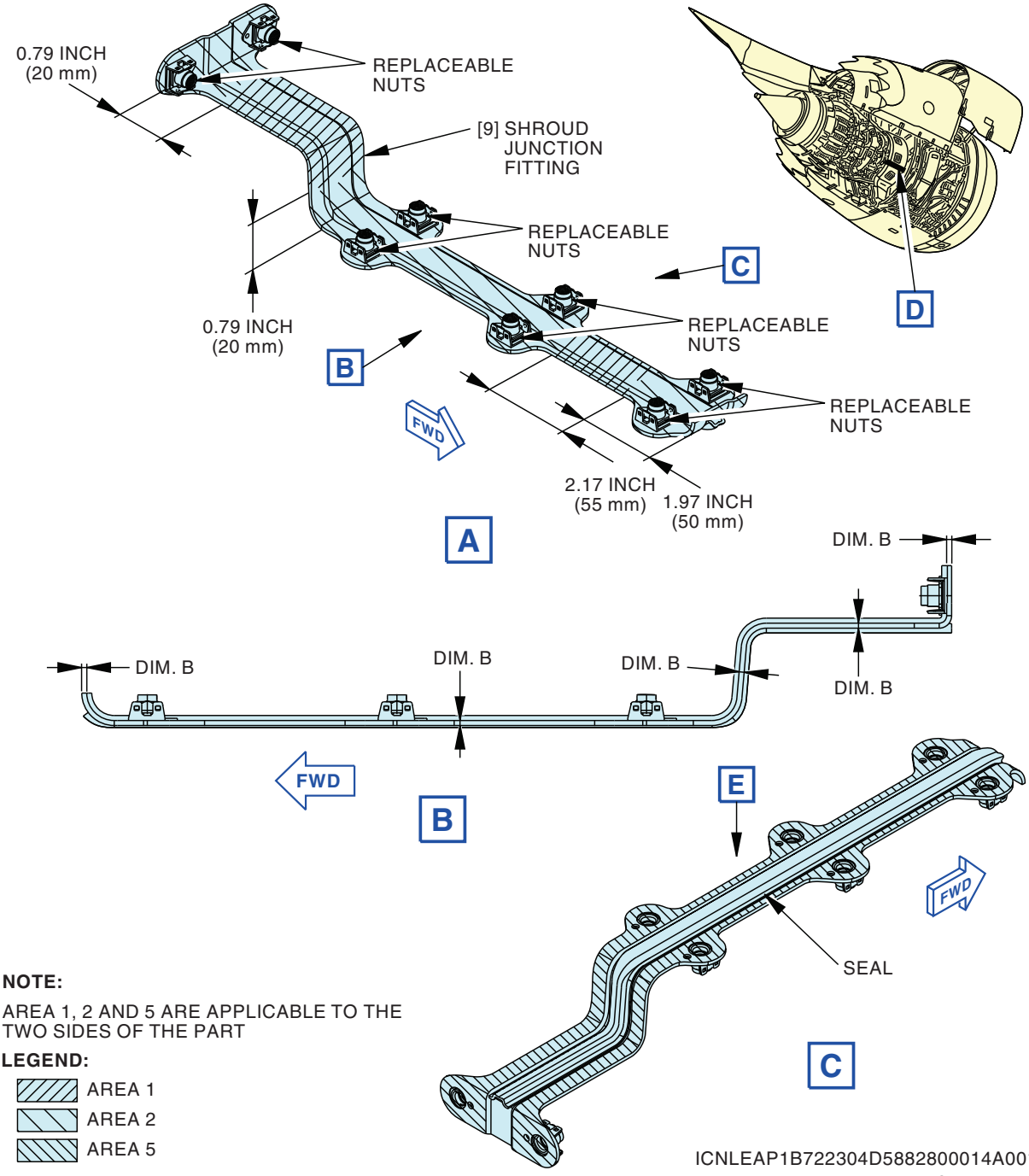
Retainer Ring Inspection
Figure 608/72-23-04-990-823-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Retainer Ring Inspection
Figure 609/72-23-04-990-835-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



NOTE:

AREA 1, 2 AND 5 ARE APPLICABLE TO THE TWO SIDES OF THE PART

LEGEND:

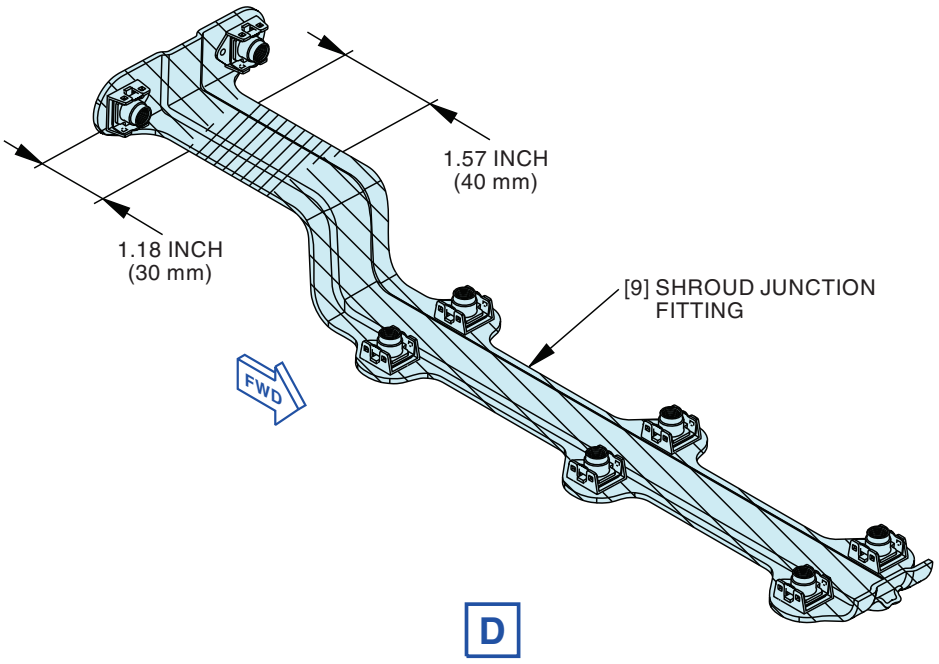
- AREA 1
- AREA 2
- AREA 5

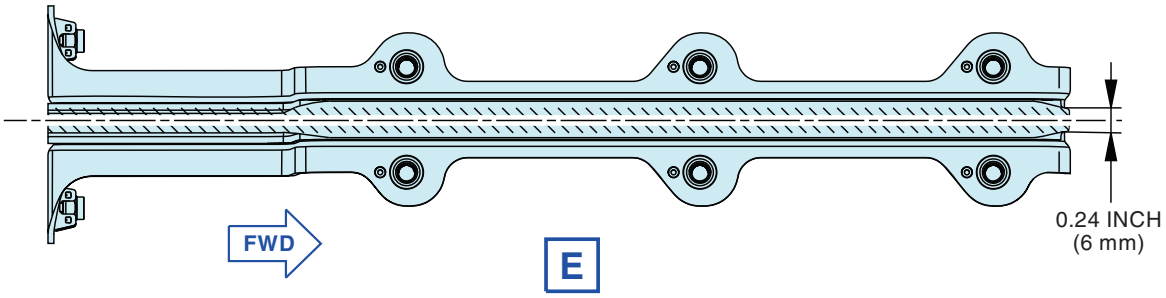
ICNLEAP1B722304D5882800014A00101

2552926 S0000607579_V2

Shroud Junction Fitting (Between Shroud Segments 2 and 3) Inspection
Figure 610/72-23-04-990-824-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





NOTE:

AREA 3 AND 4 ARE APPLICABLE TO THE TWO SIDES OF THE PART

LEGEND:

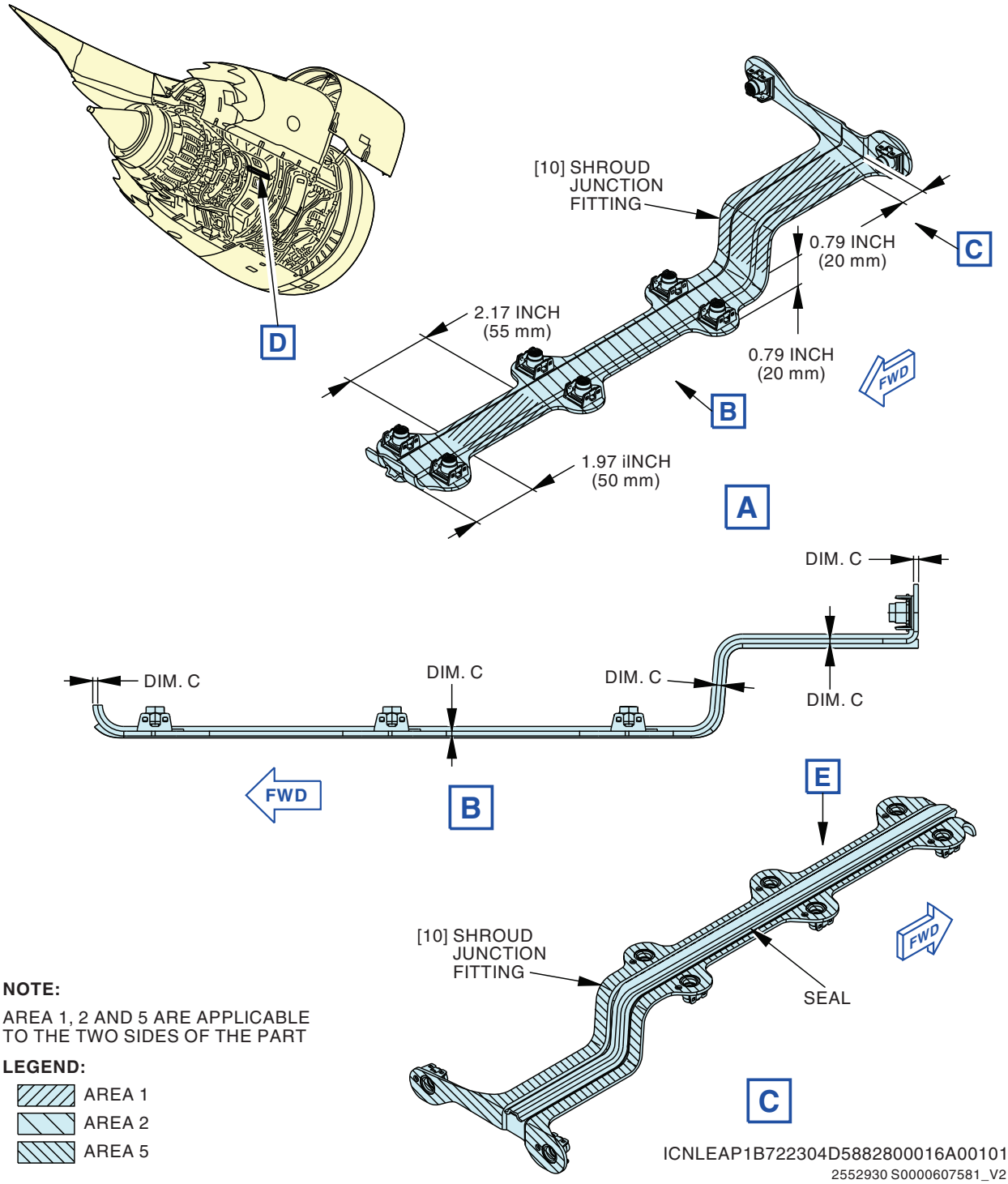
- AREA 3
- AREA 4
- AREA 6

ICNLEAP1B722304D5882800015A00101

2552927 S0000607580_V2

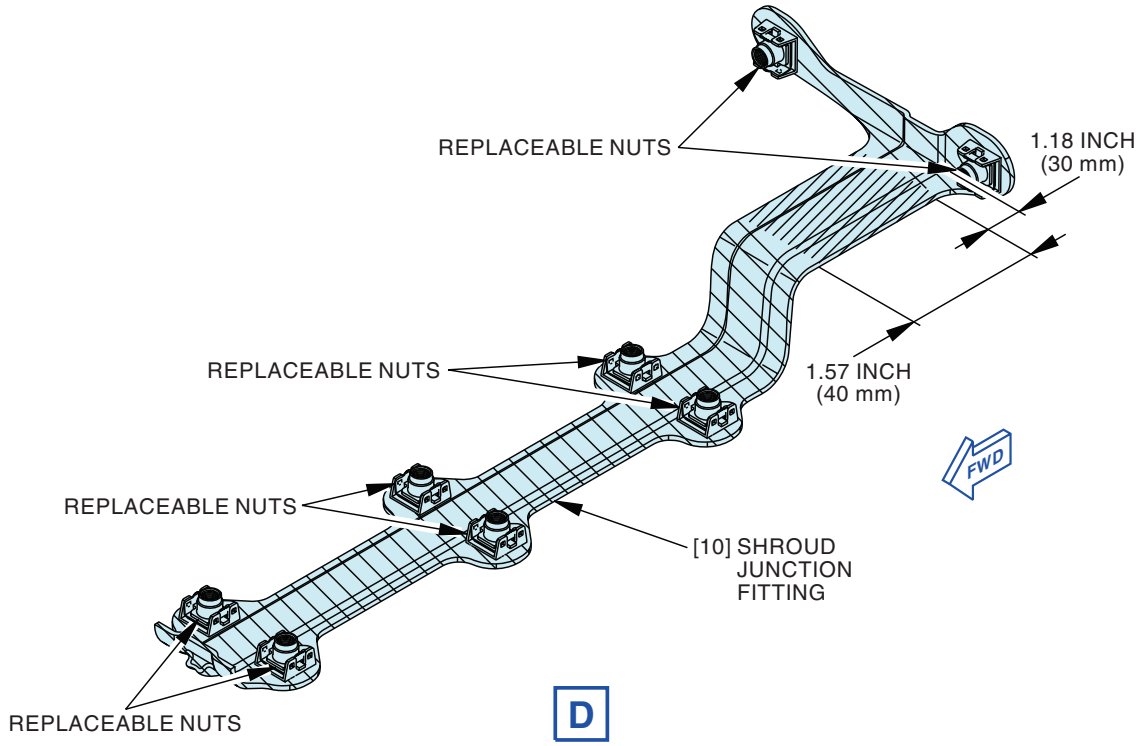
Shroud Junction Fitting (Between Shroud Segments 2 and 3) Inspection
Figure 610/72-23-04-990-824-G00 (Sheet 2 of 2)

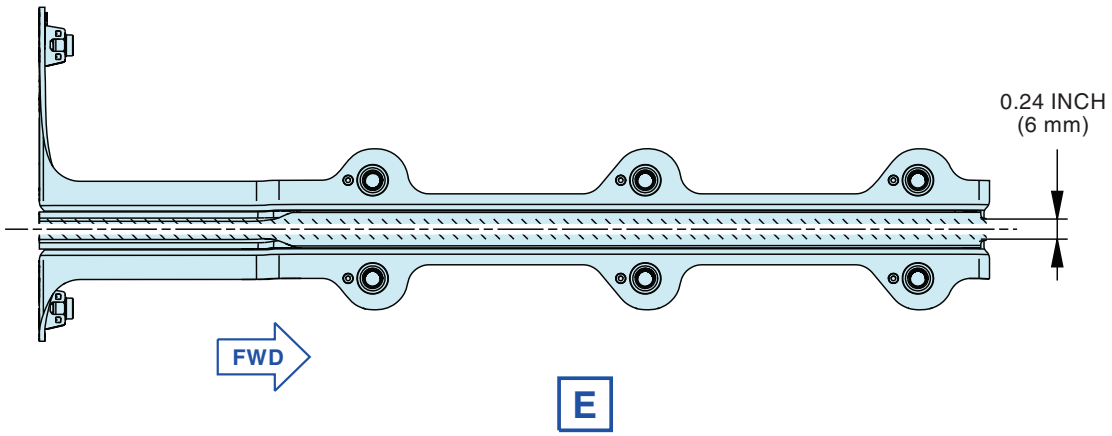
ECCN 9E991 BOEING PROPRIETARY - See title page for details



Shroud Junction Fitting (Between Shroud Segments 1 and 2) Inspection
Figure 611/72-23-04-990-826-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





NOTE:

AREA 3 AND 4 ARE APPLICABLE
TO THE TWO SIDES OF THE PART

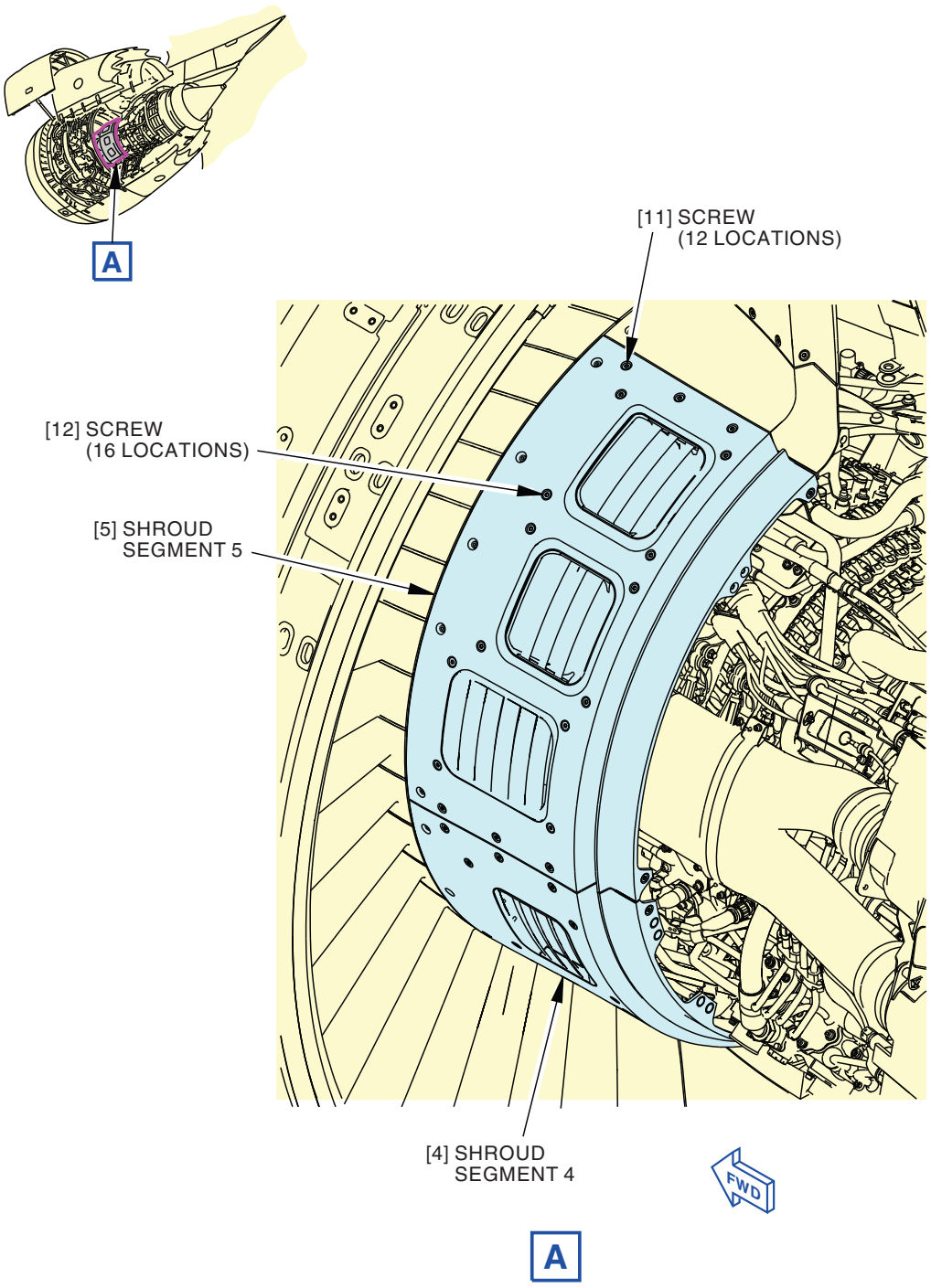
LEGEND:

AREA 3
AREA 4
AREA 6

ICNLEAP1B722304D5882800017A00101
2552931 S0000607582_V2

Shroud Junction Fitting (Between Shroud Segments 1 and 2) Inspection
Figure 611/72-23-04-990-826-G00 (Sheet 2 of 2)

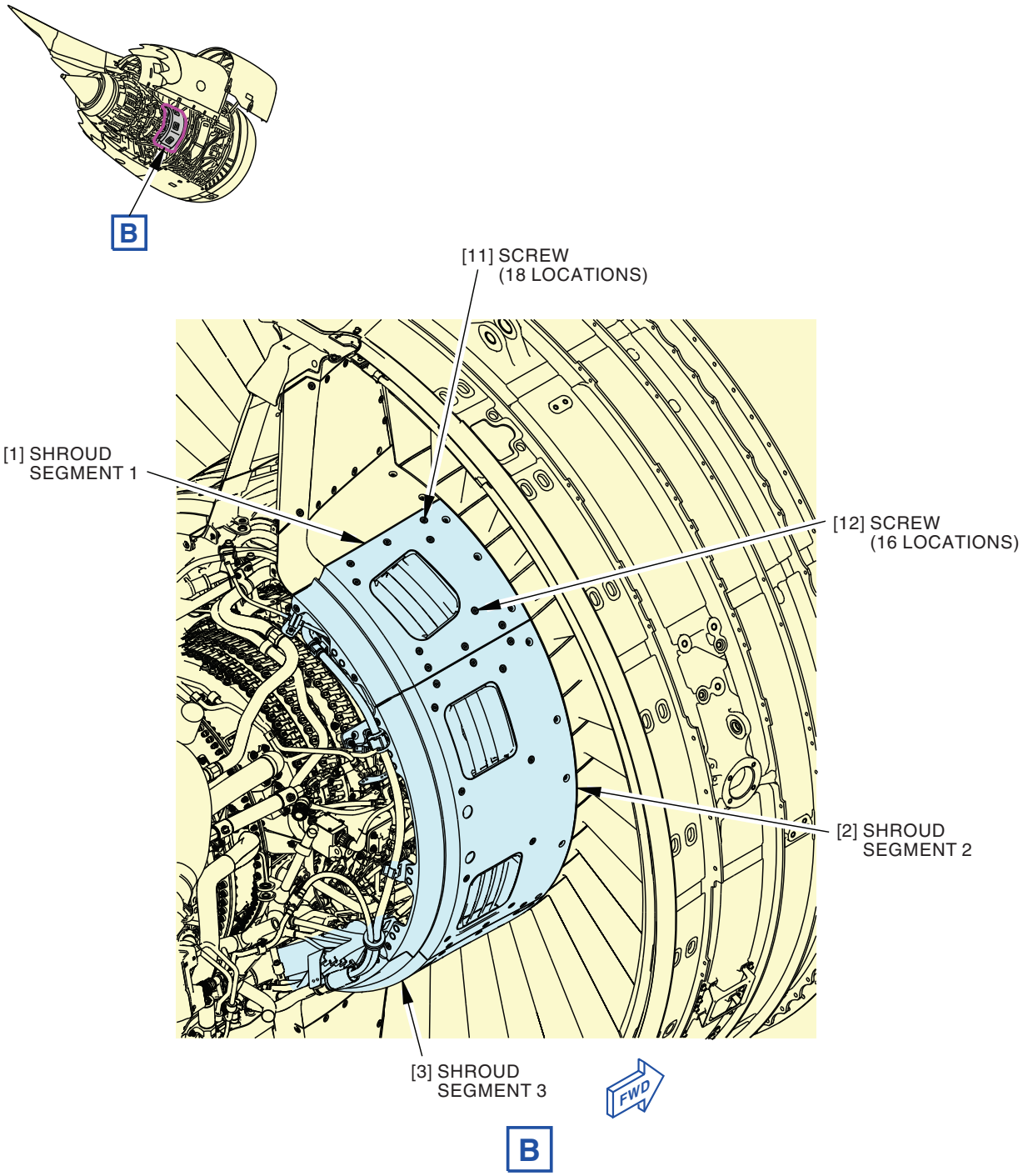
ECCN 9E991 BOEING PROPRIETARY - See title page for details



2966285 S0000745309_V1

Fan Frame Module Segment Shrouds Inspection
Figure 612/72-23-04-990-825-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



ICN-LEAP-1B-722304-B-F0301-A0002-A-001-01
2966289 S0000745312_V1

Fan Frame Module Segment Shrouds Inspection
Figure 612/72-23-04-990-825-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FAN FRAME SHROUD - REPAIRS

1. General

A. This procedure has these tasks:

(1) A replacement of the key-locked inserts in fan frame shroud

(2) A repair of the PTFE lubricant varnish in the V-groove in fan frame shroud

(3) A replacement of the bonded seal on the fittings, junction-shroud, inserts in fan frame shroud

(4) A blending of the fan frame shroud segments 1-5

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0374

(5) A replacement of the retainer ring nut

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0374

(6) A replacement of the retainer ring riveted nuts

PTF ALL

(7) A replacement of the fan frame shroud segment coating

(8) A replacement of the Variable Bleed Valve (VBV) duct nut

(9) A replacement of the shroud junction fitting nut

(10) A replacement of the shroud junction fitting nut plate two lugs.

TASK 72-23-04-300-801-G00

2. Fan Frame Shroud - Key-locked Inserts Replacement

(Figure 801)

A. General

(1) This task gives instructions to replace key-locked inserts on the fan frame shroud.

B. References

Reference	Title
70-48-13-960-801-G00	Key-Locked Insert Replacement (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
413	Engine 1 - Fan Cowl, Left
414	Engine 1 - Fan Cowl, Right
415	Engine 1 - Thrust Reverser, Left

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Zone	Area
416	Engine 1 - Thrust Reverser, Right
421	Engine 2 - Engine
423	Engine 2 - Fan Cowl, Left
424	Engine 2 - Fan Cowl, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Fan Frame Shroud - Key-locked Inserts Replacement

SUBTASK 72-23-04-860-007-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-23-04-860-008-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Fan Frame Shroud - Key-locked Inserts Replacement

SUBTASK 72-23-04-420-006-G00

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

WARNING METAL PARTICLES FROM MACHINING CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PERSONAL PROTECTION EQUIPMENT.

WARNING DO NOT BREATHE THE PARTICLES FROM GRINDING OR LET THE PARTICLES TOUCH YOU. THE PARTICLES CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PERSONAL PROTECTION EQUIPMENT. USE LOCAL MECHANICAL EXHAUST VENTILATION OR AN APPROVED RESPIRATOR.

WARNING ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. USE PERSONAL PROTECTION EQUIPMENT. USE IN A WELL-VENTILATED AREA.

NOTE: This repair should be done if accessible.

NOTE: The key-locked insert must be checked only if the associated screw is removed.

(1) Replace defective key-locked insert as follows:

(a) Replace the key-locked inserts [1], key-locked inserts [2], and key-locked inserts [3] (TASK 70-48-13-960-801-G00).

1) For the installation, do not coat the thread of the new key-locked insert and fan frame shroud tapped hole with any product.

2) Informations on key-locked inserts:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

	SIN	Internal Thread	External Thread
Key-locked insert [1]	840A0023	.250 0-28 UNJF-3B	.375 0-16 UNS-2A
Key-locked insert [2]	840A0024	.312 5-24 UNJF-3B	.437 5-16 UNS-2A
Key-locked insert [3]	840A0025	.375 0-24 UNJF-3B	.500 0-16 UNS-2A

H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-23-04-860-009-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-23-04-860-010-G00

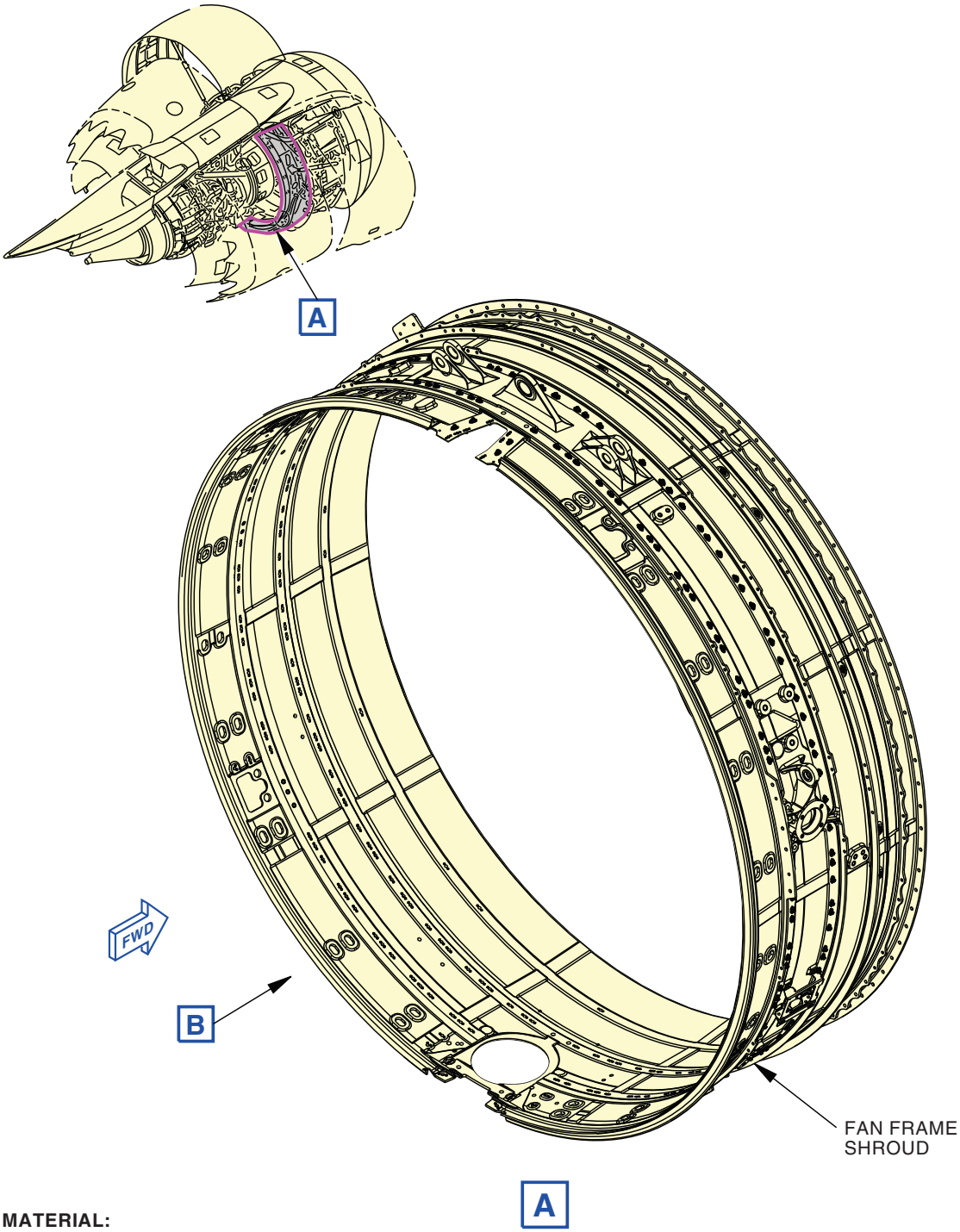
(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

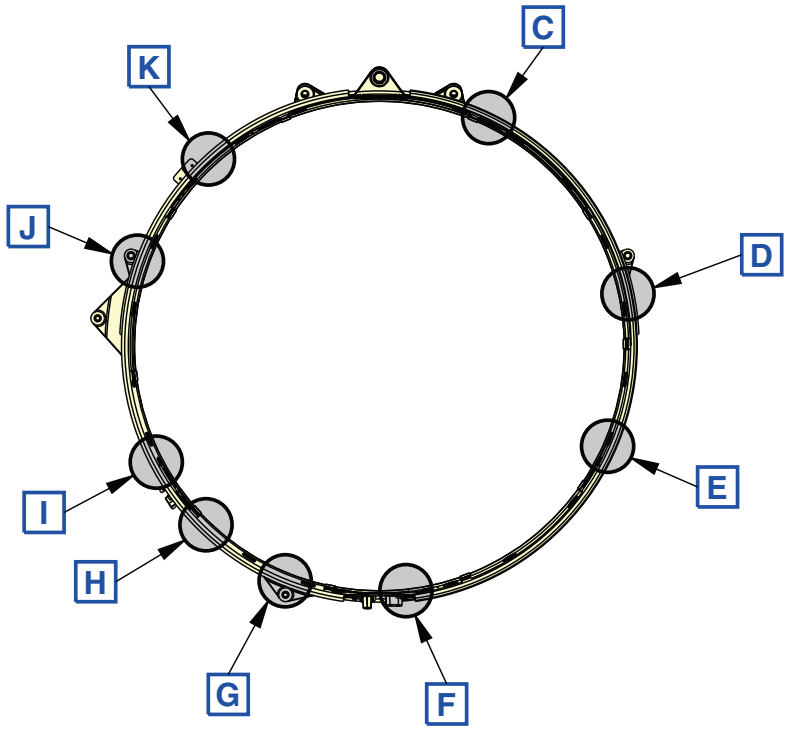


MATERIAL:
6.4 (TiAl6V)

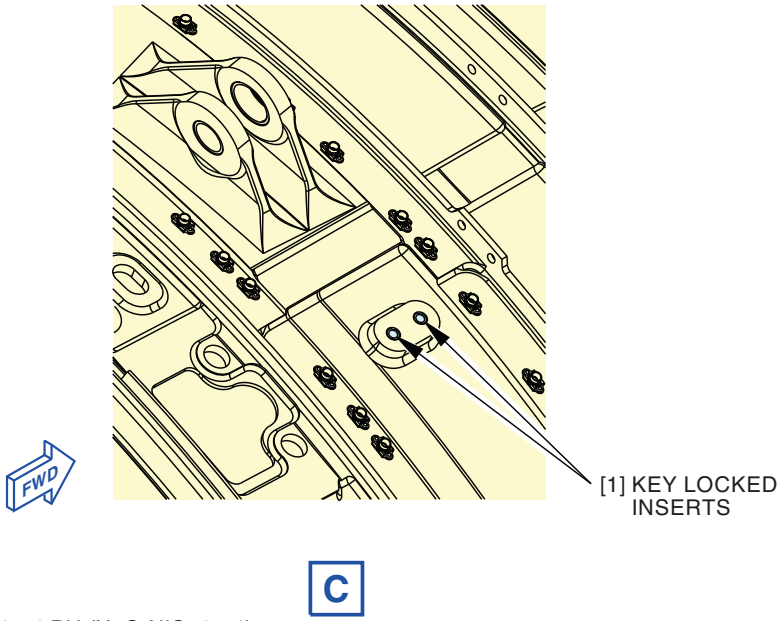
2966326 S0000745324_V1

Fan Frame Shroud
Figure 801/72-23-04-990-808-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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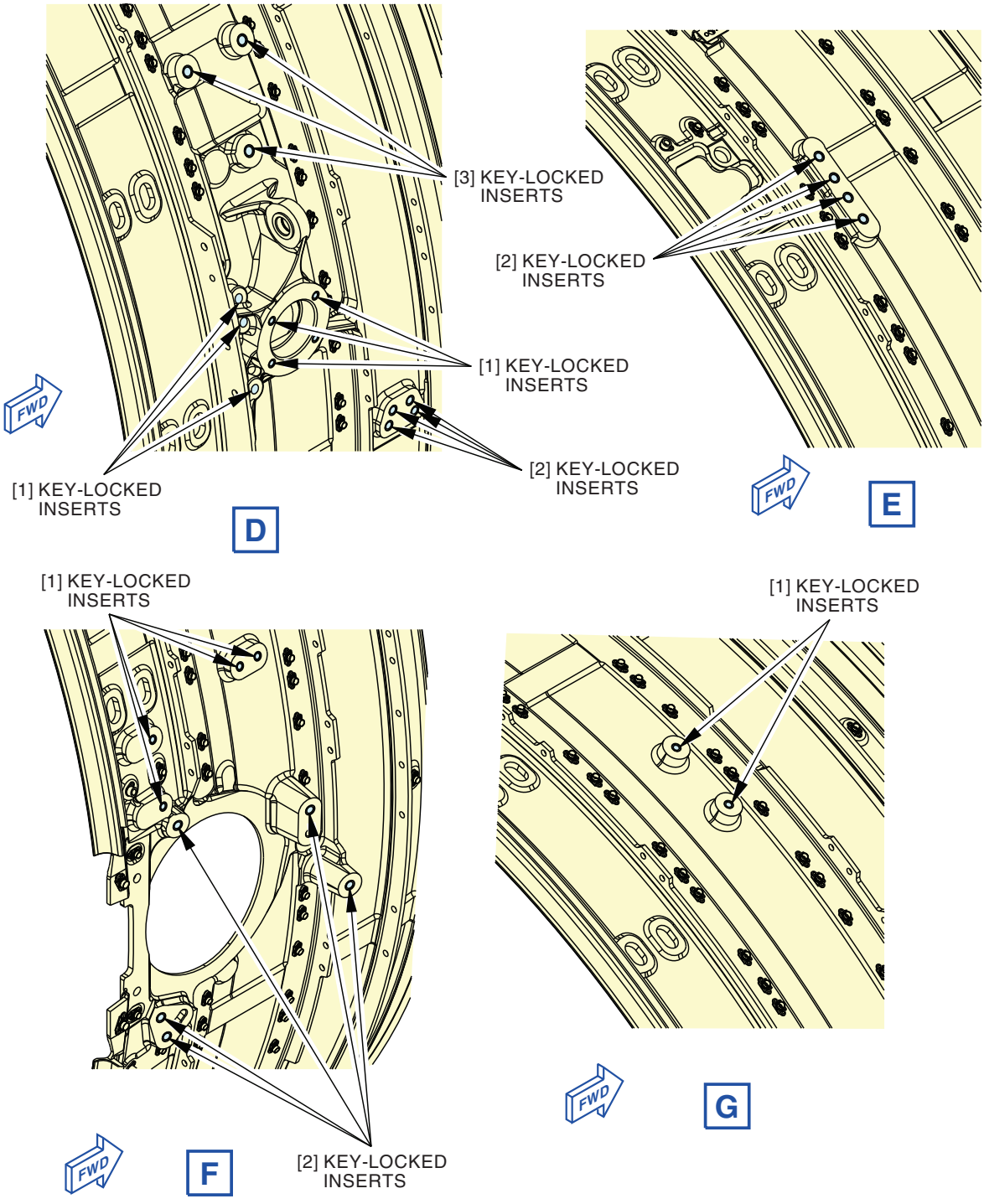


MATERIAL:
KEY-LOCKED INSERTS 17-4 PH (X5CrNiCu17-4)

2966329 S0000745325_V1

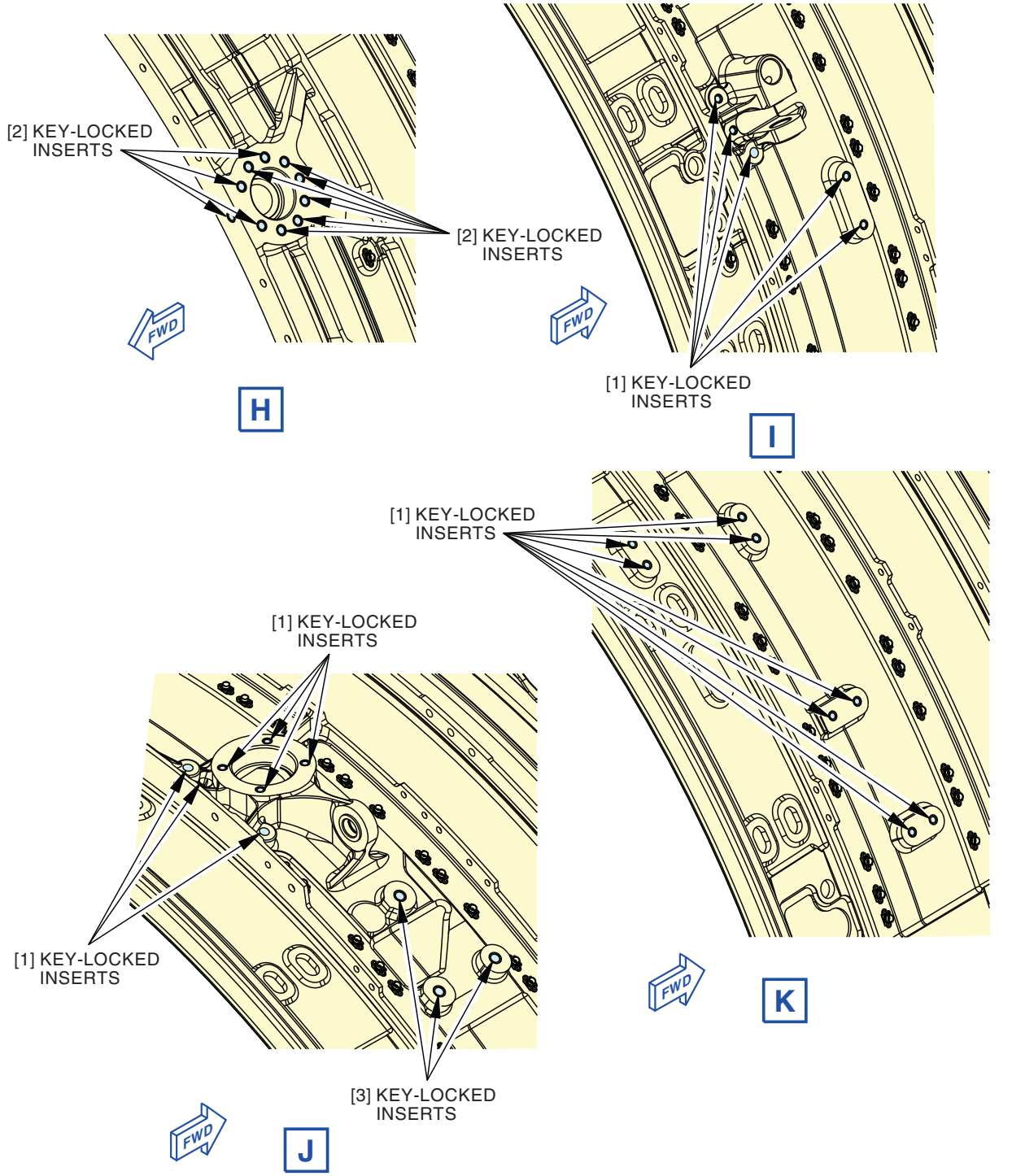
Fan Frame Shroud
Figure 801/72-23-04-990-808-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Frame Shroud
Figure 801/72-23-04-990-808-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Frame Shroud
Figure 801/72-23-04-990-808-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-23-04-300-802-G00

3. Fan Frame Shroud Repair (PTFE Lubricant Varnish in the V-groove)

(Figure 802)

A. General

(1) This task gives instructions to paint touch-up with Polytetraflouroethylene (Teflon) (PTFE) lubricant varnish.

(2) The Table 801 gives a list of the part numbers that are applicable to this repair. All part numbers are applicable to all paragraphs unless specified differently.

Table 801/72-23-04-993-808-G00

Part Number	Designation
364-049-122-0	FAN FRAME SHROUD
364-049-151-0	FAN FRAME SHROUD

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-23-00-211-804-G00	Fan Frame Module Inspection (P/B 601)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
A50568 [CP2233]	Tape - Masking - 3M 250	
B01058 [CP1039]	Solvent - Acetone, Reagent Grade	
B50190 [CP1040]	Solvent - Methyl Ethyl Ketone (MEK)	ASTM D740
B50406 [CP2945]	Degreaser - Discovert D5 Clean 2A	
G50143 [CP2208]	Tape - Glass Cloth with Silicone Adhesive, High Temperature, Abrasive Resistant - 3M 361	
G51700 [CP2968]	Tape - Masking, High-Temperature - 3M 501E	
G51856	Abrasive - Paper, 150 to 600 grit	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
413	Engine 1 - Fan Cowl, Left
414	Engine 1 - Fan Cowl, Right
415	Engine 1 - Thrust Reverser, Left

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Zone	Area
416	Engine 1 - Thrust Reverser, Right
421	Engine 2 - Engine
423	Engine 2 - Fan Cowl, Left
424	Engine 2 - Fan Cowl, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Prepare for the Repairs

SUBTASK 72-23-04-860-011-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-23-04-860-012-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Fan Frame Shroud Repairs

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

WARNING USE PROTECTIVE GLOVES TO PREVENT INJURY FROM SHARP EDGES AND ROUGH SURFACES. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

WARNING DO NOT LET TITANIUM DUST COLLECT. TITANIUM DUST IS VERY FLAMMABLE. DISCARD ALL TITANIUM WASTE IN AN APPROVED CONTAINER. TITANIUM DUST CAN CAUSE INJURY TO PERSONNEL.

WARNING DO NOT USE WATER TO EXTINGUISH A TITANIUM FIRE. EXTINGUISH WITH AN APPROVED CHEMICAL FOR METAL FIRES. IF YOU DO NOT OBEY, OPERATION OF A WATER EXTINGUISHER COULD CAUSE INJURY TO PERSONNEL.

CAUTION KEEP CHLORINATED HYDROCARBONS AND HALOGEN MATERIALS AWAY FROM TITANIUM PARTS. IF YOU DO NOT OBEY, CHLORINATED HYDROCARBONS AND HALOGEN MATERIALS CAN CAUSE THE DAMAGE TO THE TITANIUM PARTS.

CAUTION DO NOT USE TOOLS FIXTURE MADE FROM OR COATED WITH CADMIUM, BISMUTH, TIN, ZINC, GOLD OR SILVER TO TITANIUM PARTS. IF YOU DO NOT OBEY, DAMAGE TO THE TITANIUM PARTS CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION

DO NOT USE TOOLS OR MATERIALS TO MIX MATERIALS OTHER THAN TITANIUM. IF YOU DO NOT OBEY, DAMAGE TO THE TITANIUM CAN OCCUR.

CAUTION

DO NOT CAUSE HEAT THAT WILL COME IN THERMAL STRESS DURING THE OPERATION OF POWER TOOLS ON THE ENGINE. IF YOU DO NOT OBEY, THERMAL STRESS CAN CAUSE ENGINE DAMAGE.

SUBTASK 72-23-04-110-001-G00

(1) Clean the area to be repaired with Discovert D5 Clean 2A degreaser, B50406 [CP2945], or acetone solvent, B01058 [CP1039], or solvent, B50190 [CP1040].

SUBTASK 72-23-04-380-001-G00

WARNING

DO NOT BREATHE THE PARTICLES WHEN SANDING. THE PARTICLES CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PERSONAL PROTECTION EQUIPMENT. USE LOCAL MECHANICAL EXHAUST VENTILATION OR AN APPROVED RESPIRATOR.

WARNING

USE EYE PROTECTION WHEN YOU USE COMPRESSED AIR TO CLEAN, COOL, OR DRY PARTS OR TOOLS. PARTICLES CAN CAUSE AN INJURY TO YOUR EYES. DO NOT USE MORE THAN 30 PSIG (200 KPA). DO NOT POINT COMPRESSOR AIR AT YOURSELF OR OTHER PERSONS.

CAUTION

DO NOT REMOVE OR CHANGE THE SHAPE OR ROUGHNESS OF THE V-GROOVE ON THE FAN FRAM SHROUD. IF YOU DO NOT OBEY, DAMAGE TO THE FAN FRAM SHROUD CAN OCCUR.

CAUTION

DO NOT APPLY A PROCEDURE THAT CAN CHANGE THE PARENT MATERIAL PROPERTIES DURING THE REMOVAL OF THE BONDED MATERIAL. IF YOU DO NOT OBEY, DAMAGE TO THE MATERIAL CAN OCCUR.

(2) Prepare the surface where you will apply the PTFE varnish, do these steps:

(a) If it is necessary, apply masking 3M 361 tape, G50143 [CP2208], or 3M 501E tape, G51700 [CP2968], on the area around the area to be repaired.

(b) Do a soft hand sanding with 150 to 600 grit abrasive paper, G51856, on the area to be repaired.

(c) If it is necessary, remove the masking.

(d) Clean the area to be repaired with Discovert D5 Clean 2A degreaser, B50406 [CP2945], or acetone solvent, B01058 [CP1039], or solvent, B50190 [CP1040].

(e) Dry the area to be repaired with clean, dry, and oil-free air.

SUBTASK 72-23-04-220-001-G00

(3) Do a visual inspection of the area where you will apply the PTFE varnish (TASK 72-23-00-211-804-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-23-04-350-001-G00

(4) Apply the PTFE varnish as follows (Figure 802):

NOTE: Apply lubricant PTFE varnish minimum 15 minutes after the cleaning.

(a) To check quality assurance, make a test piece made of TA6V piece along with the part to be varnished.

(b) Prepare the surface of the test piece as follows:

1) Do a light hand sanding with 150 to 600 grit abrasive paper, G51856, on the area to be repaired.

2) Clean the area to be repaired with Discovert D5 Clean 2A degreaser, B50406 [CP2945], or acetone solvent, B01058 [CP1039], or solvent, B50190 [CP1040].

3) Dry the area to be repaired with clean, dry, and oil-free air.

(c) If it is necessary, apply masking 3M 361 tape, G50143 [CP2208], or 3M 501E tape, G51700 [CP2968], around the area to be repaired.

(d) Make sure to get the following viscosity:

1) AFNOR cup (consistometer) No. 4 between 20 s and 30 s, or

2) FORD cup No. 4 between 16 s and 26 s.

(e) Apply one layer of PTFE varnish.

1) Make sure that the thickness is between 0.00028 in. (0.007 mm) and 0.00118 in. (0.030 mm) on the area(s) to be repaired.

NOTE: An excess of PTFE varnish is permitted on AREAS A (View A-A, Figure 802).

(f) If it is necessary, remove the masking, overspray, or unwanted varnish with Discovert D5 Clean 2A degreaser, B50406 [CP2945], or acetone solvent, B01058 [CP1039], or solvent, B50190 [CP1040], or with a blade.

(g) Dry the lubricant varnish PTFE as follows:

1) Dry at an ambient temperature during 48 hours.

2) Dry at a temperature 131°F (55°C) - 149°F (65°C) during 2 hours minimum and 2 hours 30 minute maximum.

SUBTASK 72-23-04-220-002-G00

(5) Do a visual examination.

(a) Make sure that the varnished surfaces are smooth, uniform, and free from sags, blisters, flakes, or foreign material.

NOTE: The coating must be light green.

(b) Smooth transition area between the repaired area and adjacent areas is required.

(c) If the varnish thickness is in the permitted limits, streaks are acceptable.

(d) If the varnished surfaces fail any of these tests or examinations, remove the coating and reprocess the part.

SUBTASK 72-23-04-280-001-G00

(6) Do the following tests:

(a) On the test piece:

1) Measure the thickness of the varnish.

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a) Make sure that the thickness is between 0.00028 in. (0.007 mm) and 0.00118 in. (0.030 mm) on the area(s) to be repaired.

NOTE: Recommended means of measurement: Eddy current measuring instruments or geometrical.

NOTE: An excess of PTFE varnish is permitted on AREAS A (View A-A, Figure 802).

2) Use a sharp blade to scrape the lubricant varnish.

NOTE: Scrapings must show as a powder.

3) Criss-cross the area, then apply and press on a strip of pressure sensitive 3M 250 tape, A50568 [CP2233], to the test piece following ISO2409 norm.

NOTE: Lubricant varnish must not peel when tape is removed.

(b) On the repaired area:

1) Apply and press on a strip of pressure sensitive 3M 250 tape, A50568 [CP2233], on the repaired area.

NOTE: Lubricant varnish must not peel when tape is removed.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-23-04-860-013-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-23-04-860-014-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

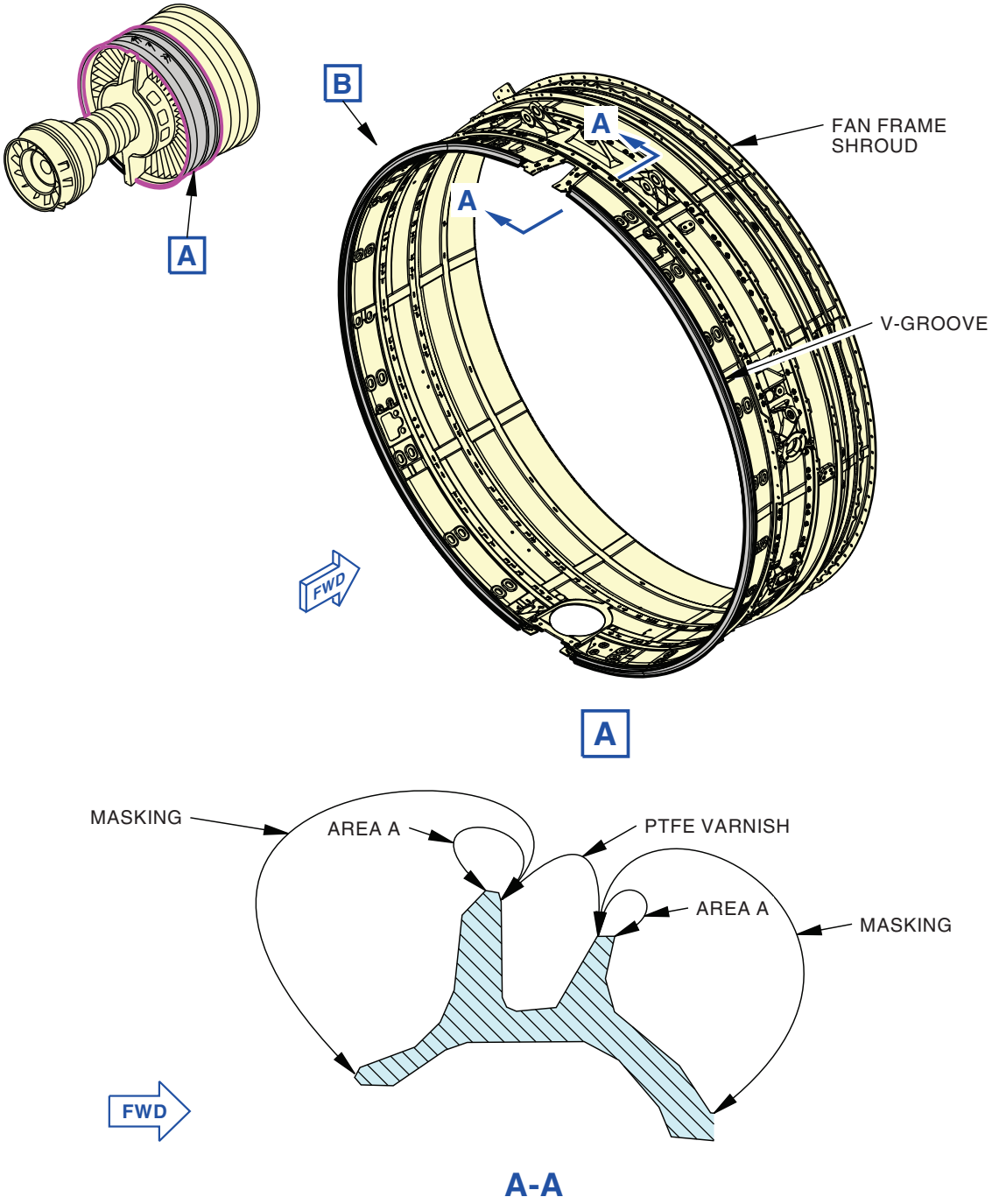
(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

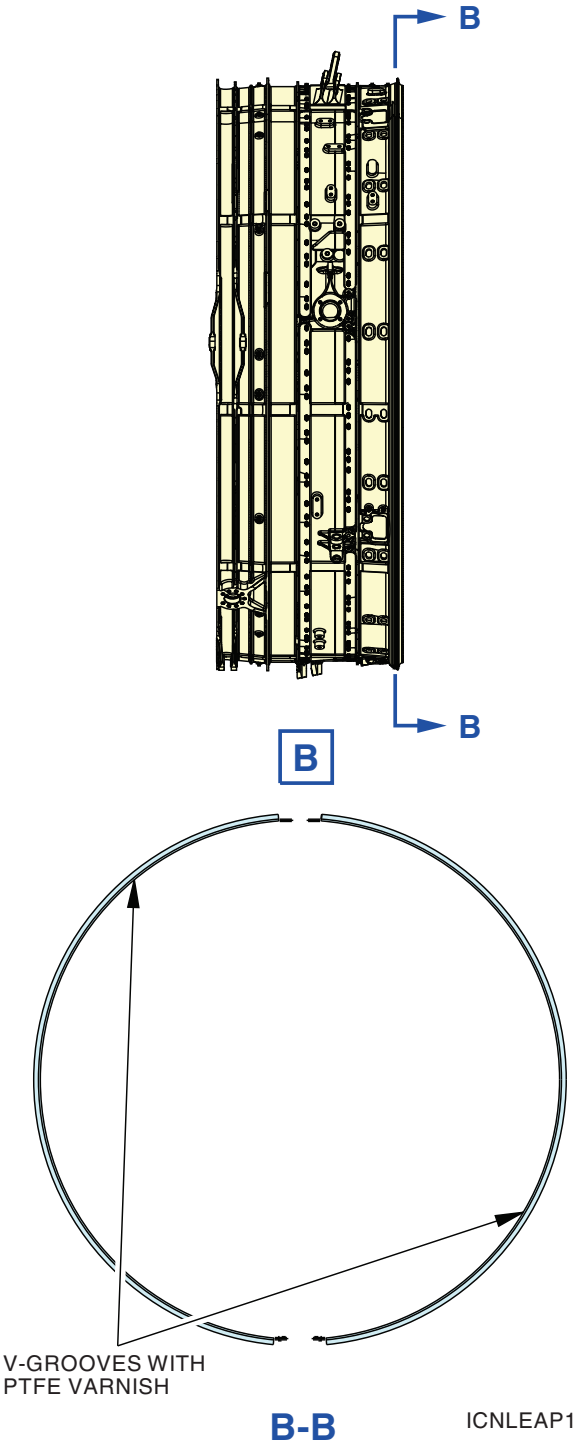
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Fan Frame Shroud Repair (PTFE Lubricant Varnish in the V-groove)
Figure 802/72-23-04-990-809-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Frame Shroud Repair (PTFE Lubricant Varnish in the V-groove)
Figure 802/72-23-04-990-809-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-23-04-960-801-G00

4. Fan Frame Shroud Fittings Seal - Replacement

(Figure 803, Figure 804, Figure 805, and Figure 806)

A. General

(1) This procedure gives instructions to replace the bonded seal on the junction shroud fitting.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
A50119	Adhesive - Silicone Rubber - RTV 103 (Black)	MIL-A-46106
B01058 [CP1039]	Solvent - Acetone, Reagent Grade	
G50077	Abrasive - Aluminum Oxide Paper, 240 grit or finer	
G51692	Cloth - 100% Cotton	AMS3819 Class 1
G51742 [CP2441]	Tape - Vinyl - 3M No. 472	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Prepare for the Replacement

SUBTASK 72-23-04-860-015-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-23-04-010-002-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Fan Frame Shroud Fittings Seal Replacement

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

WARNING USE EYE PROTECTION WHEN YOU CUT MATERIAL OR PARTS. THE PARTICLES CAN CAUSE DAMAGE TO YOUR EYES.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

WARNING USE PROTECTIVE GLOVES TO PREVENT INJURY FROM SHARP EDGES AND ROUGH SURFACES. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

SUBTASK 72-23-04-020-006-G00

(1) Remove and discard the seal as follows (Figure 803):

CAUTION MAKE SURE THAT YOU CAREFULLY REMOVE THE SEAL FROM THE FAN FRAME STRUT FAIRING. IF YOU DO NOT OBEY, DAMAGE TO THE FAN FRAME STRUT FAIRING CAN OCCUR.

(a) Carefully remove and discard the seal.

WARNING BE CAREFUL WHEN USE SHARP OR POINTED TOOLS. THESE TOOLS CAN CAUSE AN INJURY TO YOU.

CAUTION DO NOT APPLY A PROCEDURE THAT CAN CHANGE THE PARENT MATERIAL PROPERTIES DURING THE REMOVAL OF THE BONDED MATERIAL. IF YOU DO NOT OBEY, DAMAGE TO THE MATERIAL CAN OCCUR.

(b) Remove adhesive from the junction shroud fitting with a cutting tool.

(c) Make sure that you do not remove parent material from the junction shroud fitting.

SUBTASK 72-23-04-100-001-G00

(2) Prepare the area where you will bond the new seal (Figure 804).

WARNING DO NOT BREATHE THE PARTICLES FROM BLENDING OR LET THE PARTICLES TOUCH YOU. THE PARTICLES FROM BLENDING CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PROTECTION EQUIPMENT. USE LOCAL MECHANICAL EXHAUST VENTILATION OR AN APPROVED RESPIRATOR.

(a) Make the metallic part surface of the junction shroud fitting and the new seal rough with 240 grit or finer abrasive paper, G50077.

WARNING DO NOT GET SOLVENT IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE SOLVENT. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE THE SOLVENT. KEEP THE SOLVENT AWAY FROM SPARKS, FLAME AND HEAT. SOLVENT IS POISONOUS AND FLAMMABLE WHICH CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(b) Clean the metallic part area of the fitting, junction-shroud, with the clean cloth, G51692, or equivalent, soaked in acetone solvent, B01058 [CP1039].

(c) Make the surface of the new seal rough with 240 grit or finer abrasive paper, G50077.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING

DO NOT GET SOLVENT IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE SOLVENT. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE THE SOLVENT. KEEP THE SOLVENT AWAY FROM SPARKS, FLAME AND HEAT. SOLVENT IS POISONOUS AND FLAMMABLE WHICH CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(d) Make the surface of the new seal rough with acetone solvent, B01058 [CP1039].

SUBTASK 72-23-04-400-001-G00

(3) Bond the new seal (Figure 803, Figure 804, and Figure 805).

NOTE: Seals can be in 2 different shapes and colors. The seals are interchangeable.

(a) If you do not bond the material within 6 hours after surface preparation, do these steps:

1) Clean the metallic part area where you will bond the new seal with the clean cloth, G51692, or equivalent, soaked in acetone solvent, B01058 [CP1039].

2) Make surface of the new seal rough with acetone solvent, B01058 [CP1039].

(b) If it is necessary use 3M 472 tape, G51742 [CP2441], or RTV 103 adhesive, A50119, to apply mask on the adjacent areas.

(c) Apply the maximum thickness 0.019 in. (0.5 mm) of RTV 103 adhesive, A50119, on the Area 1 (Figure 804).

1) For the rectangular seal apply RTV 103 adhesive, A50119, on the longer side (Figure 805).

(d) Apply the maximum thickness 0.019 in. (0.5 mm) of RTV 103 adhesive, A50119, on the Area 2 (Figure 804).

1) For the rectangular seal apply RTV 103 adhesive, A50119, on the longer side (Figure 805).

(e) Install the seal on the Area 1 and Area 2 and apply a pressure for bonding.

(f) Cut the seal to make continuous surface with the metallic part adjacent areas (Figure 806).

(g) If it is necessary, remove the unwanted adhesive from the adjacent areas.

NOTE: Adhesive is not accepted in adjacent areas.

(h) If it is necessary, remove the mask that you apply on the adjacent areas.

I. Put Airplane Back to Its Usual Condition

SUBTASK 72-23-04-410-002-G00

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-23-04-860-016-G00

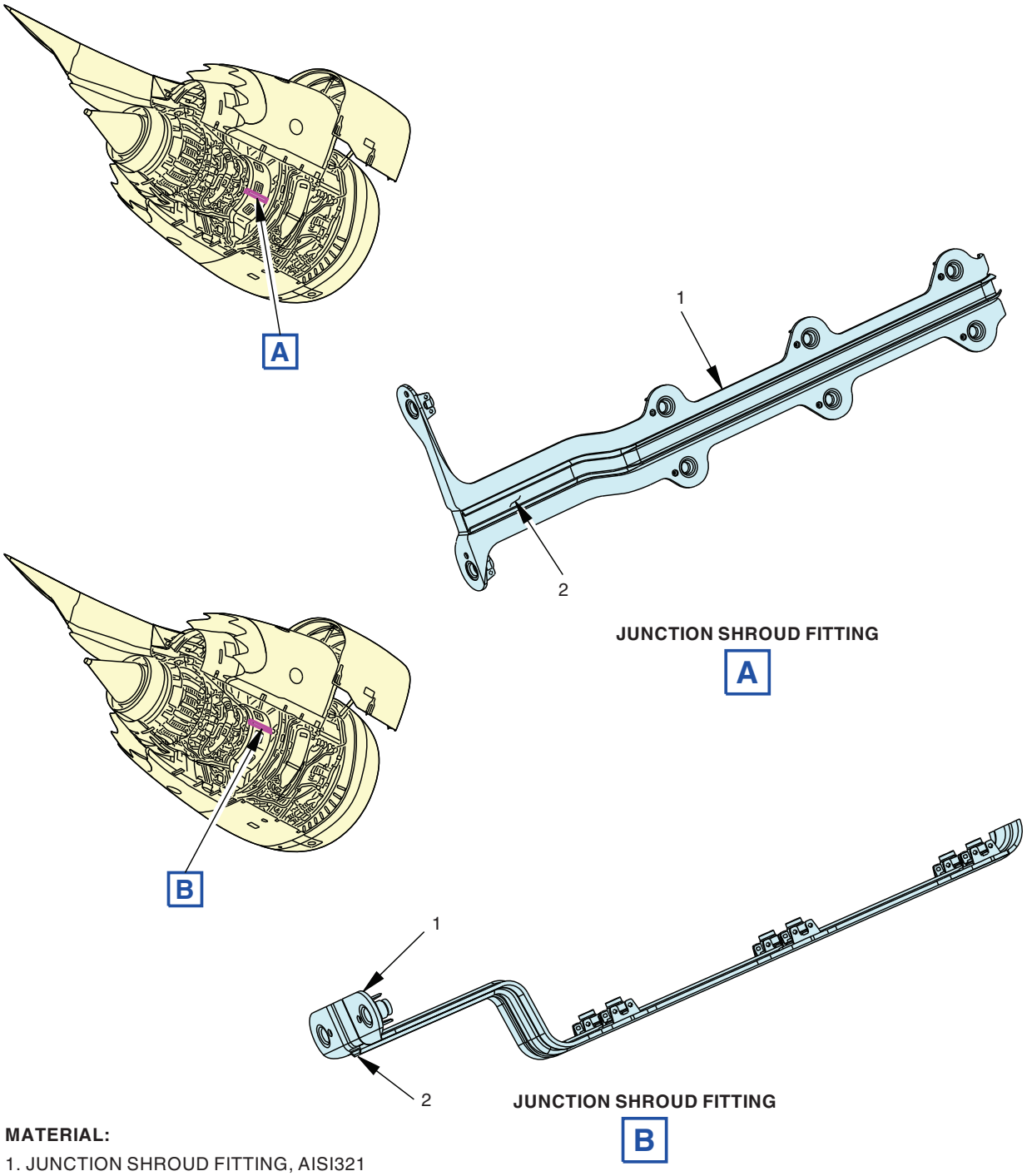
(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



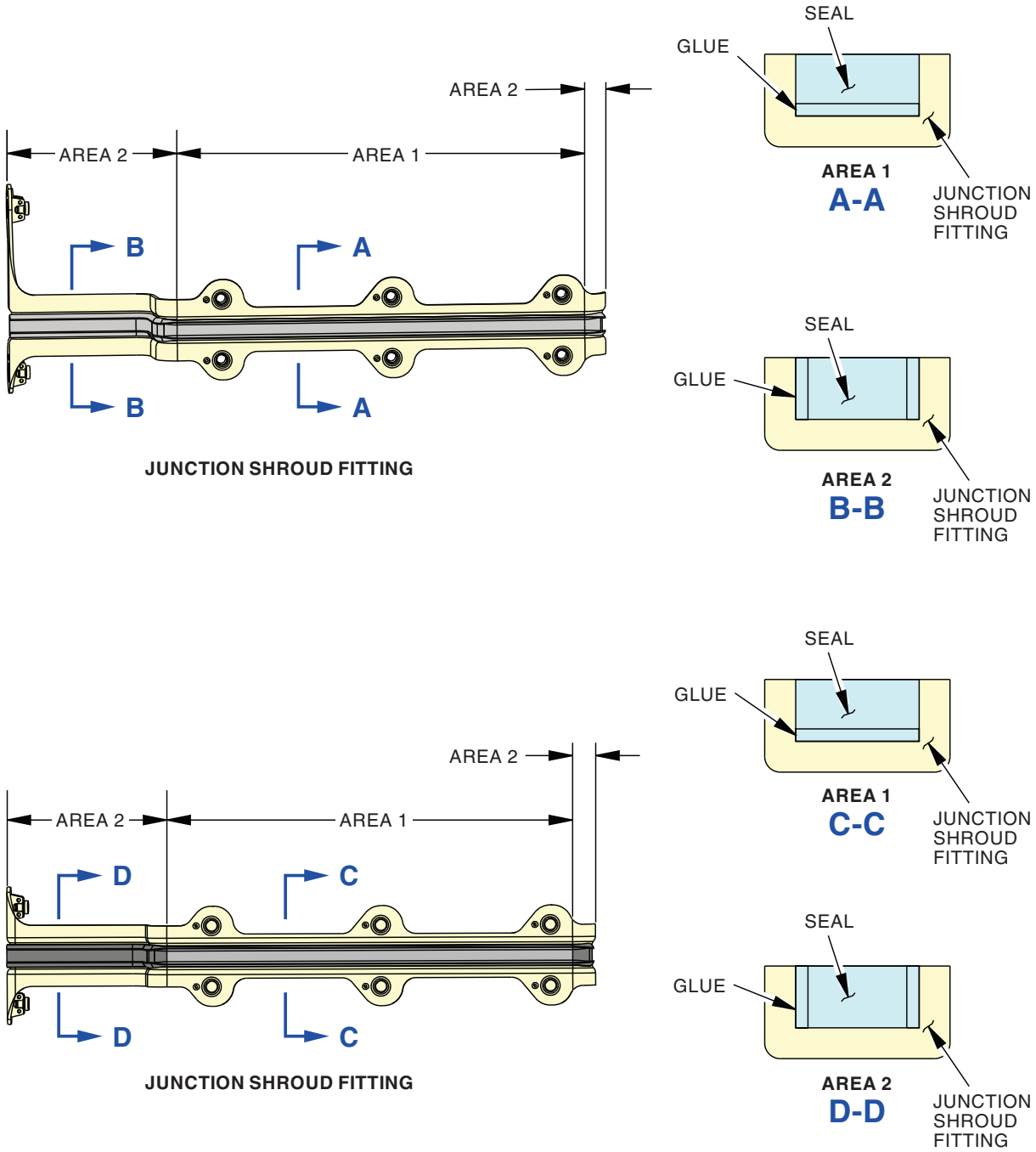
MATERIAL:

1. JUNCTION SHROUD FITTING, AISI321
2. SEAL: ELASTOMER

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Junction Shroud Fitting
Figure 803/72-23-04-990-810-G00

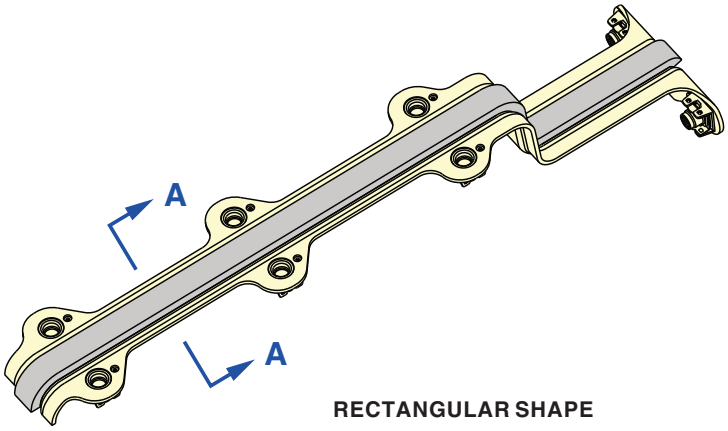
ECCN 9E991 BOEING PROPRIETARY - See title page for details



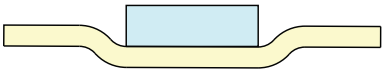
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Fitting bonding areas
Figure 804/72-23-04-990-814-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

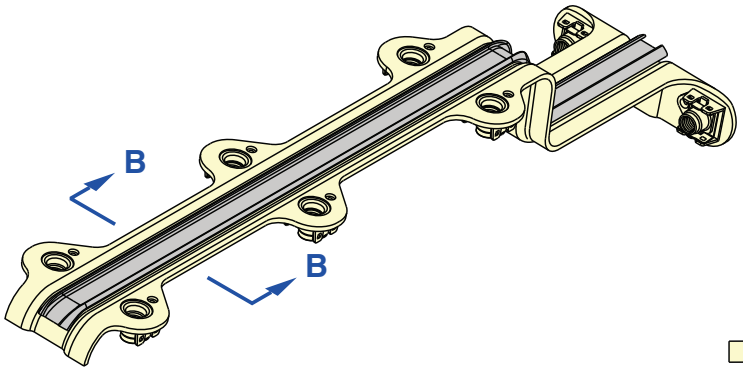


RECTANGULAR SHAPE



SEAL (COLOR OF SEAL WHITE)
(TYPICAL CROSS SECTION VIEW)

A-A



OTHER SHAPE



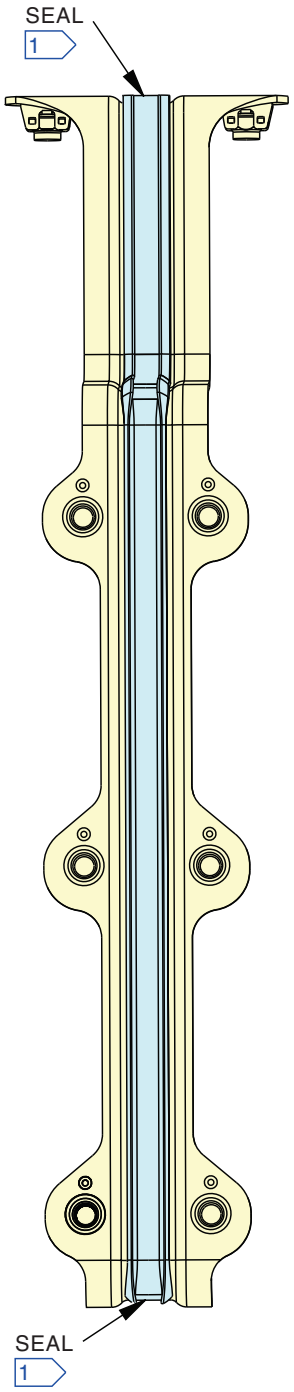
SEAL (COLOR OF SEAL BLACK)
(TYPICAL CROSS SECTION VIEW)

B-B

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Seal Types
Figure 805/72-23-04-990-815-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



1 SEAL HAVE TO BE CUT FOR FLUSH

(TYPICAL VIEW)

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Seal Cutting
Figure 806/72-23-04-990-816-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-23-04-320-801-G00

5. Fan Frame Shroud Segments 1-5 Blending

(Figure 807)

A. General

(1) This task gives the instructions to blend the shroud segments 1 - 5.

(2) The Table 802 gives a list of the part numbers that are applicable to this repair. All part numbers are applicable to all paragraphs unless specified differently.

Table 802/72-23-04-993-805-G00

Part Number	Designation
364-030-610-0	Shroud, segment 1
364-030-710-0	Shroud, segment 2
364-030-810-0	Shroud, segment 3
364-030-910-0	Shroud, segment 4
364-032-610-0	Shroud, segment 5

B. References

Reference	Title
70-11-06-230-802-G00	Fluorescent Penetrant Inspection (P/B 201)
70-11-06-230-803-G00	Fluorescent Penetrant Inspection (General) (P/B 201)
70-21-23-100-801-G00	Hand Wipe Solvent Degreasing With Chemical Agent (P/B 701)
70-43-24-100-802-G00	Swab Etching Procedure and Cleaning (P/B 201)
70-43-42-300-802-G00	Blending Procedures - Hand Blending (P/B 201)
70-43-42-300-803-G00	Blending Procedures - Power Blending (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-23-04-200-801-G00	Fan Frame Module Segment Shrouds Inspection (P/B 601)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
B00676 [CP1041]	Alcohol - Isopropyl	
G51485 [S1093]	Etchant - Titanium Alloy	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Prepare for the Blending

SUBTASK 72-23-04-860-017-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-23-04-860-018-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Fan Frame Shroud Blending

SUBTASK 72-23-04-210-011-G00

(1) Do a visual inspection of the part.

(a) Maximum three blending are permitted on shroud segments 1 and 3.

(b) Maximum five blending are permitted on shroud segments 2, 4, and 5.

(c) Make sure that the indication to be blended is on available surface and is not more than DIM. EC. as follows (Figure 807 and Figure 808):

1) Make sure that the depth DIM. EC. is less or equal to 0.023 in. (0.6 mm).

2) To locate the available areas, use the following dimensions:

- DIM. A is equal to 2.80 in. (71 mm)
- DIM. B is equal to 0.48 in. (12 mm)
- DIA C is equal to 2.37 in. (60 mm)
- DIM. D is equal to 2.09 in. (53 mm)
- DIM. E is equal to 2.49 in. (63 mm)
- DIM. F is equal to 3.59 in. (91 mm)
- DIM. G is equal to 1.38 in. (35 mm)
- DIM. H is equal to 3.63 in. (92 mm)
- DIM. J is equal to 1.66 in. (42 mm)
- DIA K is equal to 0.60 in. (15 mm)
- DIM. L is equal to 2.41 in. (61 mm)
- DIM. M is equal to 2.99 in. (76 mm)
- DIM. N is equal to 8.58 in. (218 mm)
- DIM. P is equal to 0.47 in. (12 mm)
- DIM. Q is equal to 12.00 in. (305 mm)
- DIA R is equal to 0.59 in. (15 mm)
- DIM. S is equal to 12.91 in. (328 mm)
- DIM. T is equal to 1.10 in. (28 mm)
- DIM. U is equal to 8.46 in. (215 mm)
- DIM. V is equal to 2.83 in. (72 mm)
- DIM. W is equal to 6.50 in. (165 mm)
- DIM. X is equal to 1.77 in. (45 mm)
- DIM. Y is equal to 1.46 in. (37 mm)
- DIM. Z is equal to 3.94 in. (100 mm)
- DIM. AA is equal to 3.15 in. (80 mm)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

- DIM. AB is equal to 6.06 in. (154 mm)
- DIM. AC is equal to 3.54 in. (90 mm)
- DIM. AD is equal to 2.36 in. (60 mm)
- DIM. AE is equal to 3.03 in. (77 mm)
- DIM. AF is equal to 0.35 in. (9 mm)
- DIM. AG is equal to 3.39 in. (86 mm)
- DIM. AH is equal to 2.17 in. (55 mm)
- DIM. AJ is equal to 1.89 in. (48 mm)
- DIM. AK is equal to 2.28 in. (58 mm)
- DIM. AL is equal to 3.23 in. (82 mm)
- DIM. AM is equal to 1.18 in. (30 mm)
- DIM. AN is equal to 5.67 in. (144 mm)
- DIM. AP is equal to 2.99 in. (76 mm)
- DIM. AQ is equal to 5.51 in. (140 mm)
- DIA AR is equal to 0.59 in. (15 mm)
- DIM. AS is equal to 1.26 in. (32 mm)
- DIM. AT is equal to 4.76 in. (121 mm)
- DIM. AU is equal to 3.03 in. (77 mm)
- DIM. AV is equal to 6.46 in. (164 mm)
- DIM. AW is equal to 3.94 in. (100 mm)
- DIM. AX is equal to 0.47 in. (12 mm)
- DIM. AY is equal to 6.46 in. (164 mm)
- DIM. AZ is equal to 2.76 in. (70 mm)
- DIM. BA is equal to 4.53 in. (115 mm)
- DIA BB is equal to 0.63 in. (16 mm)
- DIM. BC is equal to 2.79 in. (71 mm)
- DIM. BD is equal to 4.25 in. (108 mm)
- DIM. BE is equal to 1.06 in. (27 mm)
- DIM. BF is equal to 3.31 in. (84 mm)
- DIM. BG is equal to 2.24 in. (57 mm)
- DIM. BH is equal to 7.72 in. (196 mm)
- DIM. BJ is equal to 2.32 in. (59 mm)
- DIM. BK is equal to 3.38 in. (86 mm)
- DIM. BL is equal to 2.40 in. (61 mm)
- DIM. BM is equal to 4.09 in. (104 mm)
- DIM. BN is equal to 1.73 in. (44 mm)
- DIM. BP is equal to 4.49 in. (114 mm)
- DIM. BQ is equal to 5.59 in. (142 mm)
- DIM. BR is equal to 1.30 in. (33 mm)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

- DIM. BS is equal to 4.96 in. (126 mm)
- DIM. BT is equal to 2.40 in. (61 mm)
- DIA BU is equal to 0.63 in. (16 mm)
- DIM. BV is equal to 0.47 in. (12 mm)
- DIM. BW is equal to 12.79 in. (325 mm)
- DIM. BX is equal to 3.03 in. (77 mm)
- DIM. BY is equal to 1.06 in. (27 mm)
- DIM. BZ is equal to 5.24 in. (133 mm)
- DIM. CA is equal to 2.87 in. (73 mm)
- DIM. CB is equal to 2.36 in. (60 mm)
- DIM. CC is equal to 3.82 in. (97 mm)
- DIM. CD is equal to 2.40 in. (61 mm)
- DIM. CE is equal to 3.35 in. (85 mm)
- DIM. CF is equal to 2.32 in. (59 mm)
- DIM. CJ is equal to 2.68 in. (68 mm)
- DIM. CK is equal to 1.14 in. (29 mm)
- DIM. CL is equal to 4.13 in. (105 mm)
- DIM. CM is equal to 2.91 in. (74 mm)
- DIM. CN is equal to 3.46 in. (88 mm)
- DIM. CP is equal to 9.29 in. (236 mm)
- DIM. CQ is equal to 3.78 in. (96 mm)
- DIM. CR is equal to 0.47 in. (12 mm)
- DIM. CS is equal to 1.85 in. (47 mm)
- DIM. CT is equal to 14.37 in. (365 mm)
- DIM. CU is equal to 12.20 in. (310 mm)
- DIM. CV is equal to 10.47 in. (266 mm)
- DIM. CW is equal to 1.10 in. (28 mm)
- DIM. CX is equal to 3.03 in. (77 mm)
- DIM. CY is equal to 6.93 in. (176 mm)
- DIM. CZ is equal to 1.61 in. (41 mm)
- DIM. DA is equal to 3.46 in. (88 mm)
- DIM. DB is equal to 2.68 in. (68 mm)
- DIM. DC is equal to 3.15 in. (80 mm)
- DIA DD is equal to 0.71 in. (18 mm)
- DIM. DE is equal to 3.23 in. (82 mm)
- DIM. DF is equal to 5.00 in. (127 mm)
- DIM. DG is equal to 0.94 in. (24 mm)
- DIM. DH is equal to 1.65 in. (42 mm)
- DIM. DJ is equal to 0.63 in. (16 mm)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

- DIM. DK is equal to 3.19 in. (81 mm)
- DIM. DL is equal to 4.61 in. (117 mm)
- DIM. DM is equal to 0.91 in. (23 mm)
- DIM. DN is equal to 1.02 in. (26 mm)
- DIM. DP is equal to 1.42 in. (36 mm)
- DIM. DQ is equal to 7.05 in. (179 mm)
- DIM. DR is equal to 1.38 in. (35 mm)
- DIM. DS is equal to 2.44 in. (62 mm)
- DIM. DT is equal to 2.48 in. (63 mm)
- DIM. DU is equal to 2.40 in. (61 mm)
- DIM. DV is equal to 5.47 in. (139 mm)
- DIM. DW is equal to 1.61 in. (41 mm)
- DIM. DX is equal to 3.74 in. (95 mm)
- DIM. DY is equal to 2.91 in. (74 mm)
- DIM. DZ is equal to 1.42 in. (36 mm)
- DIM. EA is equal to 3.35 in. (85 mm)
- DIM. EB is equal to 1.57 in. (40 mm).

SUBTASK 72-23-04-320-002-G00

WARNING METAL PARTICLES FROM MACHINING CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PERSONAL PROTECTION EQUIPMENT.

WARNING DO NOT LET TITANIUM DUST COLLECT. TITANIUM DUST IS VERY FLAMMABLE. DISCARD ALL TITANIUM WASTE IN AN APPROVED CONTAINER. TITANIUM DUST CAN CAUSE INJURY TO PERSONNEL.

CAUTION REMOVE MINIMUM MATERIAL MORE THAN THE DEPTH OF DEVIATION WHEN YOU BLEND. IF YOU DO NOT OBEY, DAMAGE TO THE PART CAN OCCUR.

CAUTION DO NOT USE TOOLS ON TITANIUM THAT WERE USED ON OTHER MATERIALS. IT WILL CAUSE CONTAMINATION OF THE TITANIUM. KEEP THE BLENDING ABRASIVE TOOL USED FOR THE TITANIUM ISOLATED FROM TOOLS USED FOR OTHER METALS. IF YOU DO NOT OBEY, DAMAGE TO EQUIPMENT CAN OCCUR.

CAUTION BLENDED AREAS MUST MAKE A SMOOTH CONTOUR INTO THE ADJACENT AREAS TO KEEP CORRECT AIRFLOW. IF YOU DO NOT OBEY, DAMAGE TO PARTS CAN OCCUR.

(2) Blend the applicable shroud segment, do one of the following (Figure 807 and Figure 808):

(a) Remove the indication by hand blending (TASK 70-43-42-300-802-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Make sure that the blended area makes a smooth contour into the adjacent areas to keep correct airflow.

CAUTION

MAKE SURE THAT POWER BLENDING DOES NOT SUPPLY TOO MUCH HEAT AND THERMAL STRESS. IF YOU DO NOT OBEY, TOO MUCH HEAT CAN CAUSE DAMAGE TO THE MATERIAL.

(b) Alternative procedure. Remove the indication by power blending (TASK 70-43-42-300-803-G00).

1) Make sure that the blended area makes a smooth contour into the adjacent areas to keep correct airflow.

SUBTASK 72-23-04-320-001-G00

(3) If the minimum acceptable distance between two repaired areas is less than 40 times the DIM. EC, mix them to get only one repaired area.

SUBTASK 72-23-04-110-002-G00

WARNING

ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. USE PERSONAL PROTECTION EQUIPMENT. USE IN A WELL-VENTILATED AREA.

(4) Clean the shroud segment with alcohol, B00676 [CP1041] (TASK 70-21-23-100-801-G00).

SUBTASK 72-23-04-350-002-G00

(5) Do a local swab etching of the repaired areas with titanium alloy etchant, G51485 [S1093] (TASK 70-43-24-100-802-G00).

SUBTASK 72-23-04-230-001-G00

(6) Do a fluorescent penetrant inspection of the repaired areas (TASK 70-11-06-230-803-G00 or TASK 70-11-06-230-802-G00).

(a) Use sensitivity level 2.

(b) Indications less than 0.0118 in. (0.30 mm) are non-interpretable.

(c) Indications more than 0.0118 in. (0.30 mm) are not permitted.

(d) Linear indications are not permitted.

NOTE: Linear indications are indications that are at least four times longer than they are wide.

SUBTASK 72-23-04-210-012-G00

(7) Do this task: Fan Frame Module Segment Shrouds Inspection, TASK 72-23-04-200-801-G00.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-23-04-860-019-G00

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-23-04-860-020-G00

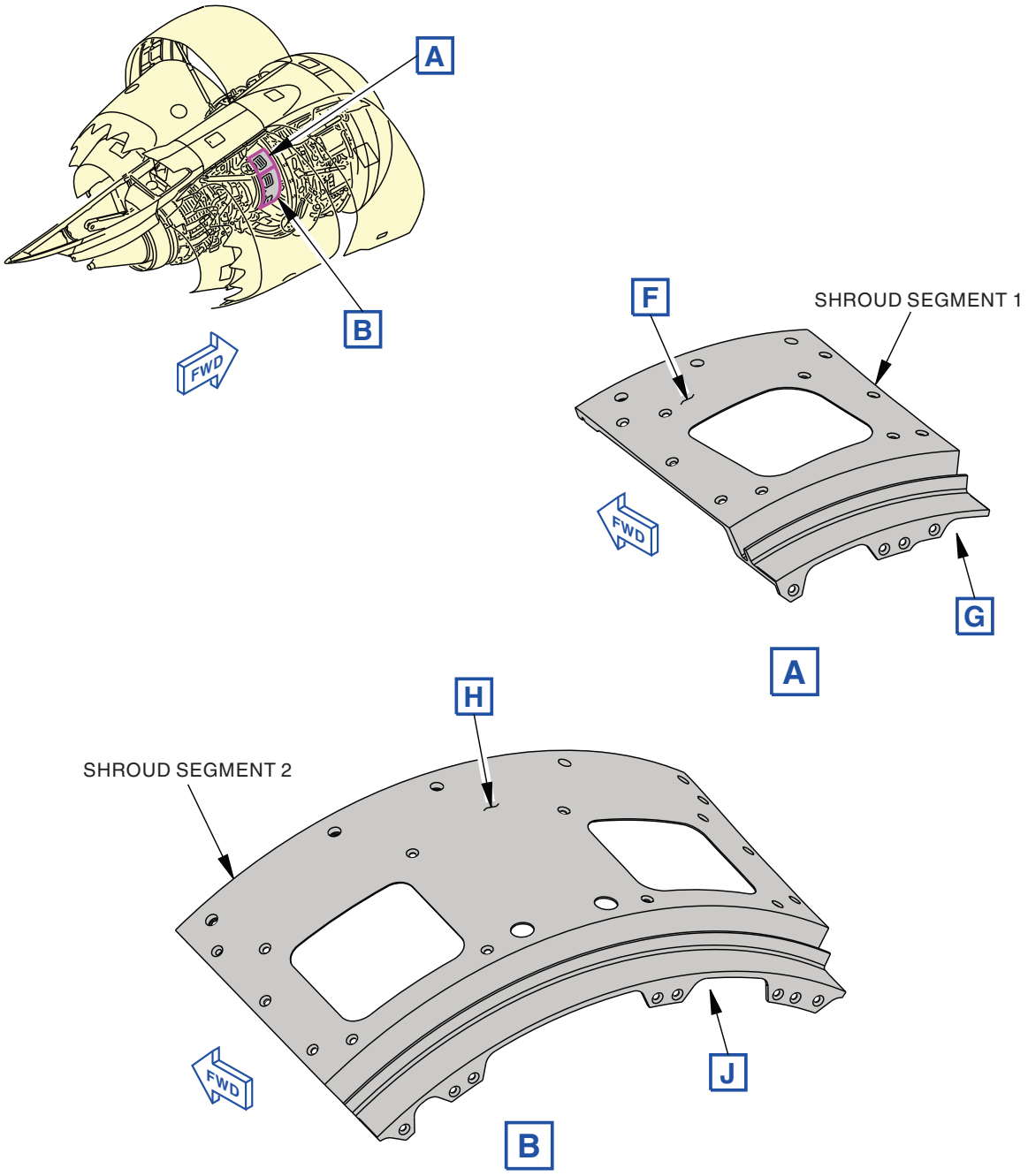
(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



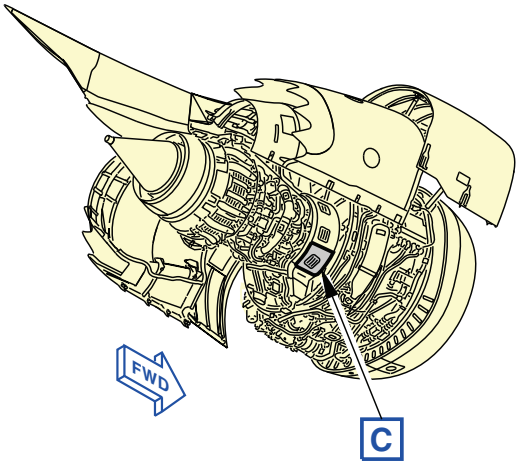
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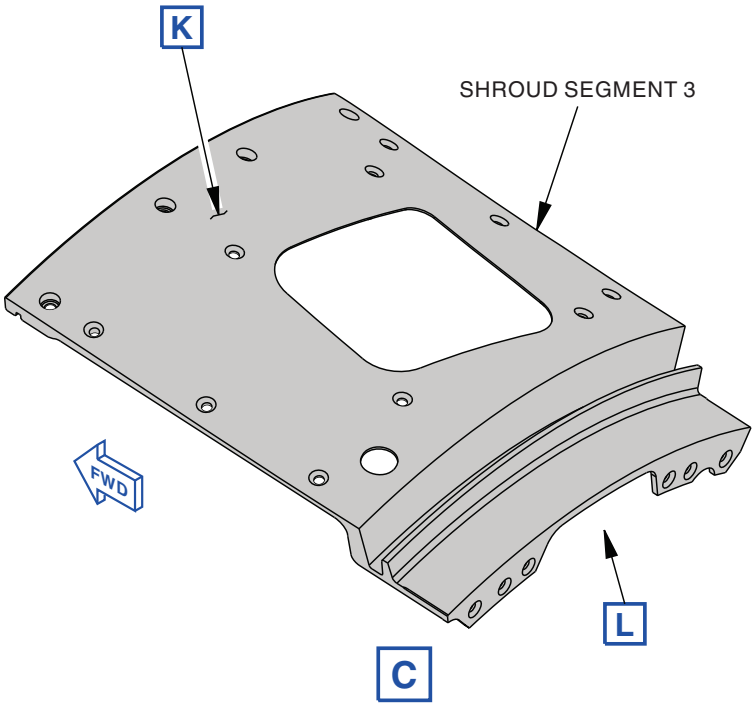
MATERIAL FOR ALL PARTS : 6.4 (TA6V)

2999339 S0000775706_V1

Fan Frame Shroud Segment Blending
Figure 807/72-23-04-990-827-G00 (Sheet 1 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



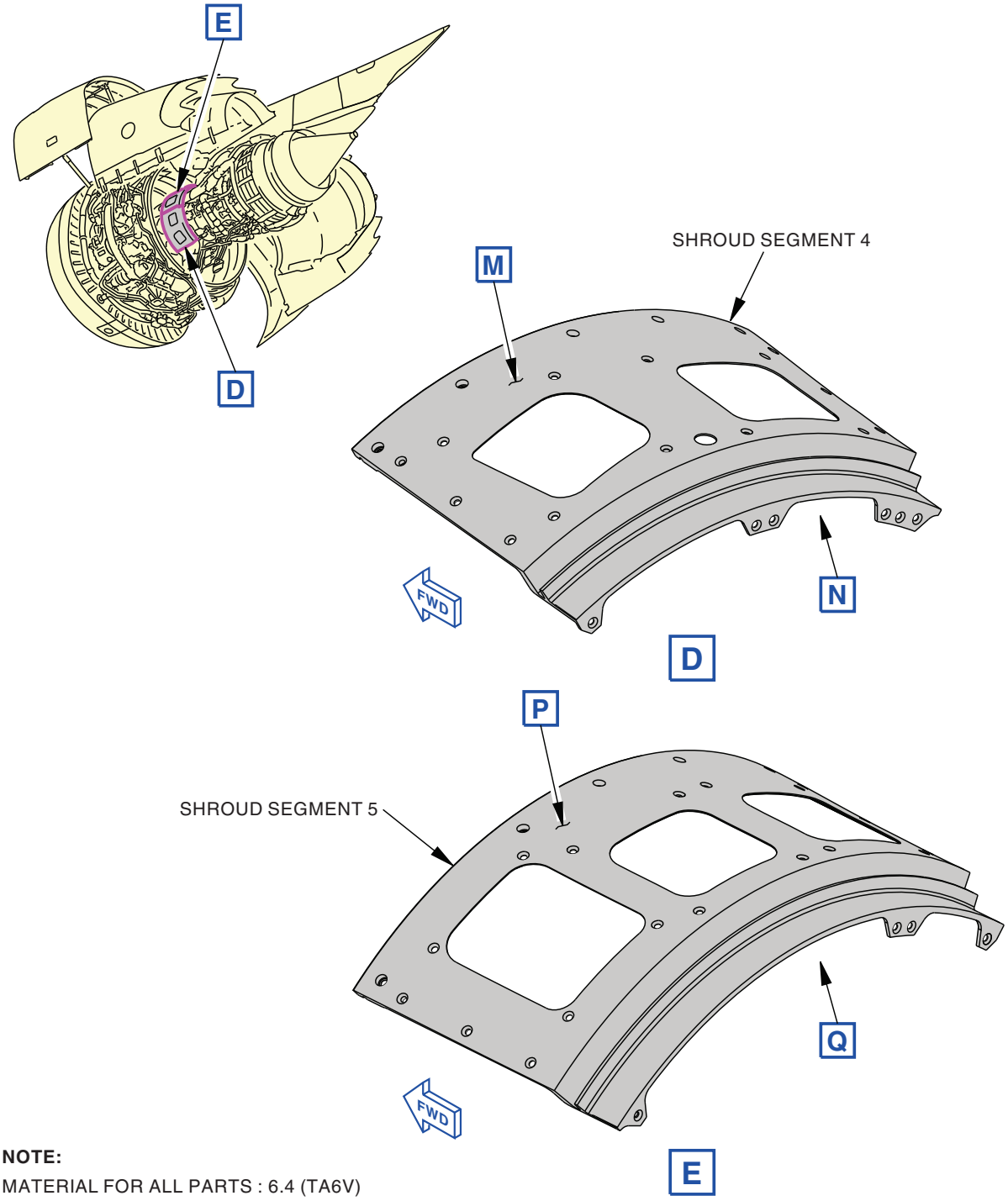


NOTE:
MATERIAL FOR ALL PARTS : 6.4 (TA6V)

3001235 S0000777244_V1

Fan Frame Shroud Segment Blending
Figure 807/72-23-04-990-827-G00 (Sheet 2 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



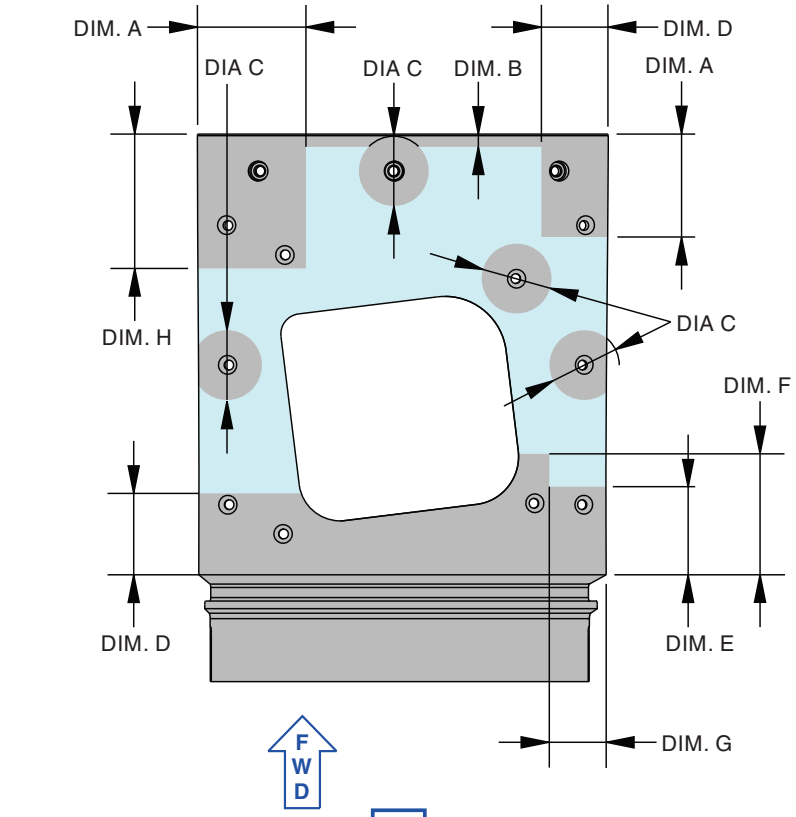
NOTE:

MATERIAL FOR ALL PARTS : 6.4 (TA6V)

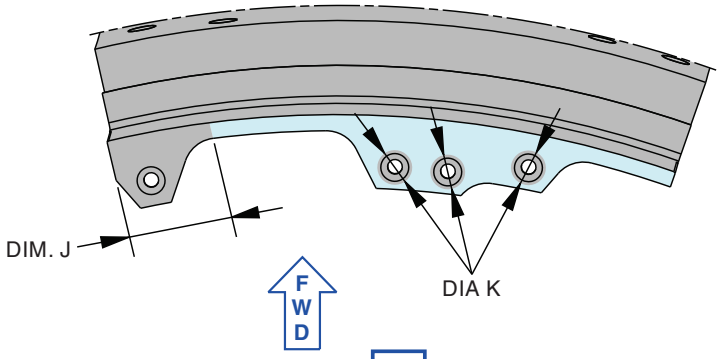
2999346 S0000775707_V1

Fan Frame Shroud Segment Blending
Figure 807/72-23-04-990-827-G00 (Sheet 3 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



F



G

NO BLENDING

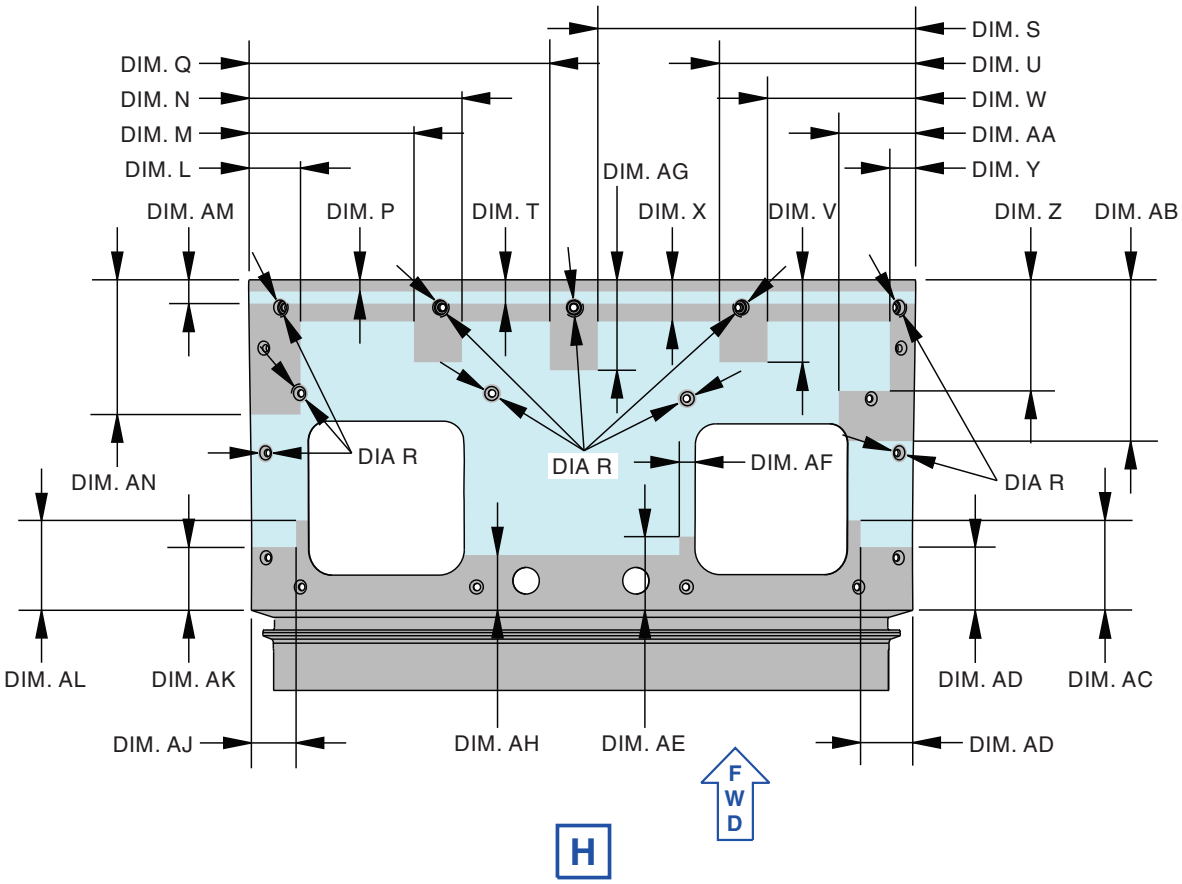
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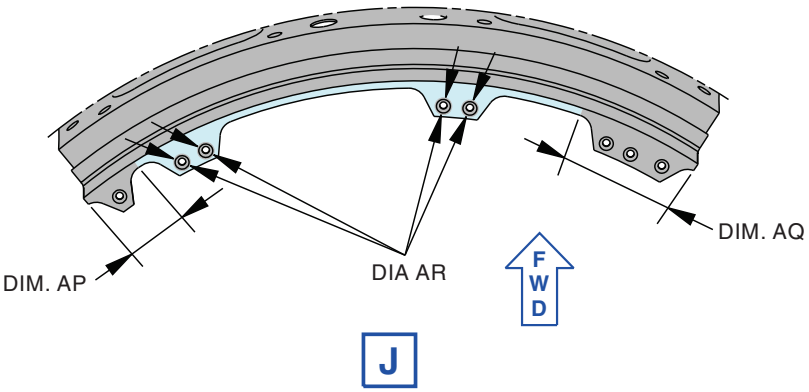
NO BLENDING PERMITTED ON INNER FACES

2999352 S0000775708_V1

Fan Frame Shroud Segment Blending
Figure 807/72-23-04-990-827-G00 (Sheet 4 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





NO BLENDING

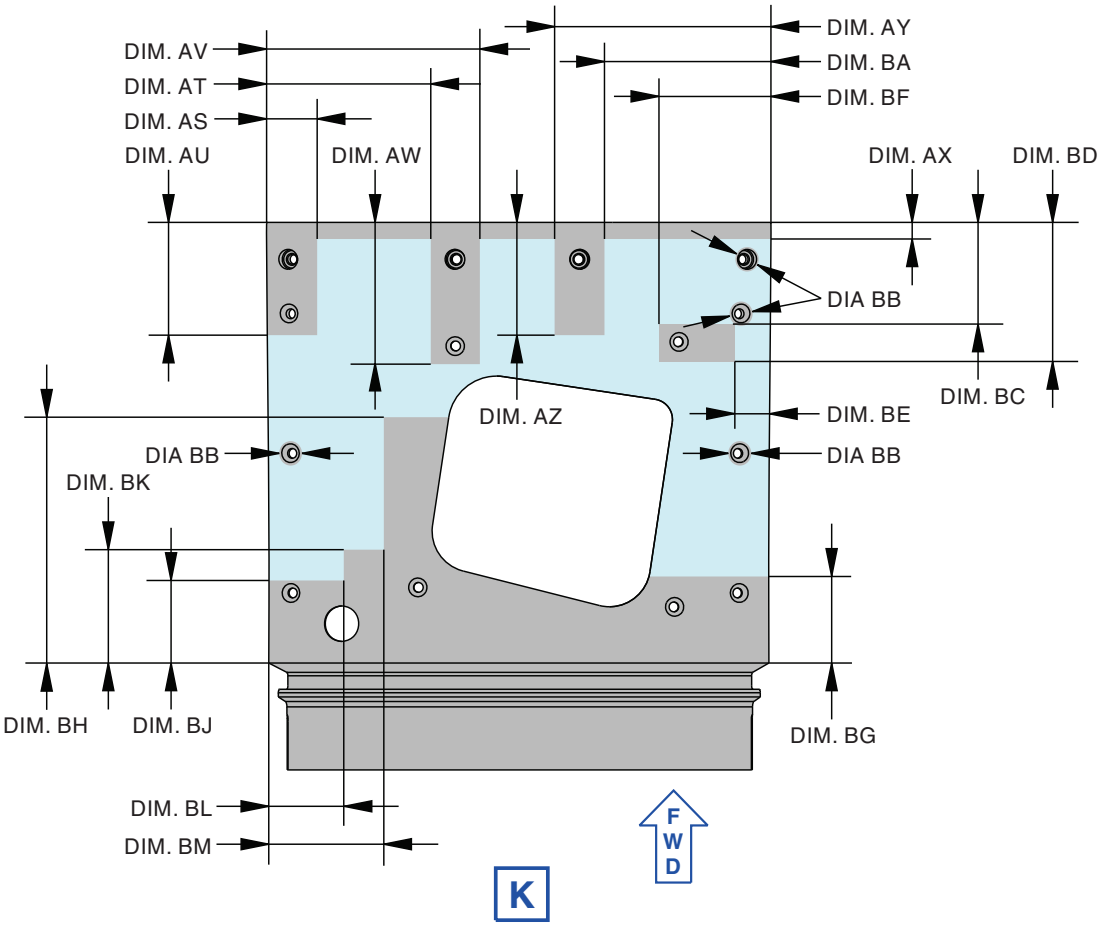
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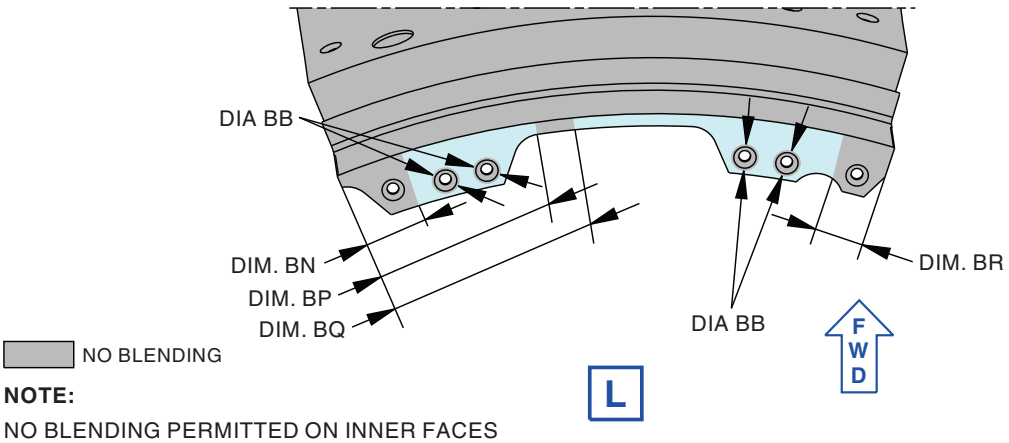
NO BLENDING PERMITTED ON INNER FACES

2999361 S0000775709_V1

Fan Frame Shroud Segment Blending
Figure 807/72-23-04-990-827-G00 (Sheet 5 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





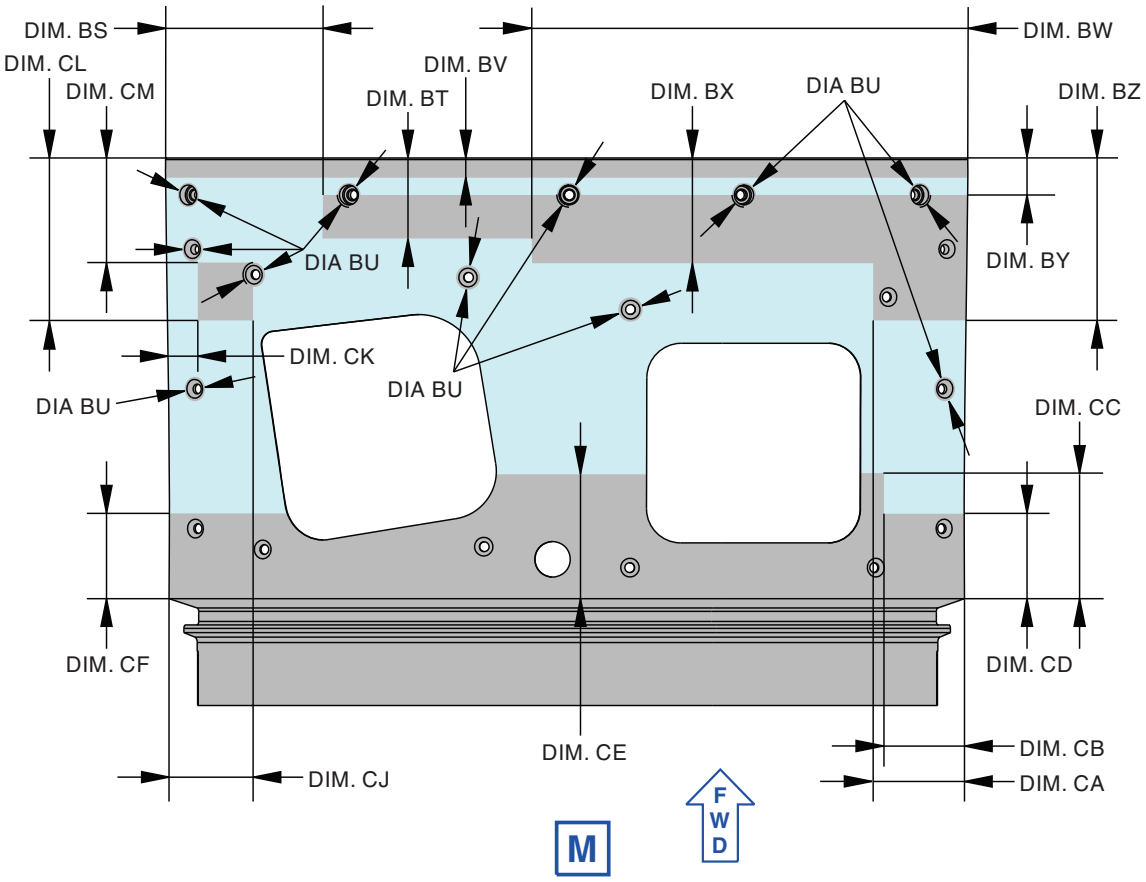
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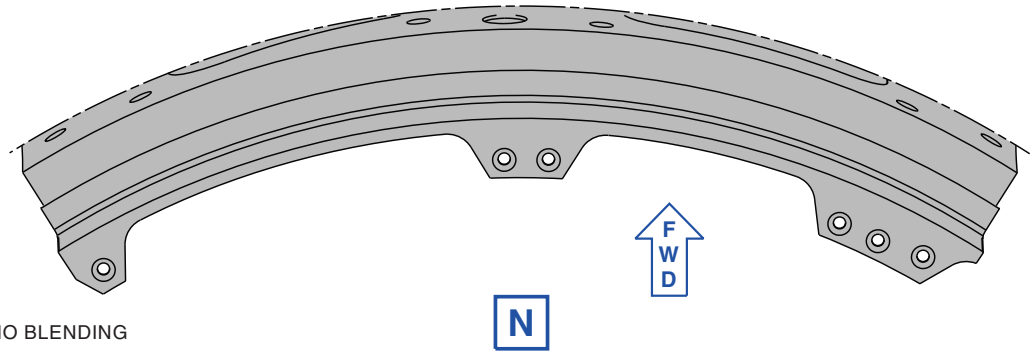
NO BLENDING PERMITTED ON INNER FACES

2999369 S0000775710_V1

Fan Frame Shroud Segment Blending
Figure 807/72-23-04-990-827-G00 (Sheet 6 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





NO BLENDING

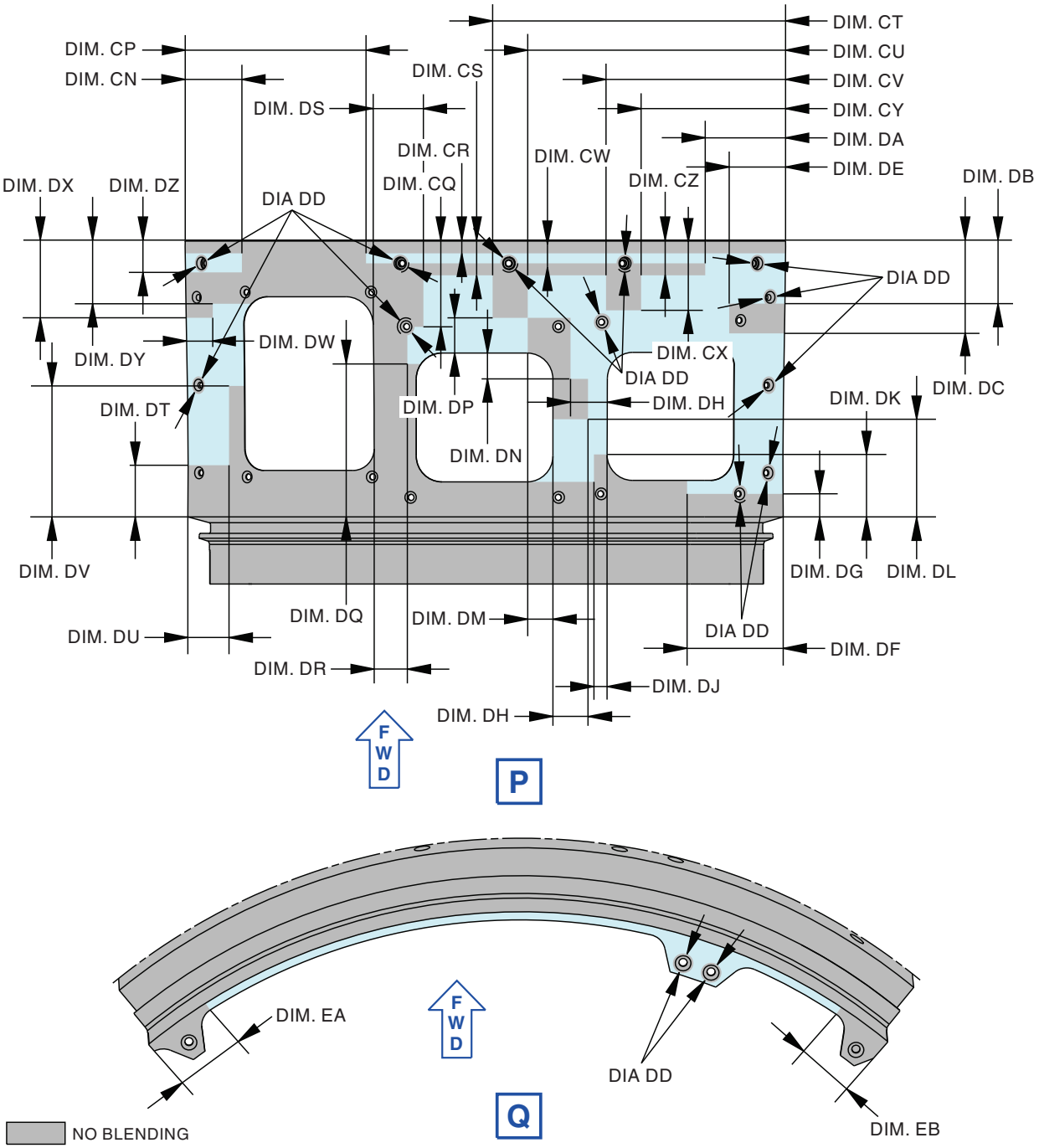
NOTE:

NO BLENDING PERMITTED ON INNER FACES

2999384 S0000775711_V1

Fan Frame Shroud Segment Blending
Figure 807/72-23-04-990-827-G00 (Sheet 7 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



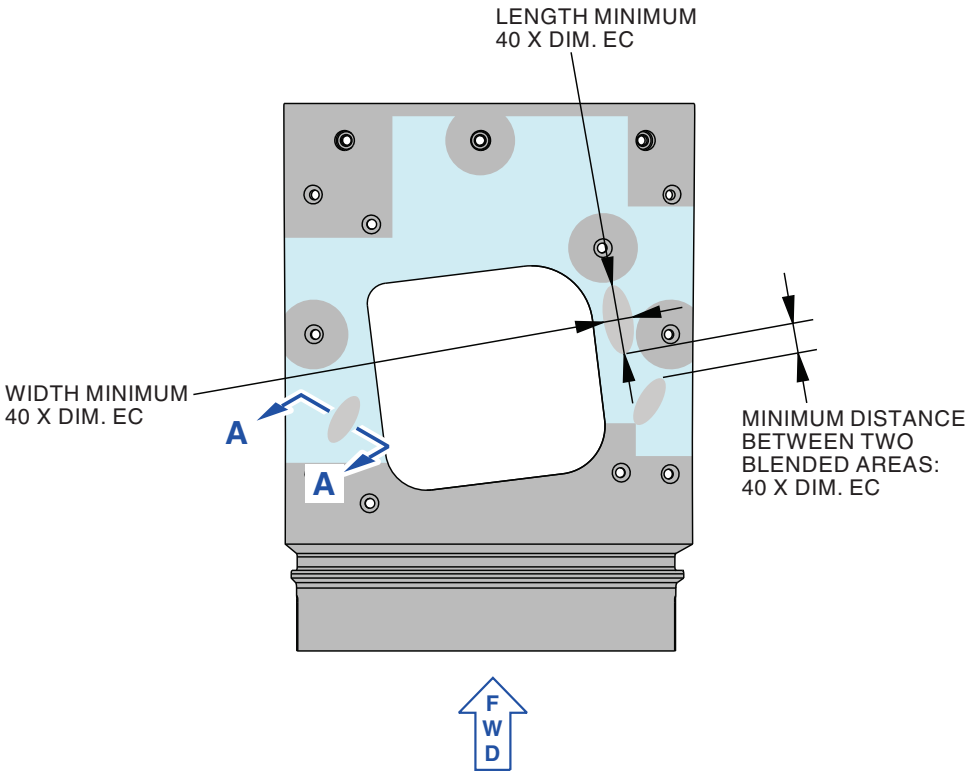
NOTE:

NO BLENDING PERMITTED ON INNER FACES

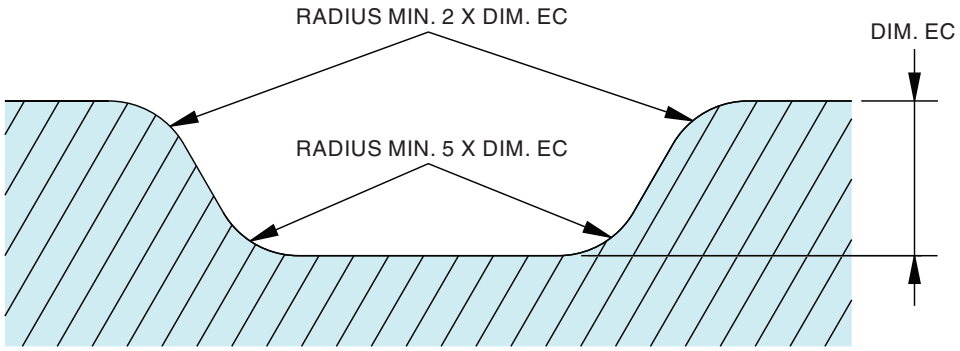
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Fan Frame Shroud Segment Blending
Figure 807/72-23-04-990-827-G00 (Sheet 8 of 8)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



SHROUD SEGMENT (EXAMPLE)



ACCEPTABLE BLENDING PROFILE

A-A

NO BLENDING

2999405 S0000775713_V1

Dimensional Information
Figure 808/72-23-04-990-828-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0374

TASK 72-23-04-960-802-G00

6. Retainer Ring Nut - Replacement

(Figure 809)

A. General

(1) This task gives the instructions to replace the nut of the Simfast floating bolt assembly or to replace Simfast floating bolt assembly of the retainer ring.

B. References

Reference	Title
70-21-23-100-801-G00	Hand Wipe Solvent Degreasing With Chemical Agent (P/B 701)
70-43-34-300-801-G00	Simfast Floating Bolt Assemblies Rivet Repair (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-23-04-200-801-G00	Fan Frame Module Segment Shrouds Inspection (P/B 601)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
B00676 [CP1041]	Alcohol - Isopropyl	

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Replacement

SUBTASK 72-23-04-860-021-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

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PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0374 (Continued)

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-23-04-010-003-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Retainer Ring Nut Replacement

SUBTASK 72-23-04-960-001-G00

(1) If it is necessary to replace the nut of the Simfast floating bolt assembly, do the following steps (Figure 809):

(a) Remove and discard the nut (TASK 70-43-34-300-801-G00).

(b) Install the nut (TASK 70-43-34-300-801-G00).

H. Simfast Floating Bolt Assembly Replacement

SUBTASK 72-23-04-960-002-G00

(1) If it is necessary to replace the Simfast floating bolt assembly, do the following steps (Figure 809):

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0374 (Continued)

WARNING METAL PARTICLES FROM MACHINING CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PERSONAL PROTECTION EQUIPMENT.

(a) Remove and discard the Simfast floating bolt assembly (TASK 70-43-34-300-801-G00).

WARNING ISOPROPYL ALCOHOL (CP 1041) IS TOXIC AND FLAMMABLE. USE PERSONAL PROTECTION EQUIPMENT. USE IN A WELL-VENTILATED AREA.

(b) Clean the area near the Simfast floating bolt assembly installation with a cloth moistened with alcohol, B00676 [CP1041] (TASK 70-21-23-100-801-G00).

(c) Do a visual inspection of the holes (TASK 72-23-04-200-801-G00).

1) Deviation is not permitted.

(d) Install the Simfast floating bolt assembly (TASK 70-43-34-300-801-G00).

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-23-04-410-003-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-23-04-860-022-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

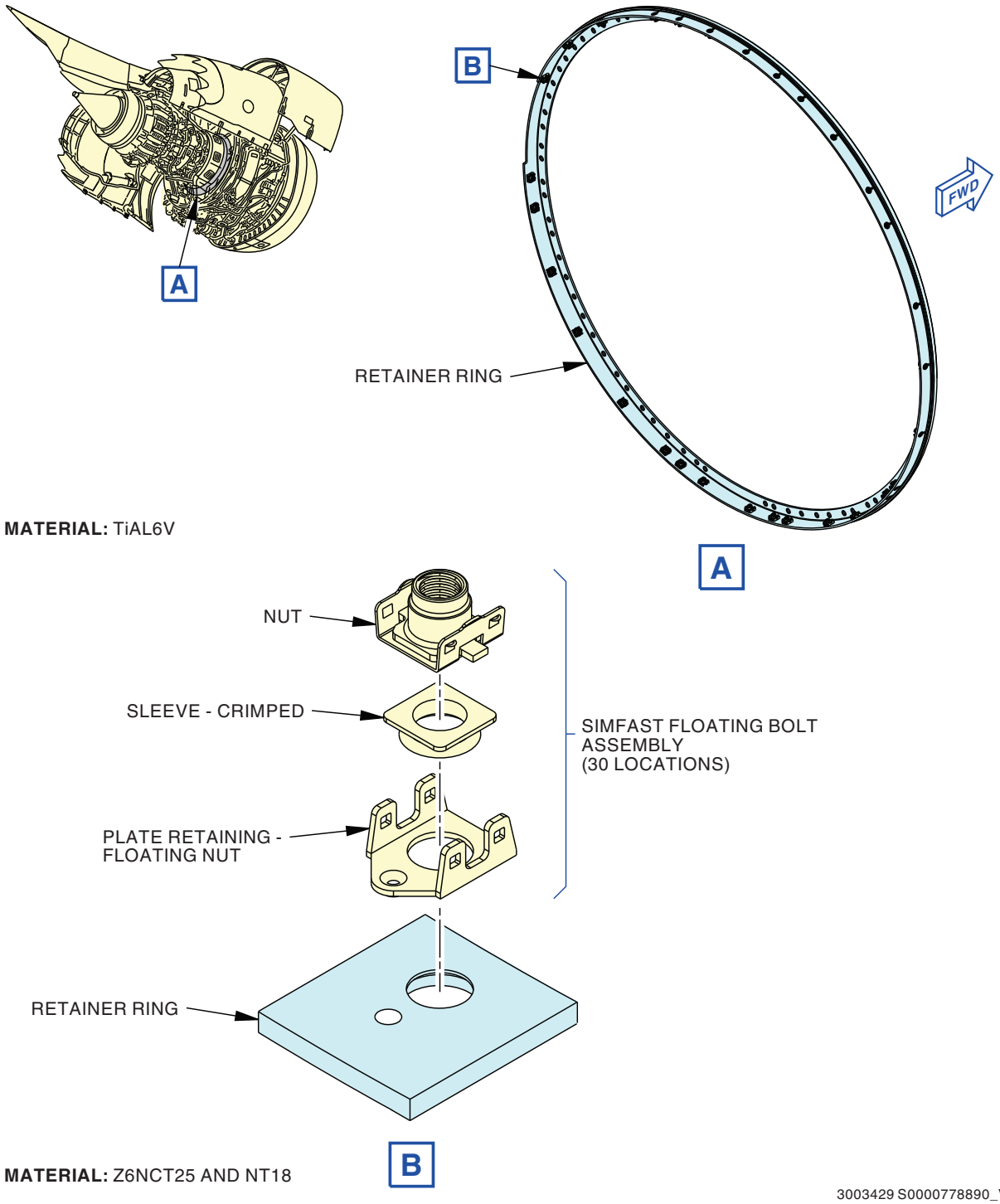
ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0374 (Continued)

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

END OF TASK

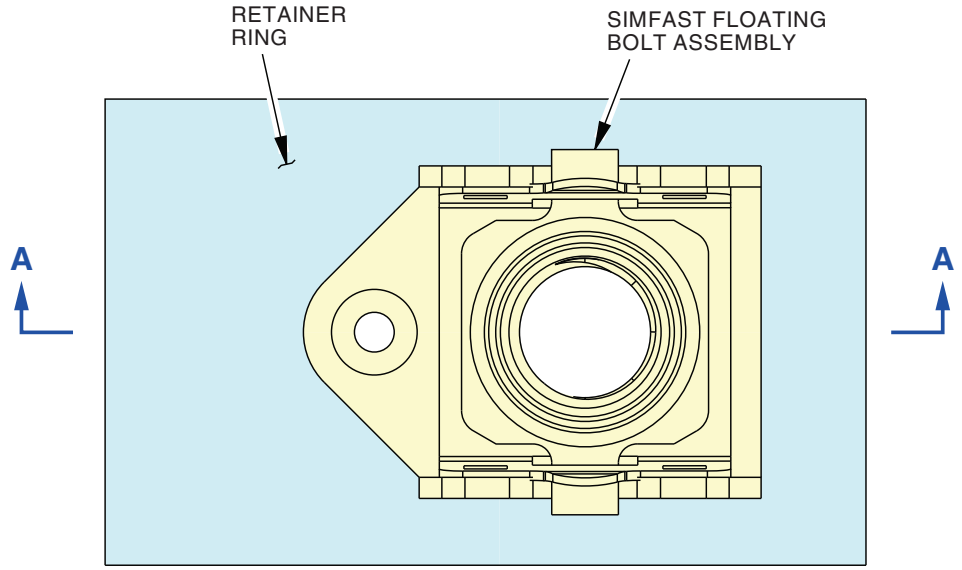
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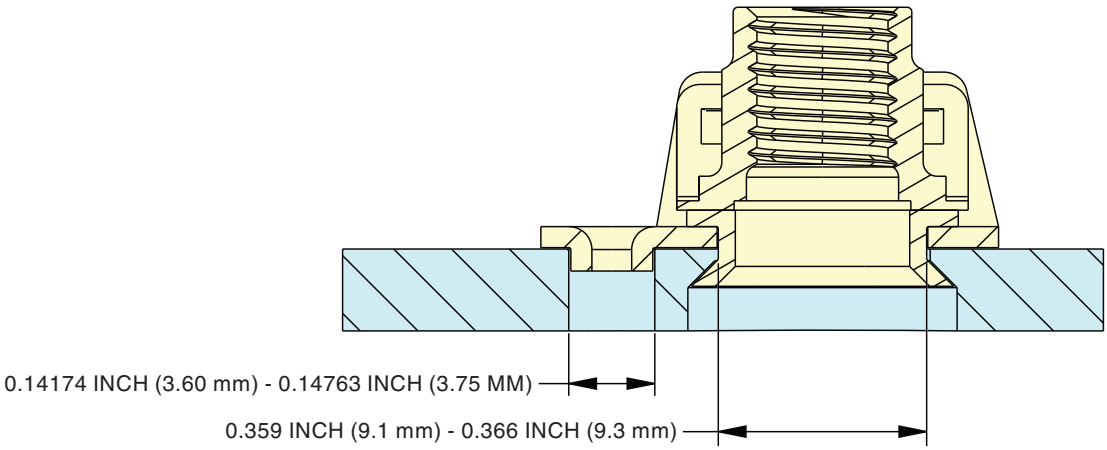
3003429 S0000778890_V1

Retainer Ring Nut Installation
Figure 809/72-23-04-990-829-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



TYPICAL VIEW:



A-A

3003435 S0000778891_V1

Retainer Ring Nut Installation
Figure 809/72-23-04-990-829-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0374

TASK 72-23-04-960-807-G00

7. Retainer Ring Riveted Nuts - Replacement

(Figure 810)

A. General

(1) This task gives instructions to replace the two-lug nut plates on the retainer ring.

(2) The table below gives a list of part numbers that are applicable to this repair.

NOTE: All part numbers are applicable to all paragraphs unless specified differently.

Table 803/72-23-04-993-812-G00 Applicable Part Numbers

Part Number	Designation
364-081-415-0	Retainer Ring

(3) The table below gives dimensional information for the repair application.

Table 804/72-23-04-993-813-G00 Dimensional Information

Description	Minimum In-Process Dimension	Maximum In-Process Dimension	Minimum Finish Dimension	Maximum Finish Dimension	Reference Dimension
DIA A	-	-	0.0969 in. (2.46 mm)	0.1007 in. (2.56 mm)	-

B. References

Reference	Title
70-21-23-100-801-G00	Hand Wipe Solvent Degreasing With Chemical Agent (P/B 701)
70-43-31-300-801-G00	Blind Rivet Assemblies Repair (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0374 (Continued)

(Continued)

Number	Name/Location
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Replacement

SUBTASK 72-23-04-860-031-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-23-04-010-008-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0374 (Continued)

(Continued)

Number	Name/Location
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Retainer Ring Riveted Nuts Replacement

SUBTASK 72-23-04-300-001-G00

WARNING USE PERSONAL PROTECTIVE EQUIPMENT WHEN YOU USE A DRILL. THE PROCEDURE MAKES METAL PARTICLES. PARTICLES CAN GET IN YOUR EYES, LUNGS, OR ON YOUR SKIN. IF YOU DO NOT OBEY, INJURY CAN OCCUR.

CAUTION DO NOT CAUSE DAMAGE TO THE RIVET HOLES AND THE ADJACENT SURFACES WHEN YOU REMOVE THE RIVETS FROM THE RETAINER RING. IF YOU DO NOT OBEY THIS INSTRUCTION, YOU CAN CAUSE DAMAGE TO THE PART.

CAUTION DO NOT CAUSE HEAT THAT CAUSES A THERMAL TENSION DURING THE OPERATION OF POWER TOOLS. IF YOU DO NOT OBEY, YOU CAN CAUSE DAMAGE TO THE PART.

(1) Remove and discard the two-lug nut plate and its fasteners (TASK 70-43-31-300-801-G00).

NOTE: The fasteners are 100° flush head blind rivets.

SUBTASK 72-23-04-110-003-G00

(2) Clean the holes, the chamfers, and the adjacent area (TASK 70-21-23-100-801-G00).

SUBTASK 72-23-04-200-001-G00

(3) Examine the holes, the chamfers, and the adjacent area as follows:

(a) Do a visual inspection (TASK 70-43-31-300-801-G00).

1) No deviation is permitted.

(b) Do a dimensional inspection of DIA A (Table 804 and TASK 70-43-31-300-801-G00).

SUBTASK 72-23-04-420-009-G00

CAUTION USE ONLY INITIAL DIMENSION RIVETS WHEN YOU INSTALL A TWO-LUG NUT PLATE ON THE RETAINER RING. DO NOT USE RIVETS THAT ARE LARGER THAN THE INITIAL DIMENSION. IF YOU DO NOT OBEY THE INSTRUCTION, YOU CAN CAUSE DAMAGE TO THE PART.

(4) Install a two-lug nut plate and its fasteners (TASK 70-43-31-300-801-G00).

NOTE: The fasteners are 100° flush head blind rivets.

(a) Make sure that the top of countersunk head of each rivet is not above the external surface of the flange.

SUBTASK 72-23-04-200-002-G00

(5) Do a visual inspection of the installed two-lug nut plate and its fasteners (TASK 70-43-31-300-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0374 (Continued)

H. Put the Airplane Back to Its Usual Condition

SUBTASK 72-23-04-410-007-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-23-04-860-032-G00

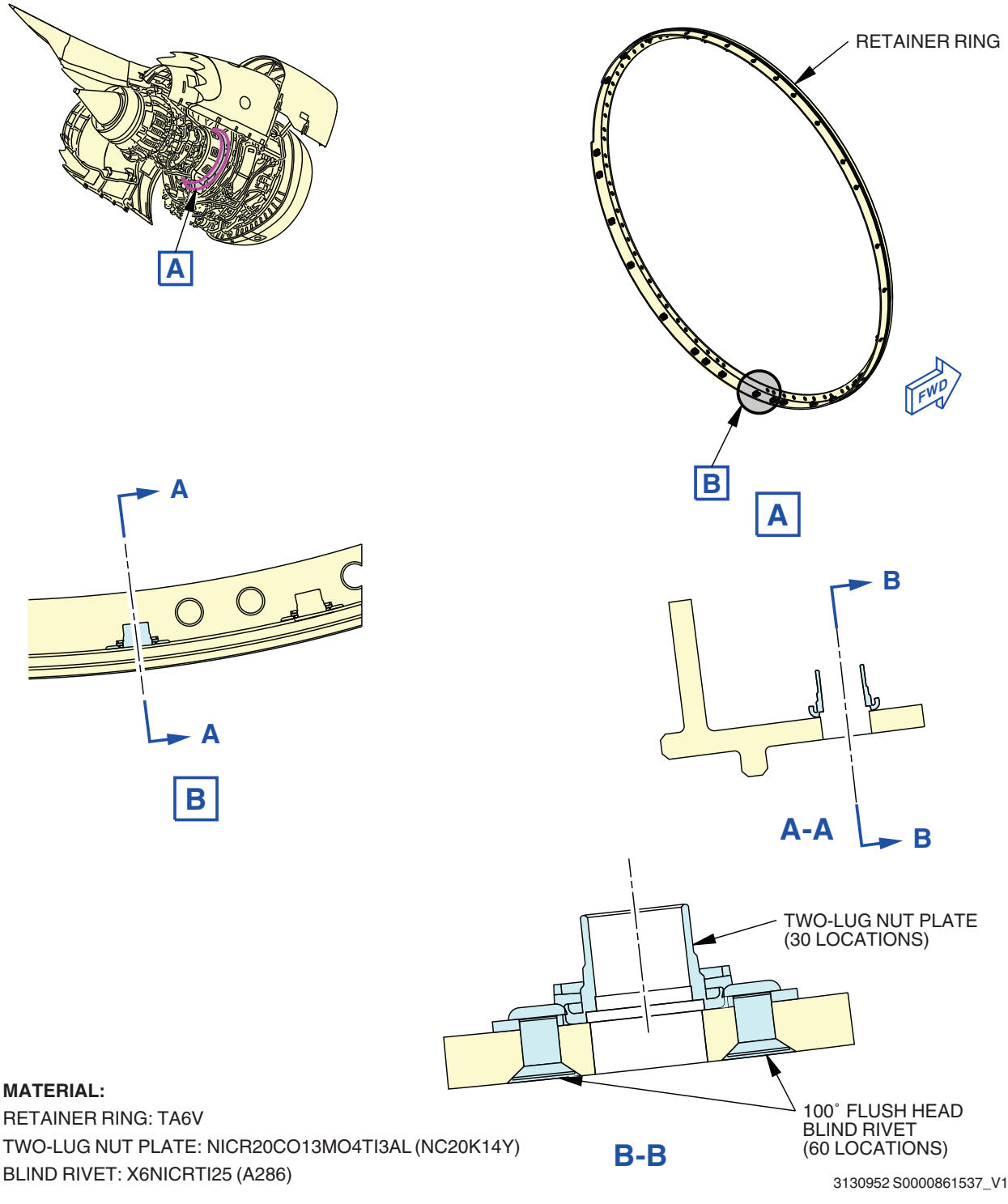
(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

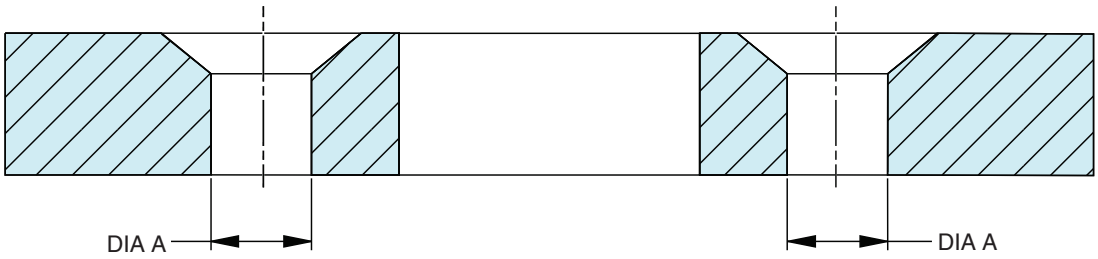


MATERIAL:
RETAINER RING: TA6V
TWO-LUG NUT PLATE: NICR20CO13MO4TI3AL (NC20K14Y)
BLIND RIVET: X6NICRTI25 (A286)

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Two-Lug Nut Plate Replacement
Figure 810/72-23-04-990-834-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(TWO-LUG NUT PLATE AND BLIND RIVETS ARE NOT SHOWN)

B-B

3130926 S0000861538_V1

Two-Lug Nut Plate Replacement
Figure 810/72-23-04-990-834-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL

TASK 72-23-04-960-803-G00

8. Fan Frame Shroud Segment Coating Replacement

(Figure 811)

A. General

(1) This task gives instructions to do a spot touch-up of the paint, or a full repair of the Polytetraflouroethylene (Teflon) (PTFE) lubricant varnish.

(2) The Table 805 gives a list of the part numbers that are applicable to this repair. All part numbers are applicable to all paragraphs unless specified differently.

Table 805/72-23-04-993-807-G00

Part Number	Designation
364-030-610-0	Shroud, segment 1
364-030-710-0	Shroud, segment 2
364-030-810-0	Shroud, segment 3
364-030-910-0	Shroud, segment 4
364-032-610-0	Shroud, segment 5

B. References

Reference	Title
70-21-23-100-801-G00	Hand Wipe Solvent Degreasing With Chemical Agent (P/B 701)
70-63-00-390-801-G00	PTFE Lubricant Varnish Dry Film Application (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-23-04-200-801-G00	Fan Frame Module Segment Shrouds Inspection (P/B 601)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
A50568 [CP2233]	Tape - Masking - 3M 250	
B00676 [CP1041]	Alcohol - Isopropyl	
B01058 [CP1039]	Solvent - Acetone, Reagent Grade	
B50190 [CP1040]	Solvent - Methyl Ethyl Ketone (MEK)	ASTM D740
G50766 [CP2240]	Sheet - Abrasive Sheet For Cleaning Up And Finishing, Scotch-Brite Type A CFS	
G51412 [CP2721]	Cloth - Abrasive, Silicone Carbide - CS 333X	

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Location Zones

Zone	Area
411	Engine 1 - Engine
413	Engine 1 - Fan Cowl, Left
414	Engine 1 - Fan Cowl, Right
415	Engine 1 - Thrust Reverser, Left
416	Engine 1 - Thrust Reverser, Right
421	Engine 2 - Engine
423	Engine 2 - Fan Cowl, Left
424	Engine 2 - Fan Cowl, Right
425	Engine 2 - Thrust Reverser, Left
426	Engine 2 - Thrust Reverser, Right

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Prepare for the Replacement

SUBTASK 72-23-04-860-023-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-23-04-860-024-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Shroud Segment Coating Replacement

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

WARNING USE PROTECTIVE GLOVES TO PREVENT INJURY FROM SHARP EDGES AND ROUGH SURFACES. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

WARNING DO NOT LET TITANIUM DUST COLLECT. TITANIUM DUST IS VERY FLAMMABLE. DISCARD ALL TITANIUM WASTE IN AN APPROVED CONTAINER. TITANIUM DUST CAN CAUSE INJURY TO PERSONNEL.

WARNING DO NOT USE WATER TO EXTINGUISH A TITANIUM FIRE. EXTINGUISH WITH AN APPROVED CHEMICAL FOR METAL FIRES. IF YOU DO NOT OBEY, OPERATION OF A WATER EXTINGUISHER COULD CAUSE INJURY TO PERSONNEL.

CAUTION KEEP CHLORINATED HYDROCARBONS AND HALOGEN MATERIALS AWAY FROM TITANIUM PARTS. IF YOU DO NOT OBEY, CHLORINATED HYDROCARBONS AND HALOGEN MATERIALS CAN CAUSE THE DAMAGE TO THE TITANIUM PARTS.

CAUTION DO NOT USE TOOLS FIXTURE MADE FROM OR COATED WITH CADMIUM, BISMUTH, TIN, ZINC, GOLD OR SILVER TO TITANIUM PARTS. IF YOU DO NOT OBEY, DAMAGE TO THE TITANIUM PARTS CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION

DO NOT USE TOOLS OR MATERIALS TO MIX MATERIALS OTHER THAN TITANIUM. IF YOU DO NOT OBEY, DAMAGE TO THE TITANIUM CAN OCCUR.

CAUTION

DO NOT CAUSE HEAT THAT WILL COME IN THERMAL STRESS DURING THE OPERATION OF POWER TOOLS ON THE ENGINE. IF YOU DO NOT OBEY, THERMAL STRESS CAN CAUSE ENGINE DAMAGE.

SUBTASK 72-23-04-100-002-G00

(1) Clean the area to be repaired with acetone solvent, B01058 [CP1039], or solvent, B50190 [CP1040], or alcohol, B00676 [CP1041] (TASK 70-21-23-100-801-G00).

SUBTASK 72-23-04-380-002-G00

WARNING

DO NOT BREATHE THE PARTICLES WHEN SANDING. THE PARTICLES CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PERSONAL PROTECTION EQUIPMENT. USE LOCAL MECHANICAL EXHAUST VENTILATION OR AN APPROVED RESPIRATOR.

CAUTION

DO NOT REMOVE OR CHANGE THE SHAPE OR ROUGHNESS OF THE V-GROOVE ON THE FAN FRAM SHROUD. IF YOU DO NOT OBEY, DAMAGE TO THE FAN FRAM SHROUD CAN OCCUR.

CAUTION

DO NOT APPLY A PROCEDURE THAT CAN CHANGE THE PARENT MATERIAL PROPERTIES DURING THE REMOVAL OF THE BONDED MATERIAL. IF YOU DO NOT OBEY, DAMAGE TO THE MATERIAL CAN OCCUR.

(2) Prepare the surface where you will apply the PTFE varnish, do these steps:

(a) If it is necessary, apply mask on the area around the area to be repaired.

(b) To do the repair, do one of the following methods:

1) For a spot touch repair: do a soft hand sanding with abrasive cloth, G51412 [CP2721] (CS 333 X grit size from 240 to 600), on the area to be repaired.

2) For a full repair: remove the remaining PTFE varnish with Scotch-Brite Type A CFS Sheet, G50766 [CP2240].

a) Do a soft hand sanding with abrasive cloth, G51412 [CP2721] (CS 333 X grit size from 240 to 600), on the area to be repaired.

(c) If it is necessary, remove the mask.

(d) Clean the area with acetone solvent, B01058 [CP1039], or solvent, B50190 [CP1040], or alcohol, B00676 [CP1041] (TASK 70-21-23-100-801-G00).

(e) Dry the area with clean, dry, and oil-free air.

(f) Do a visual inspection of the area where you will apply the PTFE varnish (TASK 72-23-04-200-801-G00).

SUBTASK 72-23-04-350-003-G00

(3) Apply the PTFE varnish as follows (View G, Figure 811) (TASK 70-63-00-390-801-G00):

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(a) To check quality assurance, make a test piece made of TA6V or stainless steel according to type Z10CNT18 piece along with the part to be varnished.

(b) Prepare the surface of the test piece as follows:

1) Do a light hand sanding with abrasive cloth, G51412 [CP2721] (CS 333 X grit size from 240 to 600), on the area to be tested.

2) Clean the area to be tested with acetone solvent, B01058 [CP1039], or solvent, B50190 [CP1040], or alcohol, B00676 [CP1041] (TASK 70-21-23-100-801-G00).

3) Dry the area to be tested with clean, dry, and oil-free air.

(c) If it is necessary, mask the area with a suitable masking tape around the area to be repaired.

(d) Make sure to get the following viscosity:

1) AFNOR cup (consistometer) No. 4 of 25±5 seconds, or

2) FORD cup No. 4 of 21±5 seconds.

(e) Apply one layer of PTFE varnish.

1) Make sure that the thickness is between 0.00078 in. (0.02 mm) to 0.00118 in. (0.03 mm) on the area(s) to be repaired and on the test piece.

(f) If it is necessary, remove masking, overspray, or unwanted varnish with a blade or with acetone solvent, B01058 [CP1039], or solvent, B50190 [CP1040], or alcohol, B00676 [CP1041] (TASK 70-21-23-100-801-G00).

(g) Dry the PTFE varnish lubricant with one of the following methods:

1) Dry the lubricant at 140 ±41°F (60 ±23°C)) during 2 hours minimum and 2 hours 30 minute maximum.

2) Dry the lubricant at ambient temperature during 48 hours minimum.

SUBTASK 72-23-04-220-003-G00

(4) Do a visual examination.

(a) Make sure that the varnished surfaces are smooth, uniform, and free from sags, blisters, flakes or foreign material.

NOTE: The coating should be light green.

(b) Smooth transition area between the repaired area and the adjacent areas is required.

(c) If the varnish thickness is in the permitted limits, streaks are acceptable

(d) If the varnished surfaces fail any of these tests or examinations, remove the coating and reprocess the part.

SUBTASK 72-23-04-280-002-G00

(5) Do the following tests:

(a) On the test piece, do these steps:

1) Measure the thickness of the varnish.

a) Make sure that the thickness is between 0.00078 in. (0.02 mm) to 0.00118 in. (0.03 mm) on the area(s) to be repaired and on the test piece.

2) Use a sharp blade to scrape the lubricant varnish.

a) Make sure that scrapings appear as a powder.

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3) Criss-cross the area, then apply and press on a strip of pressure sensitive 3M 250 tape, A50568 [CP2233], to the test piece.

NOTE: Refer to ISO2409 norm.

a) Make sure that lubricant varnish does not peel when tape is pressed and pulled off.

(b) On the repaired area, do these steps:

1) Make sure that the maximum PTFE varnish roughness of the surface is 63 microinches (1.6 micrometers).

2) Apply and press on a strip of pressure sensitive 3M 250 tape, A50568 [CP2233], on the repaired area.

NOTE: Lubricant varnish shall not peel when tape is removed.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-23-04-860-025-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-23-04-860-026-G00

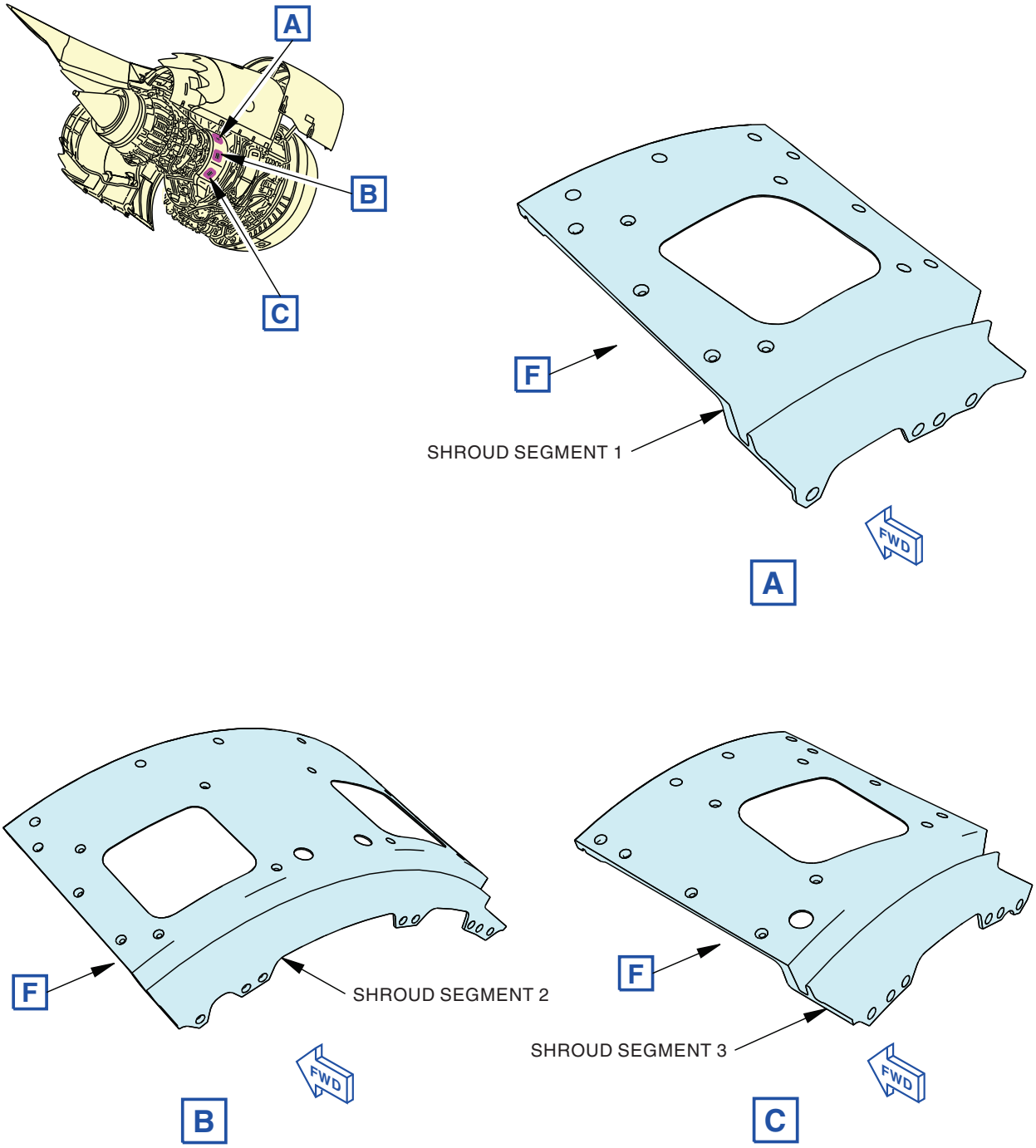
(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

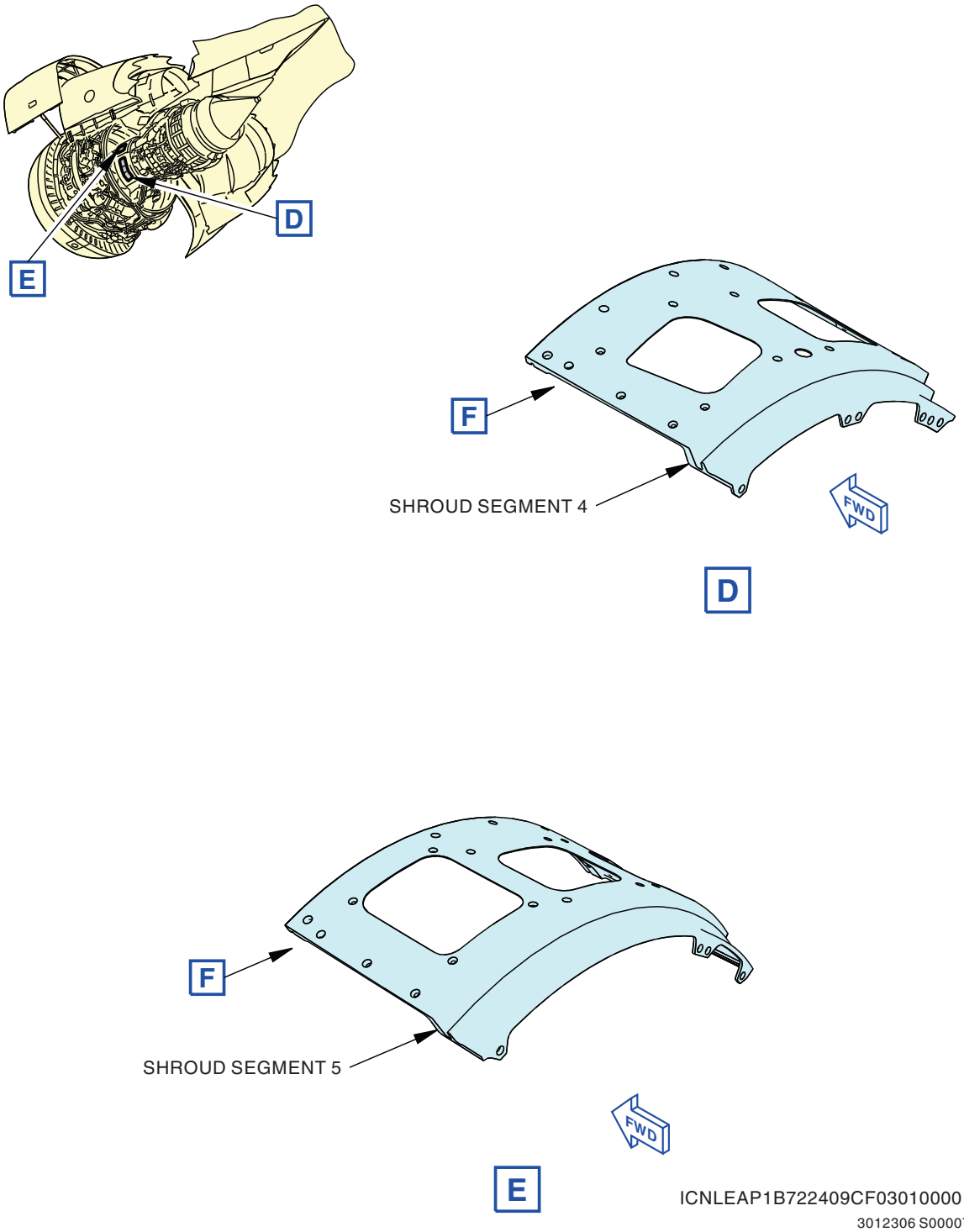


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Fan Frame Shroud Segment Coating Replacement
Figure 811/72-23-04-990-830-G00 (Sheet 1 of 3)

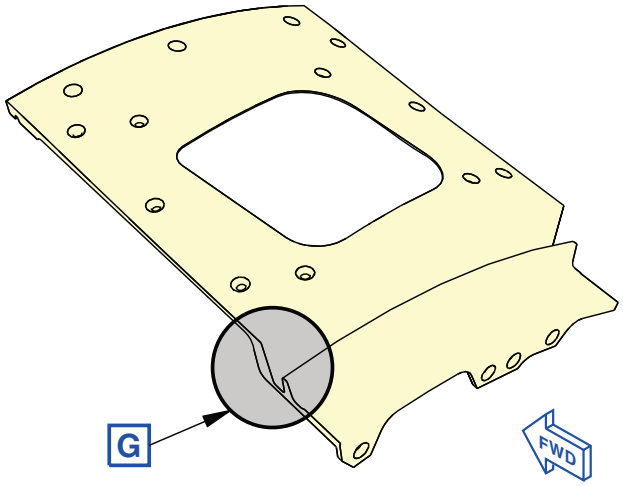
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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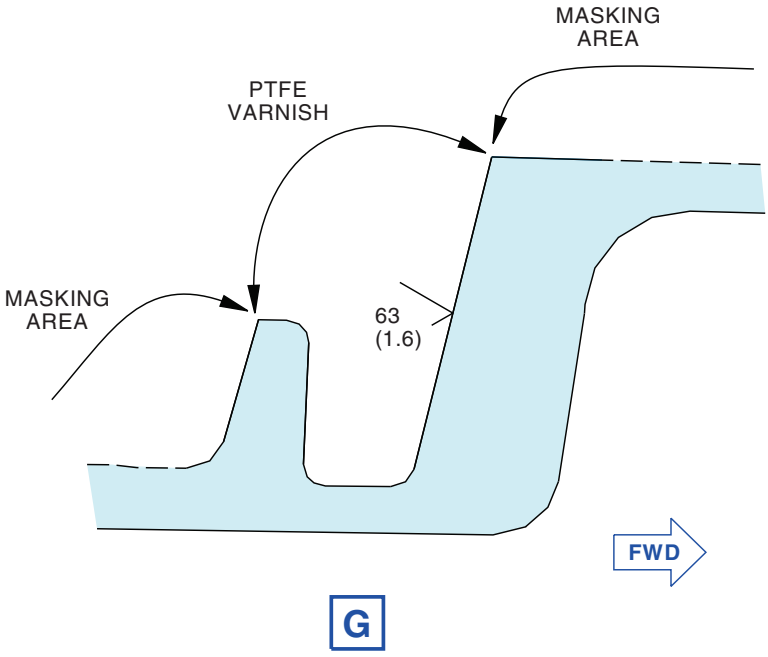
Fan Frame Shroud Segment Coating Replacement
Figure 811/72-23-04-990-830-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



(EXAMPLE)

F



G

NOTE:

ROUGHNESS IS IN MICROINCHES WITH MICROMETERS IN PARENTHESES.

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Fan Frame Shroud Segment Coating Replacement
Figure 811/72-23-04-990-830-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-23-04-960-804-G00

9. Variable Bleed Valve (VBV) Duct Nut Replacement

(Figure 812)

A. General

(1) This task gives instructions to replace the nut of the SIMFAST floating bolt assembly or to replace the SIMFAST floating bolt assembly or to replace the riveted self-locking nut assembly on the Variable Bleed Valve (VBV) duct.

(2) The table below gives a list of the part numbers that are applicable to this repair. All part numbers are applicable to all paragraphs unless specified differently.

Table 806/72-23-04-993-809-G00

Part number	Service Bulletin	CSN	Designation
364-071-511-0	-	(15–770, 72-00-00)	VBV DUCT 1
364-071-711-0	-	(15–250, 72-00-00)	VBV DUCT 2
364-079-811-0	-	(15–320, 72-00-00)	VBV DUCT 3
364-080-011-0	-	(15–610, 72-00-00)	VBV DUCT 4
364-080-111-0	-	(18–560, 72-00-00)	VBV DUCT 5
364-080-211-0	-	(18–490, 72-00-00)	VBV DUCT 6
364-080-311-0	-	(18–220, 72-00-00)	VBV DUCT 7
364-080-411-0	-	(18–130, 72-00-00)	VBV DUCT 8

B. References

Reference	Title
70-21-23-100-801-G00	Hand Wipe Solvent Degreasing With Chemical Agent (P/B 701)
70-43-11-300-801-G00	Formed Rivet Assemblies Repair (P/B 201)
70-43-34-300-801-G00	Simfast Floating Bolt Assemblies Rivet Repair (P/B 201)
70-44-50-320-801-G00	Drilling (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
B00676 [CP1041]	Alcohol - Isopropyl	

ECCN 9E991 BOEING PROPRIETARY - See title page for details

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Prepare for the Replacement

SUBTASK 72-23-04-840-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-23-04-010-004-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Variable Bleed Valve (VBV) Duct Nut Replacement

SUBTASK 72-23-04-960-003-G00

WARNING METAL PARTICLES FROM MACHINING CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PERSONAL PROTECTION EQUIPMENT.

CAUTION DO NOT TOUCH THE PART WHEN YOU REMOVE THE SIMFAST FLOATING BOLT ASSEMBLY. DAMAGE TO THE PART CAN OCCUR.

(1) Replace the SIMFAST floating bolt assembly as follows:

(a) If it is necessary to replace the nut of the SIMFAST floating bolt assembly, do these steps:

1) Remove and discard the nut (TASK 70-43-34-300-801-G00).

2) Install the nut (TASK 70-43-34-300-801-G00).

(b) If it is necessary to replace the SIMFAST floating bolt assembly, do these steps:

1) Remove and discard the SIMFAST floating bolt assembly (TASK 70-43-34-300-801-G00).

2) Remove the particles from the part.

WARNING ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. USE PERSONAL PROTECTION EQUIPMENT. USE IN A WELL-VENTILATED AREA.

3) Clean the SIMFAST floating bolt assembly installation area with a cloth moistened with alcohol, B00676 [CP1041] (TASK 70-21-23-100-801-G00).

4) Do a visual inspection of the hole.

a) Wears, tears, dents, and nicks are not permitted.

5) Install the SIMFAST floating bolt assembly (TASK 70-43-34-300-801-G00).

SUBTASK 72-23-04-960-004-G00

(2) Do these steps to replace the riveted self-locking nut:

(a) Remove and discard the rivets that attach the self-locking nut (TASK 70-43-11-300-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION

DO NOT TOUCH THE ADJACENT PARTS WHEN YOU DRILL OR GRIND THE RIVET HEAD. DAMAGE TO THE PARTS CAN OCCUR.

(b) Drill the rivet head with a 0.126 in. (3.2 mm) diameter (TASK 70-44-50-320-801-G00).

(c) Remove the particles from the part.

WARNING

ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. USE PERSONAL PROTECTION EQUIPMENT. USE IN A WELL-VENTILATED AREA.

(d) Clean the riveted self-locking nut installation area with a cloth moistened with alcohol, B00676 [CP1041] (TASK 70-21-23-100-801-G00).

(e) Do a visual inspection of the hole.

1) Wears, tears, dents, and nicks are not permitted.

(3) Install the self-locking nut (TASK 70-43-11-300-801-G00).

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-23-04-410-004-G00

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-23-04-840-002-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

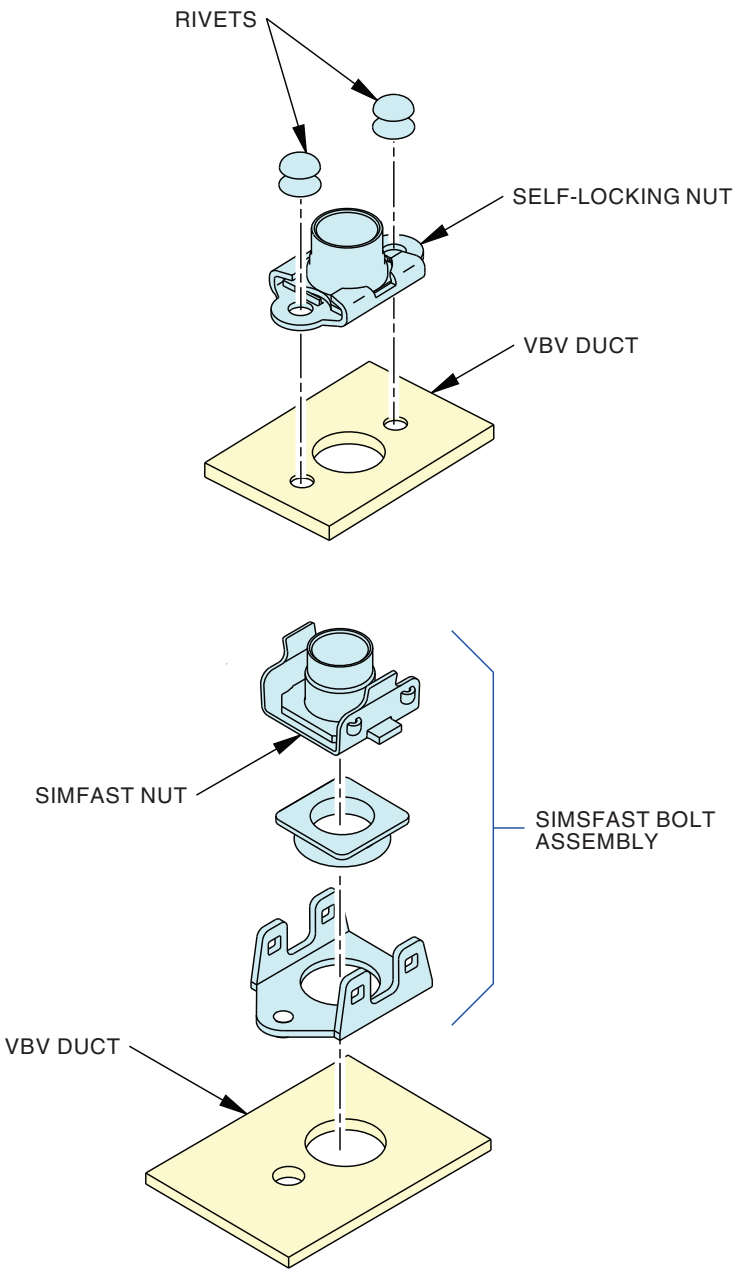
(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

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Variable Bleed Valve (VBV) Duct Nut Replacement
Figure 812/72-23-04-990-831-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-23-04-960-805-G00

10. Shroud Junction Fitting Nut - Replacement

(Figure 813)

A. General

(1) This procedure gives instructions to replace the SIMFAST floating bolt assembly or to replace the nut of the SIMFAST floating bolt assembly on the shroud junction fitting.

B. References

Reference	Title
70-11-06-230-802-G00	Fluorescent Penetrant Inspection (P/B 201)
70-11-06-230-803-G00	Fluorescent Penetrant Inspection (General) (P/B 201)
70-21-23-100-801-G00	Hand Wipe Solvent Degreasing With Chemical Agent (P/B 701)
70-43-34-300-801-G00	Simfast Floating Bolt Assemblies Rivet Repair (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-23-04-200-801-G00	Fan Frame Module Segment Shrouds Inspection (P/B 601)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
B00676 [CP1041]	Alcohol - Isopropyl	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Prepare for the Replacement

SUBTASK 72-23-04-860-027-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-23-04-010-005-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Shroud Junction Fitting Nut Replacement

SUBTASK 72-23-04-960-005-G00

(1) If it is necessary to replace the nut of the SIMFAST floating bolt assembly, do these steps (View C, Figure 813):

(a) Remove and discard the nut (TASK 70-43-34-300-801-G00).

(b) Install the nut (TASK 70-43-34-300-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

I. SIMFAST Floating Bolt Assembly Replacement

SUBTASK 72-23-04-960-006-G00

WARNING METAL PARTICLES FROM MACHINING CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PERSONAL PROTECTION EQUIPMENT.

CAUTION DO NOT TOUCH THE PART WHEN YOU REMOVE THE SIMFAST FLOATING BOLT ASSEMBLY. DAMAGE TO THE PART CAN OCCUR.

(1) If it is necessary to replace the SIMFAST floating bolt assembly, do these steps: (View C, Figure 813):

(a) Remove and discard the SIMFAST floating bolt assembly (TASK 70-43-34-300-801-G00).

(b) Remove the particles from the part.

WARNING ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. USE PERSONAL PROTECTION EQUIPMENT. USE IN A WELL-VENTILATED AREA.

(c) Clean the SIMFAST floating bolt assembly installation area with a cloth moistened with alcohol, B00676 [CP1041] (TASK 70-21-23-100-801-G00).

(d) Do the visual inspection of the holes (TASK 72-23-04-200-801-G00).

1) Deviation is not permitted.

(e) Do the dimensional inspection of the holes (View A-A, Figure 813):

1) Make sure that DIA. A is between 0.1418 in. (3.60 mm) and 0.1476 in. (3.75 mm).

2) Make sure that DIA. B is between 0.359 in. (9.12 mm) and 0.366 in. (9.30 mm).

CAUTION DO NOT APPLY FLUORESCENT PENETRANT INSPECTION PRODUCT ON THE SEAL. IF YOU DO, DAMAGE TO THE PART CAN OCCUR.

(f) Do the Fluorescent Penetrant Inspection (FPI) of the repaired areas (TASK 70-11-06-230-803-G00 or TASK 70-11-06-230-802-G00).

1) Cracks are not permitted.

(g) Install the SIMFAST floating bolt assembly (TASK 70-43-34-300-801-G00).

J. Put the Airplane Back to Its Usual Condition

SUBTASK 72-23-04-410-005-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-23-04-860-028-G00

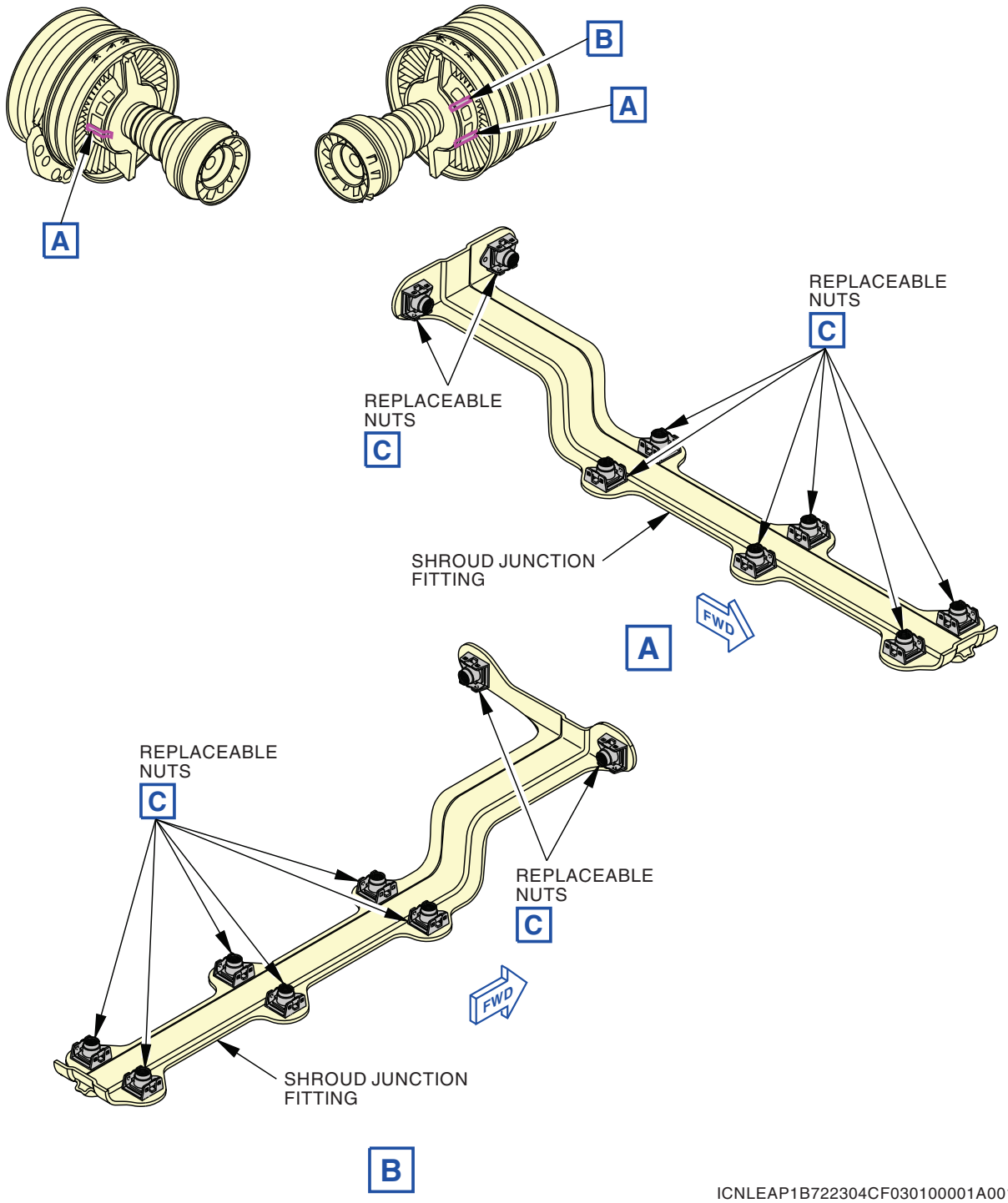
(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

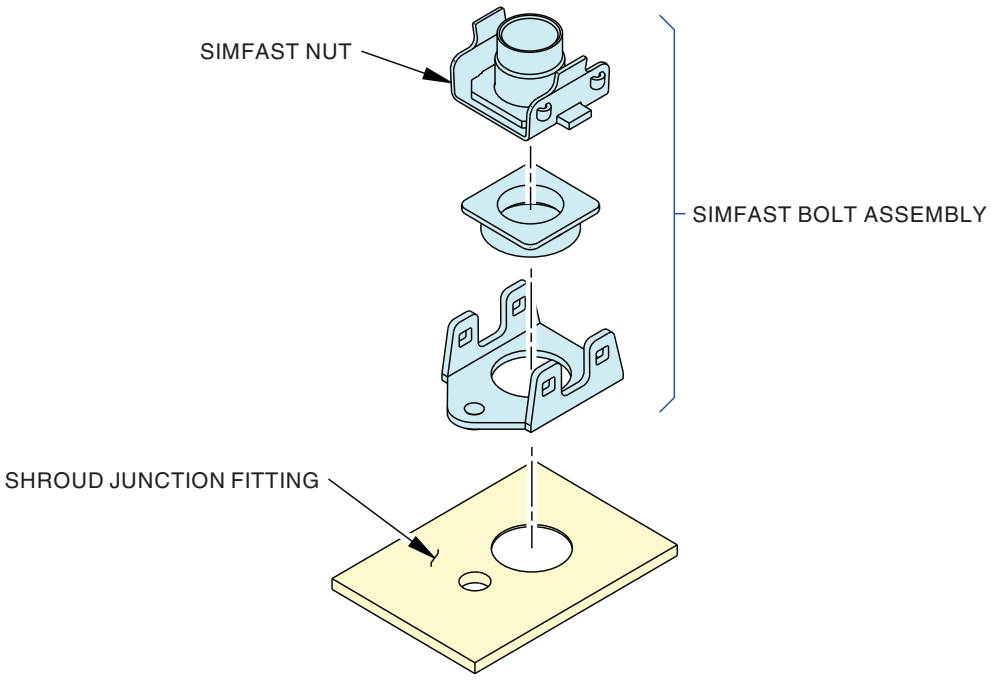
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3062109 S0000825452_V1

Shroud Junction Fitting Nut Replacement
Figure 813/72-23-04-990-832-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



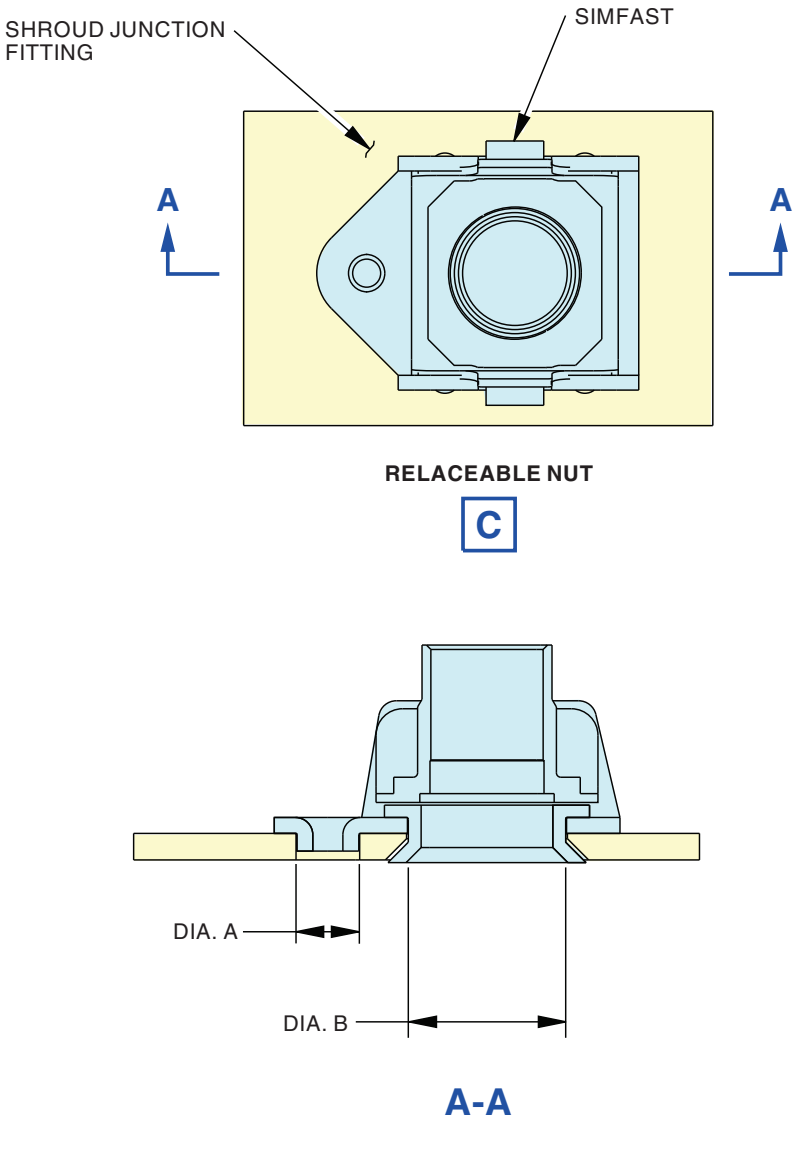
RELACEABLE NUT

C

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Shroud Junction Fitting Nut Replacement
Figure 813/72-23-04-990-832-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Shroud Junction Fitting Nut Replacement
Figure 813/72-23-04-990-832-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-23-04-960-806-G00

11. Shroud Junction Fitting - Nut Plate Two Lugs - Replacement

(Figure 814)

A. General

(1) This task gives instructions to replace the nut plate two lugs on the shroud junction fitting.

(2) The Table 807 gives a list of the part numbers that are applicable to this repair. All part numbers are applicable to all paragraphs unless specified differently.

Table 807/72-23-04-993-811-G00 Part Numbers for Nut Plate Two Lugs Replacement

Part Number	Designation
649-313-024-0	Nut plate, two lugs
NAS1399C3A3	Blind rivet
364-033-712-0	Shroud Junction Fitting
364-033-812-0	Shroud Junction Fitting

B. References

Reference	Title
70-21-23-100-801-G00	Hand Wipe Solvent Degreasing With Chemical Agent (P/B 701)
70-43-31-300-801-G00	Blind Rivet Assemblies Repair (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
B00676 [CP1041]	Alcohol - Isopropyl	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2

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(Continued)

Number	Name/Location
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Prepare for the Replacement

SUBTASK 72-23-04-860-029-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-23-04-010-007-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

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H. Shroud Junction Fitting - Nut Plate Two Lugs Replacement

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

WARNING METAL PARTICLES FROM MACHINING CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PERSONAL PROTECTION EQUIPMENT.

CAUTION DO NOT CAUSE DAMAGE TO THE RIVET HOLES WHEN YOU REMOVE THE RIVETS FROM THE SHROUD FITTING. IF THE RIVET HOLE IS OUT OF TOLERANCE, YOU CANNOT INSTALL THE RIVET CORRECTLY. IF YOU DO NOT OBEY, THIS CAN CAUSE DAMAGE TO THE ENGINE EQUIPMENT.

CAUTION DO NOT CAUSE HEAT THAT CAUSES A THERMAL TENSION DURING THE OPERATION OF POWER TOOLS. IF YOU DO NOT OBEY, YOU CAN CAUSE DAMAGE TO THE PART.

SUBTASK 72-23-04-020-008-G00

(1) Remove and discard the rivets and nut plate attached to the shroud junction fitting (Figure 814, TASK 70-43-31-300-801-G00).

SUBTASK 72-23-04-100-003-G00

WARNING ISOPROPYL ALCOHOL IS TOXIC AND FLAMMABLE. USE PERSONAL PROTECTION EQUIPMENT. USE IN A WELL-VENTILATED AREA.

(2) Clean the area where you will install a new nut plate two lugs and adjacent areas with alcohol, B00676 [CP1041] (TASK 70-21-23-100-801-G00).

SUBTASK 72-23-04-210-013-G00

(3) Do a visual inspection of the holes on the shroud junction fitting and adjacent areas (Figure 814).

(a) Deviation to a correct condition is not permitted (TASK 70-43-31-300-801-G00).

SUBTASK 72-23-04-420-008-G00

(4) Install a nut plate two lugs and its attaching parts rivets and do a check for their correct installation (TASK 70-43-31-300-801-G00).

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-23-04-410-006-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-23-04-860-030-G00

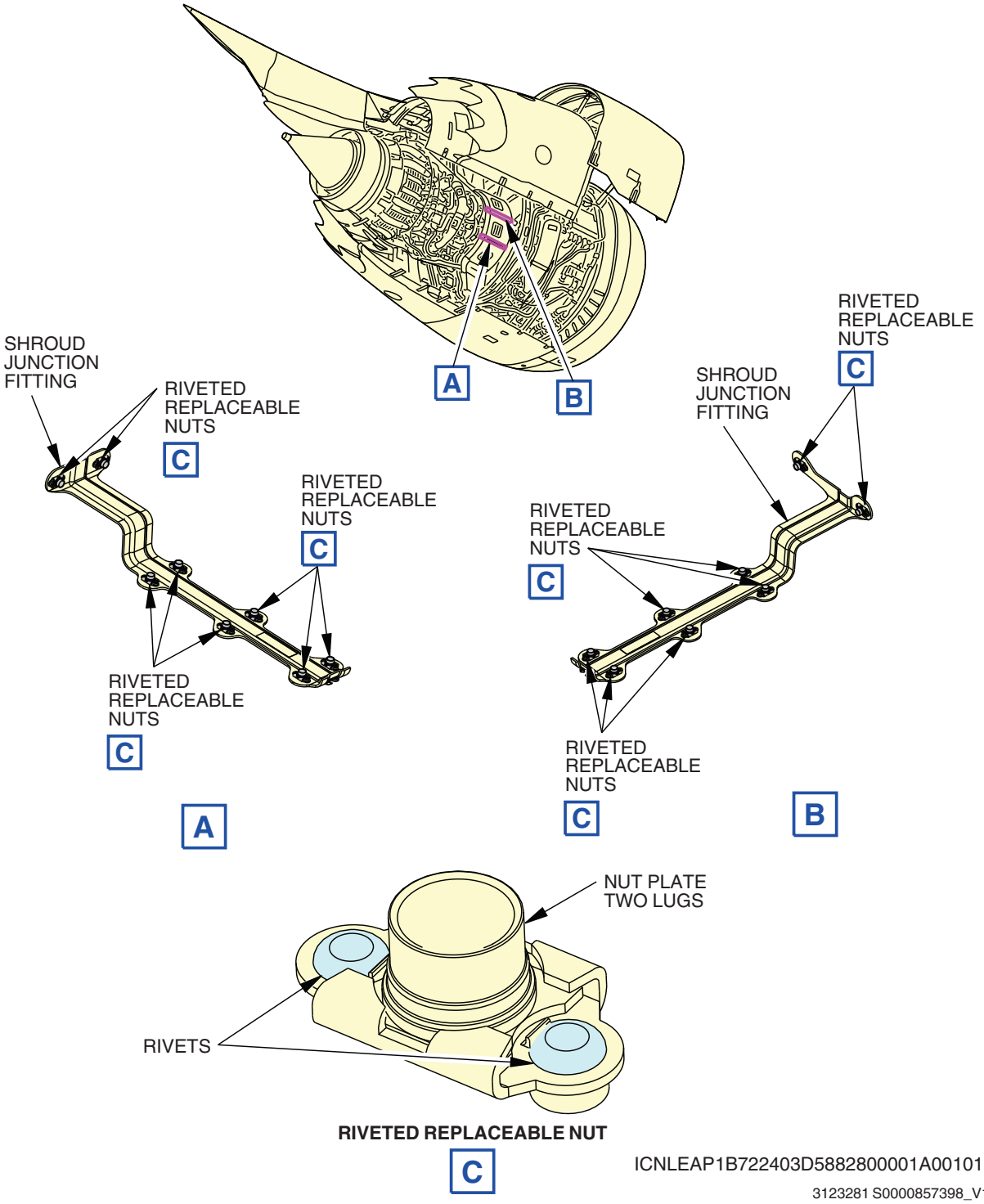
(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Shroud Junction Fitting - Nut Plate Two Lugs Replacement
Figure 814/72-23-04-990-833-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FAN FRAME STRUT PANELS - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Fan frame strut panels removal

(2) Fan frame strut panels installation.

TASK 72-23-06-000-801-G00

2. Fan Frame Strut Panels Removal

(Figure 401, Figure 402, and Figure 403)

A. General

(1) This task gives the instructions to remove the fan frame strut panels.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14947	Protector - SACOC Part #: 956A8681G02 Supplier: 58828
STD-858	Tag - DO NOT OPERATE
STD-1059	Platform - Engine and Strut Access

D. Location Zones

Zone	Area
434	Engine 1 - Aft Strut Fairing
444	Engine 2 - Aft Strut Fairing

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

F. Prepare for the Removal

SUBTASK 72-23-06-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-23-06-010-001-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

SUBTASK 72-23-06-480-001-G00

WARNING BE CAREFUL WHEN YOU WORK ON THE ENGINE COMPONENTS IMMEDIATELY AFTER THE ENGINE IS SHUTDOWN. THE ENGINE COMPONENTS CAN STAY HOT FOR UP TO ONE HOUR AND CAN CAUSE INJURY.

(3) Put the strut access platform, STD-1059 into position.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 72-23-06-480-005-G00

(4) Install the protector, SPL-14947 on the oil coolers.

G. Fan Frame Strut Panels Removal

SUBTASK 72-23-06-020-001-G00

CAUTION

MAKE SURE THAT YOU INSTALL THE SACOC PROTECTOR TO PREVENT DAMAGE TO THE OIL COOLER.

(1) Remove the fan frame strut 1 panel (right) [1] as follows:

(a) Remove the screw [2] that attaches the strut 1 panel (right) [1] to the strut 1 pad (right) [9].

(b) Remove the three screws [3] that attach the strut 1 panel (right) [1] to the strut 1 pad (right) [9].

(c) Remove the screw [4] that attaches the strut 1 panel (right) [1] to the strut 1 pad (right) [9] and the fireproof plate [18].

(d) Remove the four screws [6] that attach the strut 1 panel (right) [1] to the strut 1 panel support [5].

(e) Remove the screw [7] that attaches the strut 1 panel (right) [1] to the strut 1 outer fitting (right) [10].

(f) Remove the three screws [8] that attach the strut 1 panel (right) [1] to the strut 1 outer fitting (right) [10].

(g) Cut and remove the sealant from the strut 1 panel (right) [1].

(h) Remove the strut 1 panel (right) [1] from the engine.

SUBTASK 72-23-06-020-002-G00

CAUTION

MAKE SURE THAT YOU INSTALL THE SACOC PROTECTOR TO PREVENT DAMAGE TO THE OIL COOLER.

(2) Remove the fan frame strut 1 panel (left) [11] as follows:

(a) Remove the screw [12] that attaches the strut 1 panel (left) [11] to the strut 1 pad (left) [19].

(b) Remove the four screws [13] that attach the strut 1 panel (left) [11] to the strut 1 pad (left) [19].

(c) Remove the four screws [14] that attach the strut 1 panel (left) [11] to the strut 1 panel support [5].

(d) Remove the screw [15] that attaches the strut 1 panel (left) [11] to the strut 1 outer fitting (left) [20].

(e) Remove the two screws [16] that attach the strut 1 panel (left) [11] to the strut 1 outer fitting (left) [20].

(f) Remove the screw [17] that attaches the strut 1 panel (left) [11] to the strut 1 outer fitting (left) [20] and the fireproof plate [18].

(g) Cut and remove the sealant from the strut 1 panel (left) [11].

(h) Remove the strut 1 panel (left) [11] from the engine.

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SUBTASK 72-23-06-020-003-G00

CAUTION

MAKE SURE THAT YOU INSTALL THE SACOC PROTECTOR TO PREVENT DAMAGE TO THE OIL COOLER.

(3) Remove the strut 5 panel (right) [33] as follows:

(a) Remove the screw [37] that attaches the strut 5 panel (right) [33] to the strut 5 pad (right) [31].

(b) Remove the four screws [34] that attach the strut 5 panel (right) [33] to the strut 5 pad (right) [31].

(c) Remove the three screws [35] that attach the strut 5 panel (right) [33] to the strut 5 OGV [32].

(d) Remove the screw [38] that attaches the strut 5 panel (right) [33] to the strut 5 outer fitting [46].

(e) Remove the two screws [36] that attach the strut 5 panel (right) [33] to the strut 5 outer fitting [46].

(f) Cut and remove the sealant from the strut 5 panel (right) [33].

(g) Remove the strut 5 panel (right) [33] from the engine.

SUBTASK 72-23-06-020-004-G00

CAUTION

MAKE SURE THAT YOU INSTALL THE SACOC PROTECTOR TO PREVENT DAMAGE TO THE OIL COOLER.

(4) Remove the strut 5 panel (left) [44] as follows:

(a) Remove the screw [39] that attaches the strut 5 panel (left) [44] to the strut 5 pad (left) [45].

(b) Remove the four screws [43] that attach the strut 5 panel (left) [44] to the strut 5 pad (left) [45].

(c) Remove the three screws [42] that attach the strut 5 panel (left) [44] to the strut 5 OGV [32].

(d) Remove the screw [40] that attaches the strut 5 panel (left) [44] to the strut 5 outer fitting [46].

(e) Remove the two screws [41] that attach the strut 5 panel (left) [44] to the strut 5 outer fitting [46].

(f) Cut and remove the sealant from the strut 5 panel (left) [44].

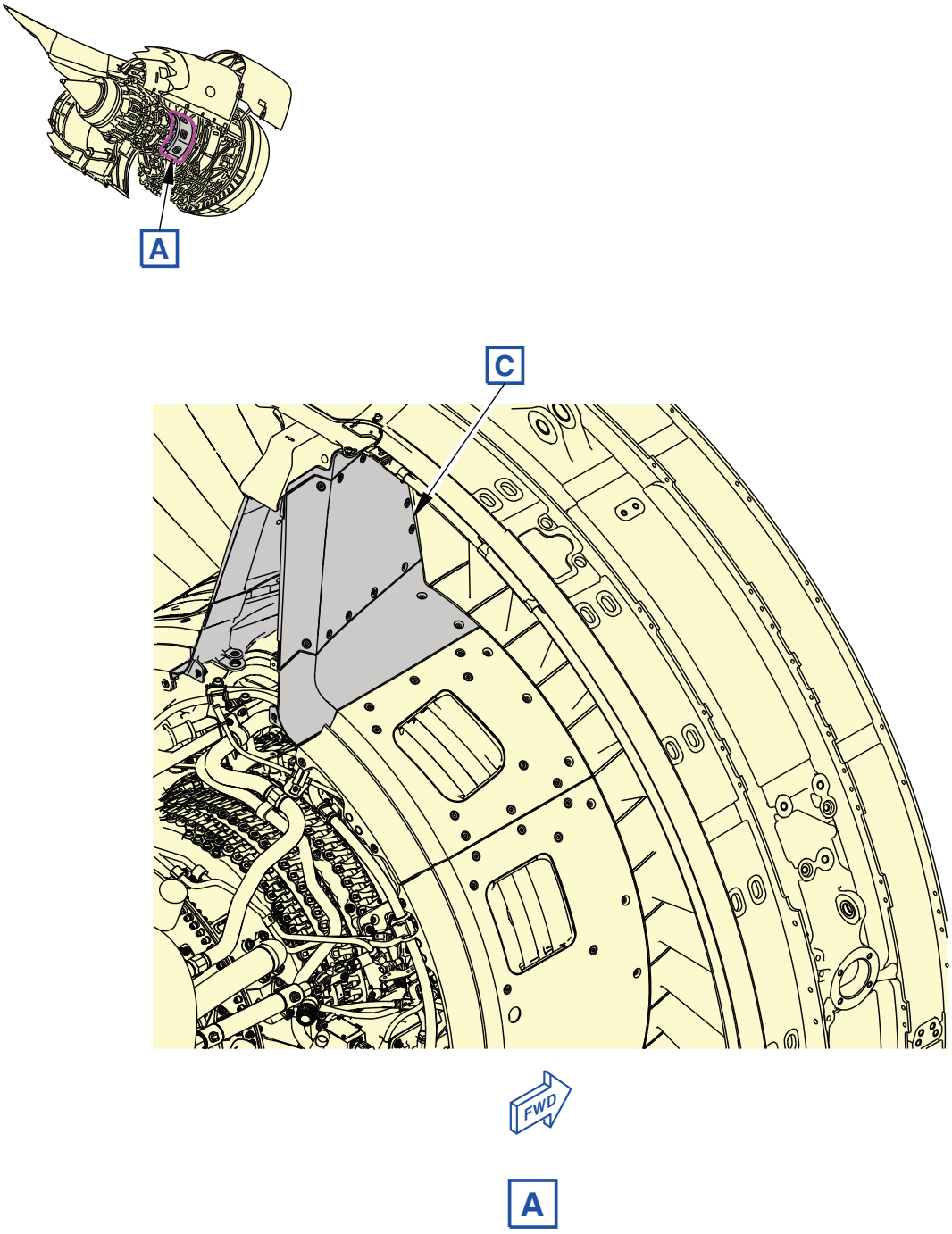
(g) Remove the strut 5 panel (left) [44] from the engine.

SUBTASK 72-23-06-480-002-G00

(5) Install the protective covers on the openings.

——— END OF TASK ———

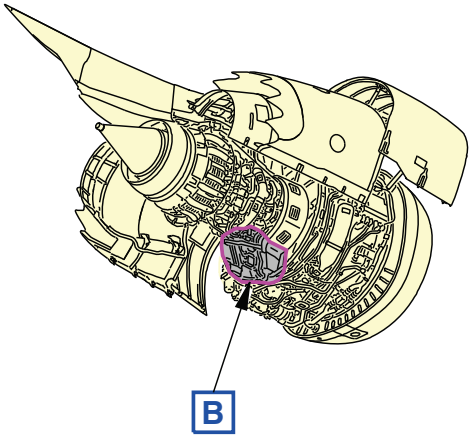
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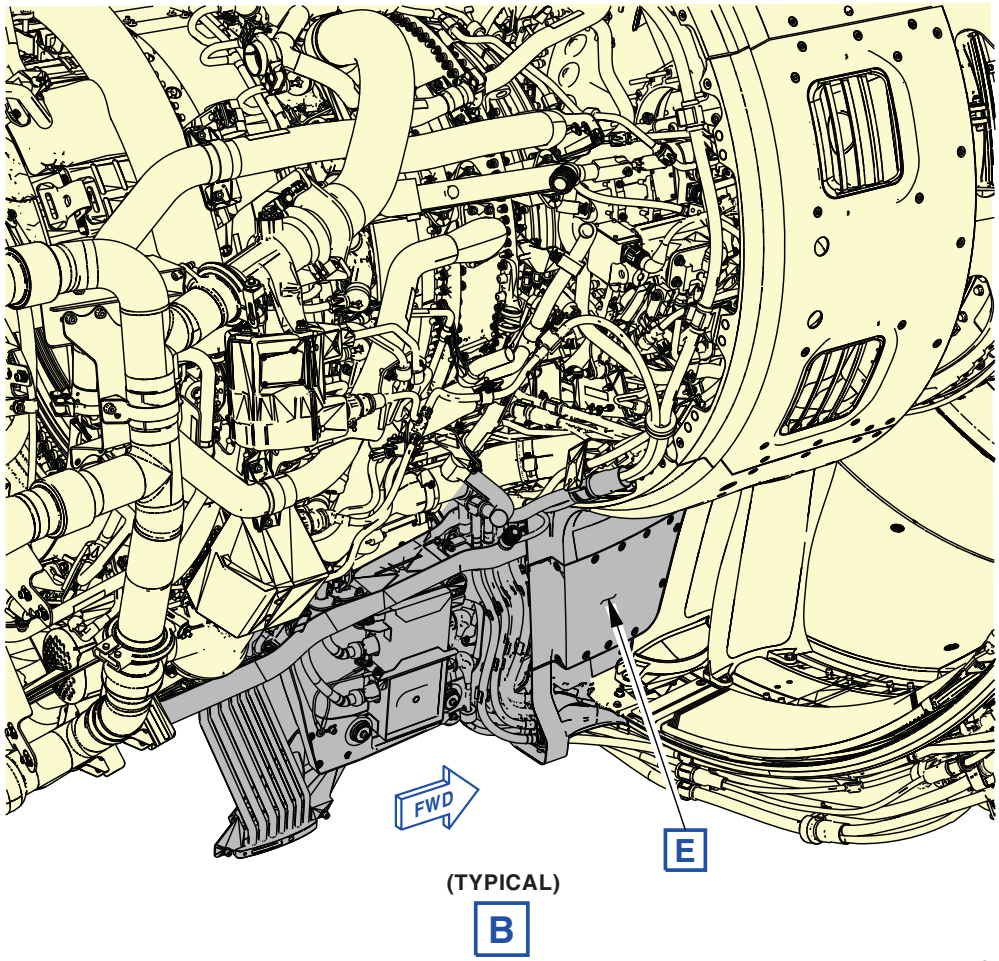


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Fan Frame Strut Panels Installation
Figure 401/72-23-06-990-801-G00 (Sheet 1 of 4)

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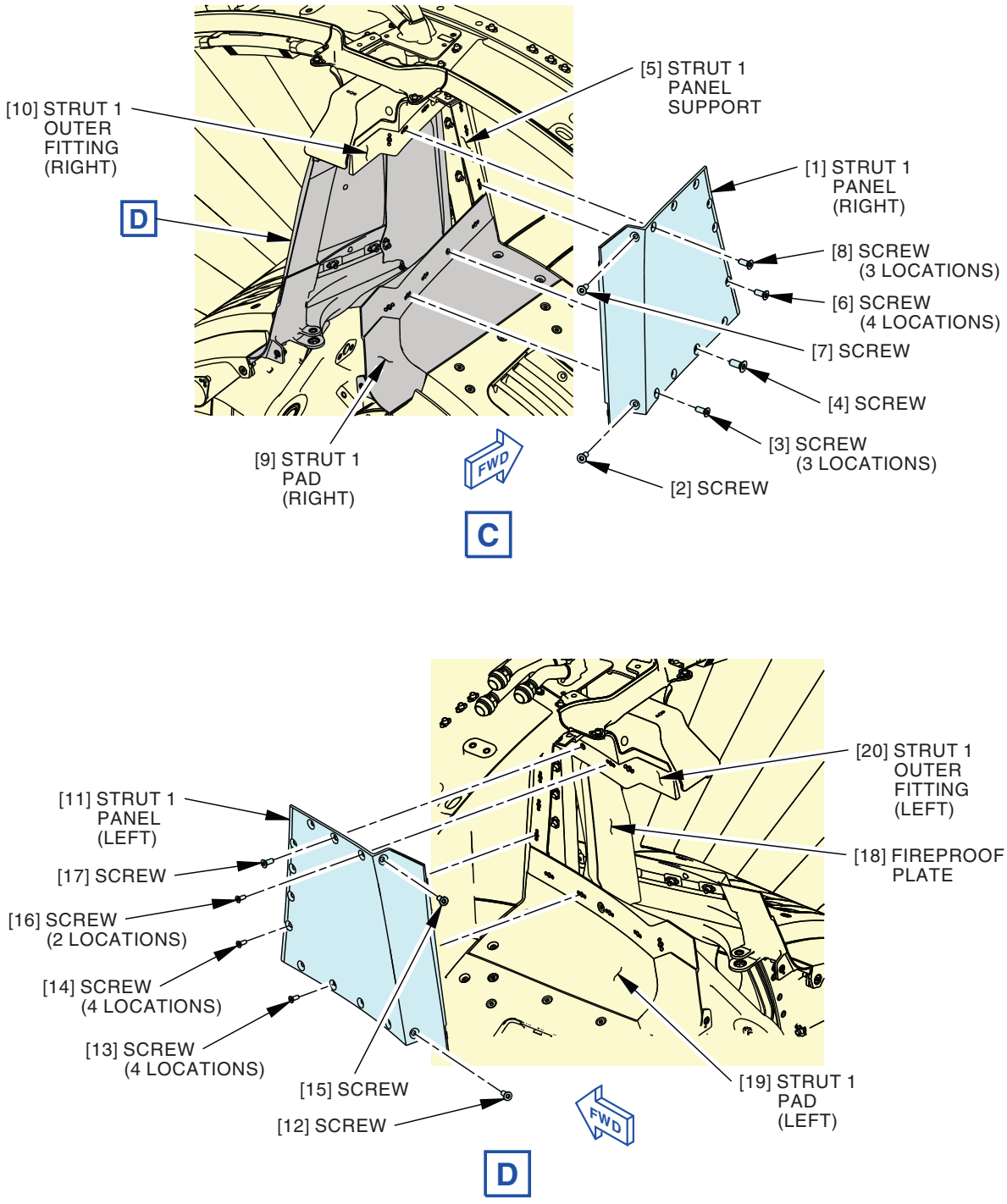




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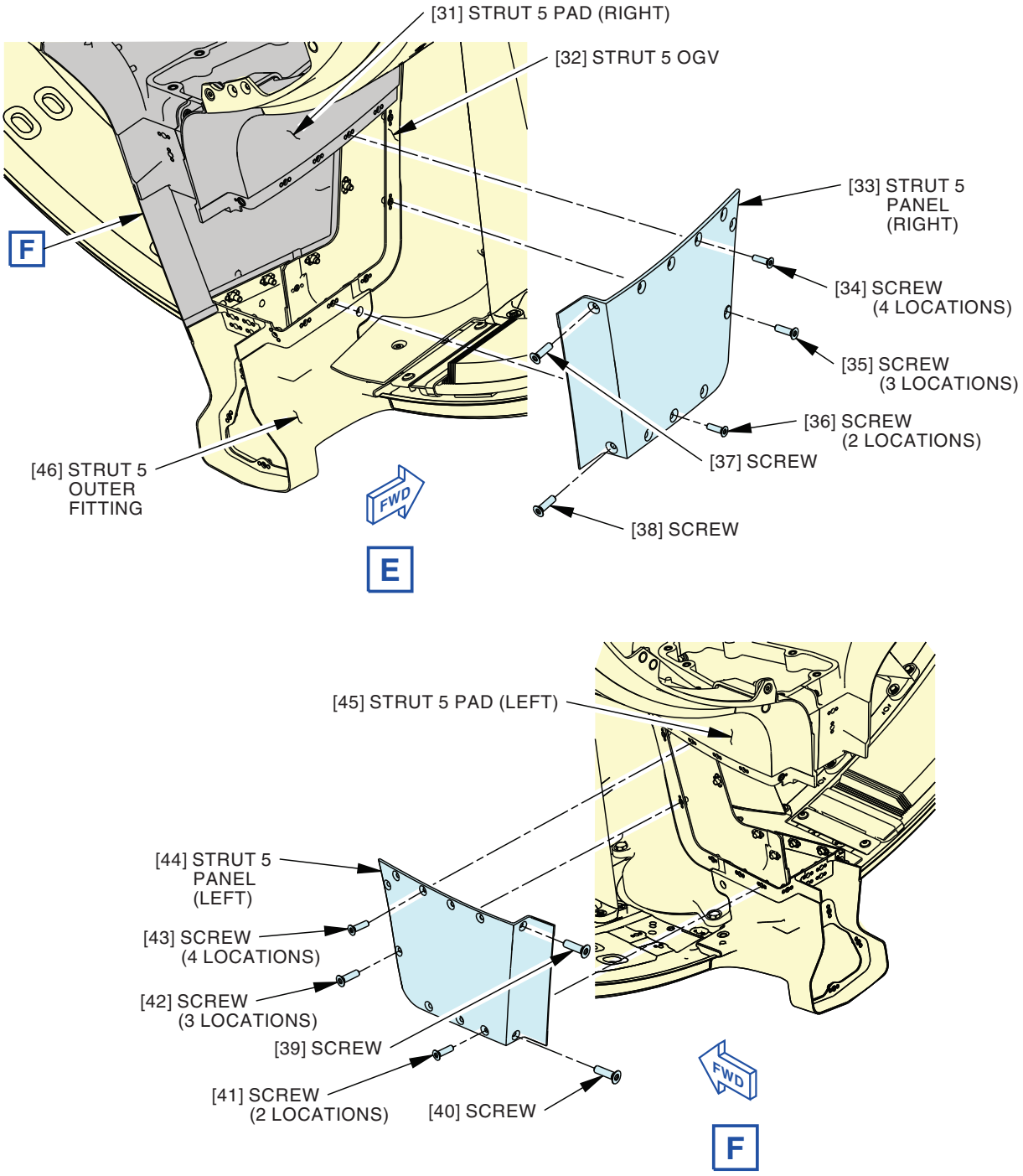
Fan Frame Strut Panels Installation
Figure 401/72-23-06-990-801-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



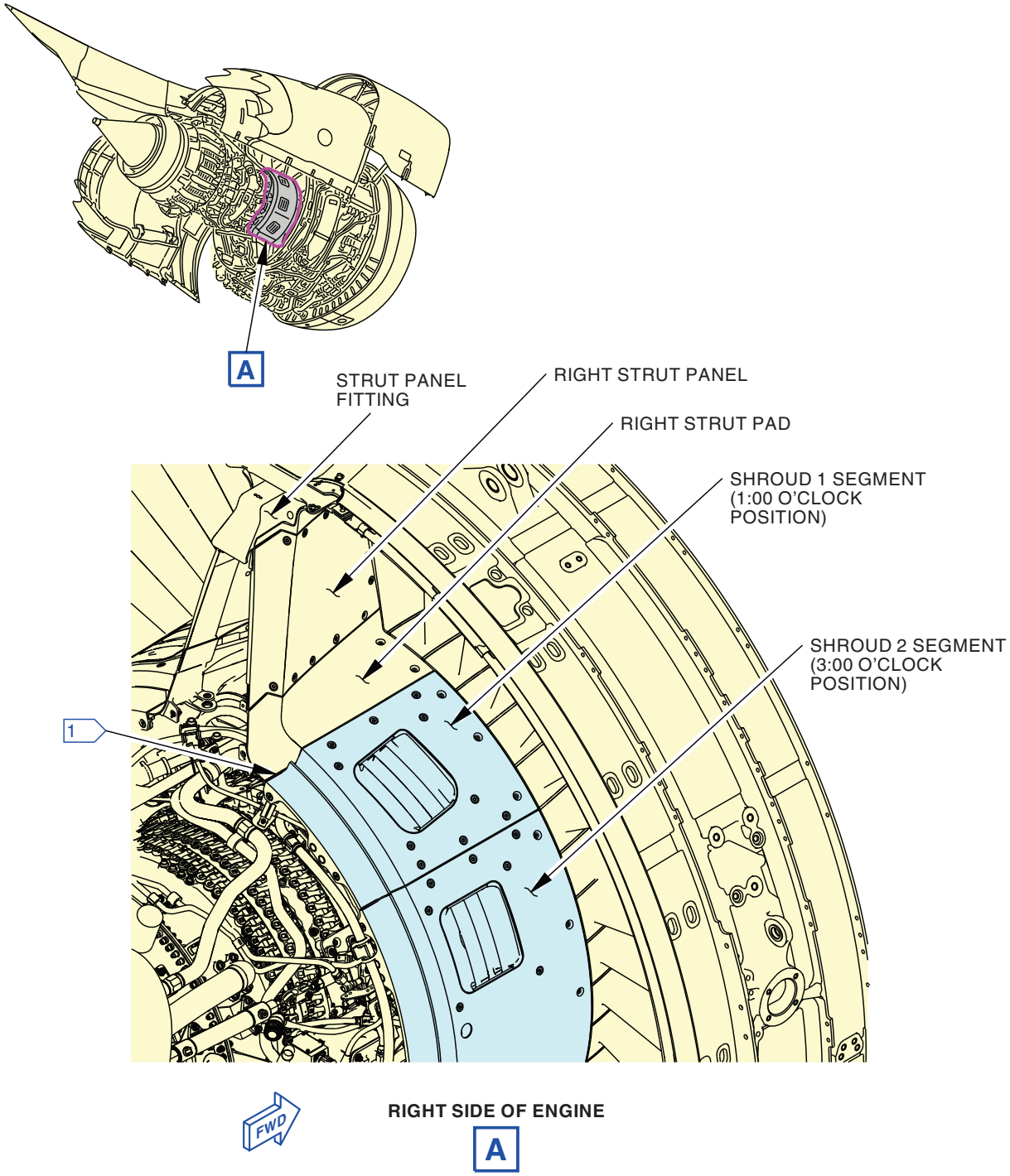
Fan Frame Strut Panels Installation
Figure 401/72-23-06-990-801-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



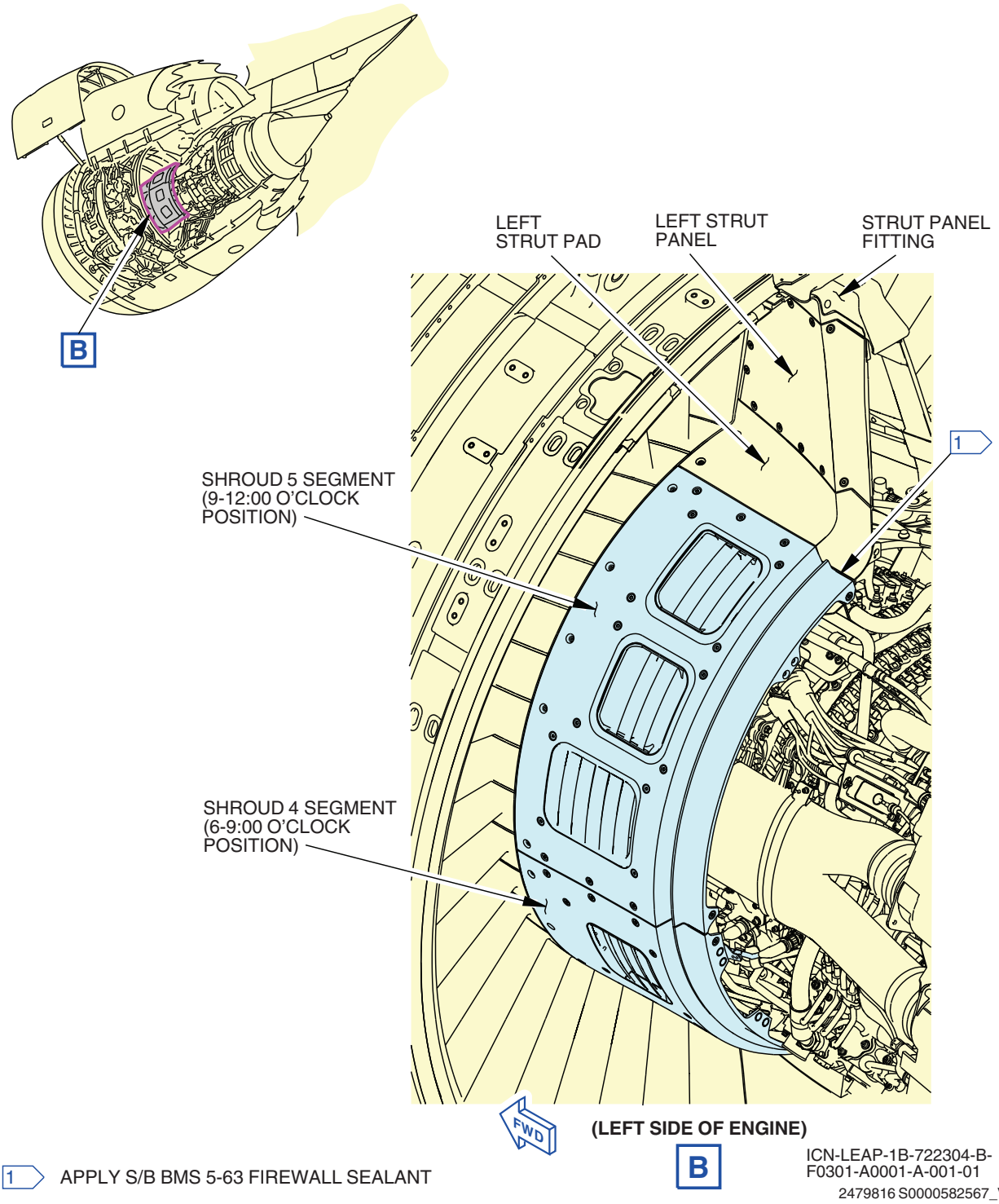
Fan Frame Strut Panels Installation
Figure 401/72-23-06-990-801-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



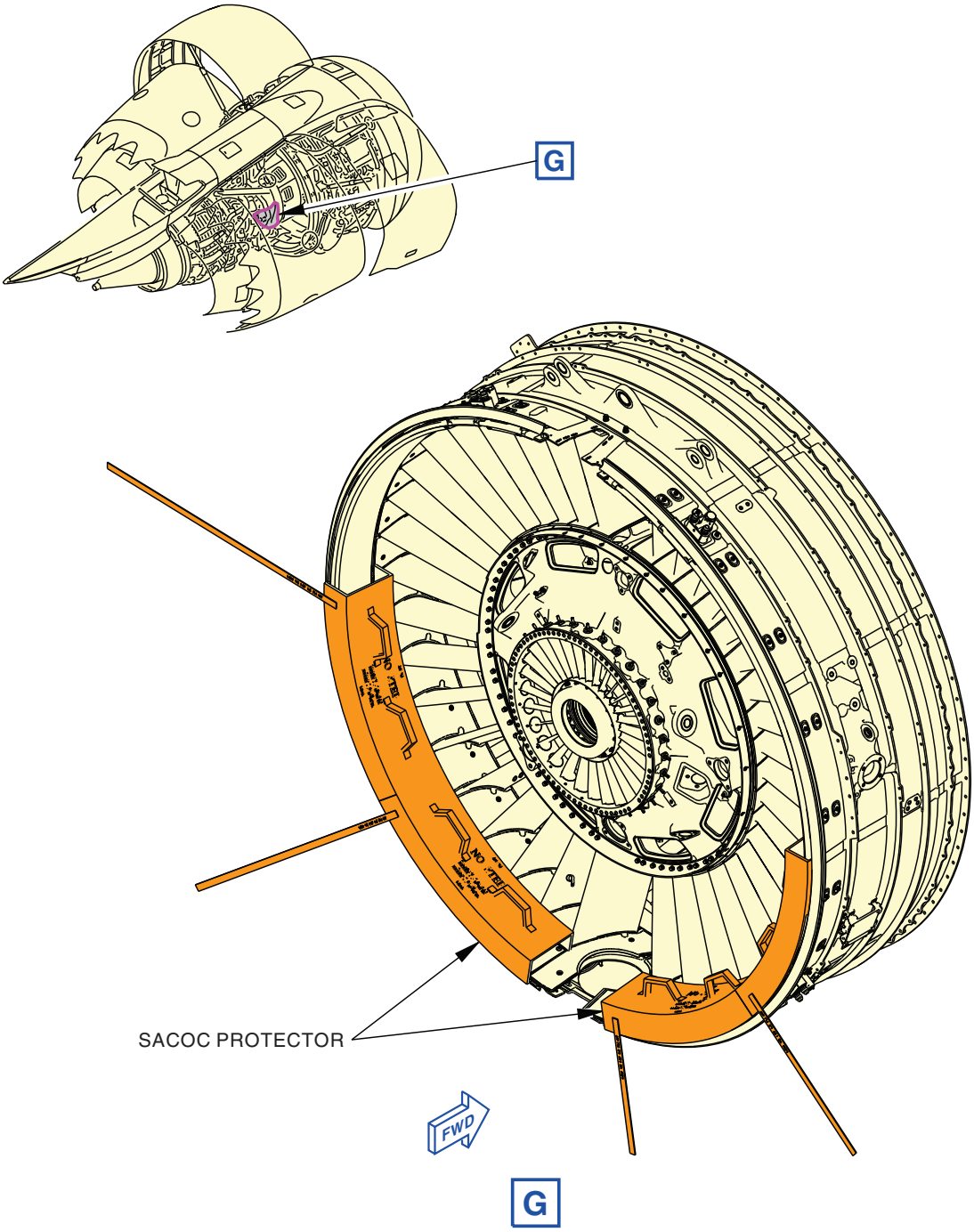
Strut Panels and Shroud Segments Firewall Sealant Installation
Figure 402/72-23-06-990-802-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Strut Panels and Shroud Segments Firewall Sealant Installation
Figure 402/72-23-06-990-802-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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SACOC Protector Installation
Figure 403/72-23-06-990-823-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-23-06-400-801-G00

3. Fan Frame Strut Panels Installation

(Figure 401, Figure 402, and Figure 403)

A. General

(1) This task gives the instructions to install the fan frame strut panels.

(2) You must apply the firewall sealant at the aft edge of the strut panel, strut panel fitting, strut pad, and the shroud 5 segment or shroud 1 segment.

(3) Make sure that SACOC protectors are installed.

B. References

Reference	Title
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14947	Protector - SACOC Part #: 956A8681G02 Supplier: 58828
STD-858	Tag - DO NOT OPERATE
STD-1059	Platform - Engine and Strut Access

D. Consumable Materials

Reference	Description	Specification
A50483 [CP2927]	Silicone Paste - Primerless, For Fire Resistance application (Dapco 2100 Firewall Sealant)	
D00601 [CP2101]	Vaseline - Graphite Mineral	
G51249	Agent - Parting, Peelable	BAC5000

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Strut 1 panel (right)	72-00-00-21-170	PTF ALL
11	Strut 1 panel (left)	72-00-00-21-070	PTF ALL
33	Strut 5 panel (right)	72-00-00-21-680	PTF ALL
44	Strut 5 panel (left)	72-00-00-21-610	PTF ALL

F. Location Zones

Zone	Area
434	Engine 1 - Aft Strut Fairing
444	Engine 2 - Aft Strut Fairing

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(Continued)

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Fan Frame Strut Panels Installation

SUBTASK 72-23-06-480-003-G00

(1) Remove the protective covers from the openings.

SUBTASK 72-23-06-160-001-G00

(2) Make sure that the mating faces are clean and clear of unwanted materials.

SUBTASK 72-23-06-420-001-G00

(3) Install the strut 5 panel (left) [44] as follows:

(a) Lubricate the threads of the screw [39], four screws [43], and three screws [42] with graphite mineral vaseline, D00601 [CP2101].

(b) Lubricate the threads of the screw [40] and two screws [41] with graphite mineral vaseline, D00601 [CP2101].

(c) Apply firewall Dapco 2100 sealant, A50483 [CP2927], on the strut 5 outer fitting [46], strut 5 OGV [32], and strut 5 pad (left) [45].

(d) Install the strut 5 panel (left) [44] in the correct position.

(e) Install the two screws [41] that attach the strut 5 panel (left) [44] to the strut 5 outer fitting [46].

(f) Install the screw [40] that attaches the strut 5 panel (left) [44] to the strut 5 pad (left) [45].

(g) Install the three screws [42] that attach the strut 5 panel (left) [44] to the strut 5 OGV [32].

(h) Install the four screws [43] that attach the strut 5 panel (left) [44] to the strut 5 pad (left) [45].

(i) Install the screw [39] that attaches the strut 5 panel (left) [44] to the strut 5 pad (left) [45].

(j) Tighten the two screws [41], screw [40], three screws [42], four screws [43], and screw [39] to 76 in-lb (8.55 N·m) - 84 in-lb (9.45 N·m).

SUBTASK 72-23-06-420-002-G00

(4) Install the strut 5 panel (right) [33] as follows:

(a) Lubricate the threads of the screw [37], four screws [34], and three screws [35] with the graphite mineral vaseline, D00601 [CP2101].

(b) Lubricate the threads of the screw [38] and two screws [36] with the graphite mineral vaseline, D00601 [CP2101].

(c) Apply firewall Dapco 2100 sealant, A50483 [CP2927], on the strut 5 outer fitting [46], strut 5 OGV [32], and strut 5 pad (right) [31].

(d) Install the strut 5 panel (right) [33] in the correct position.

(e) Install the two screws [36] that attach the strut 5 panel (right) [33] to the strut 5 outer fitting [46].

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(f) Install the screw [38] that attaches the strut 5 panel (right) [33] to the strut 5 pad (right) [31].

(g) Install the three screws [35] that attach the strut 5 panel (right) [33] to the strut 5 OGV [32].

(h) Install the four screws [34] that attach the strut 5 panel (right) [33] to the strut 5 pad (right) [31].

(i) Install the screw [37] that attaches the strut 5 panel (right) [33] to the strut 5 pad (right) [31].

(j) Tighten the screws [36], screw [38], three screws [35], four screws [34], and screw [39] to 76 in-lb (8.55 N·m) - 84 in-lb (9.45 N·m).

SUBTASK 72-23-06-420-003-G00

(5) Install the strut 1 panel (left) [11] as follows:

(a) Lubricate the threads of the screw [12], four screws [13], and four screws [14] with graphite mineral vaseline, D00601 [CP2101].

(b) Lubricate the threads of the screw [15], two screws [16], and screw [17] with graphite mineral vaseline, D00601 [CP2101].

(c) Apply firewall Dapco 2100 sealant, A50483 [CP2927], on the strut 1 outer fitting (left) [20], strut 1 panel support [5], and strut 1 pad (left) [19].

(d) Install the strut 1 panel (left) [11] in the correct position.

(e) Install the screw [17] that attaches the strut 1 panel (left) [11] to the strut 1 outer fitting (left) [20] and the fireproof plate [18].

(f) Tighten the screw [17] to 160 in-lb (18.08 N·m) - 176 in-lb (19.89 N·m).

(g) Install the two screws [16] that attach the strut 1 panel (left) [11] to the strut 1 outer fitting (left) [20].

(h) Install the screw [15] that attaches the strut 1 panel (left) [11] to the strut 1 outer fitting (left) [20].

(i) Install the four screws [14] that attach the strut 1 panel (left) [11] to the strut 1 panel support [5].

(j) Install the four screws [13] that attach the strut 1 panel (left) [11] to the strut 1 pad (left) [19].

(k) Install the screw [12] that attaches the strut 1 panel (left) [11] to the strut 1 pad (left) [19].

(l) Tighten the two screws [16], screw [15], four screws [14], four screws [13], and screw [12] to 76 in-lb (8.59 N·m) - 84 in-lb (9.49 N·m).

SUBTASK 72-23-06-640-001-G00

(6) Install the strut 1 panel (right) [1] as follows:

(a) Lubricate the threads of the screw [2], three screws [3], and screw [4] with graphite mineral vaseline, D00601 [CP2101].

(b) Lubricate the threads of the four screws [6], screw [7], and three screws [8] with graphite mineral vaseline, D00601 [CP2101].

(c) Apply firewall Dapco 2100 sealant, A50483 [CP2927] on the strut 1 outer fitting (right) [10], strut 1 panel support [5], and strut 1 pad (right) [9].

(d) Install the strut 1 panel (right) [1] in the correct position.

(e) Install the three screws [8] that attach the strut 1 panel (right) [1] to the strut 1 outer fitting (right) [10].

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(f) Install the screw [7] that attaches the strut 1 panel (right) [1] to the strut 1 outer fitting (right) [10].

(g) Install the four screws [6] that attach the strut 1 panel (right) [1] to the strut 1 panel support [5].

(h) Install the screw [4] that attaches the strut 1 panel (right) [1] to the strut 1 pad (right) [9] and fireproof plate [18].

(i) Tighten the screw [4] to 160 in-lb (18.08 N·m) - 176 in-lb (19.89 N·m).

(j) Install the three screws [3] that attach the strut 1 panel (right) [1] to the strut 1 pad (right) [9].

(k) Install the screw [2] that attaches the strut 1 panel (right) [1] to the strut 1 pad (right) [9].

(l) Tighten the three screws [8], screw [7], screws [6], three screws [3], and screw [2] to 76 in-lb (8.59 N·m) - 84 in-lb (9.49 N·m).

SUBTASK 72-23-06-390-001-G00

(7) Do the steps that follow to apply the firewall sealant on the strut 1 panel (right) [1] or strut 1 panel (left) [11]:

(a) Clean the surfaces of the aft edge of the strut panel, the strut panel fitting, the strut pad, and the shroud 5 segment or the shroud 1 segment.

(b) Apply peelable parting agent, G51249, to the gaps between the strut fairing and engine kit fittings.

(c) For the left strut 1 panel (left) [11], apply Dapco 2100 sealant, A50483 [CP2927], to the aft edge areas as follows:

NOTE: The Dapco 2100 sealant, A50483 [CP2927], is to be flush between adjacent surfaces. Apply peelable parting agent, G51249, to shroud surfaces prior applying sealant.

1) Between the left upper kit panel and left strut pad

2) Between the left strut pad and bottom of the left 12:00 o'clock strut panel

3) Between the left strut pad and shroud 5 segment.

(d) For the right strut 1 panel (right) [1], apply Dapco 2100 sealant, A50483 [CP2927] to the aft edge as follows:

NOTE: The Dapco 2100 sealant, A50483 [CP2927], is to be flush between the adjacent surfaces. Apply peelable parting agent, G51249, to shroud surfaces prior applying sealant.

1) Between the right upper kit panel and right strut pad

2) Between the right strut pad and bottom of the right 12:00 o'clock strut panel

3) Between the right strut pad and shroud 1 segment.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-23-06-080-001-G00

(1) Remove the protector, SPL-14947, from the engine.

SUBTASK 72-23-06-480-004-G00

(2) Remove the strut access platform, STD-1059, from the work area.

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SUBTASK 72-23-06-410-001-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(3) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-23-06-860-002-G00

(4) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

END OF TASK

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FAN FRAME STRUT PANELS - INSPECTION/CHECK

1. General

A. This procedure has one task:

(1) Fan Frame Strut Panels Inspection.

TASK 72-23-06-200-801-G00

2. Fan Frame Strut Panels Inspection

(Figure 601, Figure 602, Figure 603, Figure 604, Figure 605, Figure 606, Figure 607, Figure 608, Figure 609, Figure 610, and Figure 611)

A. General

(1) This task gives the instructions to examine the fan frame strut panels.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-23-04-960-804-G00	Variable Bleed Valve (VBV) Duct Nut Replacement (P/B 801)
72-23-06-300-801-G00	Fan Frame Strut 1 and 5 Fairings Blending (P/B 801)
72-23-06-300-802-G00	Fan Frame Strut Pad Blending (P/B 801)
72-23-06-960-801-G00	Fan Frame Strut 1 Fairing Seal Replacement (P/B 801)
72-23-06-960-802-G00	Fan Frame Strut Pad Seal Replacement (P/B 801)
72-23-06-960-803-G00	Fan Frame Strut Pad Nut Replacement (P/B 801)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
A50472 [CP2734]	Product - Silicone Elastomer Sealing	
A50483 [CP2927]	Silicone Paste - Primerless, For Fire Resistance application (Dapco 2100 Firewall Sealant)	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1

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(Continued)

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Prepare for the Inspection

SUBTASK 72-23-06-860-003-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-23-06-010-002-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1

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(Continued)

Number	Name/Location
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Fan Frame Strut Panels Inspection

SUBTASK 72-23-06-211-001-G00

(1) Do these steps to examine the RH-strut 1 pad [1] (Figure 601):

(a) Examine all areas for these conditions:

1) Cracks

a) Not permitted.

2) Nicks and scratches

a) Permitted up to 0.003 in. (0.1 mm) in depth.

b) Maximum repairable limit: in the limits of applicable repair.

<1> If the depth is less than 0.023 in. (0.584 mm).

<2> If it can be blended to a finished blend depth of 0.023 in. (0.584 mm).

c) Repair method: Fan Frame Strut Pad Blending, TASK 72-23-06-300-802-G00.

3) Dents

a) Permitted up to 0.019 in. (0.5 mm) in depth.

b) Maximum repairable limit: in the limits of applicable repair.

<1> If the depth is less than 0.023 in. (0.584 mm).

<2> If it can be blended to a finished blend depth of 0.023 in. (0.584 mm).

c) Repair method: Fan Frame Strut Pad Blending, TASK 72-23-06-300-802-G00.

(b) Examine the Area A for these conditions (Figure 601 (Sheet 1) and Figure 601 (Sheet 2)):

1) Fretting, erosion, or wear

a) Permitted if dimension A is between 0.036 in. (0.9 mm) and 0.122 in. (3.1 mm).

(c) Examine the Area B for these conditions (Figure 601 (Sheet 1) and Figure 601 (Sheet 2)):

1) Fretting, erosion, or wear

a) Permitted if dimension B is between 0.1201 in. (3.05 mm) and 0.1948 in. (4.95 mm).

(d) Examine the Area C for these conditions (Figure 601 (Sheet 1) and Figure 601 (Sheet 2)):

1) Fretting, erosion, or wear

a) Permitted if dimension C is between 0.170 in. (4.3 mm) and 0.185 in. (4.7 mm).

(e) Examine the Area D for these conditions (Figure 601 (Sheet 1) and Figure 601 (Sheet 2)):

1) Fretting, erosion, or wear

a) Permitted if dimension D is between 0.130 in. (3.3 mm) and 0.145 in. (3.7 mm).

(f) Examine the Area E for these conditions (Figure 601 (Sheet 1) and Figure 601 (Sheet 2)):

1) Fretting, erosion, or wear

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a) Permitted if dimension E is 0.0925 in. (2.35 mm) or more.

(g) Examine the inserts for these conditions (Figure 601 (Sheet 3)):

1) Missing

a) Not permitted.

2) Looseness

a) Not permitted.

(h) Examine the replaceable nuts for these conditions (Figure 601 (Sheet 3)):

1) Missing

a) Not permitted.

b) Maximum repairable limit: Any amount.

c) To repair, do this task: Fan Frame Strut Pad Nut Replacement, TASK 72-23-06-960-803-G00.

2) Looseness

a) Not permitted.

b) Maximum repairable limit: Any amount.

c) To repair, do this task: Fan Frame Strut Pad Nut Replacement, TASK 72-23-06-960-803-G00.

(i) For the configuration 1, with the white rectangular cross section, examine the rectangular cross section seal for these conditions (Figure 601 (Sheet 3)):

1) Missing

a) Not permitted.

b) Maximum repairable limit: any amount.

c) To repair, do this task: Fan Frame Strut Pad Seal Replacement, TASK 72-23-06-960-802-G00.

2) Looseness

a) Not permitted.

b) Maximum repairable limit: any amount.

c) To repair, do this task: Fan Frame Strut Pad Seal Replacement, TASK 72-23-06-960-802-G00.

3) Tear

a) Permitted up to 0.19 in. (5 mm) in width and 0.39 in. (10 mm) in length.

b) Maximum repairable limit: any amount.

c) To repair, do this task: Fan Frame Strut Pad Seal Replacement, TASK 72-23-06-960-802-G00.

4) Bubbles

a) Not permitted.

b) Maximum repairable limit: any amount.

c) To repair, do this task: Fan Frame Strut Pad Seal Replacement, TASK 72-23-06-960-802-G00.

5) Permanent strain and excessive strain

a) Not permitted.

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b) Maximum repairable limit: any amount.

c) To repair, do this task: Fan Frame Strut Pad Seal Replacement, TASK 72-23-06-960-802-G00.

(j) For the configuration 2, with the two lips cross section, examine the two lips cross section seal for these conditions (Figure 601 (Sheet 3)):

1) Missing

a) Not permitted.

b) Maximum repairable limit: any amount.

c) To repair, do this task: Fan Frame Strut Pad Seal Replacement, TASK 72-23-06-960-802-G00.

2) Looseness

a) Not permitted.

b) Maximum repairable limit: any amount.

c) To repair, do this task: Fan Frame Strut Pad Seal Replacement, TASK 72-23-06-960-802-G00.

3) Tear

a) Permitted up to 0.39 in. (10 mm) on one of the two seal lips.

b) Maximum repairable limit: any amount.

c) To repair, do this task: Fan Frame Strut Pad Seal Replacement, TASK 72-23-06-960-802-G00.

4) Bubbles

a) Not permitted.

b) Maximum repairable limit: any amount.

c) To repair, do this task: Fan Frame Strut Pad Seal Replacement, TASK 72-23-06-960-802-G00.

5) Permanent strain and excessive strain

a) Not permitted.

b) Maximum repairable limit: any amount.

c) To repair, do this task: Fan Frame Strut Pad Seal Replacement, TASK 72-23-06-960-802-G00.

(k) Examine the black silicone sealing elastomer (RTV 103 adhesive, A50472 [CP2734]) for these conditions:

1) Missing

a) Permitted.

2) Looseness

a) Permitted.

3) Wear

a) Permitted.

(l) Examine the screws of the RH-strut 1 pad [1] (View D, Figure 601).

1) Surface oxidation on screw head

a) Permitted.

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SUBTASK 72-23-06-211-002-G00

(2) Do these steps to examine the LH-strut 1 pad [2] (Figure 602):

(a) Examine all areas for these conditions:

1) Cracks

a) Not permitted.

2) Nicks and scratches

a) Permitted up to 0.0019 in. (0.05 mm) in depth.

b) Maximum repairable limit: in the limits of applicable repair.

<1> If the depth is less than 0.023 in. (0.584 mm).

<2> If it can be blended to a finished blend depth of 0.023 in. (0.584 mm).

c) Repair method: Fan Frame Strut Pad Blending, TASK 72-23-06-300-802-G00.

3) Dents

a) Permitted up to 0.019 in. (0.5 mm) in depth.

b) Maximum repairable limit: in the limits of applicable repair.

<1> If the depth is less than 0.023 in. (0.584 mm).

<2> If it can be blended to a finished blend depth of 0.023 in. (0.584 mm).

c) Repair method: Fan Frame Strut Pad Blending, TASK 72-23-06-300-802-G00.

(b) Examine the Area A for these conditions (Figure 602 (Sheet 1) and Figure 602 (Sheet 2)):

1) Fretting, erosion, or wear

a) Permitted if dimension A is between 0.036 in. (0.9 mm) and 0.122 in. (3.1 mm).

(c) Examine the Area B for these conditions (Figure 602 (Sheet 1) and Figure 602 (Sheet 2)):

1) Fretting, erosion, or wear

a) Permitted if dimension B is between 0.1201 in. (3.05 mm) and 0.1948 in. (4.95 mm).

(d) Examine the Area C for these conditions (Figure 602 (Sheet 1) and Figure 602 (Sheet 2)):

1) Fretting, erosion, or wear

a) Permitted if dimension C is between 0.170 in. (4.3 mm) and 0.185 in. (4.7 mm).

(e) Examine the Area D for these conditions (Figure 602 (Sheet 1) and Figure 602 (Sheet 2)):

1) Fretting, erosion, or wear

a) Permitted if dimension D is between 0.130 in. (3.3 mm) and 0.145 in. (3.7 mm).

(f) Examine the Area E for these conditions (Figure 602 (Sheet 1) and Figure 602 (Sheet 2)):

1) Fretting, erosion, or wear

a) Permitted if dimension E is 0.0885 in. (2.25 mm) or more.

(g) Examine the inserts for these conditions (Figure 602 (Sheet 3)):

1) Missing

a) Not permitted.

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2) Looseness

a) Not permitted.

(h) Examine the replaceable nuts for these conditions (Figure 601 (Sheet 3)):

1) Missing

a) Not permitted.

b) Maximum repairable limit: Any amount.

c) To repair, do this task: Fan Frame Strut Pad Nut Replacement, TASK 72-23-06-960-803-G00.

2) Looseness

a) Not permitted.

b) Maximum repairable limit: Any amount.

c) To repair, do this task: Fan Frame Strut Pad Nut Replacement, TASK 72-23-06-960-803-G00.

(i) For the configuration 1, with the white rectangular cross section, examine the rectangular cross section seal for these conditions (Figure 602 (Sheet 3)):

1) Missing

a) Not permitted.

b) Maximum repairable limit: any amount.

c) To repair, do this task: Fan Frame Strut Pad Seal Replacement, TASK 72-23-06-960-802-G00.

2) Looseness

a) Not permitted.

b) Maximum repairable limit: any amount.

c) To repair, do this task: Fan Frame Strut Pad Seal Replacement, TASK 72-23-06-960-802-G00.

3) Tear

a) Permitted up to 0.19 in. (5 mm) in width and 0.39 in. (10 mm) in length.

b) Maximum repairable limit: any amount.

c) To repair, do this task: Fan Frame Strut Pad Seal Replacement, TASK 72-23-06-960-802-G00.

4) Bubbles

a) Not permitted.

b) Maximum repairable limit: any amount.

c) To repair, do this task: Fan Frame Strut Pad Seal Replacement, TASK 72-23-06-960-802-G00.

5) Permanent strain and excessive strain

a) Not permitted.

b) Maximum repairable limit: any amount.

c) To repair, do this task: Fan Frame Strut Pad Seal Replacement, TASK 72-23-06-960-802-G00.

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(j) For the configuration 2, with the two lips cross section, examine the two lips cross section seal for these conditions (Figure 602 (Sheet 3)):

1) Missing
   a) Not permitted.
   b) Maximum repairable limit: any amount.
   c) To repair, do this task: Fan Frame Strut Pad Seal Replacement, TASK 72-23-06-960-802-G00.
2) Looseness
   a) Not permitted.
   b) Maximum repairable limit: any amount.
   c) To repair, do this task: Fan Frame Strut Pad Seal Replacement, TASK 72-23-06-960-802-G00.
3) Tear
   a) Permitted up to 0.39 in. (10 mm) on one of the two seal lips.
   b) Maximum repairable limit: any amount.
   c) To repair, do this task: Fan Frame Strut Pad Seal Replacement, TASK 72-23-06-960-802-G00.
4) Bubbles
   a) Not permitted.
   b) Maximum repairable limit: any amount.
   c) To repair, do this task: Fan Frame Strut Pad Seal Replacement, TASK 72-23-06-960-802-G00.
5) Permanent strain and excessive strain
   a) Not permitted.
   b) Maximum repairable limit: any amount.
   c) To repair, do this task: Fan Frame Strut Pad Seal Replacement, TASK 72-23-06-960-802-G00.

(k) Examine the black silicone sealing elastomer (RTV 103 adhesive, A50472 [CP2734]) for these conditions:

1) Missing
   a) Permitted.
2) Looseness
   a) Permitted.
3) Wear
   a) Permitted.

(l) Examine the screws of the LH-strut 1 pad [2] (View D, Figure 602).

1) Surface oxidation on screw head
   a) Permitted.

SUBTASK 72-23-06-211-003-G00

(3) Do these steps to examine the RH-strut 5 pad [3] (Figure 603):

(a) Examine all areas for these conditions:

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1) Cracks
   a) Not permitted.
2) Nicks and scratches
   a) Not permitted.
   b) Maximum repairable limit: in the limits of applicable repair.
      <1> If the depth is less than 0.023 in. (0.584 mm).
      <2> If it can be blended to a finished blend depth of 0.023 in. (0.584 mm).
   c) Repair method: Fan Frame Strut Pad Blending, TASK 72-23-06-300-802-G00.

(b) Examine the Area 1 for these conditions (Figure 603 (Sheet 1)):
1) Dents
   a) Permitted up to 0.023 in. (0.6 mm) in depth.
   b) Maximum repairable limit: in the limits of applicable repair.
      <1> In the limits of applicable repair, if the depth is less than 0.023 in. (0.584 mm).
      <2> In the limits of applicable repair, if it can be blended to a finished blend depth of 0.023 in. (0.584 mm).
   c) Repair method: Fan Frame Strut Pad Blending, TASK 72-23-06-300-802-G00.

(c) Examine the Area 2 for these conditions (Figure 603 (Sheet 1)):
1) Dents
   a) Not permitted.
   b) Maximum repairable limit: in the limits of applicable repair.
      <1> If the depth is less than 0.023 in. (0.584 mm).
      <2> If it can be blended to a finished blend depth of 0.023 in. (0.584 mm).
   c) Repair method: Fan Frame Strut Pad Blending, TASK 72-23-06-300-802-G00.

(d) Examine the Area A for these conditions (Figure 603 (Sheet 2) and Figure 603 (Sheet 3)):
1) Fretting, erosion, or wear
   a) Permitted if dimension A is between 0.036 in. (0.9 mm) and 0.173 in. (4.4 mm).

(e) Examine the Area B for these conditions (Figure 603 (Sheet 2) and Figure 603 (Sheet 3)):
1) Fretting, erosion, or wear
   a) Permitted if dimension B is between 0.1201 in. (3.05 mm) and 0.1948 in. (4.95 mm).

(f) Examine the Area C for these conditions (Figure 603 (Sheet 2) and Figure 603 (Sheet 3)):
1) Fretting, erosion, or wear
   a) Permitted if dimension C is between 0.170 in. (4.3 mm) and 0.185 in. (4.7 mm).

(g) Examine the Area D for these conditions (Figure 603 (Sheet 2) and Figure 603 (Sheet 3)):
1) Fretting, erosion, or wear
   a) Permitted if dimension D is between 0.134 in. (3.4 mm) and 0.228 in. (5.8 mm).

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(h) Examine the Area 4 for these conditions (Figure 603 (Sheet 2)):

1) Wear

a) Permitted up to 0.0019 in. (0.05 mm) in depth.

(i) Examine the inserts for these conditions (Figure 603 (Sheet 4)):

1) Missing

a) Not permitted.

2) Looseness

a) Not permitted.

(j) Examine the replaceable nuts for these conditions (Figure 601 (Sheet 3)):

1) Missing

a) Not permitted.

b) Maximum repairable limit: Any amount.

c) To repair, do this task: Fan Frame Strut Pad Nut Replacement, TASK 72-23-06-960-803-G00.

2) Looseness

a) Not permitted.

b) Maximum repairable limit: Any amount.

c) To repair, do this task: Fan Frame Strut Pad Nut Replacement, TASK 72-23-06-960-803-G00.

(k) For the configuration 1, with the white rectangular cross section and the Dapco 2100 sealant, A50483 [CP2927] (Figure 603 (Sheet 5)):

1) Examine the white rectangular cross section seal for these conditions:

a) Missing

<1> Not permitted.

<2> Maximum repairable limit: any amount.

<3> To repair, do this task: Fan Frame Strut Pad Seal Replacement, TASK 72-23-06-960-802-G00.

b) Looseness

<1> Not permitted.

<2> Maximum repairable limit: any amount.

<3> To repair, do this task: Fan Frame Strut Pad Seal Replacement, TASK 72-23-06-960-802-G00.

c) Tear

<1> Permitted up to 0.19 in. (5 mm) in width and 0.39 in. (10 mm) in length.

<2> Maximum repairable limit: any amount.

<3> To repair, do this task: Fan Frame Strut Pad Seal Replacement, TASK 72-23-06-960-802-G00.

d) Bubbles

<1> Not permitted.

<2> Maximum repairable limit: any amount.

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<3> To repair, do this task: Fan Frame Strut Pad Seal Replacement, TASK 72-23-06-960-802-G00.

e) Permanent strain and excessive strain

<1> Not permitted.

<2> Maximum repairable limit: any amount.

<3> To repair, do this task: Fan Frame Strut Pad Seal Replacement, TASK 72-23-06-960-802-G00.

2) Examine the Dapco 2100 sealant, A50483 [CP2927], located next to the white rectangular cross section seal for these conditions:

a) Missing

<1> Not permitted.

b) Looseness

<1> Not permitted.

c) Disbonding

<1> Not permitted.

d) Bubbles

<1> Not permitted.

e) Permanent strain and excessive strain

<1> Not permitted.

f) Lack of materials or through cavity

<1> Permitted up to 0.012 in. (0.3 mm).

(l) For the configuration 2, with the black two lips cross section seal and the Dapco 2100 sealant, A50483 [CP2927] (Figure 603 (Sheet 6)):

1) Examine the two lips cross section seal for these conditions:

a) Missing

<1> Not permitted.

<2> Maximum repairable limit: any amount.

<3> To repair, do this task: Fan Frame Strut Pad Seal Replacement, TASK 72-23-06-960-802-G00.

b) Looseness

<1> Not permitted.

<2> Maximum repairable limit: any amount.

<3> To repair, do this task: Fan Frame Strut Pad Seal Replacement, TASK 72-23-06-960-802-G00.

c) Tear

<1> Permitted up to 0.39 in. (10 mm) on one of the two seal lips.

<2> Maximum repairable limit: any amount.

<3> To repair, do this task: Fan Frame Strut Pad Seal Replacement, TASK 72-23-06-960-802-G00.

d) Bubbles

<1> Not permitted.

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<2> Maximum repairable limit: any amount.

<3> To repair, do this task: Fan Frame Strut Pad Seal Replacement, TASK 72-23-06-960-802-G00.

e) Permanent strain and excessive strain

<1> Not permitted.

<2> Maximum repairable limit: any amount.

<3> To repair, do this task: Fan Frame Strut Pad Seal Replacement, TASK 72-23-06-960-802-G00.

2) Examine the Dapco 2100 sealant, A50483 [CP2927], located next to the white rectangular cross section seal for these conditions:

a) Missing

<1> Not permitted.

b) Looseness

<1> Not permitted.

c) Disbonding

<1> Not permitted.

d) Bubbles

<1> Not permitted.

e) Permanent strain and excessive strain

<1> Not permitted.

f) Lack of materials or through cavity

<1> Permitted up to 0.012 in. (0.3 mm).

(m) Examine the black silicone sealing elastomer (RTV 103 adhesive, A50472 [CP2734]) for these conditions:

1) Missing

a) Permitted.

2) Looseness

a) Permitted.

3) Wear

a) Permitted.

SUBTASK 72-23-06-211-004-G00

(4) Do these steps to examine the LH-strut 5 pad [4] (Figure 604):

(a) Examine all areas for these conditions:

1) Cracks

a) Not permitted.

2) Nicks and scratches

a) Not permitted.

b) Maximum repairable limit: in the limits of applicable repair.

<1> If the depth is less than 0.023 in. (0.584 mm).

<2> If it can be blended to a finished blend depth of 0.023 in. (0.584 mm).

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c) Repair method: Fan Frame Strut Pad Blending, TASK 72-23-06-300-802-G00.

(b) Examine the Area 1 for these conditions (Figure 604 (Sheet 1)):

1) Dents

a) Permitted up to 0.019 in. (0.5 mm) in depth.

b) Maximum repairable limit: in the limits of applicable repair.

<1> If the depth is less than 0.023 in. (0.584 mm).

<2> If it can be blended to a finished blend depth of 0.023 in. (0.584 mm).

c) Repair method: Fan Frame Strut Pad Blending, TASK 72-23-06-300-802-G00.

(c) Examine the Area 2 for these conditions (Figure 604 (Sheet 1)):

1) Dents

a) Not permitted.

b) Maximum repairable limit: in the limits of applicable repair.

<1> If the depth is less than 0.023 in. (0.584 mm).

<2> If it can be blended to a finished blend depth of 0.023 in. (0.584 mm).

c) Repair method: Fan Frame Strut Pad Blending, TASK 72-23-06-300-802-G00.

(d) Examine the Area A for these conditions (Figure 604 (Sheet 2) and Figure 604 (Sheet 3)):

1) Fretting, erosion, or wear

a) Permitted if dimension A is between 0.036 in. (0.9 mm) and 0.173 in. (4.4 mm).

(e) Examine the Area B for these conditions (Figure 604 (Sheet 2) and Figure 604 (Sheet 3)):

1) Fretting, erosion, or wear

a) Permitted if dimension B is between 0.1201 in. (3.05 mm) and 0.1948 in. (4.95 mm)).

(f) Examine the Area C for these conditions (Figure 604 (Sheet 2) and Figure 604 (Sheet 3)):

1) Fretting, erosion, or wear

a) Permitted if dimension C is between 0.170 in. (4.3 mm) and 0.185 in. (4.7 mm).

(g) Examine the Area D for these conditions (Figure 604 (Sheet 2) and Figure 604 (Sheet 3)):

1) Fretting, erosion, or wear

a) Permitted if dimension D is between 0.134 in. (3.4 mm) and 0.228 in. (5.8 mm).

(h) Examine the Area 4 for these conditions (Figure 604 (Sheet 2) and Figure 604 (Sheet 3)):

1) Wear

a) Permitted up to 0.0019 in. (0.05 mm) in depth.

(i) Examine the inserts for these conditions (Figure 604 (Sheet 4)):

1) Missing

a) Not permitted.

2) Looseness

a) Not permitted.

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(j) Examine the replaceable nuts for these conditions (Figure 601 (Sheet 3)):

1) Missing

a) Not permitted.

b) Maximum repairable limit: Any amount.

c) To repair, do this task: Fan Frame Strut Pad Nut Replacement, TASK 72-23-06-960-803-G00.

2) Looseness

a) Not permitted.

b) Maximum repairable limit: Any amount.

c) To repair, do this task: Fan Frame Strut Pad Nut Replacement, TASK 72-23-06-960-803-G00.

(k) For the configuration 1, with the white rectangular cross section and the Dapco 2100 sealant, A50483 [CP2927] (Figure 604 (Sheet 5)):

1) Examine the rectangular cross section seal for these conditions:

a) Missing

<1> Not permitted.

<2> Maximum repairable limit: any amount.

<3> To repair, do this task: Fan Frame Strut Pad Seal Replacement, TASK 72-23-06-960-802-G00.

b) Looseness

<1> Not permitted.

<2> Maximum repairable limit: any amount.

<3> To repair, do this task: Fan Frame Strut Pad Seal Replacement, TASK 72-23-06-960-802-G00.

c) Tear

<1> Permitted up to 0.19 in. (5 mm) in width and 0.39 in. (10 mm) in length.

<2> Maximum repairable limit: any amount.

<3> To repair, do this task: Fan Frame Strut Pad Seal Replacement, TASK 72-23-06-960-802-G00.

d) Bubbles

<1> Not permitted.

<2> Maximum repairable limit: any amount.

<3> To repair, do this task: Fan Frame Strut Pad Seal Replacement, TASK 72-23-06-960-802-G00.

e) Permanent strain and excessive strain

<1> Not permitted.

<2> Maximum repairable limit: any amount.

<3> To repair, do this task: Fan Frame Strut Pad Seal Replacement, TASK 72-23-06-960-802-G00.

2) Examine the Dapco 2100 sealant, A50483 [CP2927], located next to the white rectangular cross section seal for these conditions:

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a) Missing

<1> Not permitted.

b) Looseness

<1> Not permitted.

c) Disbonding

<1> Not permitted.

d) Bubbles

<1> Not permitted.

e) Permanent strain and excessive strain

<1> Not permitted.

f) Lack of materials or through cavity

<1> Permitted up to 0.012 in. (0.3 mm).

(l) For the configuration 2, with the black two lips cross section seal and the Dapco 2100 sealant, A50483 [CP2927] (Figure 604 (Sheet 6)):

1) Examine the two lips cross section seal for these conditions:

a) Missing

<1> Not permitted.

<2> Maximum repairable limit: any amount.

<3> To repair, do this task: Fan Frame Strut Pad Seal Replacement, TASK 72-23-06-960-802-G00.

b) Looseness

<1> Not permitted.

<2> Maximum repairable limit: any amount.

<3> To repair, do this task: Fan Frame Strut Pad Seal Replacement, TASK 72-23-06-960-802-G00.

c) Tear

<1> Permitted up to 0.39 in. (10 mm) on one of the two seal lips.

<2> Maximum repairable limit: any amount.

<3> To repair, do this task: Fan Frame Strut Pad Seal Replacement, TASK 72-23-06-960-802-G00.

d) Bubbles

<1> Not permitted.

<2> Maximum repairable limit: any amount.

<3> To repair, do this task: Fan Frame Strut Pad Seal Replacement, TASK 72-23-06-960-802-G00.

e) Permanent strain and excessive strain

<1> Not permitted.

<2> Maximum repairable limit: any amount.

<3> To repair, do this task: Fan Frame Strut Pad Seal Replacement, TASK 72-23-06-960-802-G00.

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2) Examine the Dapco 2100 sealant, A50483 [CP2927], located next to the white rectangular cross section seal for these conditions:
   a) Missing
      <1> Not permitted.
   b) Looseness
      <1> Not permitted.
   c) Disbonding
      <1> Not permitted.
   d) Bubbles
      <1> Not permitted.
   e) Permanent strain and excessive strain
      <1> Not permitted.
   f) Lack of materials or through cavity
      <1> Permitted up to 0.012 in. (0.3 mm).

(m) Examine the black silicone sealing elastomer (RTV 103 adhesive, A50472 [CP2734]) for these conditions:
   1) Missing
      a) Permitted.
   2) Looseness
      a) Permitted.
   3) Wear
      a) Permitted.

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0189

SUBTASK 72-23-06-211-005-G00

(5) Do these steps to examine the RH-strut 1 fairing [5] (Figure 605):
   (a) Examine all areas for these conditions:
      1) Cracks
         a) Not permitted.
   (b) Examine the Area 1 for these conditions (Figure 605 (Sheet 1)):
      1) Dents
         a) Permitted up to 0.023 in. (0.6 mm) in depth.
         b) Maximum repairable limit: in the limits of applicable repair.
            <1> If the depth is less than 0.023 in. (0.584 mm).
            <2> If it can be blended to a finished blend depth of 0.023 in. (0.584 mm).
         c) Repair method: Fan Frame Strut 1 and 5 Fairings Blending, TASK 72-23-06-300-801-G00.
   (c) Examine the Area 2 for these conditions (Figure 605 (Sheet 1)):
      1) Dents
         a) Not permitted.
         b) Maximum repairable limit: in the limits of applicable repair.

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<1> If the depth is less than 0.023 in. (0.584 mm).

<2> If it can be blended to a finished blend depth of 0.023 in. (0.584 mm).

c) Repair method: Fan Frame Strut 1 and 5 Fairings Blending, TASK 72-23-06-300-801-G00.

(d) Examine the Area 3 for these conditions (Figure 605 (Sheet 2)):

1) Nicks and scratches

a) Permitted up to 0.023 in. (0.6 mm) in depth.

b) Maximum repairable limit: in the limits of applicable repair.

<1> If the depth is less than 0.023 in. (0.584 mm).

<2> If it can be blended to a finished blend depth of 0.023 in. (0.584 mm).

c) Repair method: Fan Frame Strut 1 and 5 Fairings Blending, TASK 72-23-06-300-801-G00.

(e) Examine the Area 4 for these conditions (Figure 605 (Sheet 2)):

1) Nicks and scratches

a) Not permitted.

b) Maximum repairable limit: in the limits of applicable repair.

<1> If the depth is less than 0.023 in. (0.584 mm).

<2> If it can be blended to a finished blend depth of 0.023 in. (0.584 mm).

c) Repair method: Fan Frame Strut 1 and 5 Fairings Blending, TASK 72-23-06-300-801-G00.

(f) Examine the Area 5 for these conditions (Figure 605 (Sheet 3)):

1) Wear

a) Permitted up to 0.0007 in. (0.02 mm) in depth.

(g) Examine the seals for these conditions (Figure 605 (Sheet 3)):

1) Missing

a) Not permitted.

b) Maximum repairable limit: any amount.

c) To repair, do this task: Fan Frame Strut 1 Fairing Seal Replacement, TASK 72-23-06-960-801-G00.

2) Looseness

a) Not permitted.

b) Maximum repairable limit: any amount.

c) To repair, do this task: Fan Frame Strut 1 Fairing Seal Replacement, TASK 72-23-06-960-801-G00.

3) Break, crack, scratch, and color change

a) Not permitted.

b) Maximum repairable limit: any amount.

c) To repair, do this task: Fan Frame Strut 1 Fairing Seal Replacement, TASK 72-23-06-960-801-G00.

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PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0189 (Continued)

4) Bubbles
   a) Not permitted.
   b) Maximum repairable limit: any amount.
   c) To repair, do this task: Fan Frame Strut 1 Fairing Seal Replacement, TASK 72-23-06-960-801-G00.
5) Permanent strain and excessive strain
   a) Not permitted.
   b) Maximum repairable limit: any amount.
   c) To repair, do this task: Fan Frame Strut 1 Fairing Seal Replacement, TASK 72-23-06-960-801-G00.

(h) Examine the screws of the RH-strut 1 fairing [5] (Figure 605 (Sheet 4)).
1) Surface oxidation on screw head
   a) Permitted.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0189

SUBTASK 72-23-06-211-013-G00

(6) Do these steps to examine the RH-strut 1 fairing [5] (Figure 605):

NOTE: Contact CFM if necessary.

(a) Do a visual examination of all areas.
1) Nicks, dents, scratches, 211–010 cracks, fretting, erosion, or wear
   a) Not permitted.

(b) Do a visual examination of replaceable nuts.
1) Missing
   a) Not permitted.
2) Looseness
   a) Not permitted.

(c) Do a visual examination of the black silicone sealing elastomer RTV 103 adhesive, A50472 [CP2734].
1) Missing
   a) Not permitted.
2) Looseness
   a) Not permitted.
3) Wear
   a) Not permitted.

(d) Examine the screws of the RH-strut 1 fairing [5] (Figure 605 (Sheet 4)).
1) Surface oxidation on screw head
   a) Permitted.

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PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0189

SUBTASK 72-23-06-211-006-G00

(7) Do these steps to examine the LH-strut 1 fairing [6] (Figure 606):

(a) Examine all areas for these conditions:

1) Cracks

a) Not permitted.

(b) Examine the Area 1 for these conditions (Figure 606 (Sheet 1)):

1) Dents

a) Permitted up to 0.023 in. (0.6 mm) in depth.

b) Maximum repairable limit: in the limits of applicable repair.

<1> If the depth is less than 0.023 in. (0.584 mm).

<2> If it can be blended to a finished blend depth of 0.023 in. (0.584 mm).

c) Repair method: Fan Frame Strut 1 and 5 Fairings Blending, TASK 72-23-06-300-801-G00.

(c) Examine the Area 2 for these conditions (Figure 606 (Sheet 1)):

1) Dents

a) Not permitted.

b) Maximum repairable limit: in the limits of applicable repair.

<1> If the depth is less than 0.023 in. (0.584 mm).

<2> If it can be blended to a finished blend depth of 0.023 in. (0.584 mm).

c) Repair method: Fan Frame Strut 1 and 5 Fairings Blending, TASK 72-23-06-300-801-G00.

(d) Examine the Area 3 for these conditions (Figure 606 (Sheet 2)):

1) Nicks and scratches

a) Permitted up to 0.023 in. (0.6 mm) in depth.

b) Maximum repairable limit: in the limits of applicable repair.

<1> If the depth is less than 0.023 in. (0.584 mm).

<2> If it can be blended to a finished blend depth of 0.023 in. (0.584 mm).

c) Repair method: Fan Frame Strut 1 and 5 Fairings Blending, TASK 72-23-06-300-801-G00.

(e) Examine the Area 4 for these conditions (Figure 606 (Sheet 2)):

1) Nicks and scratches

a) Not permitted.

b) Maximum repairable limit: in the limits of applicable repair.

<1> If the depth is less than 0.023 in. (0.584 mm).

<2> If it can be blended to a finished blend depth of 0.023 in. (0.584 mm).

c) Repair method: Fan Frame Strut 1 and 5 Fairings Blending, TASK 72-23-06-300-801-G00.

(f) Examine the Area 5 for these conditions (Figure 606 (Sheet 3)):

1) Wear

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PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0189 (Continued)

a) Permitted up to 0.0007 in. (0.02 mm) in depth.

(g) Examine the seals for these conditions (Figure 606 (Sheet 3)):

1) Missing

a) Not permitted.

b) Maximum repairable limit: any amount.

c) To repair, do this task: Fan Frame Strut 1 Fairing Seal Replacement, TASK 72-23-06-960-801-G00.

2) Looseness

a) Not permitted.

b) Maximum repairable limit: any amount.

c) To repair, do this task: Fan Frame Strut 1 Fairing Seal Replacement, TASK 72-23-06-960-801-G00.

3) Break, crack, scratch, and color change

a) Not permitted.

b) Maximum repairable limit: any amount.

c) To repair, do this task: Fan Frame Strut 1 Fairing Seal Replacement, TASK 72-23-06-960-801-G00.

4) Bubbles

a) Not permitted.

b) Maximum repairable limit: any amount.

c) To repair, do this task: Fan Frame Strut 1 Fairing Seal Replacement, TASK 72-23-06-960-801-G00.

5) Permanent strain and excessive strain

a) Not permitted.

b) Maximum repairable limit: any amount.

c) To repair, do this task: Fan Frame Strut 1 Fairing Seal Replacement, TASK 72-23-06-960-801-G00.

(h) Examine the screws of the LH-strut 1 fairing [6] (Figure 606 (Sheet 4)).

1) Surface oxidation on screw head

a) Permitted.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0189

SUBTASK 72-23-06-211-014-G00

(8) Do these steps to examine the LH-strut 1 fairing [6] (Figure 606):

NOTE: Contact CFM if necessary.

(a) Do a visual examination of all areas.

1) Nicks, dents, scratches, cracks, fretting, erosion, or wear

a) Not permitted.

(b) Do a visual examination of replaceable nuts.

1) Missing

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0189 (Continued)

a) Not permitted.

2) Looseness

a) Not permitted.

(c) Do a visual examination of the black silicone sealing elastomer RTV 103 adhesive, A50472 [CP2734].

1) Missing

a) Not permitted.

2) Looseness

a) Not permitted.

3) Wear

a) Not permitted.

(d) Examine the screws of the LH-strut 1 fairing [6] (Figure 606 (Sheet 4)).

1) Surface oxidation on screw head

a) Permitted.

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0177

SUBTASK 72-23-06-211-007-G00

(9) Do these steps to examine the RH-strut 5 fairing [7] (Figure 607 (Sheet 1), Figure 607 (Sheet 2), and Figure 607 (Sheet 5)):

(a) Examine all areas for these conditions:

1) Cracks

a) Not permitted.

(b) Examine the Area 1 for these conditions (Figure 607 (Sheet 1)):

1) Dents

a) Permitted up to 0.023 in. (0.6 mm) in depth.

b) Maximum repairable limit: in the limits of applicable repair.

<1> If the depth is less than 0.023 in. (0.584 mm).

<2> If it can be blended to a finished blend depth of 0.023 in. (0.584 mm).

c) Repair method: Fan Frame Strut 1 and 5 Fairings Blending, TASK 72-23-06-300-801-G00.

2) Nicks and scratches

a) Permitted up to 0.003 in. (0.1 mm) in depth.

b) Maximum repairable limit: in the limits of applicable repair.

<1> If the depth is less than 0.023 in. (0.584 mm).

<2> If it can be blended to a finished blend depth of 0.023 in. (0.584 mm).

c) Repair method: Fan Frame Strut 1 and 5 Fairings Blending, TASK 72-23-06-300-801-G00.

(c) Examine the Area 2 for these conditions (Figure 607 (Sheet 1)):

1) Dents

a) Not permitted.

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PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0177 (Continued)

b) Maximum repairable limit: in the limits of applicable repair.

<1> If the depth is less than 0.023 in. (0.584 mm).

<2> If it can be blended to a finished blend depth of 0.023 in. (0.584 mm).

c) Repair method: Fan Frame Strut 1 and 5 Fairings Blending, TASK 72-23-06-300-801-G00.

(d) Examine the Area 3 for these conditions (Figure 607 (Sheet 2)):

1) Wear.

a) Permitted up to 0.0007 in. (0.02 mm) in depth.

(e) Examine the black silicone sealing elastomer (RTV 103 adhesive, A50472 [CP2734]) for these conditions:

1) Missing

a) Permitted.

2) Looseness

a) Permitted.

3) Wear

a) Permitted.

(f) Examine the screws of the RH-strut 5 fairing [7] (Figure 607 (Sheet 5)).

1) Surface oxidation on screw head

a) Permitted.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0177

SUBTASK 72-23-06-211-011-G00

(10) Do these steps to examine the RH-strut 5 fairing [7] (Figure 607 (Sheet 3), Figure 607 (Sheet 4), Figure 607 (Sheet 5), and Figure 611):

(a) Examine all areas for these conditions except Area 5:

1) Cracks/ broken fibers

a) Not permitted.

(b) Examine the Area 1 for these conditions:

1) Dents, nicks, and scratches

a) Not permitted.

(c) Examine the Area 2 for these conditions:

1) Dents, nicks, and scratches

a) Not permitted.

2) Wear

a) Permitted up to 0.0007 in. (0.02 mm) in depth.

3) Flaking

a) Not permitted.

(d) Examine the Area 3 except trailing edge for these conditions:

1) Dents, nicks, and scratches

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) Permitted up to 0.0035 in. (0.09 mm) in depth.

2) Eroded glass film coating

a) Permitted up to 50% cumulated surface of the Area 3 surface.

3) Eroded carbon composite fiber

a) Not permitted.

(e) Examine the Area 4 for these conditions:

1) Dents, nicks, and scratches

a) Not permitted.

2) Wear of glass film coating

a) Permitted up to 0.0035 in. (0.09 mm) in depth.

3) Worn carbon composite fiber

a) Not permitted.

(f) Examine the Area 5 for these conditions:

1) Dents, nicks, scratches, and cracks/broken fibers

a) Permitted up to 0.024 in. (0.6 mm) in depth.

2) Flaking

a) Permitted with the following conditions:

- Up to 0.024 in. (0.6 mm) in depth
- Up to 1.00 in. (25.4 mm) diameter around the countersinks axis.

3) Eroded glass film coating

a) Permitted up to 50% cumulated surface of the Area 5 surface.

4) Eroded carbon composite fiber

a) Not permitted.

(g) Examine the countersinks and the cylindrical part of the holes for these conditions:

1) Wear

a) Permitted up to 0.0007 in. (0.02 mm) in depth.

2) Flaking of carbon fiber

a) Not permitted.

(h) Examine the contour of the fairing (except the corner) and the trailing edge (Figure 607 (Sheet 4)) for these conditions:

1) Dents, nicks, and scratches

a) Not permitted.

(i) Examine the corner of the fairing for these conditions:

1) Flaking, nicks, dents, scratches, and cracks/broken fiber

a) Not permitted.

(j) Examine the black silicone sealing elastomer (RTV 103 adhesive, A50472 [CP2734]) for these conditions:

1) Missing

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0177 (Continued)

a) Permitted.

2) Looseness

a) Permitted.

3) Wear

a) Permitted.

(k) Examine the trailing edge (Figure 607 (Sheet 4) and Figure 611).

1) Migration of the gray gap filler through the plies.

a) Any amount on the whole edge and on the area 0.59 in. (15 mm) on both sides of the edge.

2) Porosities

a) Up to 30 if dimensions within 0.039 in. (1.0 mm) and no sharp edges.

b) Up to 5 if dimensions between 0.039 in. (1.0 mm) and 0.590 in. (15.0 mm) and no sharp edges.

(l) Examine the screws of the RH-strut 5 fairing [7] (Figure 607 (Sheet 5)).

1) Surface oxidation on screw head

a) Permitted.

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0177

SUBTASK 72-23-06-211-008-G00

(11) Do these steps to examine the LH-strut 5 fairing [8] (Figure 608 (Sheet 1), Figure 608 (Sheet 2), and Figure 608 (Sheet 5)):

(a) Examine all areas for these conditions:

1) Cracks

a) Not permitted.

(b) Examine the Area 1 for these conditions (Figure 608 (Sheet 1)):

1) Dents

a) Permitted up to 0.023 in. (0.6 mm) in depth.

b) Maximum repairable limit: in the limits of applicable repair.

<1> If the depth is less than 0.023 in. (0.584 mm).

<2> If it can be blended to a finished blend depth of 0.023 in. (0.584 mm).

c) Repair method: Fan Frame Strut 1 and 5 Fairings Blending, TASK 72-23-06-300-801-G00.

2) Nicks and scratches

a) Permitted up to 0.003 in. (0.1 mm) in depth.

b) Maximum repairable limit: in the limits of applicable repair.

<1> If the depth is less than 0.023 in. (0.584 mm).

<2> If it can be blended to a finished blend depth of 0.023 in. (0.584 mm).

c) Repair method: Fan Frame Strut 1 and 5 Fairings Blending, TASK 72-23-06-300-801-G00.

(c) Examine the Area 2 for these conditions (Figure 608 (Sheet 1)):

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0177 (Continued)

1) Dents
   a) Not permitted.
   b) Maximum repairable limit: in the limits of applicable repair.
      <1> If the depth is less than 0.023 in. (0.584 mm).
      <2> If it can be blended to a finished blend depth of 0.023 in. (0.584 mm).
   c) Repair method: Fan Frame Strut 1 and 5 Fairings Blending, TASK 72-23-06-300-801-G00.

(d) Examine the Area 3 for these conditions (Figure 608 (Sheet 2)):
1) Wear.
   a) Permitted up to 0.0007 in. (0.02 mm) in depth.

(e) Examine the Area 4 for these conditions (Figure 608 (Sheet 2)):
1) Contact marks and wear
   a) Permitted up to 0.0007 in. (0.02 mm) in depth.

(f) Examine the black silicone sealing elastomer (RTV 103 adhesive, A50472 [CP2734]) for these conditions:
1) Missing
   a) Permitted.
2) Looseness
   a) Permitted.
3) Wear
   a) Permitted.

(g) Examine the screws of the LH-strut 5 fairing [8] (Figure 608 (Sheet 5)).
1) Surface oxidation on screw head
   a) Permitted.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0177

SUBTASK 72-23-06-211-012-G00

(12) Do these steps to examine the LH-strut 5 fairing [8] (Figure 608 (Sheet 3), Figure 608 (Sheet 4), Figure 608 (Sheet 5), and Figure 611):
(a) Examine all areas for these conditions except Area 5:
1) Cracks/ broken fibers
   a) Not permitted.

(b) Examine the Area 1 for these conditions:
1) Dents, nicks, and scratches
   a) Not permitted.

(c) Examine the Area 2 for these conditions:
1) Dents, nicks, and scratches
   a) Not permitted.
2) Wear.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) Permitted up to 0.0007 in. (0.02 mm) in depth.

3) Flaking

a) Not permitted.

(d) Examine the Area 3 except trailing edge for these conditions:

1) Dents, nicks, and scratches

a) Permitted up to 0.0035 in. (0.09 mm) in depth.

2) Eroded glass film coating

a) Permitted up to 50% cumulated surface of the Area 3 surface

3) Eroded carbon composite fiber

a) Not permitted.

(e) Examine the Area 4 for these conditions:

1) Dents, nicks, and scratches

a) Not permitted.

2) Wear of glass film coating

a) Permitted up to 0.0035 in. (0.09 mm) in depth.

3) Worn carbon composite fiber

a) Not permitted.

(f) Examine the Area 5 for these conditions:

1) Dents, nicks, scratches, and cracks/broken fibers

a) Permitted up to 0.024 in. (0.6 mm) in depth.

2) Flaking

a) Permitted with the following conditions:

- Up to 0.024 in. (0.6 mm) in depth
- Up to 1.00 in. (25.4 mm) diameter around the countersinks axis.

3) Eroded glass film coating

a) Permitted up to 50% cumulated surface of the Area 5 surface.

4) Eroded carbon composite fiber

a) Not permitted.

(g) Examine the countersinks and the cylindrical part of the holes for these conditions:

1) Wear

a) Permitted up to 0.0007 in. (0.02 mm) in depth

2) Flaking of carbon fiber

a) Not permitted.

(h) Examine the contour of the fairing (except the corner) and the trailing edge (Figure 608 (Sheet 4)) for these conditions:

1) Dents, nicks, and scratches

a) Not permitted.

(i) Examine the corner of the fairing for these conditions:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Flaking, nicks, dents, scratches, and cracks/broken fibers
   a) Not permitted.

(j) Examine the black silicone sealing elastomer (RTV 103 adhesive, A50472 [CP2734]) for these conditions:
1) Missing
   a) Permitted.
2) Looseness
   a) Permitted.
3) Wear
   a) Permitted.

(k) Examine the trailing edge (Figure 608 (Sheet 4) and Figure 611).
1) Migration of the gray gap filler through the plies.
   a) Any amount on the whole edge and on the area 0.59 in. (15 mm) on both sides of the edge.
2) Porosities
   a) Up to 30 if dimensions within 0.039 in. (1.0 mm) and no sharp edges.
   b) Up to 5 if dimensions between 0.039 in. (1.0 mm) and 0.590 in. (15.0 mm) and no sharp edges.

(l) Examine the screws of the LH-strut 5 fairing [8] (Figure 608 (Sheet 5)).
1) Surface oxidation on screw head
   a) Permitted.

PTF ALL

SUBTASK 72-23-06-211-009-G00

(13) Do these steps to examine the VBV ducts [9] (Figure 609 (Sheet 1) and Figure 609 (Sheet 2)):
(a) Examine all areas for these conditions:
1) Cracks
   a) Not permitted.
2) Dent
   a) Not permitted.
3) Nicks and scratches
   a) Not permitted.

(b) Examine the Area 3 for these conditions (Figure 609 (Sheet 1)):
1) Wear
   a) Permitted up to 0.0007 in. (0.02 mm) in depth.

(c) Examine the Area 4 for these conditions (Figure 609 (Sheet 1)):
1) Fretting, erosion, or wear
   a) Permitted if dimension A is more or equal than 0.023 in. (0.6 mm).

(d) Examine the replaceable nuts for these conditions (Figure 609 (Sheet 2)):
1) Missing

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) Not permitted.

b) Maximum repairable limit: Any amount.

c) To repair, do this task: Variable Bleed Valve (VBV) Duct Nut Replacement, TASK 72-23-04-960-804-G00.

2) Looseness

a) Not permitted.

b) Maximum repairable limit: Any amount.

c) To repair, do this task: Variable Bleed Valve (VBV) Duct Nut Replacement, TASK 72-23-04-960-804-G00.

SUBTASK 72-23-06-211-010-G00

(14) Do these steps to examine the VBV duct seals [10] (Figure 610 (Sheet 1)):

(a) Examine the area A (seal profile) for these conditions:

1) Wear, textile delamination, and color change

a) Not permitted.

2) Break, crack, and scratch

a) Not permitted.

3) Bubbles

a) Not permitted.

4) Permanent strain, excessive strain

a) Not permitted.

(b) Examine the Area B for these conditions:

1) Missing

a) Permitted, but not two or more consecutive pins.

2) Looseness

a) Not permitted.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-23-06-410-002-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-23-06-860-004-G00

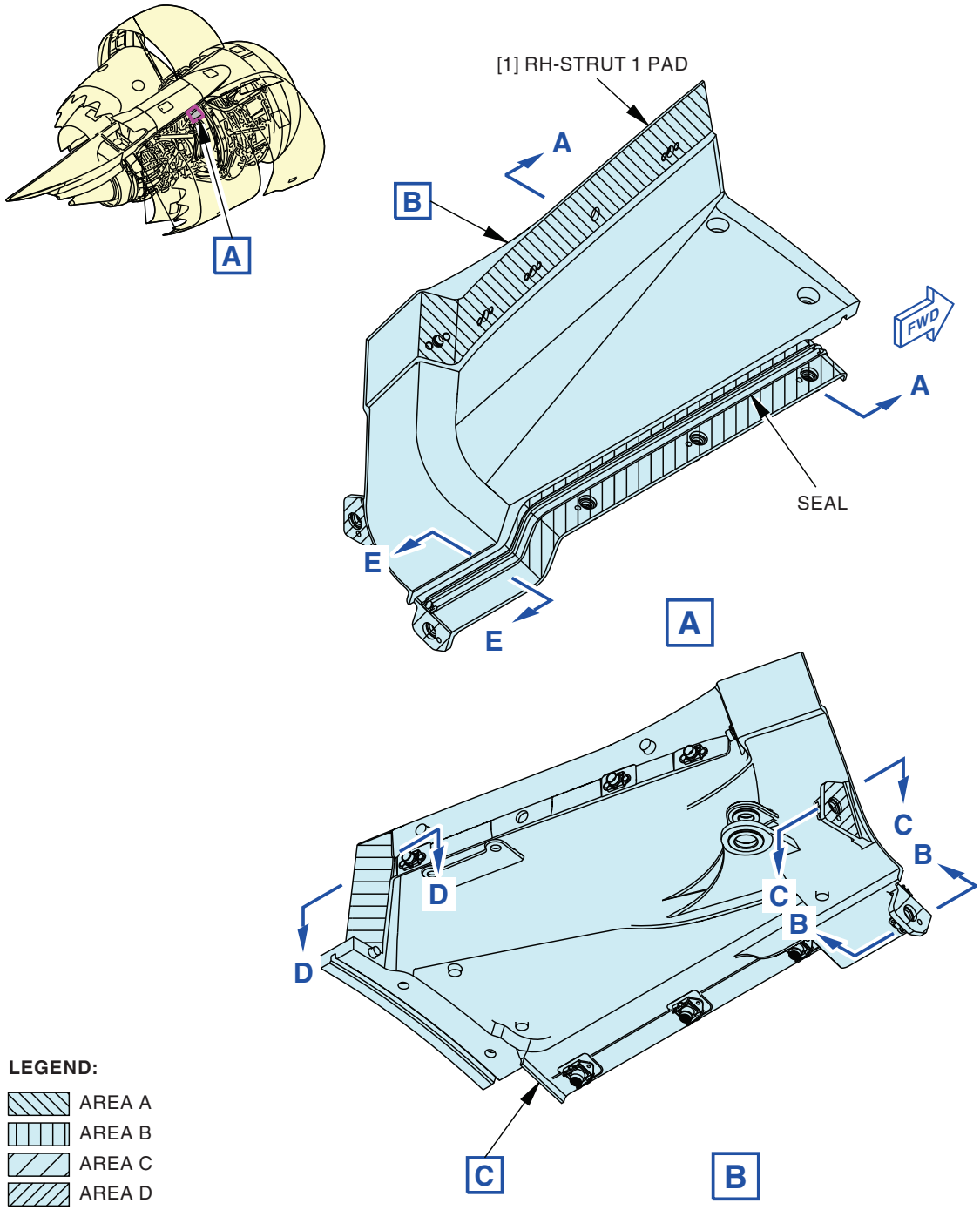
(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

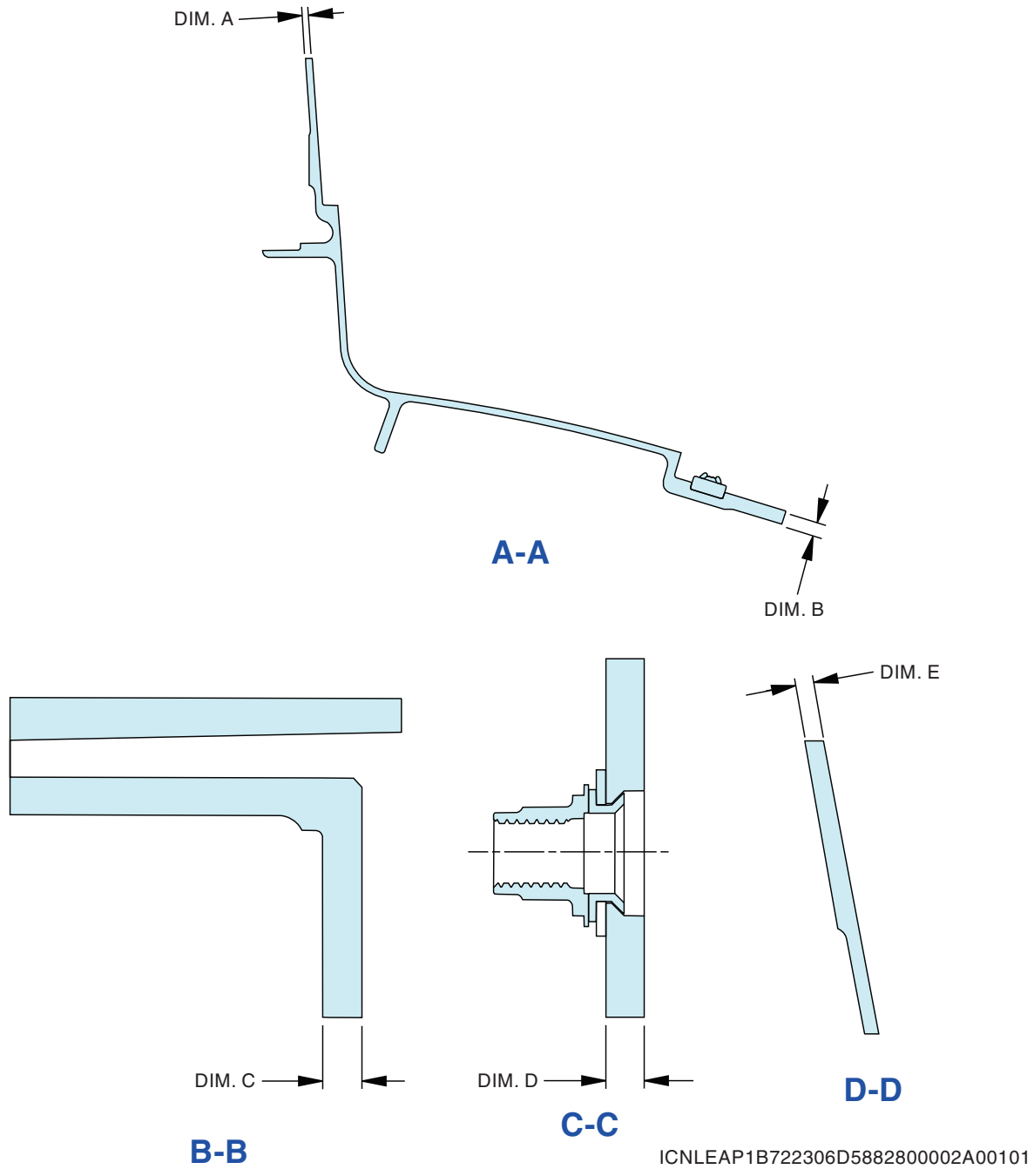
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



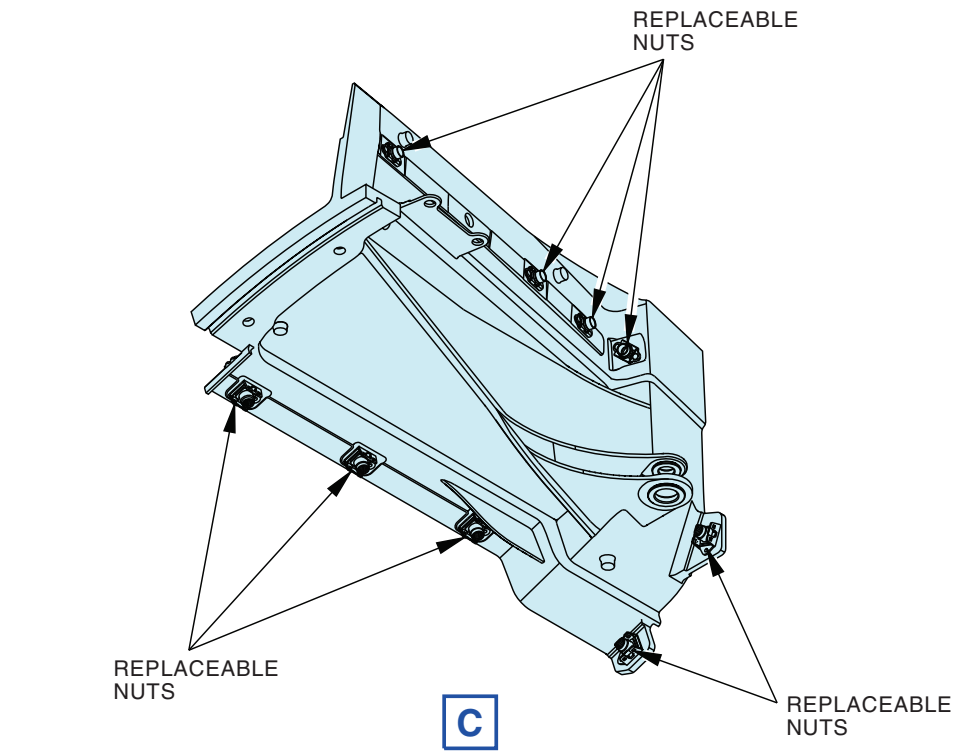
Fan Frame Strut Panels (RH-Strut 1 Pad) Inspection
Figure 601/72-23-06-990-803-G00 (Sheet 1 of 4)

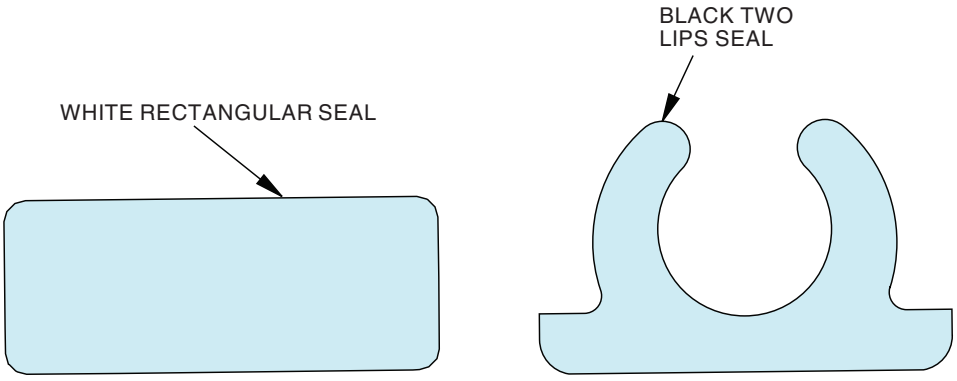
ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Frame Strut Panels (RH-Strut 1 Pad) Inspection
Figure 601/72-23-06-990-803-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



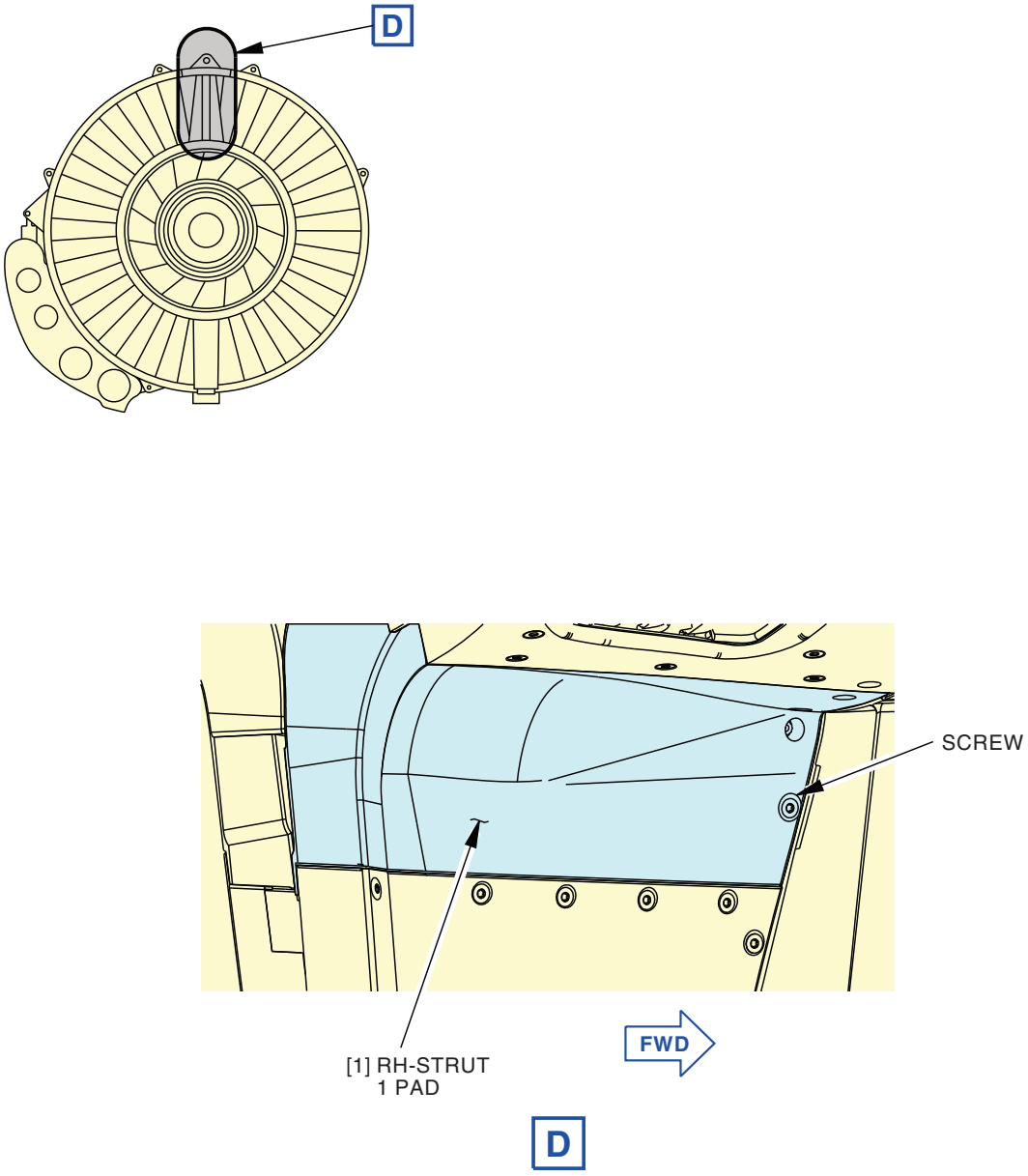


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Fan Frame Strut Panels (RH-Strut 1 Pad) Inspection
Figure 601/72-23-06-990-803-G00 (Sheet 3 of 4)

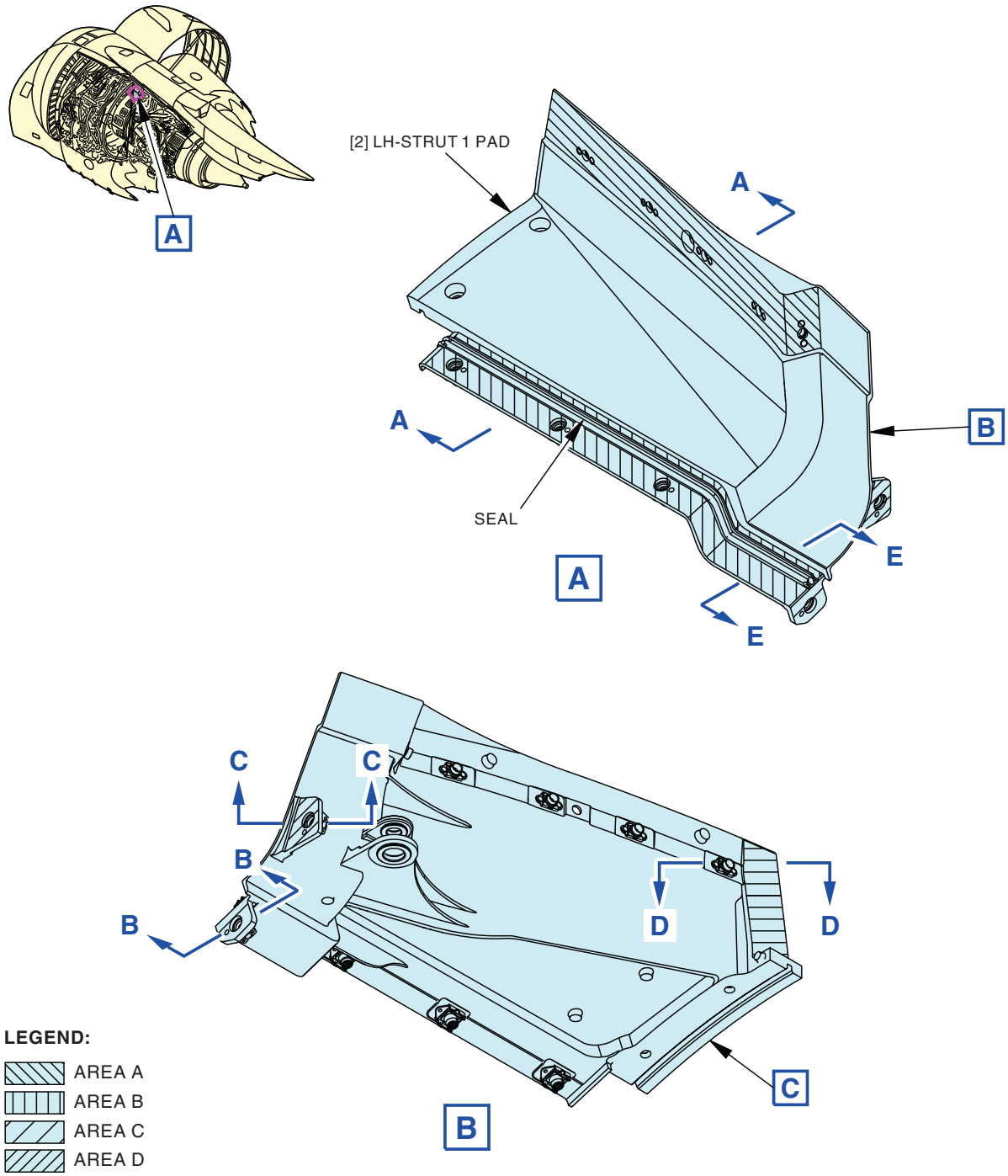
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Fan Frame Strut Panels (RH-Strut 1 Pad) Inspection
Figure 601/72-23-06-990-803-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

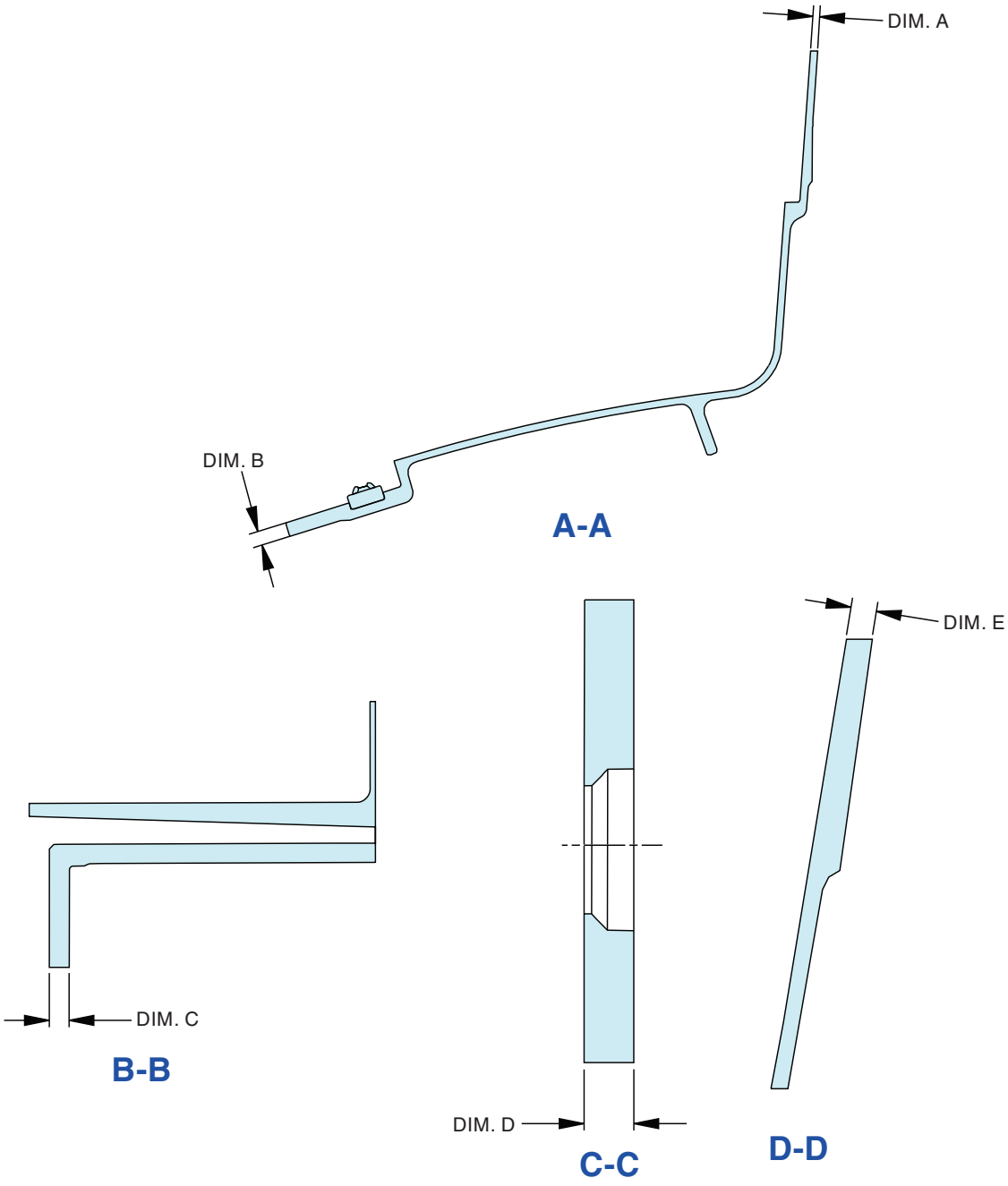


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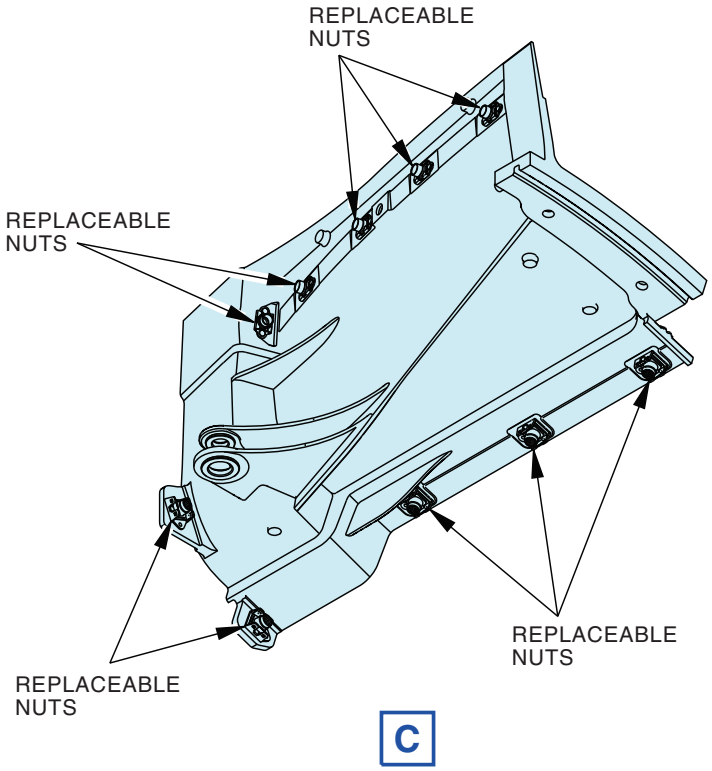
Fan Frame Strut Panels (LH-Strut 1 Pad) Inspection
Figure 602/72-23-06-990-820-G00 (Sheet 1 of 4)

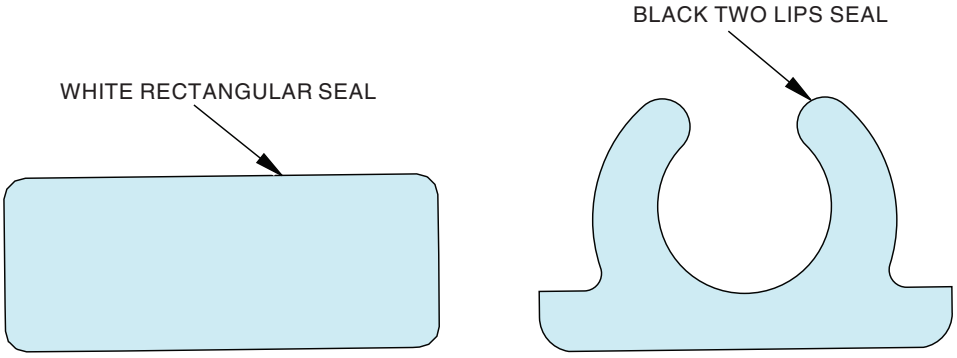
ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Frame Strut Panels (LH-Strut 1 Pad) Inspection
Figure 602/72-23-06-990-820-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



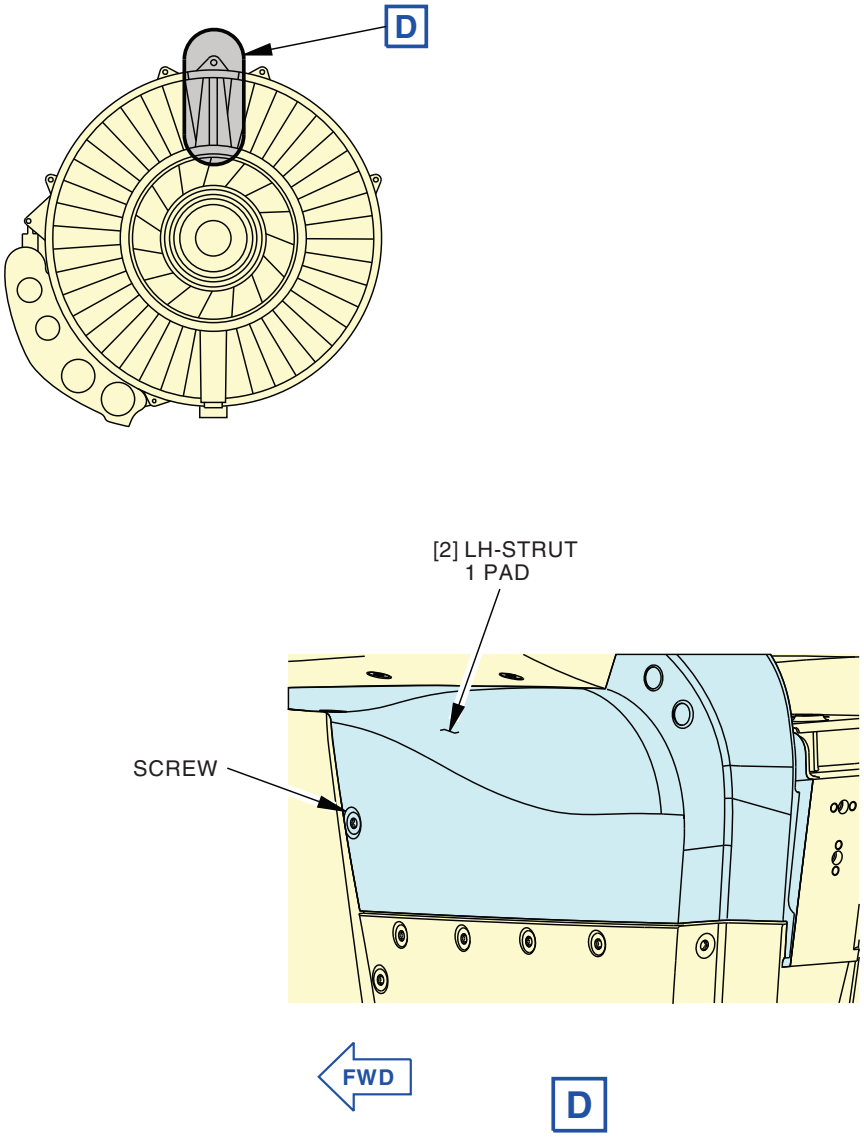


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Fan Frame Strut Panels (LH-Strut 1 Pad) Inspection
Figure 602/72-23-06-990-820-G00 (Sheet 3 of 4)

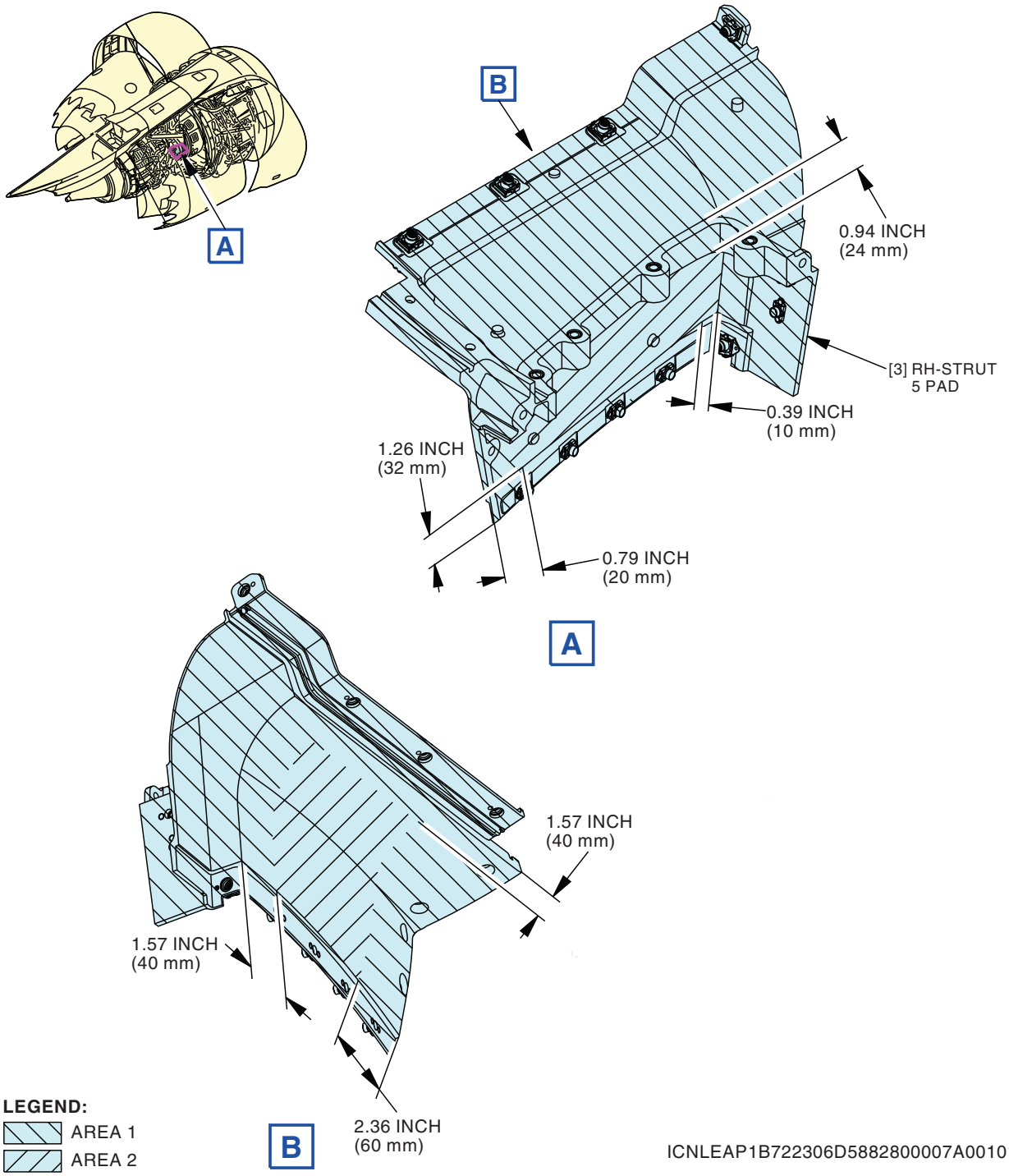
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Fan Frame Strut Panels (LH-Strut 1 Pad) Inspection
Figure 602/72-23-06-990-820-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

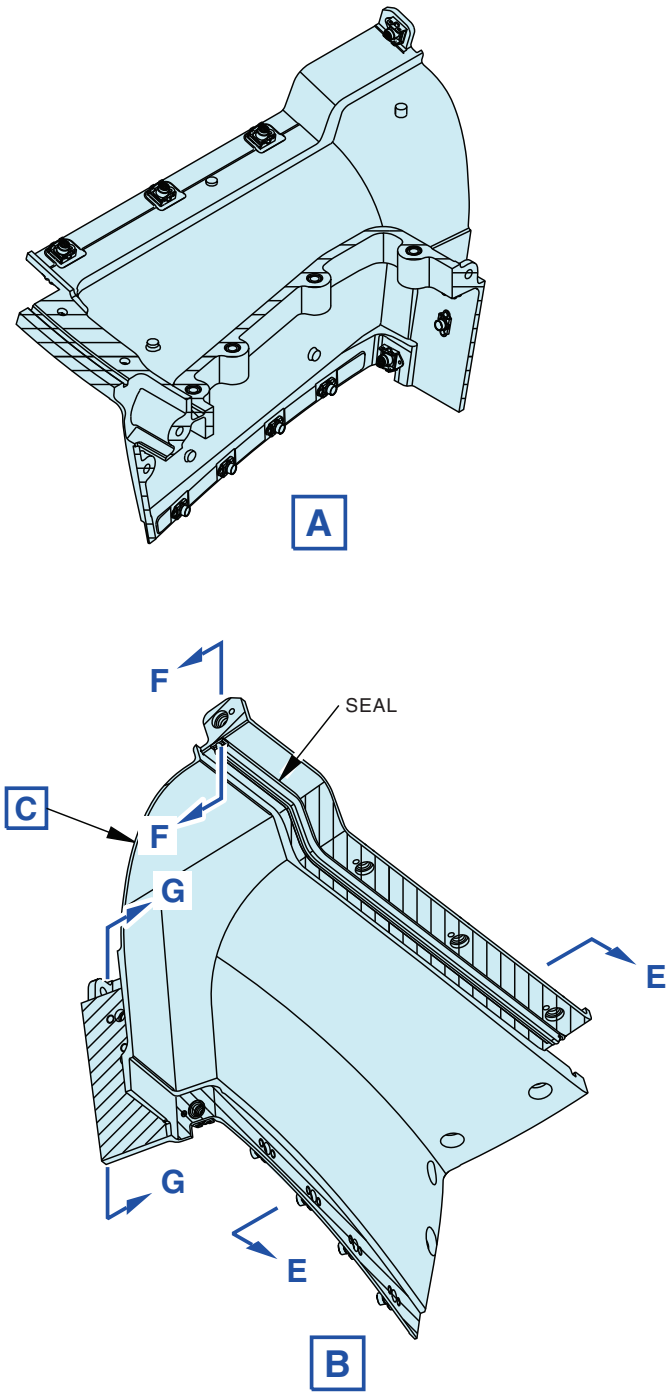


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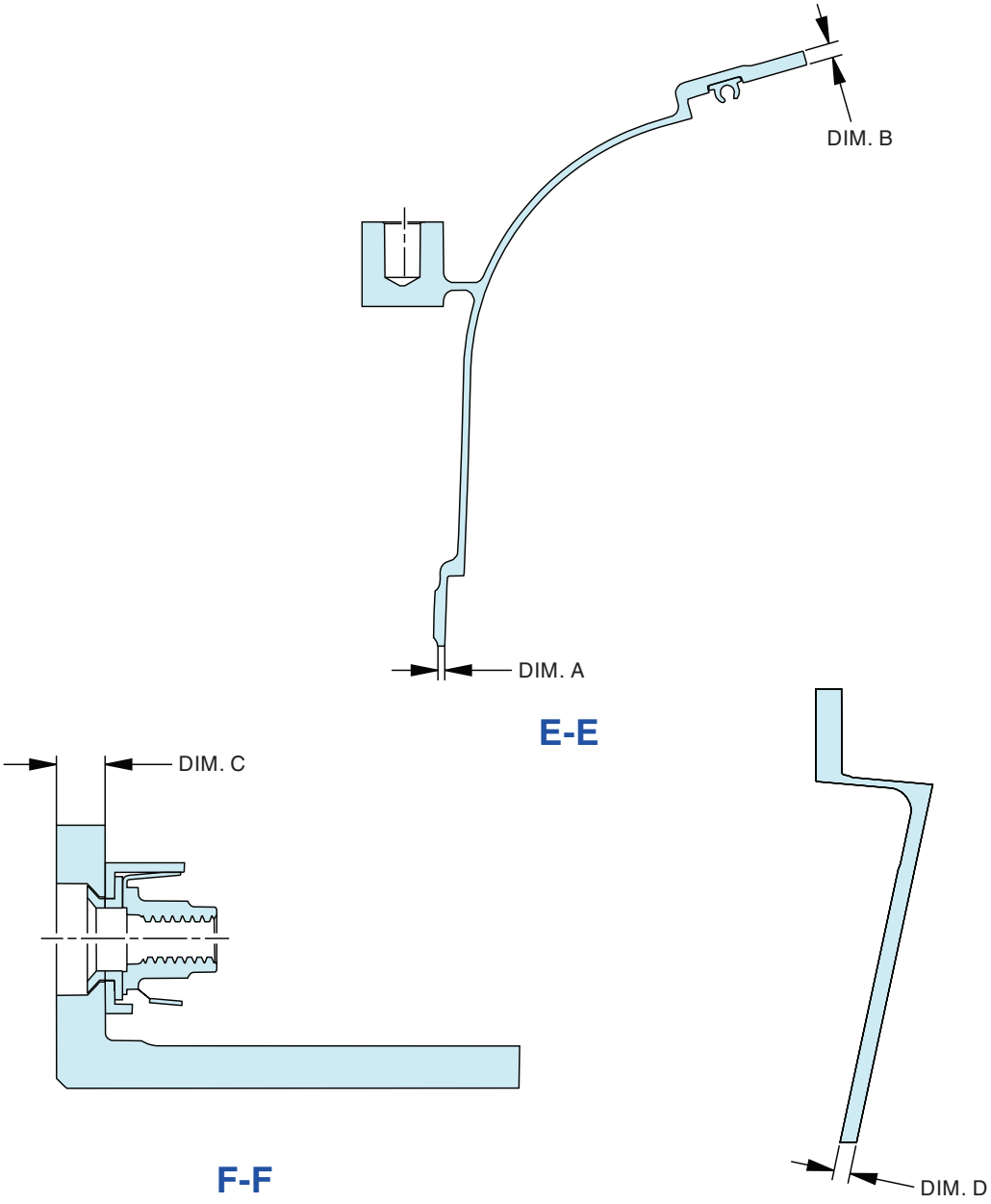
Fan Frame Strut Panels (RH-Strut 5 Pad) Inspection
Figure 603/72-23-06-990-813-G00 (Sheet 1 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Frame Strut Panels (RH-Strut 5 Pad) Inspection
Figure 603/72-23-06-990-813-G00 (Sheet 2 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

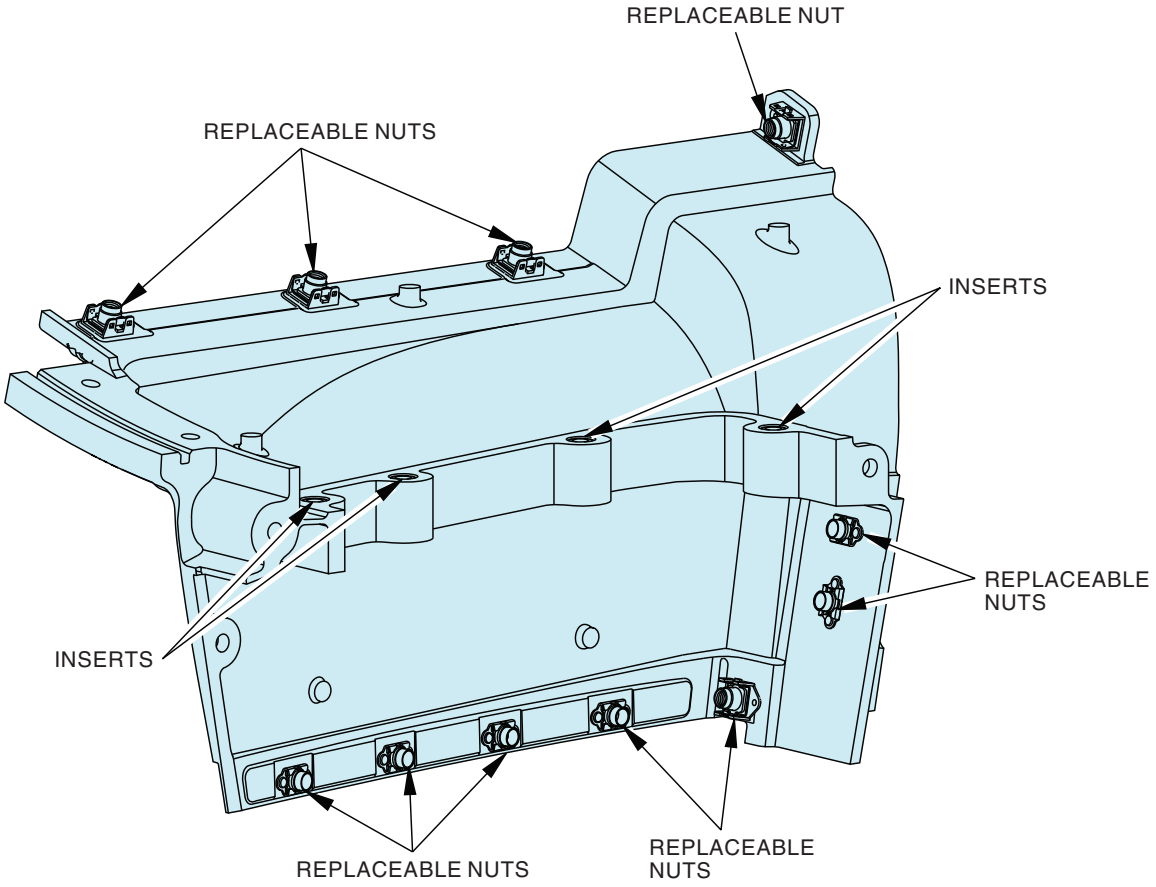


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Fan Frame Strut Panels (RH-Strut 5 Pad) Inspection
Figure 603/72-23-06-990-813-G00 (Sheet 3 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



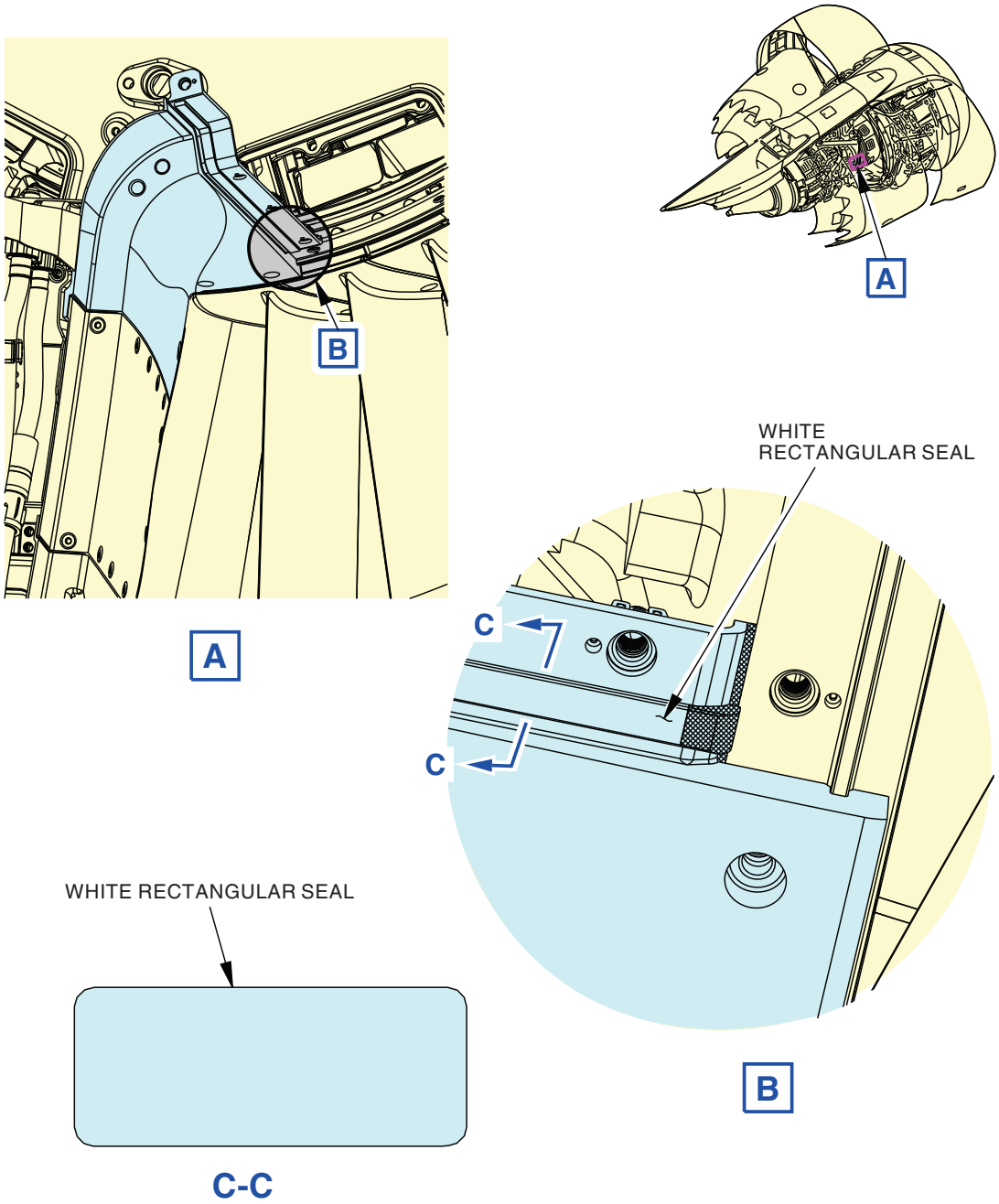
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Fan Frame Strut Panels (RH-Strut 5 Pad) Inspection
Figure 603/72-23-06-990-813-G00 (Sheet 4 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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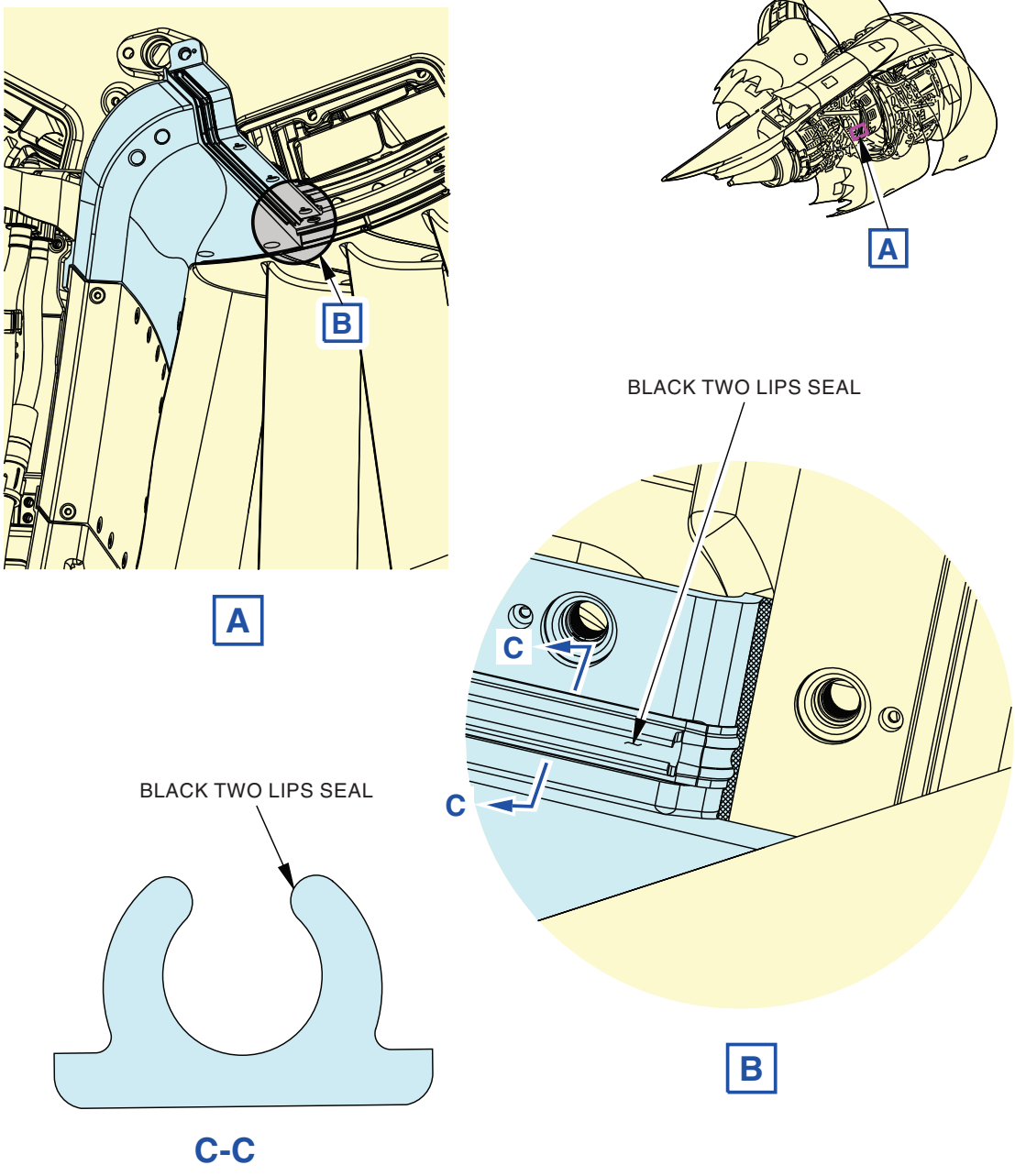
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Fan Frame Strut Panels (RH-Strut 5 Pad) Inspection
Figure 603/72-23-06-990-813-G00 (Sheet 5 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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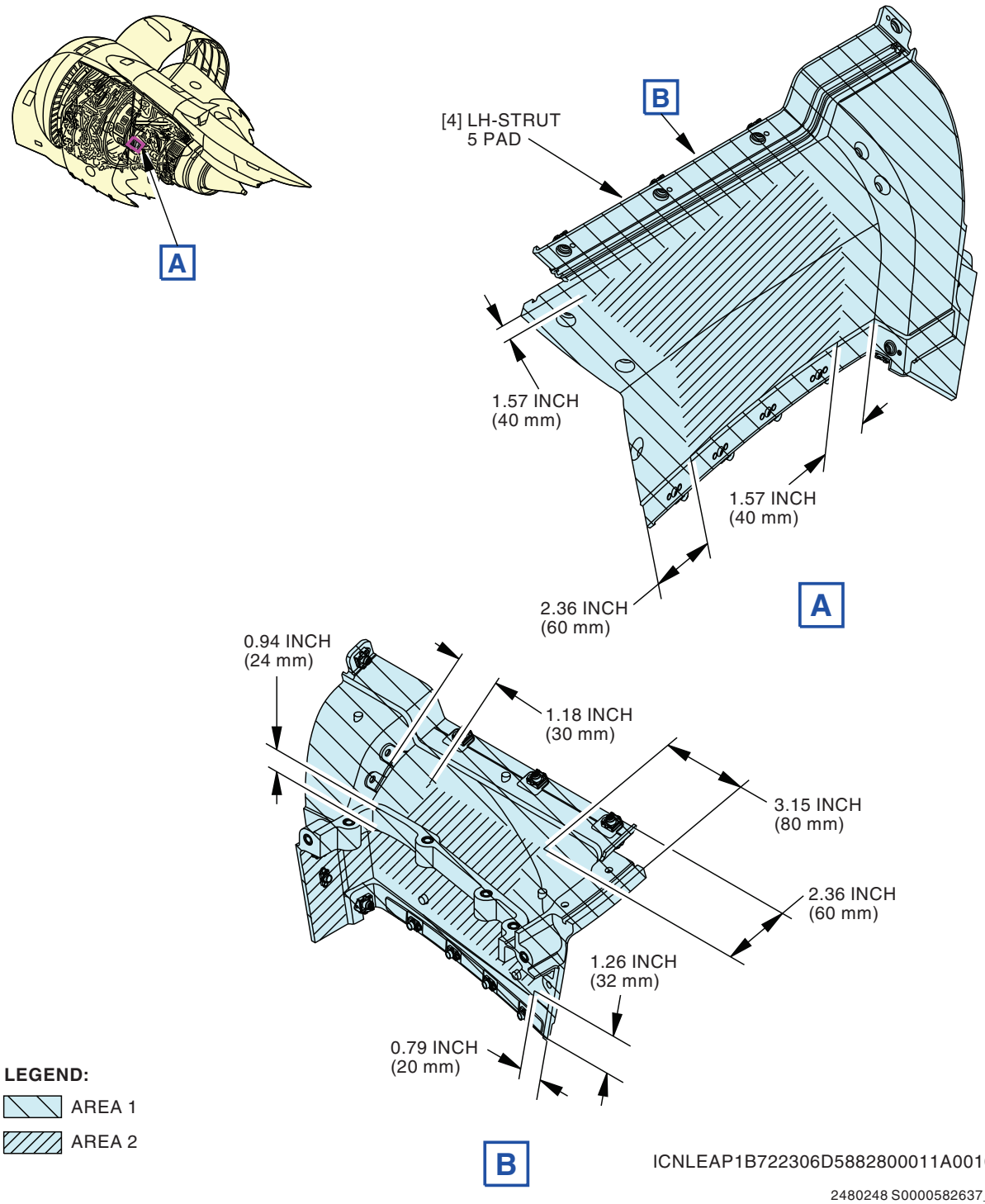
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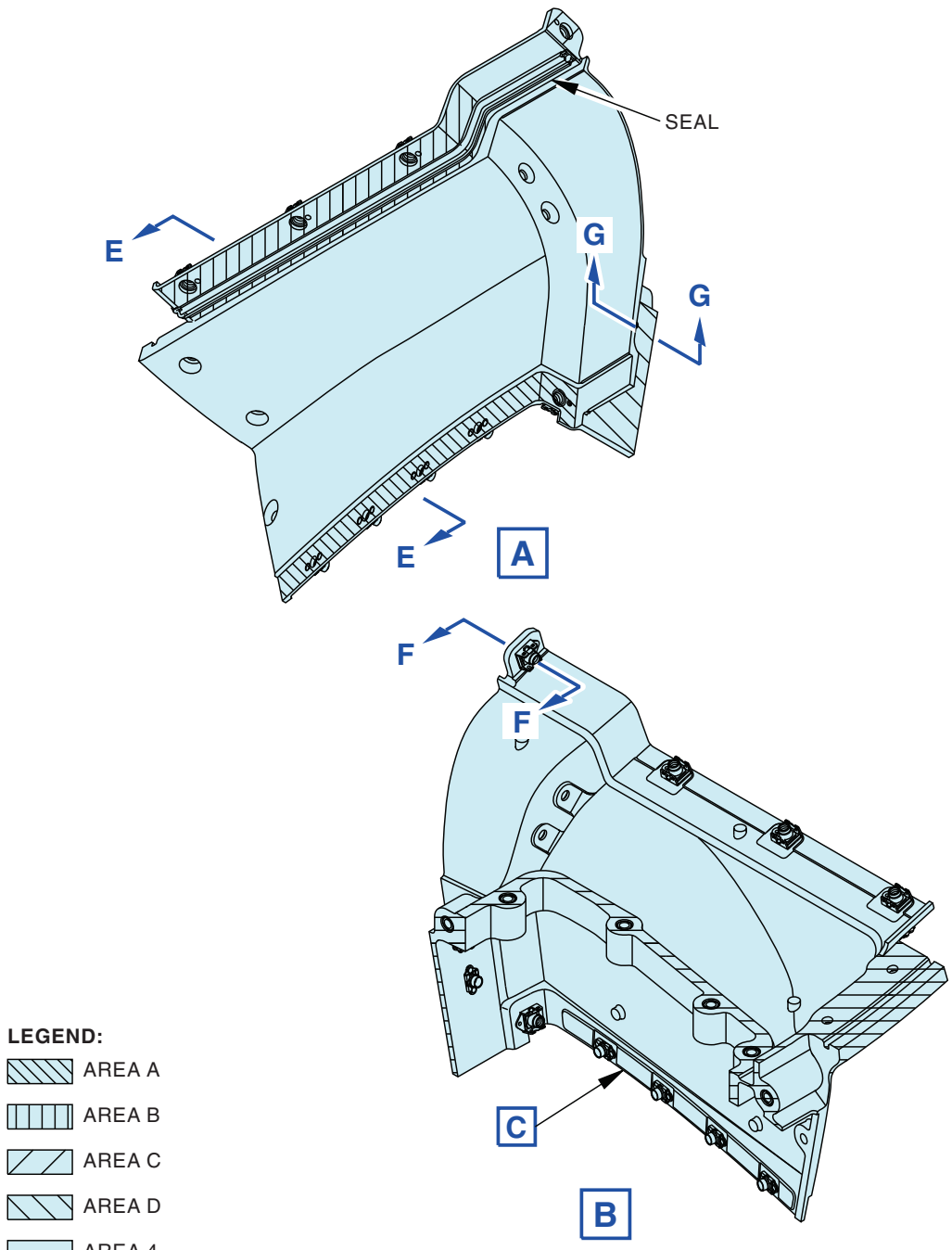
Fan Frame Strut Panels (RH-Strut 5 Pad) Inspection
Figure 603/72-23-06-990-813-G00 (Sheet 6 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Frame Strut Panels (LH-Strut 5 Pad) Inspection
Figure 604/72-23-06-990-814-G00 (Sheet 1 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

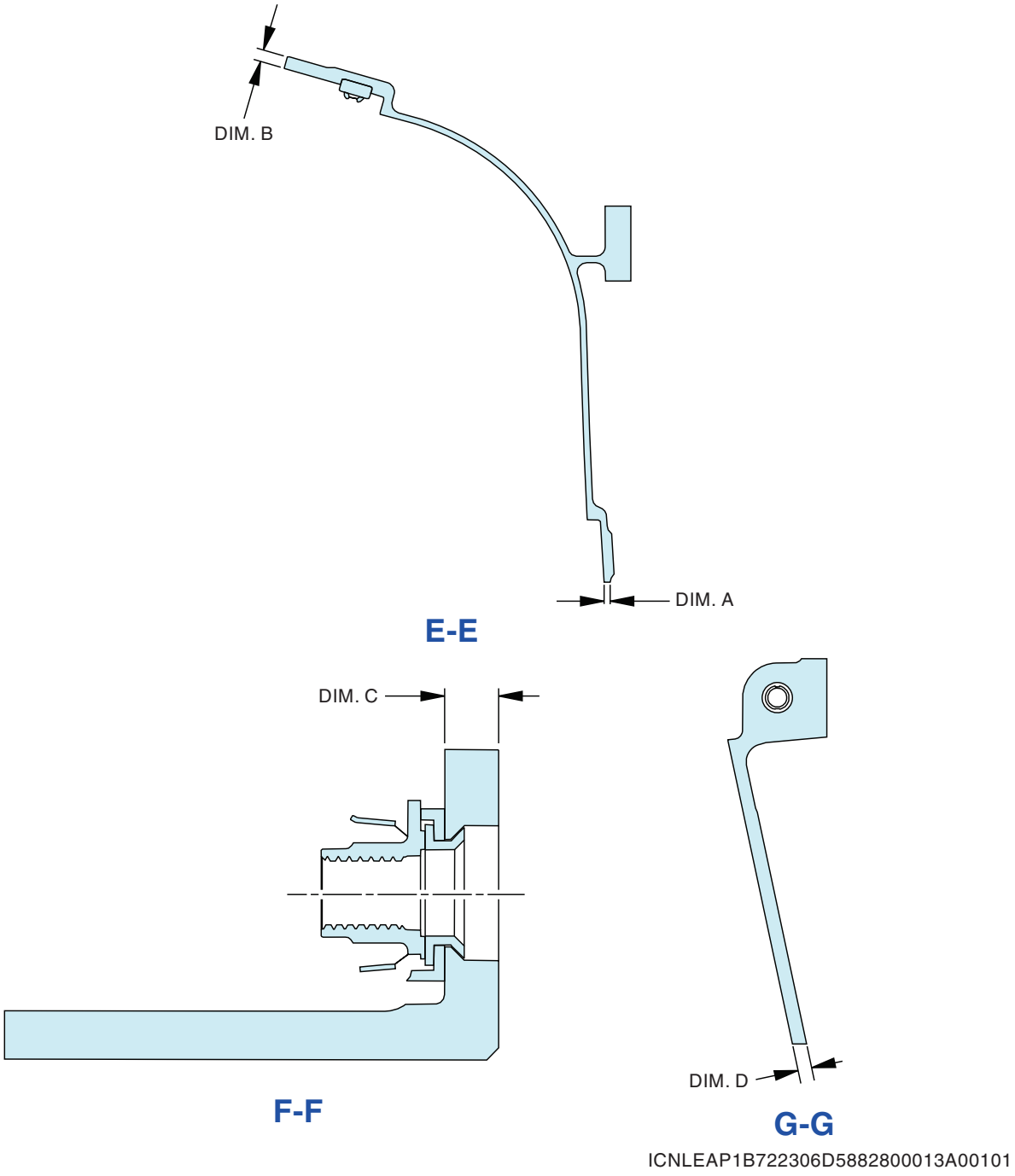


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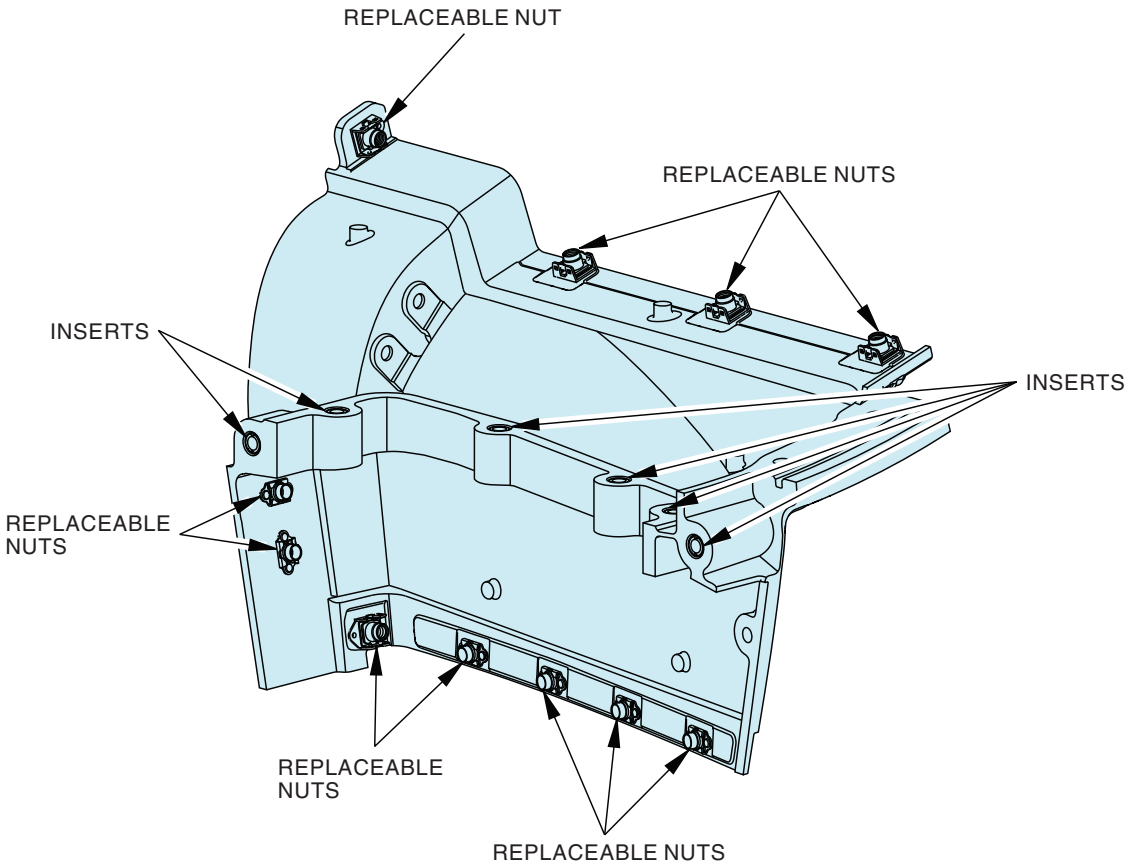
Fan Frame Strut Panels (LH-Strut 5 Pad) Inspection
Figure 604/72-23-06-990-814-G00 (Sheet 2 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Frame Strut Panels (LH-Strut 5 Pad) Inspection
Figure 604/72-23-06-990-814-G00 (Sheet 3 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

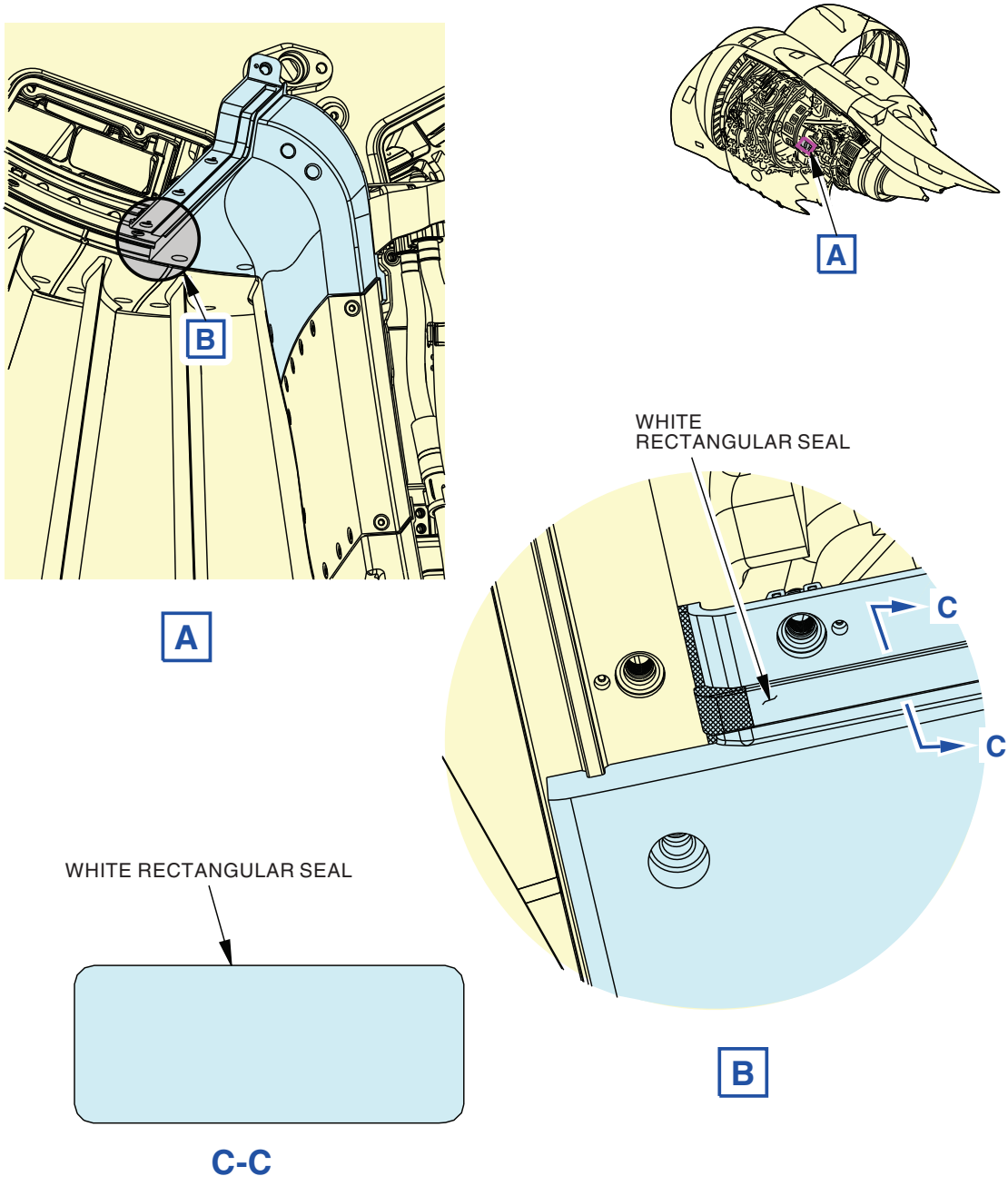


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Fan Frame Strut Panels (LH-Strut 5 Pad) Inspection
Figure 604/72-23-06-990-814-G00 (Sheet 4 of 6)

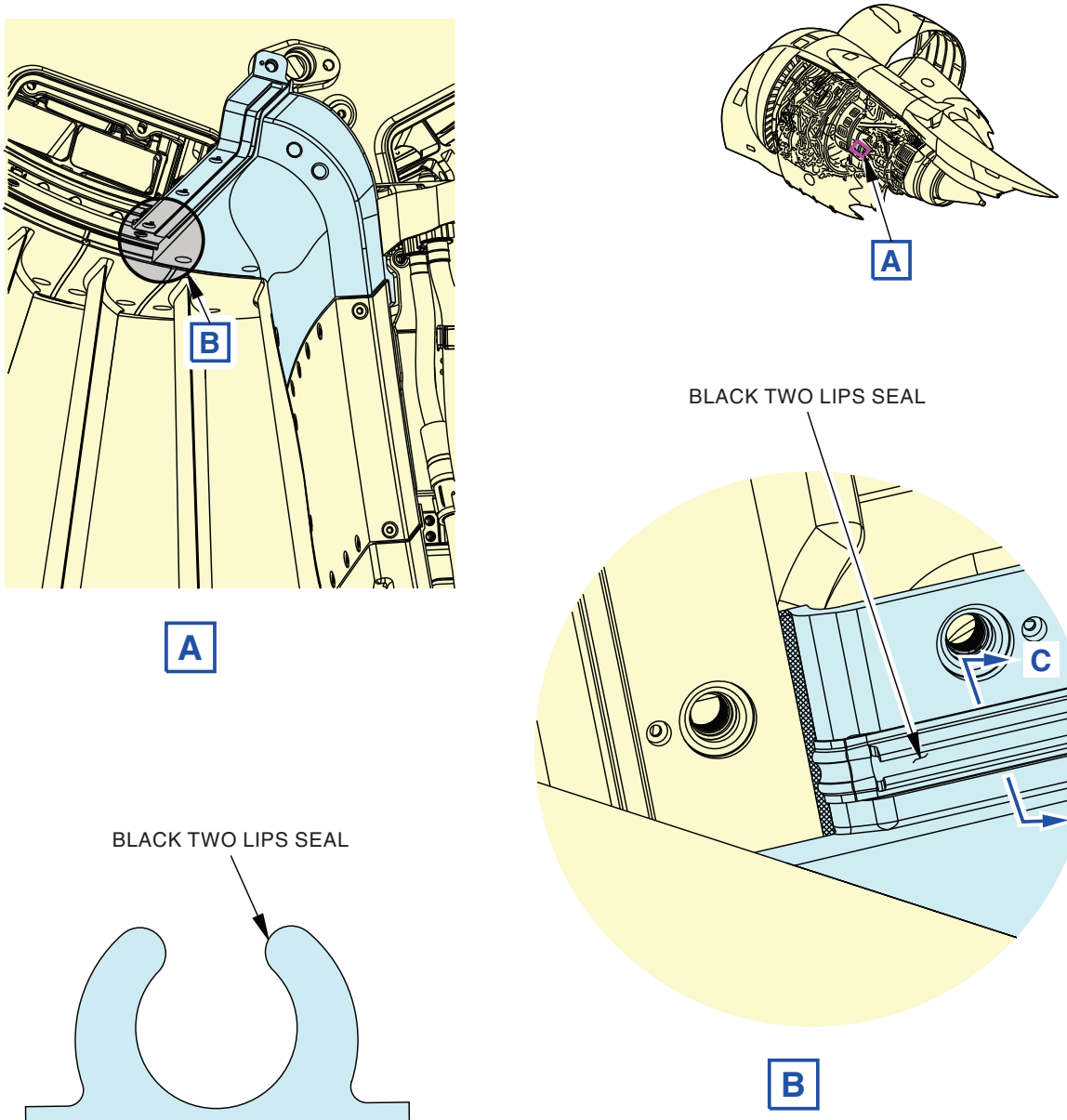
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Fan Frame Strut Panels (LH-Strut 5 Pad) Inspection
Figure 604/72-23-06-990-814-G00 (Sheet 5 of 6)

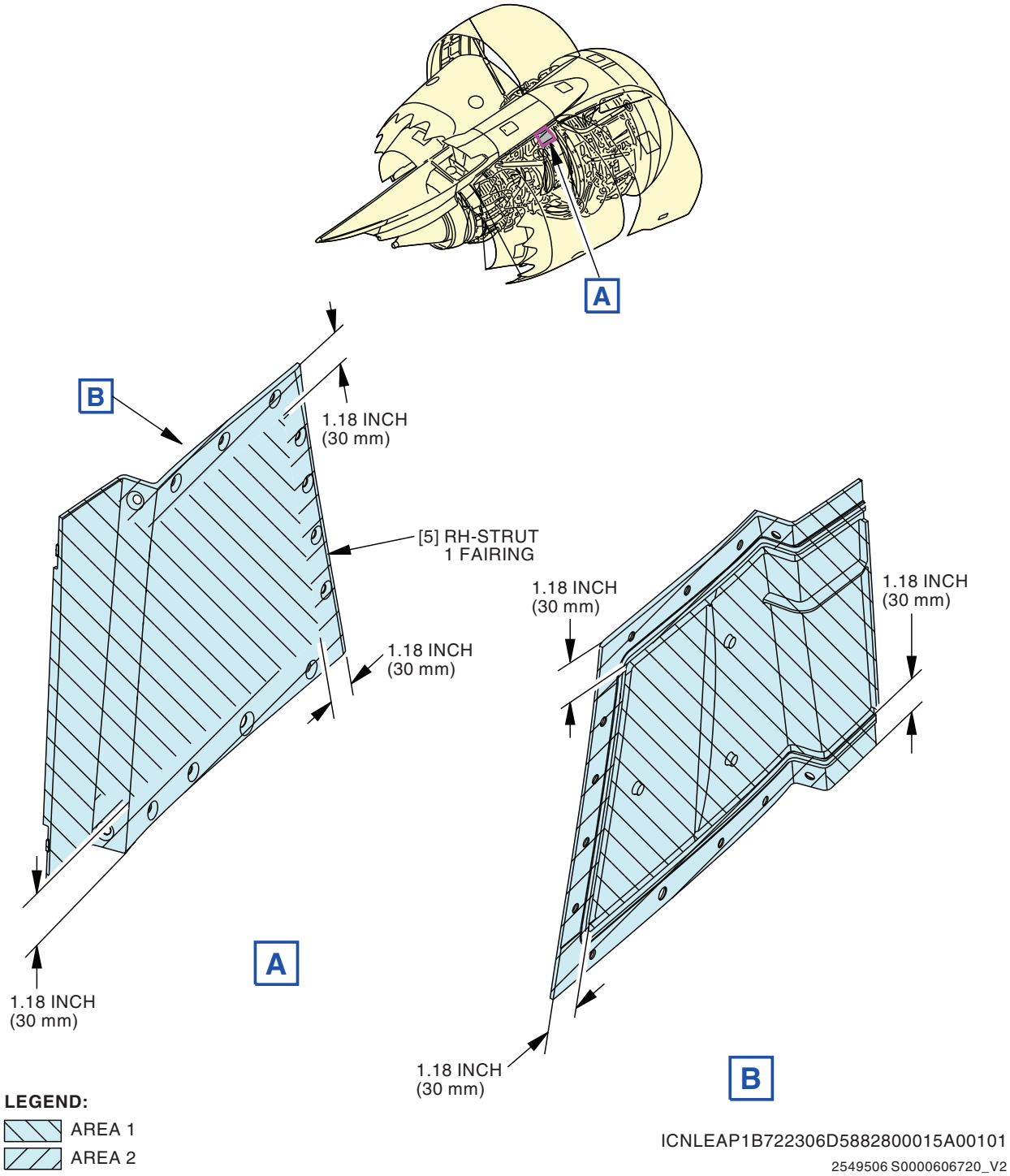
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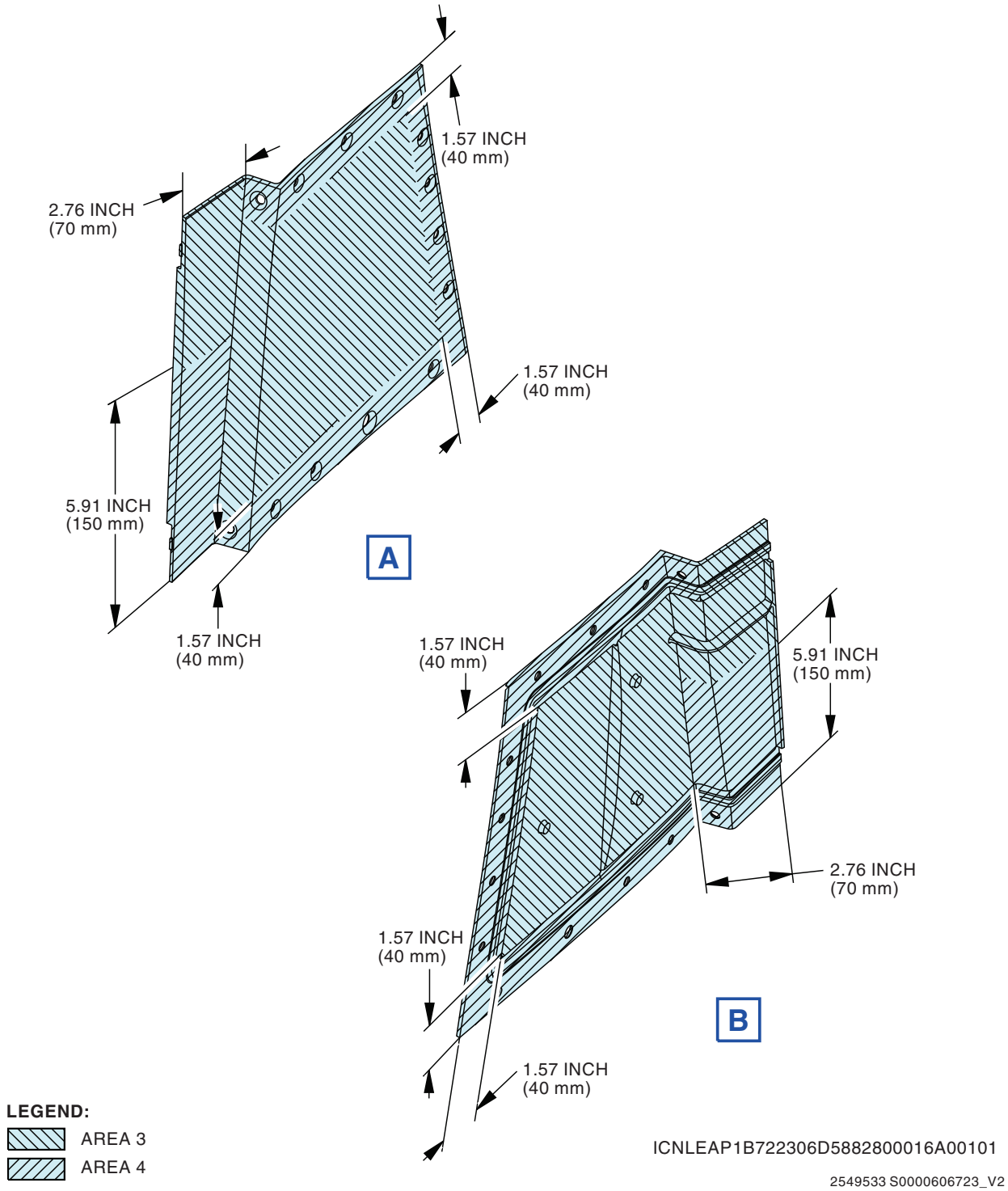
Fan Frame Strut Panels (LH-Strut 5 Pad) Inspection
Figure 604/72-23-06-990-814-G00 (Sheet 6 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



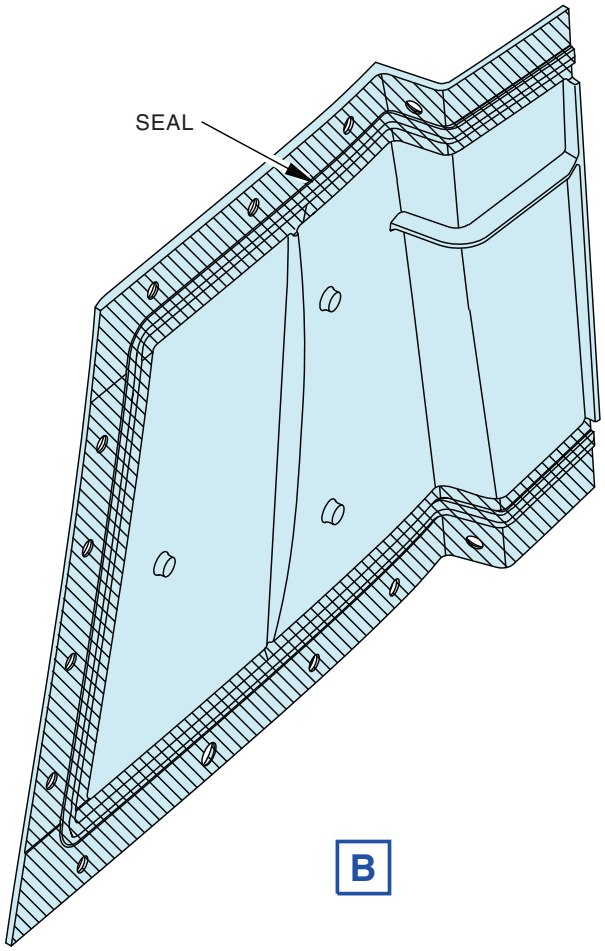
Fan Frame Strut Panels (RH-Strut 1 Fairing) Inspection
Figure 605/72-23-06-990-815-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Frame Strut Panels (RH-Strut 1 Fairing) Inspection
Figure 605/72-23-06-990-815-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

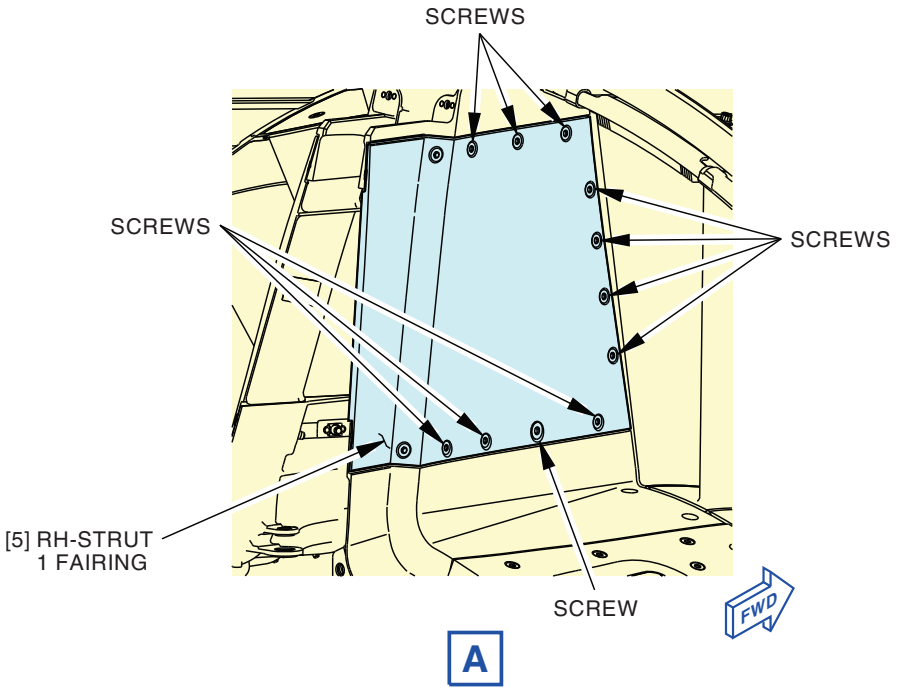
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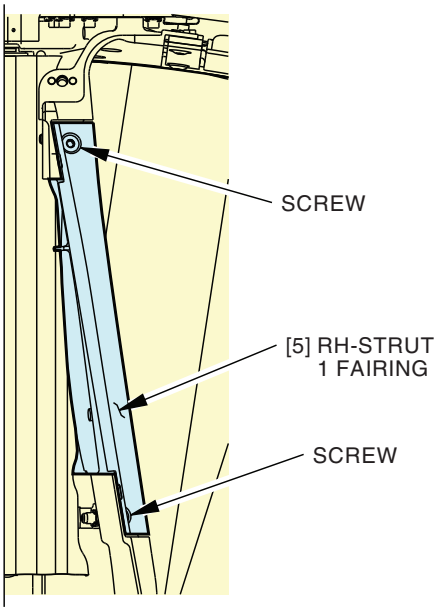
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Fan Frame Strut Panels (RH-Strut 1 Fairing) Inspection
Figure 605/72-23-06-990-815-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





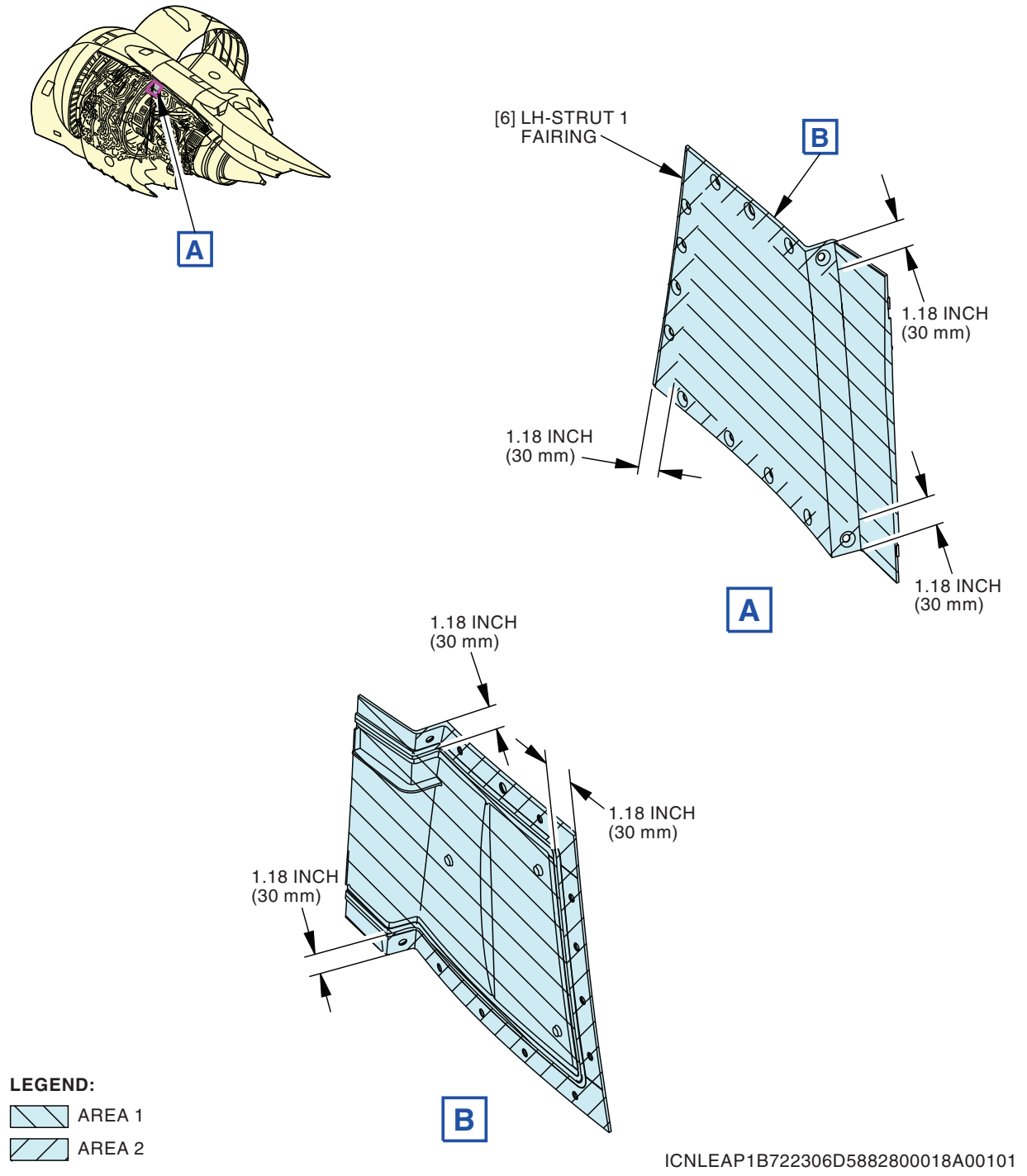
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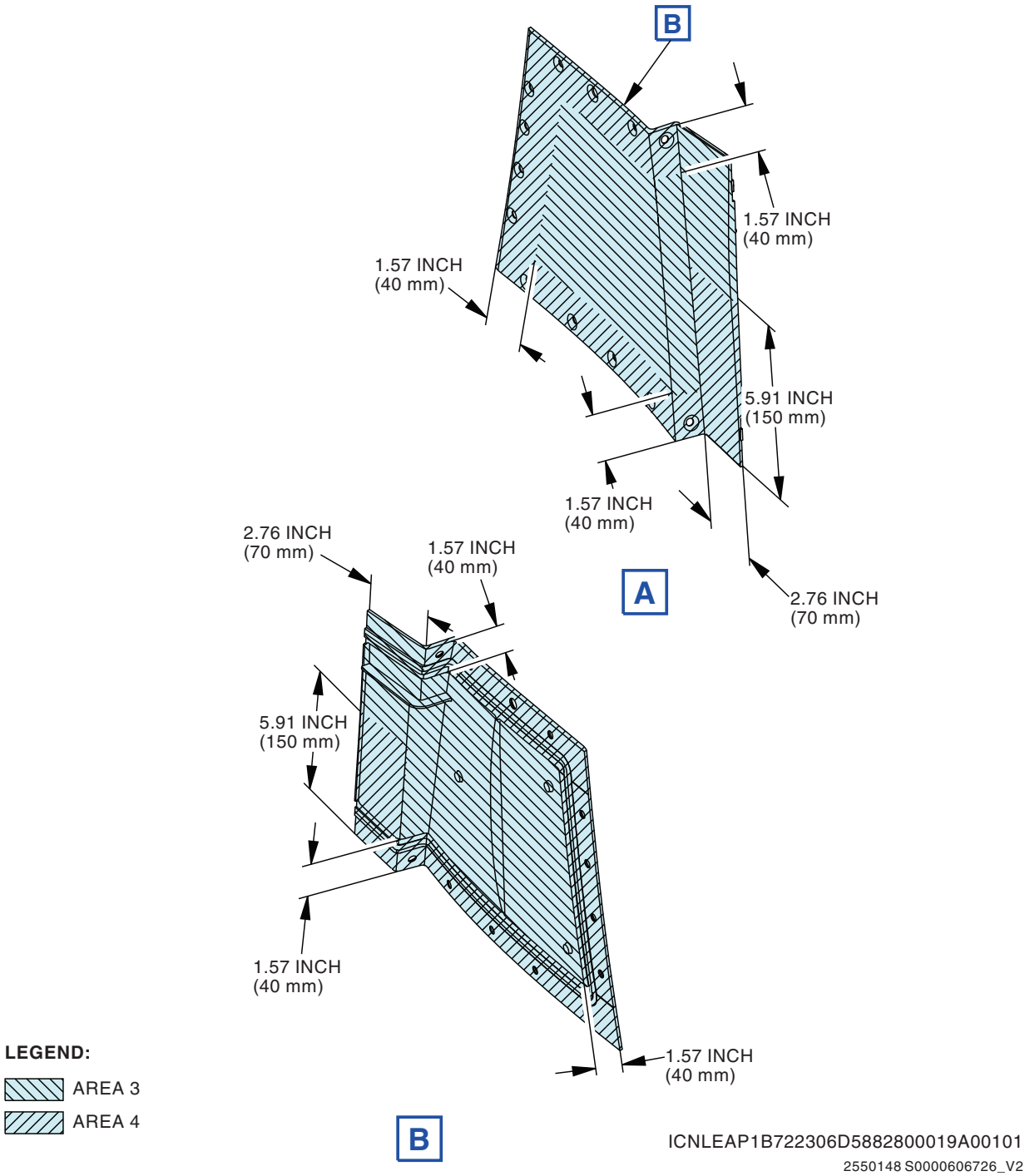
Fan Frame Strut Panels (RH-Strut 1 Fairing) Inspection
Figure 605/72-23-06-990-815-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



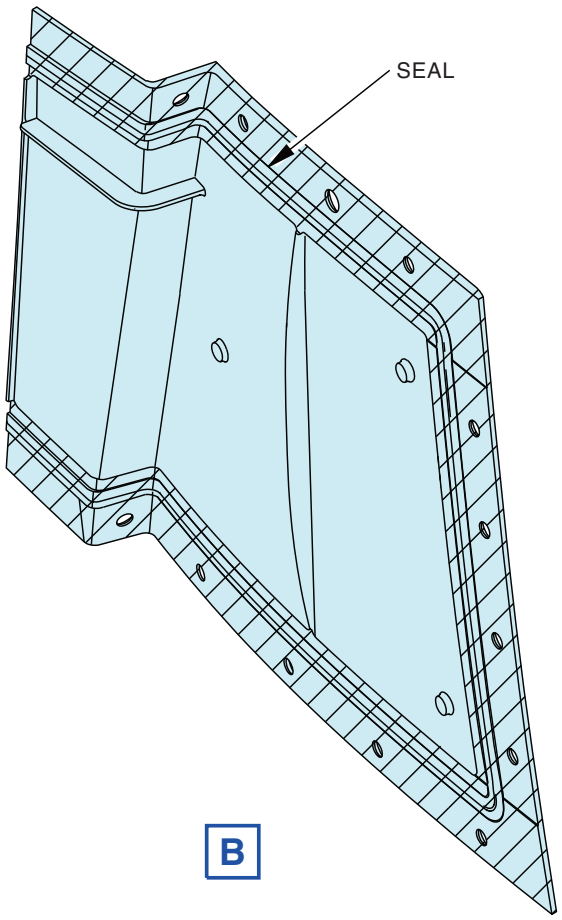
Fan Frame Strut Panels (LH-Strut 1 Fairing) Inspection
Figure 606/72-23-06-990-816-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Frame Strut Panels (LH-Strut 1 Fairing) Inspection
Figure 606/72-23-06-990-816-G00 (Sheet 2 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



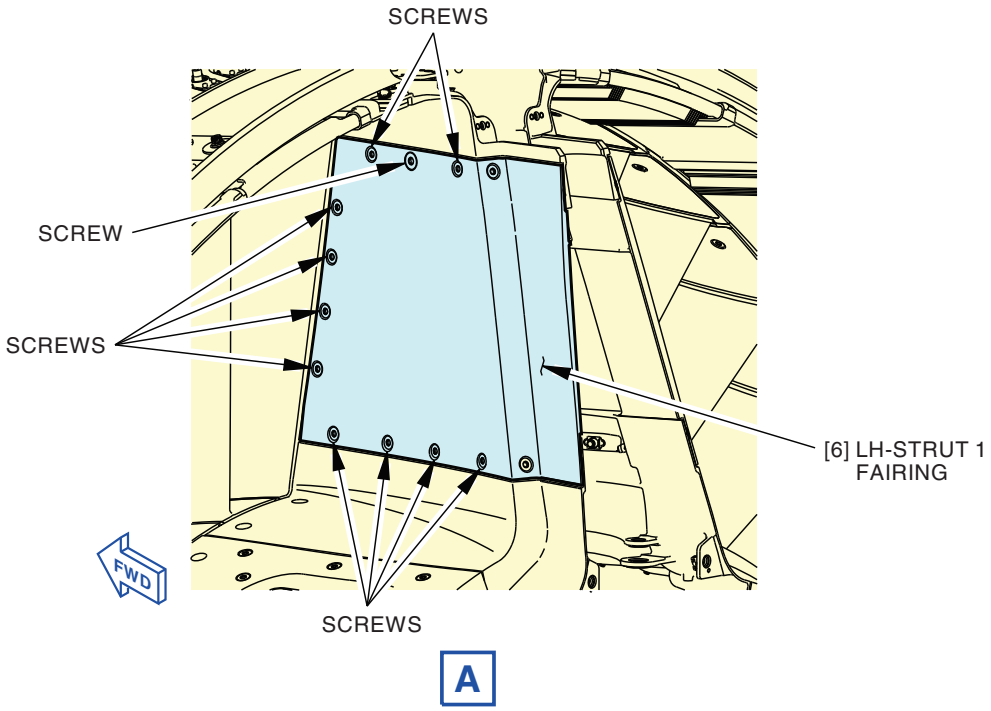
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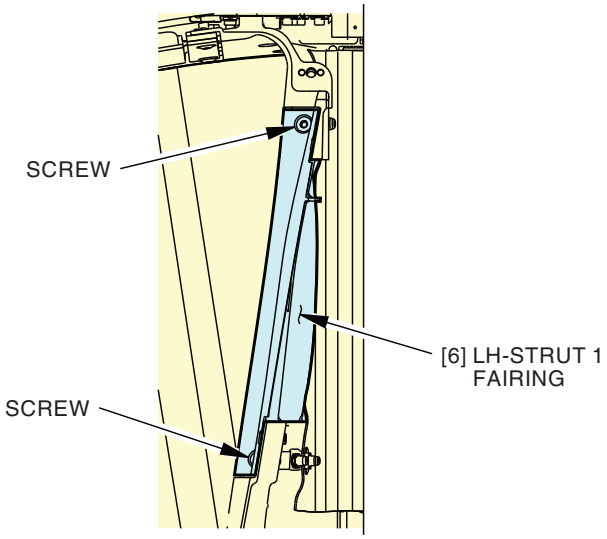
AREA 5

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Fan Frame Strut Panels (LH-Strut 1 Fairing) Inspection
Figure 606/72-23-06-990-816-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





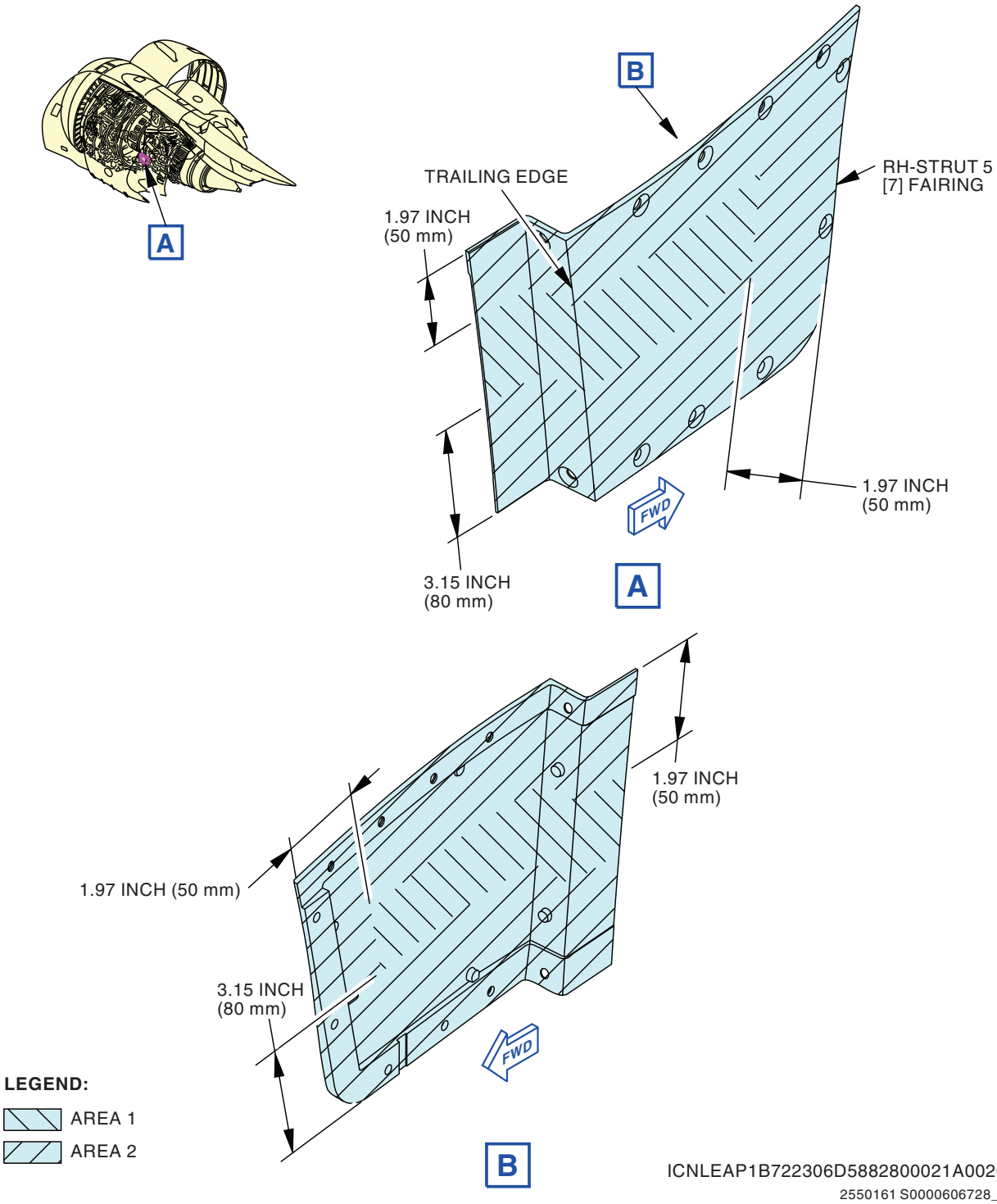
(VIEW IN THE FORWARD DIRECTION)

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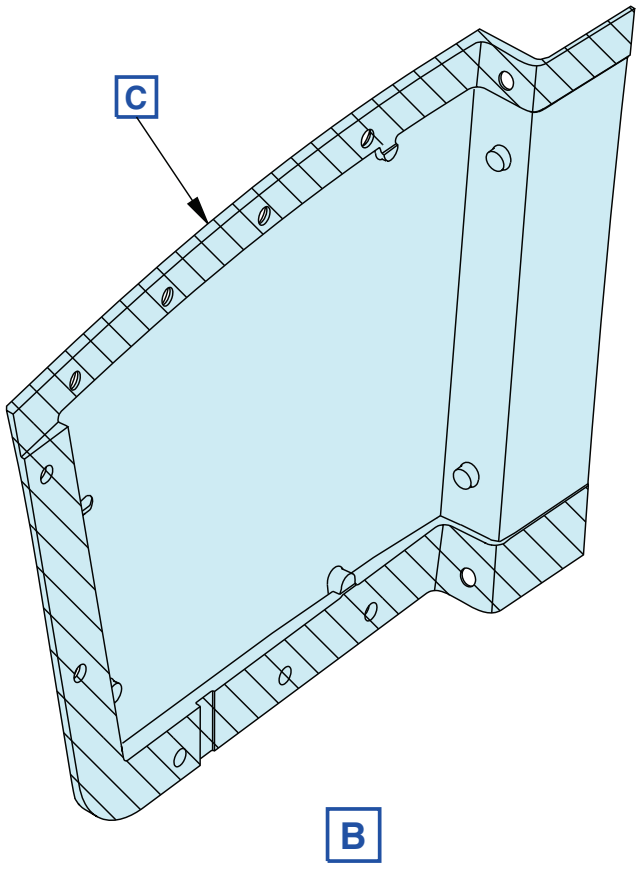
Fan Frame Strut Panels (LH-Strut 1 Fairing) Inspection
Figure 606/72-23-06-990-816-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fan Frame Strut Panels (RH-Strut 5 Fairing) Inspection
Figure 607/72-23-06-990-817-G00 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



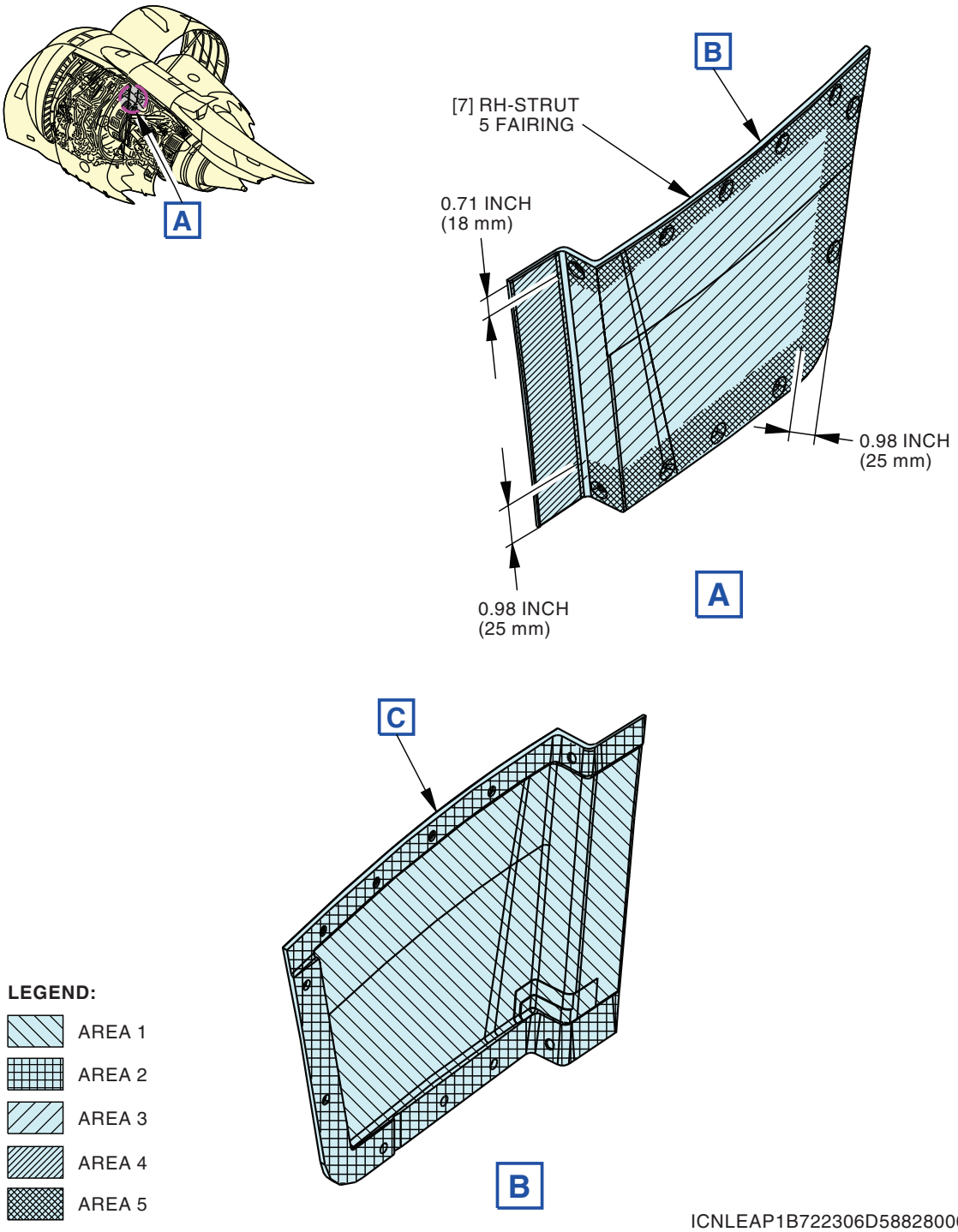
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AREA 3

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Fan Frame Strut Panels (RH-Strut 5 Fairing) Inspection
Figure 607/72-23-06-990-817-G00 (Sheet 2 of 5)

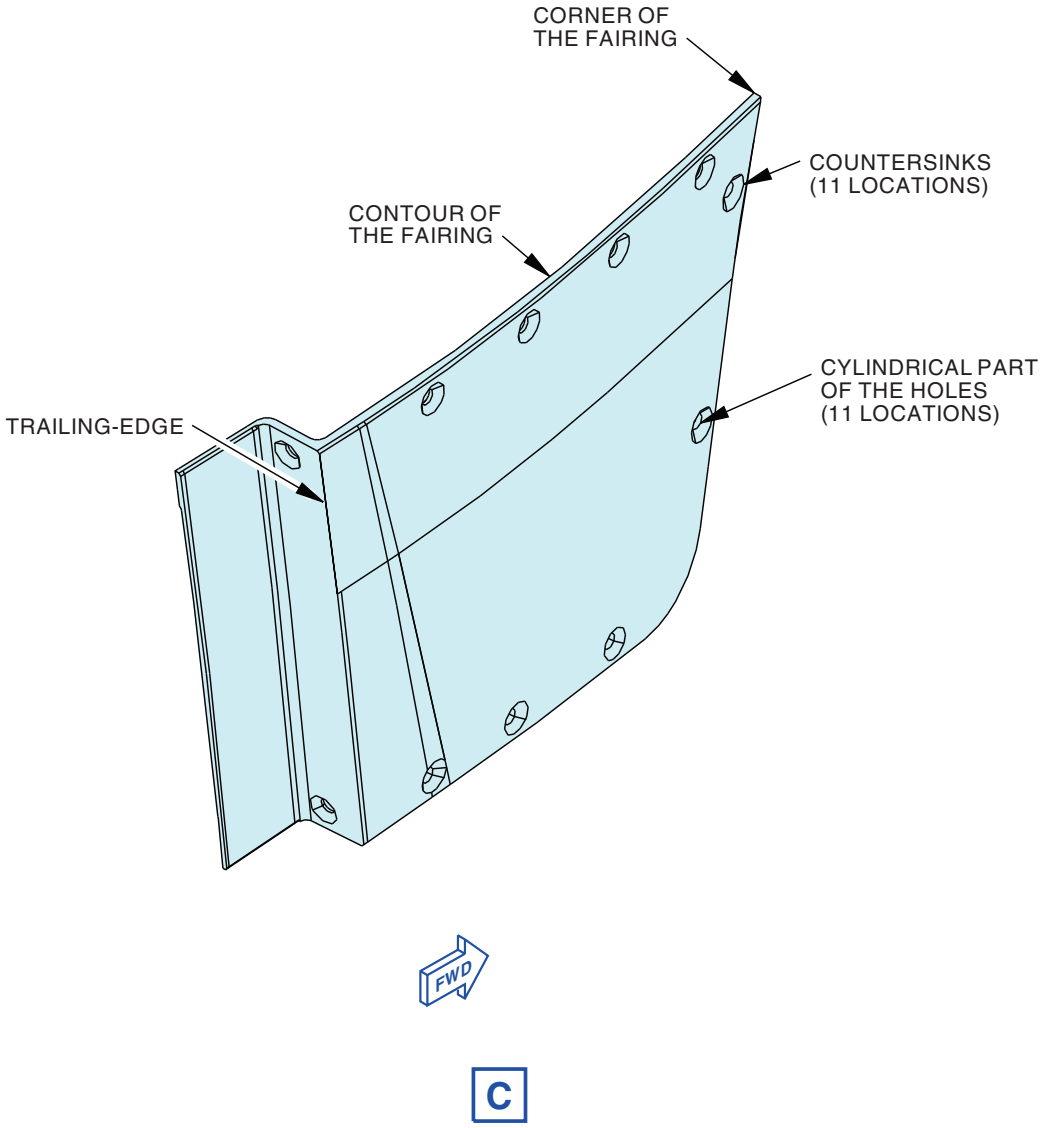
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Fan Frame Strut Panels (RH-Strut 5 Fairing) Inspection
Figure 607/72-23-06-990-817-G00 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

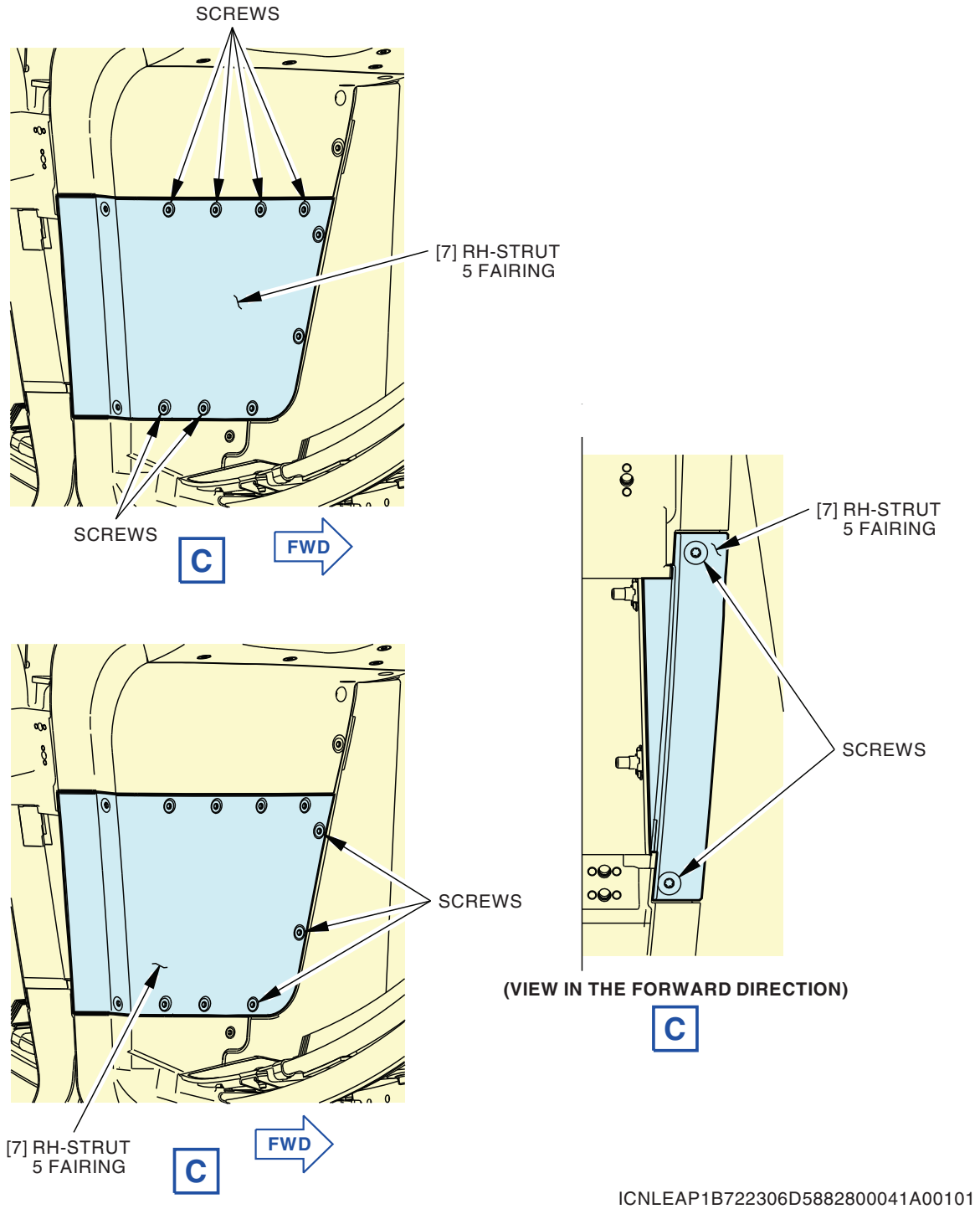


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Fan Frame Strut Panels (RH-Strut 5 Fairing) Inspection
Figure 607/72-23-06-990-817-G00 (Sheet 4 of 5)

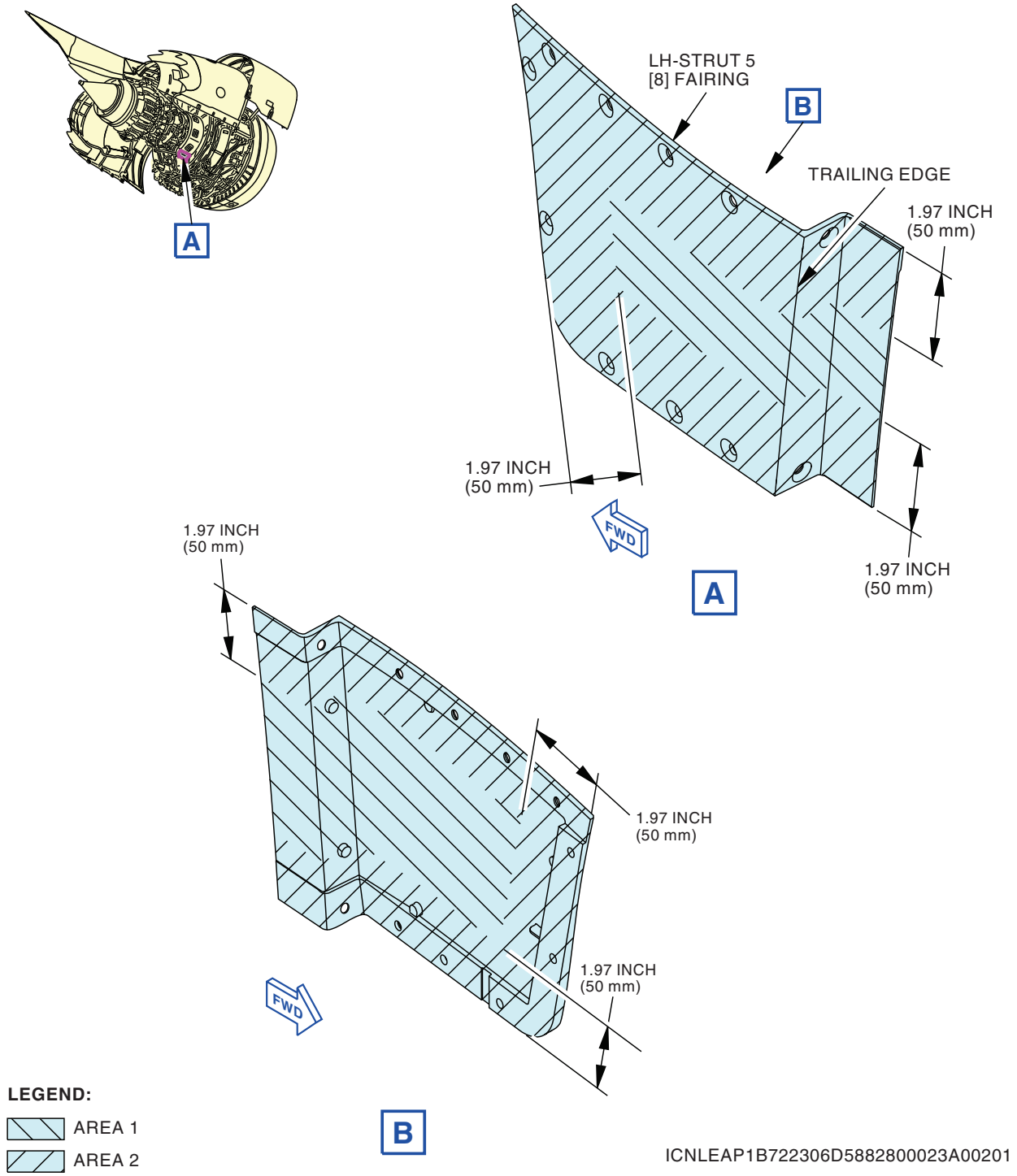
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Fan Frame Strut Panels (RH-Strut 5 Fairing) Inspection
Figure 607/72-23-06-990-817-G00 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

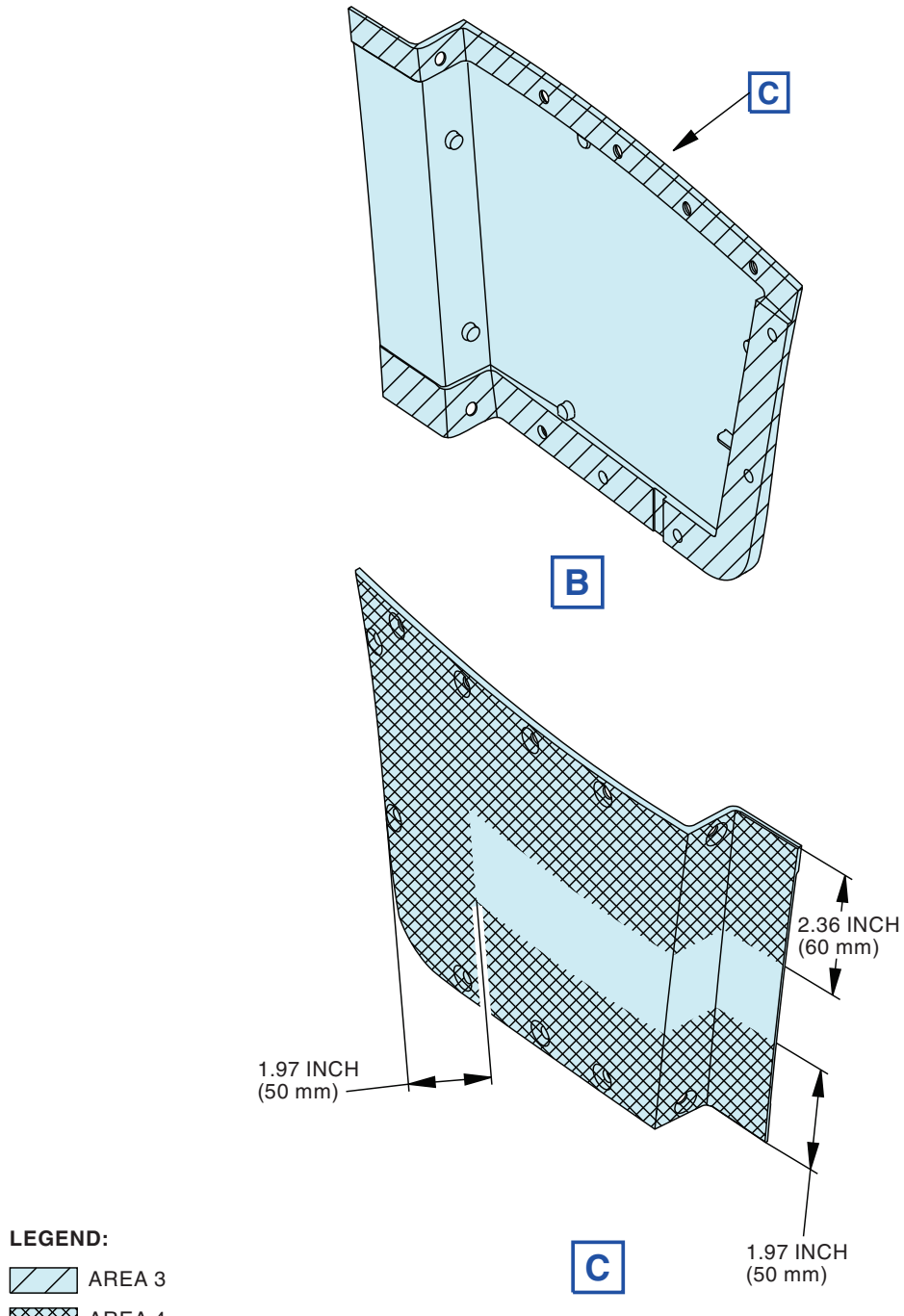


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Fan Frame Strut Panels (LH-Strut 5 Fairing) Inspection
Figure 608/72-23-06-990-818-G00 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

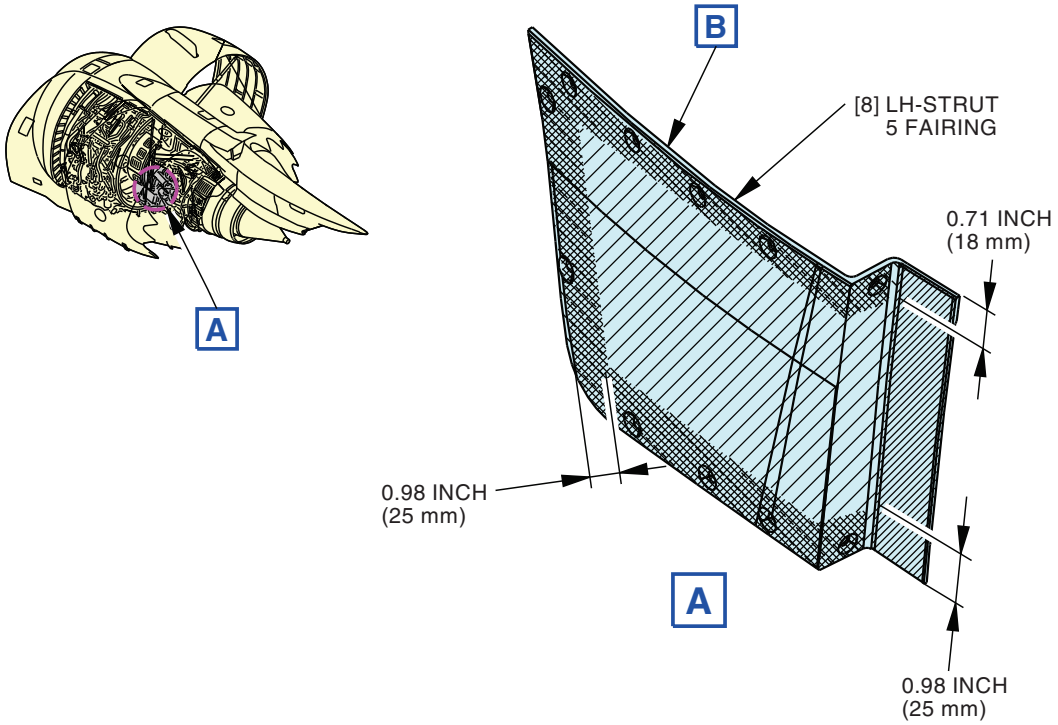


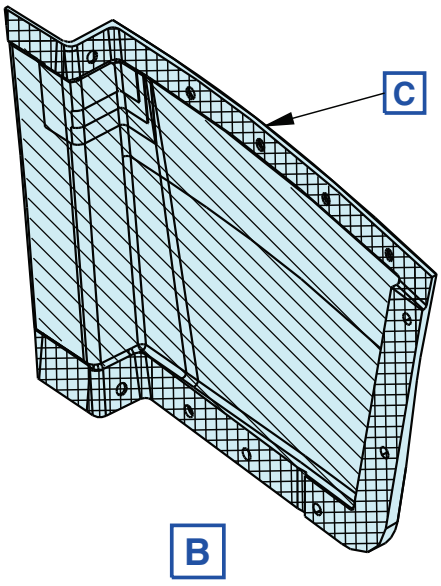
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Fan Frame Strut Panels (LH-Strut 5 Fairing) Inspection
Figure 608/72-23-06-990-818-G00 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





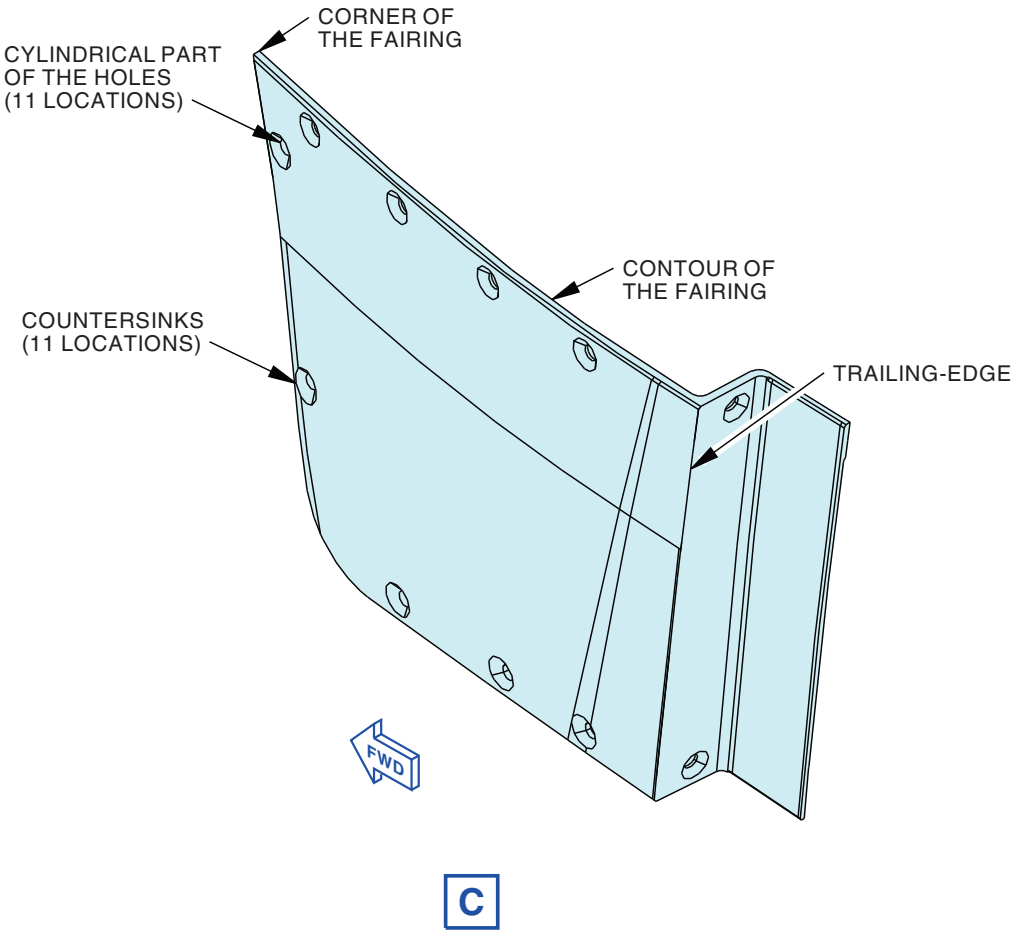
LEGEND:

- AREA 1
- AREA 2
- AREA 3
- AREA 4
- AREA 5

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Fan Frame Strut Panels (LH-Strut 5 Fairing) Inspection
Figure 608/72-23-06-990-818-G00 (Sheet 3 of 5)

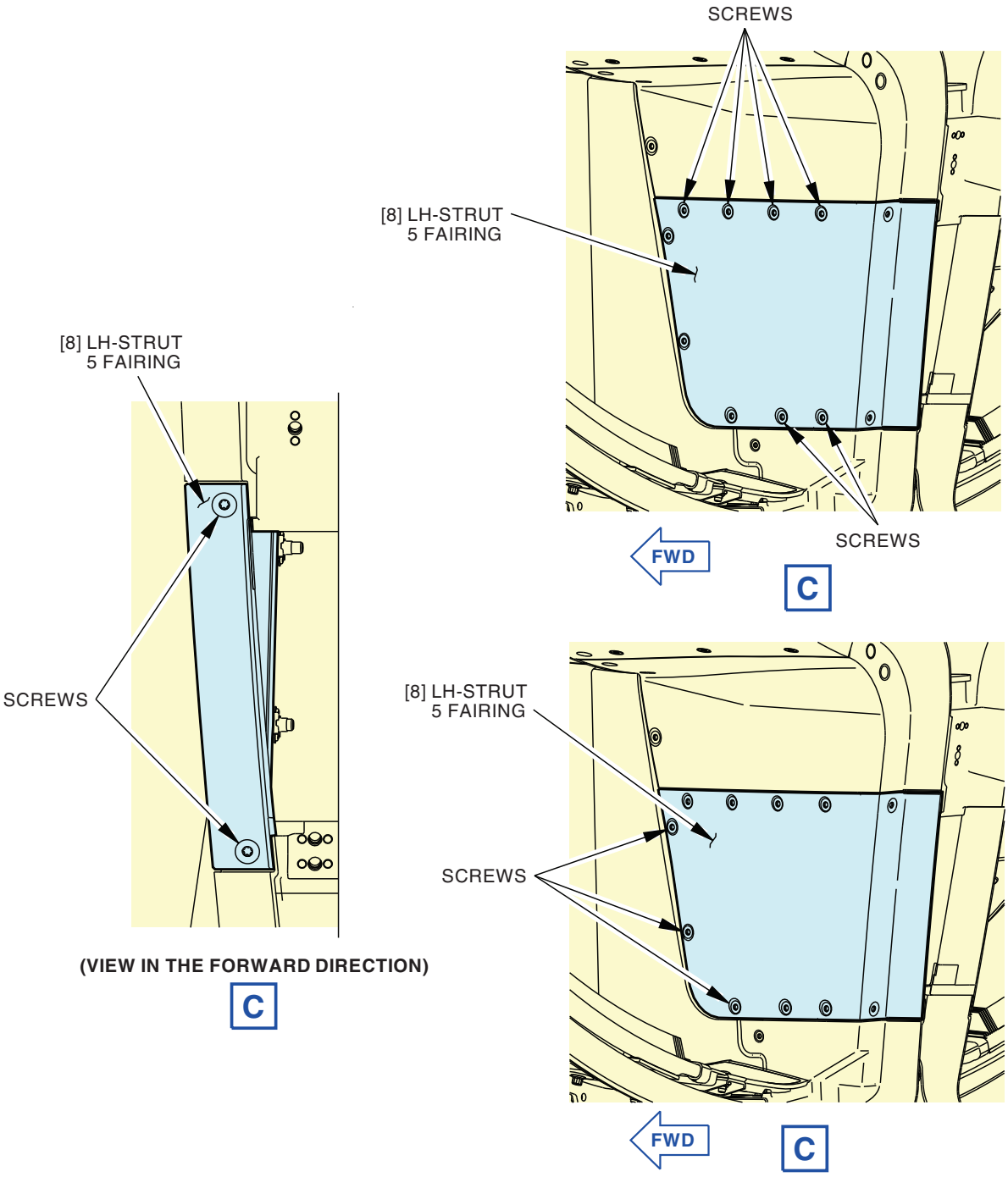
ECCN 9E991 BOEING PROPRIETARY - See title page for details



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Fan Frame Strut Panels (LH-Strut 5 Fairing) Inspection
Figure 608/72-23-06-990-818-G00 (Sheet 4 of 5)

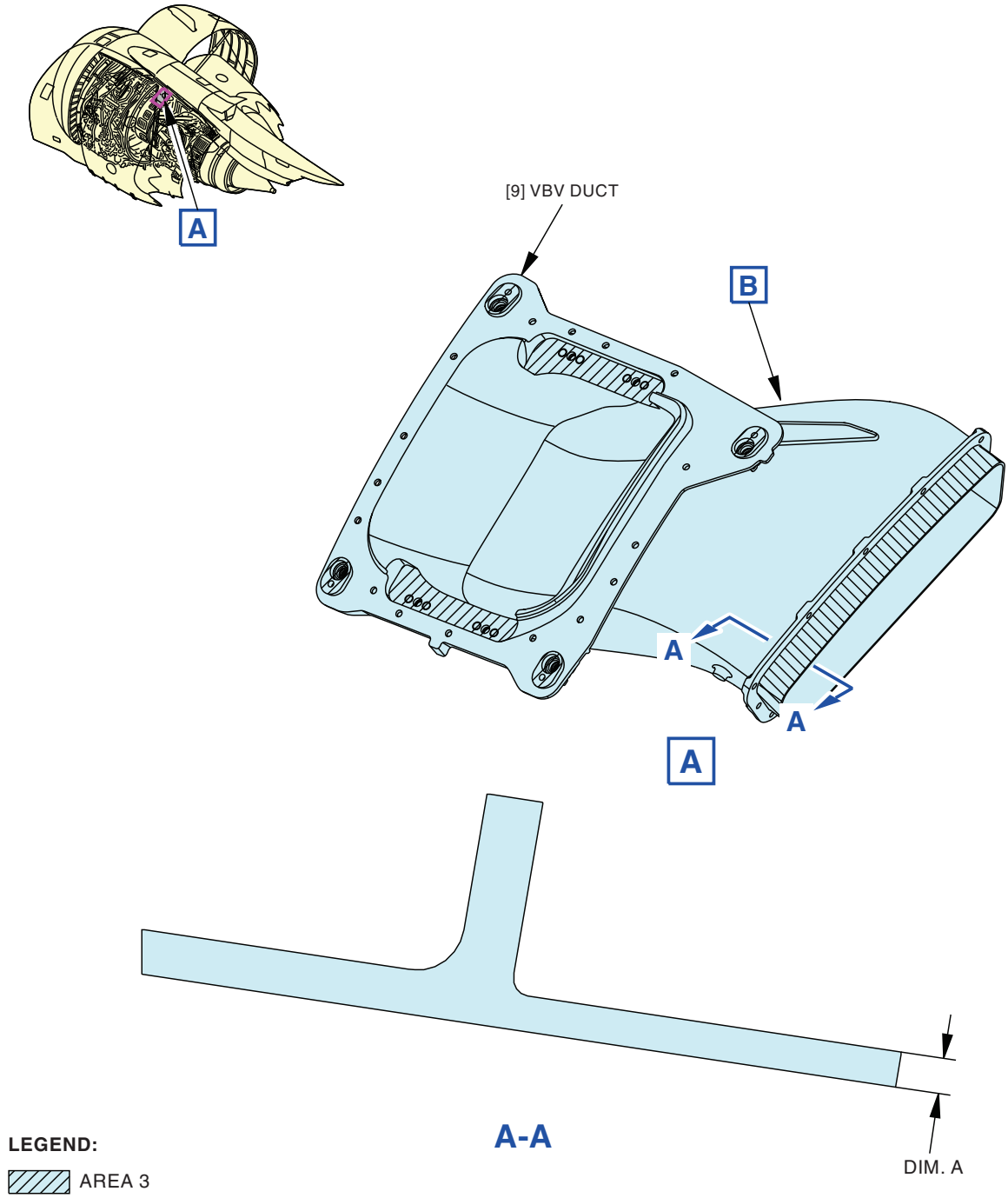
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2965338 S0000743814_V1

Fan Frame Strut Panels (LH-Strut 5 Fairing) Inspection
Figure 608/72-23-06-990-818-G00 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

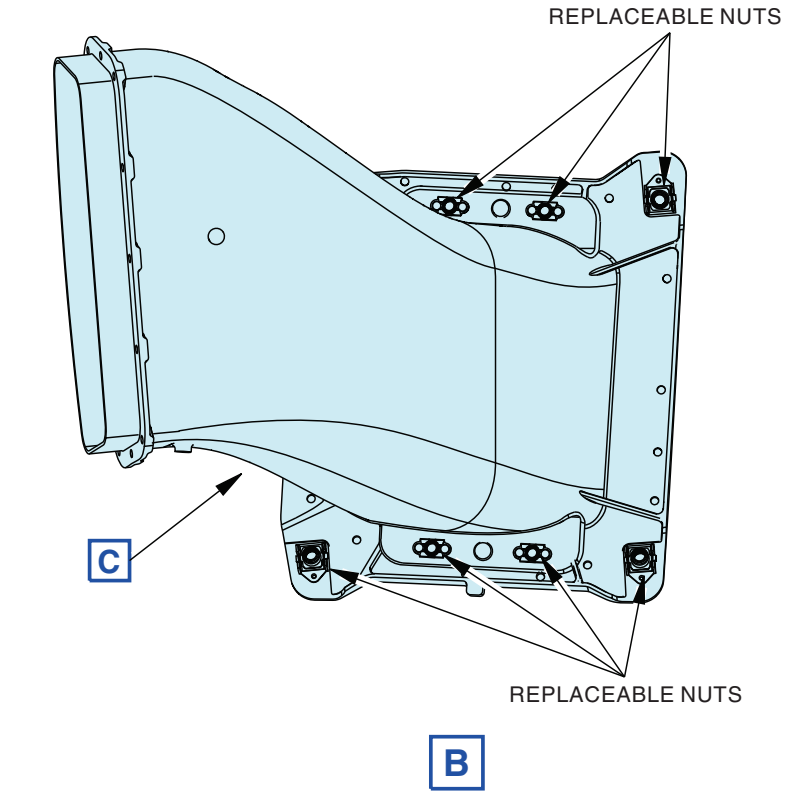


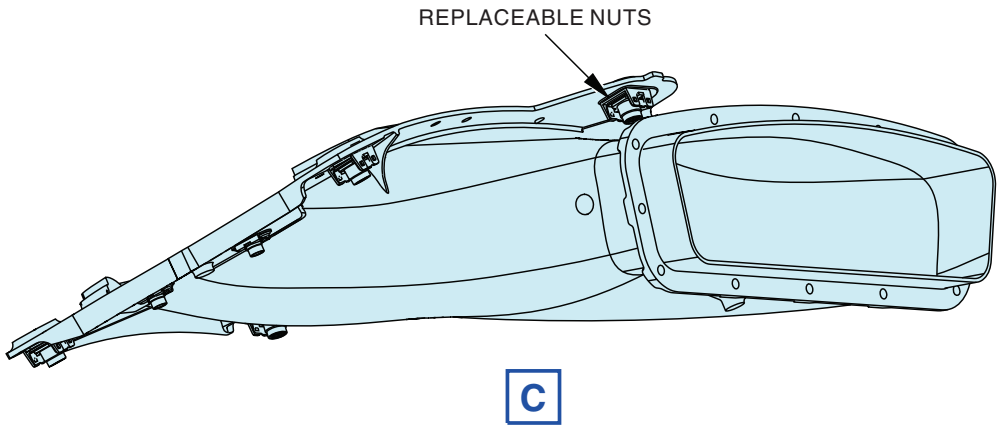
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Fan Frame Strut Panels (VBV Duct) Inspection
Figure 609/72-23-06-990-819-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



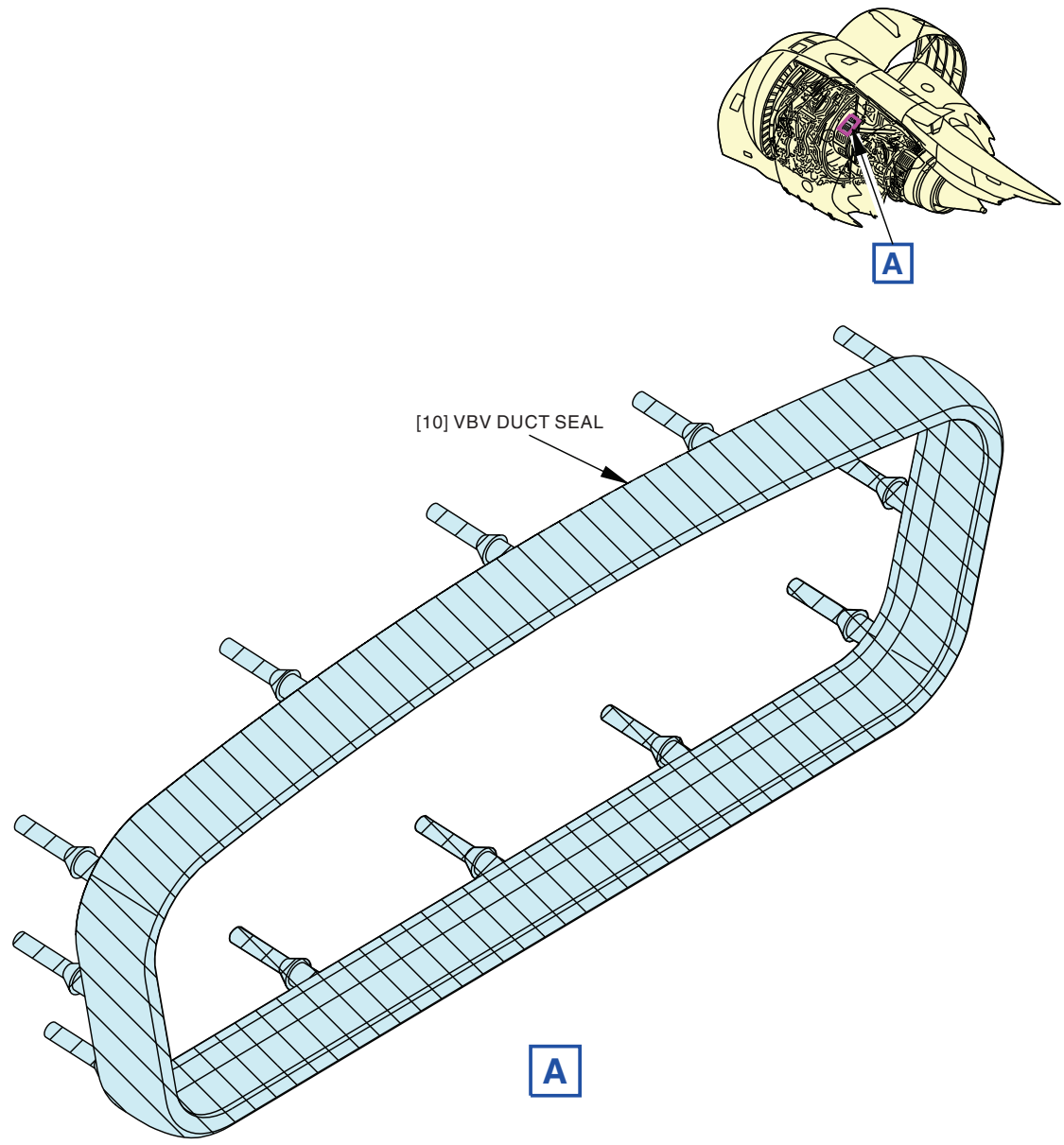


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Fan Frame Strut Panels (VBV Duct) Inspection
Figure 609/72-23-06-990-819-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



LEGEND:

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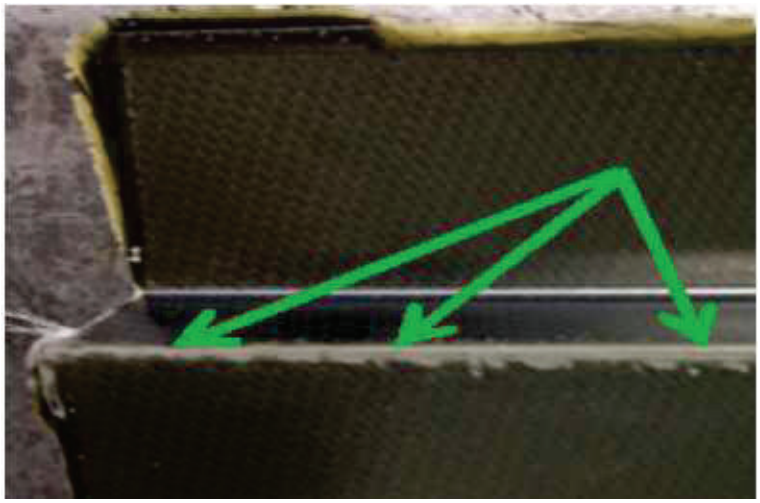
AREA B (SEAL RUBBER PIN)

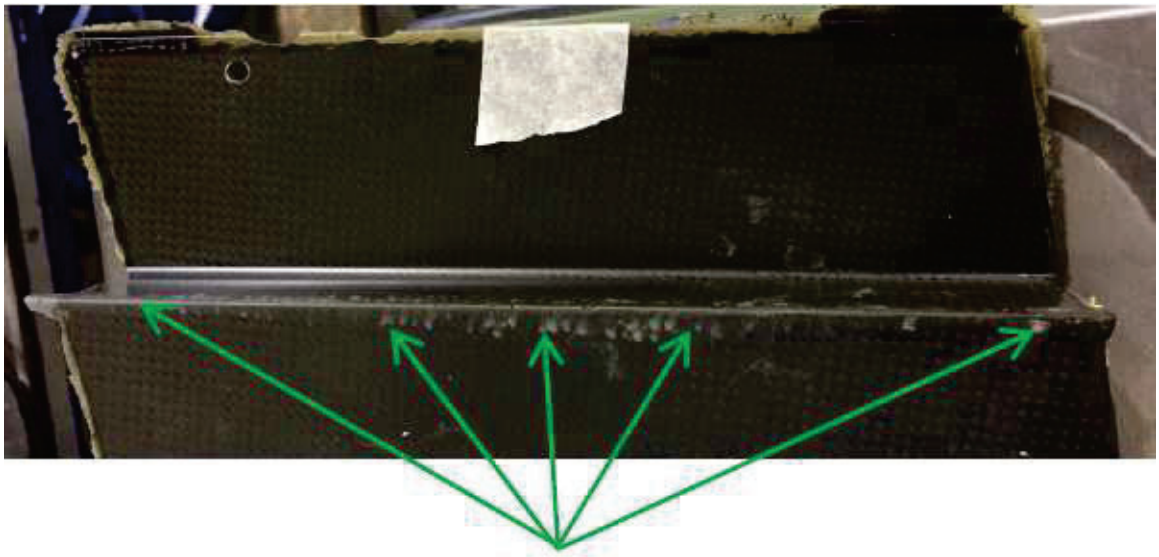
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VBV Duct Seal Inspection
Figure 610/72-23-06-990-821-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



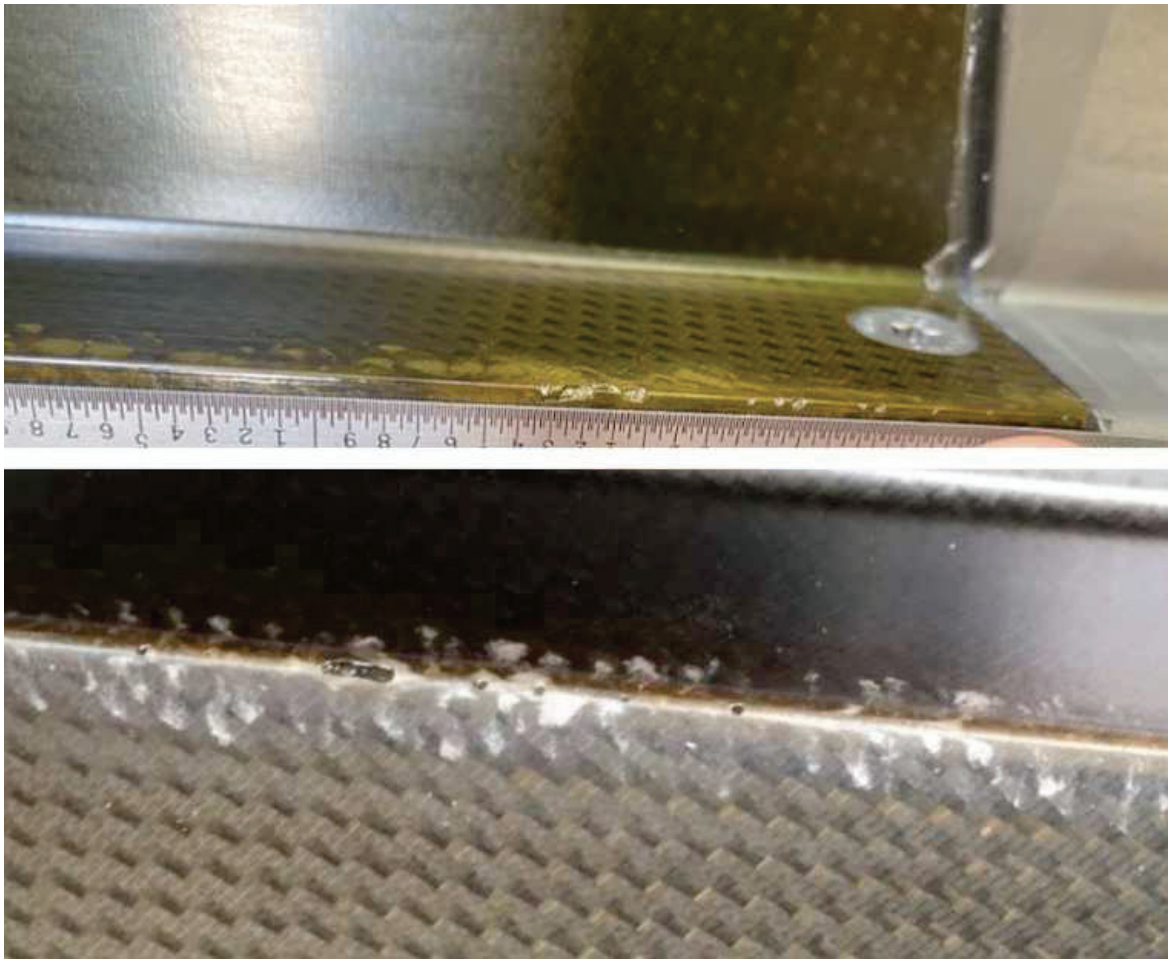


MIGRATION OF THE GRAY GAP FILLER THROUGH THE PLIES

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2886337 S0000685436_V1

Strut 5 LH and RH Fairing Trailing Edge
Figure 611/72-23-06-990-824-G00 (Sheet 1 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



POROSITIES ON TRAILING EDGE

ICNLEAP1B722306D5882800043A00101
2965340 S0000743815_V1

Strut 5 LH and RH Fairing Trailing Edge
Figure 611/72-23-06-990-824-G00 (Sheet 2 of 2)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FAN FRAME STRUT PANELS - REPAIR

1. General

A. This procedure has these tasks:

(1) Fan Frame Strut 1 Fairing Seal Replacement

(2) Fan Frame Strut Pad Seal Replacement

(3) Fan Frame Strut 1 and 5 Fairings Blending

(4) Fan Frame Strut Pad Blending

(5) Fan Frame Strut Pad Nut Replacement.

TASK 72-23-06-960-801-G00

2. Fan Frame Strut 1 Fairing Seal Replacement

(Figure 801, Figure 802, and Figure 803)

A. General

(1) This task gives instructions to replace the seals on the left and right strut 1 fairings.

(2) The tables below give a list of the part numbers that are applicable to the fan frame strut 1 fairing seal replacement.

Table 801/72-23-06-993-801-G00

Part number	Configuration	Service Bulletin	Designation
364-031-801-0	1	72-0001	FAIRING, RH-STRUT 1
364-031-810-0	2	72-0189	FAIRING, RH-STRUT 1
364-031-901-0	1	72-0001	FAIRING, LH-STRUT 1
364-031-910-0	2	72-0189	FAIRING, LH-STRUT 1

Table 802/72-23-06-993-803-G00

Name	Part Number	Quantity
SEAL FAIRING RH-STRUT 1	364-868-100-0	As Required
SEAL FAIRING RH-STRUT 1	364-868-100-0	As Required
SEAL FAIRING LH-STRUT 1	364-868-105-0	As Required
SEAL FAIRING LH-STRUT 1	364-868-105-0	As Required

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
A01077 [CP2266]	Adhesive - Silicone Rubber - RTV 102	
A50012 [CP2242]	Adhesive - Silicone, Thixotropic Paste, One-Part RTV - RTV 732	MIL-A-46106 Type I
A50119	Adhesive - Silicone Rubber - RTV 103 (Black)	MIL-A-46106
B01058 [CP1039]	Solvent - Acetone, Reagent Grade	
G50077	Abrasive - Aluminum Oxide Paper, 240 grit or finer	
G50143 [CP2208]	Tape - Glass Cloth with Silicone Adhesive, High Temperature, Abrasive Resistant - 3M 361	
G51692	Cloth - 100% Cotton	AMS3819 Class 1
G51700 [CP2968]	Tape - Masking, High-Temperature - 3M 501E	
G51742 [CP2441]	Tape - Vinyl - 3M No. 472	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Prepare for the Replacement

SUBTASK 72-23-06-840-003-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-23-06-010-003-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Fan Frame Strut 1 Fairing Seal Replacement

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

WARNING USE EYE PROTECTION WHEN YOU CUT MATERIAL OR PARTS. THE PARTICLES CAN CAUSE DAMAGE TO YOUR EYES.

WARNING USE PROTECTIVE GLOVES TO PREVENT INJURY FROM SHARP EDGES AND ROUGH SURFACES. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

SUBTASK 72-23-06-020-005-G00

(1) Remove the seal [1] as follows (View A, Figure 801 or Figure 802):

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CAUTION MAKE SURE THAT YOU CAREFULLY REMOVE THE SEAL FROM THE FAN FRAME STRUT FAIRING. IF YOU DO NOT OBEY, DAMAGE TO THE FAN FRAME STRUT FAIRING CAN OCCUR.

(a) Carefully remove and discard the seal [1].

WARNING BE CAREFUL WHEN USE SHARP OR POINTED TOOLS. THESE TOOLS CAN CAUSE AN INJURY TO YOU.

CAUTION DO NOT APPLY A PROCEDURE THAT CAN CHANGE THE PARENT MATERIAL PROPERTIES DURING THE REMOVAL OF THE BONDED MATERIAL. IF YOU DO NOT OBEY, DAMAGE TO THE MATERIAL CAN OCCUR.

(b) Remove adhesive from the fan frame strut 1 fairing with a cutting tool.

(c) Make sure that you do not remove parent material from the fan frame strut 1 fairing.

SUBTASK 72-23-06-100-001-G00

(2) Prepare the area where you will bond the new seal [1] (View A-A and View B-B, Figure 801 or Figure 802).

NOTE: The seal material is elastomer.

WARNING DO NOT BREATHE THE PARTICLES FROM BLENDING OR LET THE PARTICLES TOUCH YOU. THE PARTICLES FROM BLENDING CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PROTECTION EQUIPMENT. USE LOCAL MECHANICAL EXHAUST VENTILATION OR AN APPROVED RESPIRATOR.

(a) Make the metallic part surface of the fan frame strut 1 fairing rough with 240 grit or finer abrasive paper, G50077.

NOTE: The fan frame strut 1 fairing material is TA6V or A286.

WARNING DO NOT GET SOLVENT IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE SOLVENT. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE THE SOLVENT. KEEP THE SOLVENT AWAY FROM SPARKS, FLAME AND HEAT. SOLVENT IS POISONOUS AND FLAMMABLE WHICH CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(b) Clean the metallic part area of fan frame strut 1 fairing with clean cloth, G51692, or equivalent, soaked in acetone solvent, B01058 [CP1039].

(c) Make the surface of the new seal [1] rough with 240 grit or finer abrasive paper, G50077.

WARNING DO NOT GET SOLVENT IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE SOLVENT. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE THE SOLVENT. KEEP THE SOLVENT AWAY FROM SPARKS, FLAME AND HEAT. SOLVENT IS POISONOUS AND FLAMMABLE WHICH CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(d) Make the surface of the new seal [1] rough with acetone solvent, B01058 [CP1039].

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SUBTASK 72-23-06-400-001-G00

(3) Bond the new seal [1] (Figure 801, Figure 802, and Figure 803).

NOTE: Seals can be in 2 different shapes and colors. The seals are interchangeable.

(a) If you do not bond material within 6 hours after surface preparation, do these steps:

1) Clean the metallic part area where you will bond the new seal [1] with clean cloth, G51692, or equivalent, soaked in acetone solvent, B01058 [CP1039].

2) Make the surface of the new seal [1] rough with acetone solvent, B01058 [CP1039].

(b) If it is necessary, for the configuration 1 parts use 3M 361 tape, G50143 [CP2208], or 3M 501E tape, G51700 [CP2968], to apply mask on the adjacent areas (Table 801).

(c) If it is necessary, for the configuration 2 parts use 3M 472 tape, G51742 [CP2441], to apply mask on the adjacent areas (Table 801).

NOTE: Do not apply primer.

(d) Apply the maximum thickness 0.007 in. (0.2 mm) of RTV 732 adhesive, A50012 [CP2242] (preferred), or RTV 102 adhesive, A01077 [CP2266] (alternate), or RTV 103 adhesive, A50119 (alternate), on the Area 2.

1) For the rectangular seal [1] apply RTV 732 adhesive, A50012 [CP2242] (preferred), RTV 102 adhesive, A01077 [CP2266] (alternate), or RTV 103 adhesive, A50119 (alternate), on the longer side (Figure 803).

(e) Apply the maximum thickness 0.019 in. (0.5 mm) of RTV 732 adhesive, A50012 [CP2242] (preferred), or RTV 102 adhesive, A01077 [CP2266] (alternate), or RTV 103 adhesive, A50119 (alternate), on the Area 3.

1) For the rectangular seal [1] apply RTV 732 adhesive, A50012 [CP2242] (preferred), RTV 102 adhesive, A01077 [CP2266] (alternate), or RTV 103 adhesive, A50119 (alternate), on the longer side (Figure 803).

(f) Install the seal [1] on the Area 2 and Area 3 and apply a pressure for bonding.

(g) Cut the seal [1] to make a continuous surface with the metallic part adjacent areas.

(h) If it is necessary, remove the unwanted adhesive from the adjacent areas.

NOTE: Adhesive is not accepted in adjacent areas.

(i) If it is necessary, remove the mask that you apply on the adjacent areas.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-23-06-840-001-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

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(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 72-23-06-840-002-G00

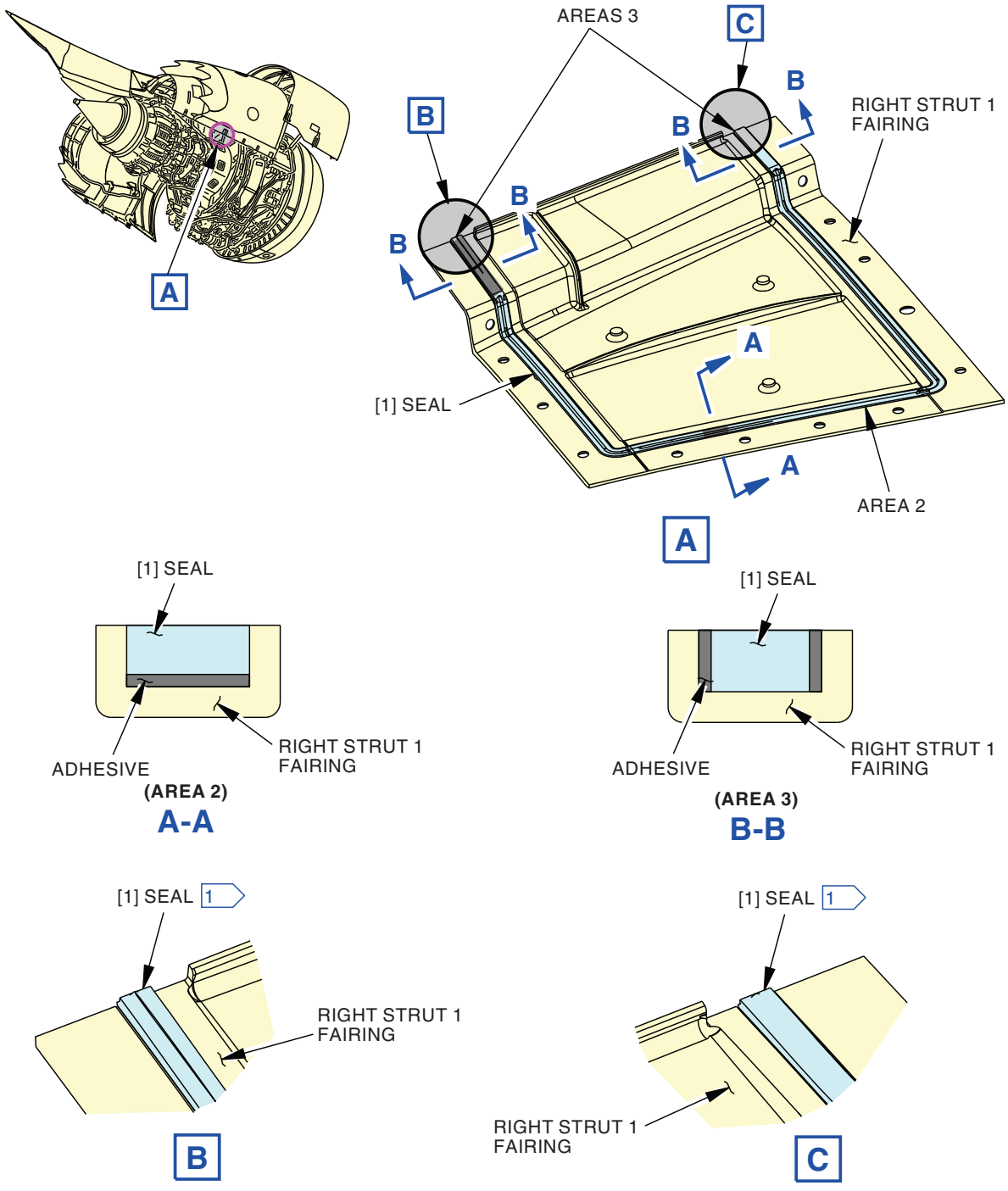
(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

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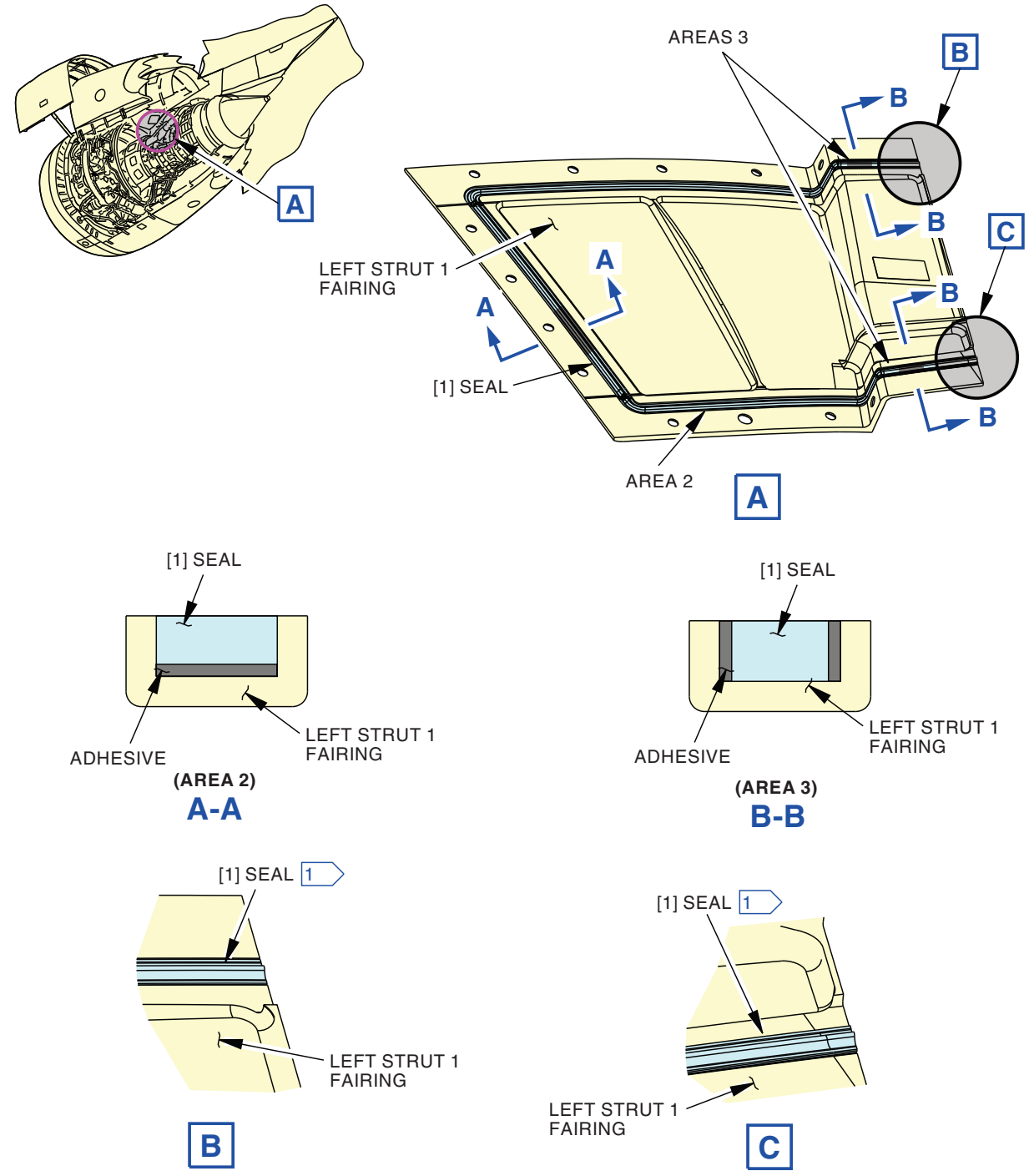


1 IT IS NECESSARY TO CUT THE SEAL FLUSH

2979797 S0000758003_V1

Fan Frame Right Strut 1 Fairing Seal Replacement
Figure 801/72-23-06-990-825-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

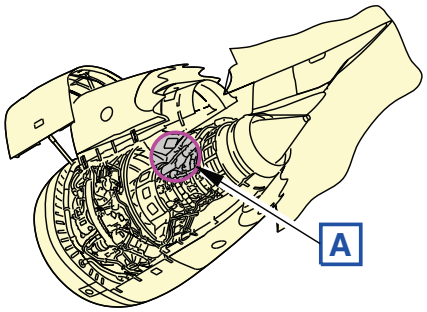


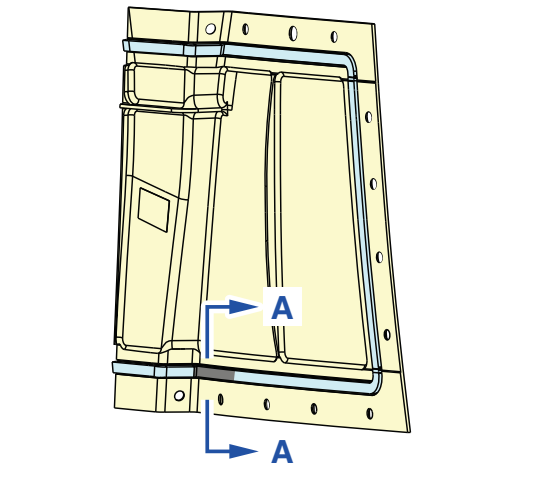
1 IT IS NECESSARY TO CUT THE SEAL FLUSH

2979801 S0000758029_V1

Fan Frame Left Strut 1 Fairing Seal Replacement
Figure 802/72-23-06-990-826-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details





2979800 S0000758031_V1

Fan Frame Strut 1 Fairing Seal Types
Figure 803/72-23-06-990-827-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 72-23-06-960-802-G00

3. Fan Frame Strut Pad Seal Replacement

(Figure 804, Figure 805, and Figure 806)

A. General

(1) This task gives the instruction to replace the bonded seal on the right and left pad strut 1 and right and left pad strut 5.

(2) The tables below give a list of the part numbers that applicable to the fan frame strut pad seal replacement.

Table 803/72-23-06-993-802-G00

Part number	Service Bulletin	Designation
364-080-510-0	72-0001	PAD, RH-STRUT 1
364-082-810-0	72-0001	PAD, LH-STRUT 1
364-082-910-0	72-0001	PAD, RH-STRUT 5
364-083-010-0	72-0001	PAD, LH-STRUT 5

Table 804/72-23-06-993-804-G00

Name	Part Number	Quantity
SEAL PAD	364-868-080-0	As Required
SEAL PAD	364-868-080-0	As Required
SEAL PAD	364-868-080-0	As Required
SEAL PAD	364-868-080-0	As Required

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
A01077 [CP2266]	Adhesive - Silicone Rubber - RTV 102	
A50012 [CP2242]	Adhesive - Silicone, Thixotropic Paste, One-Part RTV - RTV 732	MIL-A-46106 Type I
A50119	Adhesive - Silicone Rubber - RTV 103 (Black)	MIL-A-46106
B01058 [CP1039]	Solvent - Acetone, Reagent Grade	

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(Continued)

Reference	Description	Specification
G50077	Abrasive - Aluminum Oxide Paper, 240 grit or finer	
G50143 [CP2208]	Tape - Glass Cloth with Silicone Adhesive, High Temperature, Abrasive Resistant - 3M 361	
G51692	Cloth - 100% Cotton	AMS3819 Class 1
G51700 [CP2968]	Tape - Masking, High-Temperature - 3M 501E	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Prepare for the Replacement

SUBTASK 72-23-06-840-004-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 72-23-06-840-005-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

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1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Fan Frame Strut Pad Seal Replacement

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

WARNING USE EYE PROTECTION WHEN YOU CUT MATERIAL OR PARTS. THE PARTICLES CAN CAUSE DAMAGE TO YOUR EYES.

WARNING USE PROTECTIVE GLOVES TO PREVENT INJURY FROM SHARP EDGES AND ROUGH SURFACES. IF YOU DO NOT OBEY, INJURY TO PERSONNEL CAN OCCUR.

CAUTION KEEP CHLORINATED HYDROCARBONS AND HALOGEN MATERIALS AWAY FROM TITANIUM PARTS. IF YOU DO NOT OBEY, CHLORINATED HYDROCARBONS AND HALOGEN MATERIALS CAN CAUSE THE DAMAGE TO THE TITANIUM PARTS.

CAUTION DO NOT USE TOOLS FIXTURE MADE FROM OR COATED WITH CADMIUM, BISMUTH, TIN, ZINC, GOLD OR SILVER TO TITANIUM PARTS. IF YOU DO NOT OBEY, DAMAGE TO THE TITANIUM PARTS CAN OCCUR.

SUBTASK 72-23-06-020-006-G00

(1) Remove the seal [2] as follows (View A, Figure 804 or Figure 805):

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CAUTION MAKE SURE THAT YOU CAREFULLY REMOVE THE SEAL FROM THE STRUT PAD. IF YOU DO NOT OBEY, DAMAGE TO THE STRUT PAD CAN OCCUR.

(a) Carefully remove and discard the seal [2].

WARNING BE CAREFUL WHEN USE SHARP OR POINTED TOOLS. THESE TOOLS CAN CAUSE AN INJURY TO YOU.

CAUTION DO NOT APPLY A PROCEDURE THAT CAN CHANGE THE PARENT MATERIAL PROPERTIES DURING THE REMOVAL OF THE BONDED MATERIAL. IF YOU DO NOT OBEY, DAMAGE TO THE MATERIAL CAN OCCUR.

(b) Remove adhesive from the strut pad with cutting tool.

NOTE: Small remaining bonding material is acceptable if still bonded, of small area and non adjacent areas.

NOTE: The thickness of remaining particles have to be less than 0.007 in. (0.2 mm).

(c) Make sure that you do not remove parent material from the strut pad.

SUBTASK 72-23-06-100-002-G00

(2) Prepare the area where you will bond the new seal [2] (View A, Figure 804 or Figure 805).

NOTE: The seal material is elastomer.

WARNING DO NOT BREATHE THE PARTICLES FROM BLENDING OR LET THE PARTICLES TOUCH YOU. THE PARTICLES FROM BLENDING CAN CAUSE DAMAGE, INJURY, OR IRRITATION TO YOU. USE PROTECTION EQUIPMENT. USE LOCAL MECHANICAL EXHAUST VENTILATION OR AN APPROVED RESPIRATOR.

(a) Make the metallic part surfaces of the strut pad rough with 240 grit or finer abrasive paper, G50077.

NOTE: The strut pad material is TA6V.

WARNING DO NOT GET SOLVENT IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE SOLVENT. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE THE SOLVENT. KEEP THE SOLVENT AWAY FROM SPARKS, FLAME AND HEAT. SOLVENT IS POISONOUS AND FLAMMABLE WHICH CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(b) Clean the metallic part area of the strut pad with clean cloth, G51692, or equivalent soaked in acetone solvent, B01058 [CP1039].

(c) Make the surfaces of the new seal [2] rough with 240 grit or finer abrasive paper, G50077.

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WARNING

DO NOT GET SOLVENT IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE SOLVENT. PUT ON A PROTECTIVE SPLASH GOGGLE AND GLOVES WHEN YOU USE THE SOLVENT. KEEP THE SOLVENT AWAY FROM SPARKS, FLAME AND HEAT. SOLVENT IS POISONOUS AND FLAMMABLE WHICH CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(d) Make the surface of the new seal [2] rough with acetone solvent, B01058 [CP1039].

SUBTASK 72-23-06-420-004-G00

(3) Bond the new seal [2] (Figure 804, Figure 805, and Figure 806).

NOTE: Seals can be in 2 different shapes and colors. The seals are interchangeable.

(a) If you do not bond material within 6 hours after surface preparation, do these steps:

1) Clean the metallic part area where you will bond the new seal [2] with clean cloth, G51692, or equivalent soaked in acetone solvent, B01058 [CP1039].

2) Make the surfaces of the new seal [2] rough with 240 grit or finer abrasive paper, G50077.

(b) If it is necessary, use 3M 361 tape, G50143 [CP2208], or 3M 501E tape, G51700 [CP2968], to apply mask on the adjacent areas.

NOTE: Do not apply primer.

(c) Apply the maximum thickness 0.007 in. (0.2 mm) of RTV 732 adhesive, A50012 [CP2242] (preferred), RTV 102 adhesive, A01077 [CP2266] (alternate), or RTV 103 adhesive, A50119 (alternate).

1) For the rectangular seal [2], apply RTV 732 adhesive, A50012 [CP2242] (preferred), RTV 103 adhesive, A50119 (alternate), or RTV 102 adhesive, A01077 [CP2266] (alternate), on the longer side (View A-A, Figure 805).

(d) Put the seal [2] and apply a pressure for bonding.

(e) Cut the seal [2] to make a continuous surface with the metallic part adjacent areas (Figure 806).

(f) If it is necessary, remove the unwanted adhesive from the adjacent areas.

NOTE: Adhesive is not accepted in adjacent areas.

(g) If it is necessary, remove the mask that you apply on the adjacent areas.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 72-23-06-840-006-G00

WARNING

OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details